



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br>< |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|----|--|----------------|--|-----------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|---|--|---|--|----------------|--|------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
|    |  | 1              |  | 2                                                                                                                           |  | 3 |  | 4 |  | 5 |  | 6 |  | 7              |  | 8                                                                                                                            |  |  |  |  |  |  |  |  |  |
|    |  | Blatt<br>Sheet |  | Benennungen<br>Titles                                                                                                       |  |   |  |   |  |   |  |   |  | Blatt<br>Sheet |  | Benennungen<br>Titles                                                                                                        |  |  |  |  |  |  |  |  |  |
|    |  | 51             |  | HAUPTFAHRSTAND, MOTOR 1: STROMVERSORGUNG 24VDC RCS (CPP)<br>MAIN CONTROL STAND, ENGINE 1: POWER SUPPLY 24VDC RCS (CPP)      |  |   |  |   |  |   |  |   |  | 76             |  | HAUPTFAHRSTAND, MOTOR 1/2: ANZEIGEinSTRUMENTE/CPP (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: IND. INSTR./CPP (HORN EAD)   |  |  |  |  |  |  |  |  |  |
|    |  | 52             |  | HAUPTFAHRSTAND, MOTOR 1: STROMVERSORGUNG 24VDC RCS (WJ)<br>MAIN CONTROL STAND, ENGINE 1: POWER SUPPLY 24VDC RCS (WJ)        |  |   |  |   |  |   |  |   |  | 77             |  | HAUPTFAHRSTAND, MOTOR 1/2: ANZEIGEinSTRUMENTE/WJ (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: IND. INSTR./WJ (HORN EAD)     |  |  |  |  |  |  |  |  |  |
|    |  | 53             |  | HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (FPP)<br>MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (FPP)      |  |   |  |   |  |   |  |   |  | 78             |  | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHlinSTRUMENTE (HORN EA)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (HORN EA)      |  |  |  |  |  |  |  |  |  |
|    |  | 54             |  | HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (CPP)<br>MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (CPP)      |  |   |  |   |  |   |  |   |  | 79             |  | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHlinSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (VDO)              |  |  |  |  |  |  |  |  |  |
|    |  | 55             |  | HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (WJ)<br>MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (WJ)        |  |   |  |   |  |   |  |   |  | 80             |  | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHlinSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (VDO)              |  |  |  |  |  |  |  |  |  |
|    |  | 56             |  | HAUPTFAHRSTAND, MOTOR 1/2: BEDIENSTATION ROS7<br>MAIN CONTROL STAND, ENGINE 1/2: OPERATING STATION ROS7                     |  |   |  |   |  |   |  |   |  | 81             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUMENTS                  |  |  |  |  |  |  |  |  |  |
|    |  | 57             |  | HAUPTFAHRSTAND, PROZESSBUS (MCU8/11/12)<br>MAIN CONTROL STAND, PROCESS BUS (MCU8/11/12)                                     |  |   |  |   |  |   |  |   |  | 82             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE (GIORGIO)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM. (GIORGIO) |  |  |  |  |  |  |  |  |  |
|    |  | 58             |  | HAUPTFAHRSTAND, PROZESSBUS (MCU14)<br>MAIN CONTROL STAND, PROCESS BUS (MCU14)                                               |  |   |  |   |  |   |  |   |  | 83             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE (GIORGIO)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM. (GIORGIO) |  |  |  |  |  |  |  |  |  |
|    |  | 59             |  | HAUPTFAHRSTAND, MOTOR 1: FELDBUS<br>MAIN CONTROL STAND, ENGINE 1: FIELD BUS                                                 |  |   |  |   |  |   |  |   |  | 84             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMEN. (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM.(HORN EAD) |  |  |  |  |  |  |  |  |  |
|    |  | 60             |  | HAUPTFAHRSTAND, MOTOR 1: FELDBUS (WJ)<br>MAIN CONTROL STAND, ENGINE 1: FIELD BUS (WJ)                                       |  |   |  |   |  |   |  |   |  | 85             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMEN. (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM.(HORN EAD) |  |  |  |  |  |  |  |  |  |
|    |  | 61             |  | HAUPTFAHRSTAND, MOTOR 2: FELDBUS<br>MAIN CONTROL STAND, ENGINE 2: FIELD BUS                                                 |  |   |  |   |  |   |  |   |  | 86             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUMENTS (VDO)      |  |  |  |  |  |  |  |  |  |
|    |  | 62             |  | HAUPTFAHRSTAND, MOTOR 2: FELDBUS (WJ)<br>MAIN CONTROL STAND, ENGINE 2: FIELD BUS (WJ)                                       |  |   |  |   |  |   |  |   |  | 87             |  | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUMENTS (VDO)      |  |  |  |  |  |  |  |  |  |
|    |  | 63             |  | HAUPTFAHRSTAND, MOTOR 1/2: KOORDINATIONSBUS RCS-5FPP/CPP/WJ<br>MAIN CTR.STAND, ENGINE 1/2: COORDINATION BUS RCS-5FPP/CPP/WJ |  |   |  |   |  |   |  |   |  | 88             |  | HAUPTFAHRSTAND, MOTOR 1/2: STEUERRAD (WJ)<br>MAIN CONTROL STAND, ENGINE 1/2: STEERING WHEEL (WJ)                             |  |  |  |  |  |  |  |  |  |
|    |  | 64             |  | HAUPTFAHRSTAND, MOTOR 1/2: ANSCHLUSS HANDSTEUERGERÄT ROS8<br>MAIN CTR. STAND, ENGINE 1/2:CONNECTING PORT.REMOTE CTR. ROS8   |  |   |  |   |  |   |  |   |  | 89             |  | HAUPTFAHRSTAND, MOTOR 1/2: JOYSTICK (WJ)<br>MAIN CONTROL STAND, ENGINE 1/2: JOYSTICK (WJ)                                    |  |  |  |  |  |  |  |  |  |
|    |  | 65             |  | HAUPTFAHRSTAND, MOTOR 1-4: KOORDINATIONSBUS RCS-5DUO/TRIO<br>MAIN CTR. STAND, ENGINE 1-4:COORDINATION BUS RCS-5DUO/TRIO     |  |   |  |   |  |   |  |   |  | 90             |  | NEBENF.ST.(VOLL), MOTOR 1/2: STROMVER. 24VDC (RCS VIA DIODE)<br>SL.CTR.ST.(FULL), ENG.1/2: POWER SUPPLY 24VDC(RCS VIA DIODE) |  |  |  |  |  |  |  |  |  |
|    |  | 66             |  | HAUPTFAHRSTAND, HILFSFAHRSTÄNDE 1-4, KOORDINATIONSBUS<br>MAIN CTR. STAND, AUX. CTR. STANDS 1-4, COORDINATION BUS            |  |   |  |   |  |   |  |   |  | 91             |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STROMVERSORGUNG 24VDC<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: POWER SUPPLY 24VDC          |  |  |  |  |  |  |  |  |  |
|    |  | 67             |  | HAUPTFAHRSTAND, ANSCHLUSSSCHNITTSTELLE (ANKER, ...)<br>MAIN CONTROL STAND, CONNECTION INTERFACE (ANCHOR, ...)               |  |   |  |   |  |   |  |   |  | 92             |  | NEBENFAHRST.(VOLL), MOTOR 1/2: 24VDC EIN/AUS (RCS VIA DIODE)<br>SL.CTR.STAND(FULL), ENGINE 1/2: 24VDC ON/OFF (RCS VIA DIODE) |  |  |  |  |  |  |  |  |  |
|    |  | 68             |  | HAUPTFAHRSTAND, MOTOR 1/2: SENSOR-/AKTORBUS<br>MAIN CONTROL STAND, ENGINE 1/2: SENSOR/ACTUATOR BUS                          |  |   |  |   |  |   |  |   |  | 93             |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: 24VDC EIN/AUS<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: 24VDC ON/OFF                        |  |  |  |  |  |  |  |  |  |
|    |  | 69             |  | HAUPTFAHRSTAND, MOTOR 1/2: SENSOR-/AKTORBUS (CPP)<br>MAIN CONTROL STAND, ENGINE 1/2: SENSOR/ACTUATOR BUS (CPP)              |  |   |  |   |  |   |  |   |  | 94             |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: POWER SUPPLY 24VDC MCS   |  |  |  |  |  |  |  |  |  |
|    |  | 70             |  | HAUPTFAHRSTAND, MOTOR 1/2: SENSOR-/AKTORBUS (WJ)<br>MAIN CONTROL STAND, ENGINE 1/2: SENSOR/ACTUATOR BUS (WJ)                |  |   |  |   |  |   |  |   |  | 95             |  | NEBENF.ST.(VOLL), MOTOR 1/2: STROMVER. 24VDC RCS (VIA DIODE)<br>SL.CTR.ST.(FULL), ENG.1/2: POWER SUPPLY 24VDC RCS(VIA DIODE) |  |  |  |  |  |  |  |  |  |
|    |  | 71             |  | HAUPTFAHRSTAND, MOTOR 1: NOTSTOPPTASTER<br>MAIN CONTROL STAND, ENGINE 1: EMERG.-STOP PUSHBUTTON                             |  |   |  |   |  |   |  |   |  | 96             |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: POWER SUPPLY 24VDC RCS   |  |  |  |  |  |  |  |  |  |
|    |  | 72             |  | HAUPTFAHRSTAND, MOTOR 2: NOTSTOPPTASTER<br>MAIN CONTROL STAND, ENGINE 2: EMERG.-STOP PUSHBUTTON                             |  |   |  |   |  |   |  |   |  | 97             |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: POWER SUPPLY 24VDC MCS   |  |  |  |  |  |  |  |  |  |
|    |  | 73             |  | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHlinSTRUMENTE (GIORGIO)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (GIORGIO)     |  |   |  |   |  |   |  |   |  | 98             |  | NEBENFAHRST.(VOLL),STROMVERS.24VDC, GRAFIKST.1&2(MCU8/11/12)<br>SLAVE CTR.ST.(FULL), P.S.24VDC, GRAPHIC STAT.1&2(MCU8/11/12) |  |  |  |  |  |  |  |  |  |
|    |  | 74             |  | HAUPTFAHRSTAND, MOTOR 1/2: RESERVE<br>MAIN CONTROL STAND, ENGINE 1/2: SPARE                                                 |  |   |  |   |  |   |  |   |  | 99             |  | NEBENFAHRST.(VOLL),STROMVERS.24VDC, GRAFIKST.1&2(MCU14)<br>SLAVE CTR.ST.(FULL), P.S.24VDC, GRAPHIC STAT.1&2(MCU14)           |  |  |  |  |  |  |  |  |  |
|    |  | 75             |  | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHlinSTRUMENTE (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (HORN EAD)   |  |   |  |   |  |   |  |   |  | 100            |  | NEBENFAHRSTAND(VOLL), GRAPHIKSTATIONEN<br>SLAVE CONTROL STAND(FULL), GRAPHIC STATIONS                                        |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
|    |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |
| </ |  |                |  |                                                                                                                             |  |   |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |  |  |  |  |  |  |  |  |

|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|--|--|----------------|--|-------------------------------------------------------------------------------------------------------------------------------|--|---|--|---|--|---|--|----------------|--|------------------------------------------------------------------------------------------------------------------------------|--|---|--|--|--|--|--|
|  |  | 1              |  | 2                                                                                                                             |  | 3 |  | 4 |  | 5 |  | 6              |  | 7                                                                                                                            |  | 8 |  |  |  |  |  |
|  |  | Blatt<br>Sheet |  | Benennungen<br>Titles                                                                                                         |  |   |  |   |  |   |  | Blatt<br>Sheet |  | Benennungen<br>Titles                                                                                                        |  |   |  |  |  |  |  |
|  |  | 101            |  | NEBENFAHRSTAND(VOLL), MOTOR 1: STROMVERSORG. 24VDC RCS (FPP)<br>SLAVE CONTR. STAND(FULL), ENGINE 1: POW.SUP. 24VDC RCS (FPP)  |  |   |  |   |  |   |  | 126            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INSTRUMENTE<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INSTRUMENTS        |  |   |  |  |  |  |  |
|  |  | 102            |  | NEBENFAHRSTAND(VOLL), MOTOR 1: STROMVERSORG. 24VDC RCS (CPP)<br>SLAVE CONTR. STAND(FULL), ENGINE 1: POW.SUP. 24VDC RCS (CPP)  |  |   |  |   |  |   |  | 127            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST. (GIORGIO)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INST.(GIORGIO) |  |   |  |  |  |  |  |
|  |  | 103            |  | NEBENFAHRSTAND(VOLL), MOTOR 1: STROMVERSORG. 24VDC RCS (WJ)<br>SLAVE CONTR. STAND(FULL), ENGINE 1: POW.SUP. 24VDC RCS (WJ)    |  |   |  |   |  |   |  | 128            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST. (GIORGIO)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INST.(GIORGIO) |  |   |  |  |  |  |  |
|  |  | 104            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (FPP)<br>SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (FPP)  |  |   |  |   |  |   |  | 129            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  |  |   |  |  |  |  |  |
|  |  | 105            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (CPP)<br>SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (CPP)  |  |   |  |   |  |   |  | 130            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  |  |   |  |  |  |  |  |
|  |  | 106            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (WJ)<br>SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (WJ)    |  |   |  |   |  |   |  | 131            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INSTR. (VDO)    |  |   |  |  |  |  |  |
|  |  | 107            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: BEDIENSTATION ROS7<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: OPERATING STATION ROS7           |  |   |  |   |  |   |  | 132            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INSTR. (VDO)    |  |   |  |  |  |  |  |
|  |  | 108            |  | NEBENFAHRSTAND(VOLL), PROZESSBUS (MCU8/11/12)<br>SLAVE CONTROL STAND(FULL), PROCESS BUS (MCU8/11/12)                          |  |   |  |   |  |   |  | 133            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STEUERRAD (WJ)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: STEERING WHEEL (WJ)                |  |   |  |  |  |  |  |
|  |  | 109            |  | NEBENFAHRSTAND(VOLL), PROZESSBUS (MCU14)<br>SLAVE CONTROL STAND(FULL), PROCESS BUS (MCU14)                                    |  |   |  |   |  |   |  | 134            |  | NEBENFAHRSTAND(VOLL) 1, MOTOR 1/2: JOYSTICK (WJ)<br>SLAVE CONTROL STAND(FULL) 1, ENGINE 1/2: JOYSTICK (WJ)                   |  |   |  |  |  |  |  |
|  |  | 110            |  | NEBENFAHRSTAND(VOLL), MOTOR 1: FELDBUS<br>SLAVE CONTROL STAND(FULL), ENGINE 1: FIELD BUS                                      |  |   |  |   |  |   |  | 135            |  | NEBENFAHRSTAND(VOLL) 2, MOTOR 1/2: JOYSTICK (WJ)<br>SLAVE CONTROL STAND(FULL) 2, ENGINE 1/2: JOYSTICK (WJ)                   |  |   |  |  |  |  |  |
|  |  | 111            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: FELDBUS<br>SLAVE CONTROL STAND(FULL), ENGINE 2: FIELD BUS                                      |  |   |  |   |  |   |  | 136            |  | NEBENF.ST.(RED.), MOTOR 1/2: STROMVER. 24VDC (RCS VIA DIODE)<br>SL.CTR.ST.(RED.), ENG.1/2: POWER SUPPLY 24VDC(RCS VIA DIODE) |  |   |  |  |  |  |  |
|  |  | 112            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: KOORDINATIONSBUS RCS<br>SLAVE CTR.STAND(FULL), ENGINE 1/2: COORDINATION BUS RCS              |  |   |  |   |  |   |  | 137            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC          |  |   |  |  |  |  |  |
|  |  | 113            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: SENSOR-/AKTORBUS<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SENSOR/ACTUATOR BUS               |  |   |  |   |  |   |  | 138            |  | NEBENFAHRST.(RED.), MOTOR 1/2: 24VDC EIN/AUS (RCS VIA DIODE)<br>SL.CTR.STAND(RED.), ENGINE 1/2: 24VDC ON/OFF (RCS VIA DIODE) |  |   |  |  |  |  |  |
|  |  | 114            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: SENSOR-/AKTORBUS (CPP)<br>SLAVE CTR.STAND(FULL), ENGINE 1/2: SENSOR/ACTUATOR BUS (CPP)       |  |   |  |   |  |   |  | 139            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: 24VDC EIN/AUS<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: 24VDC ON/OFF                        |  |   |  |  |  |  |  |
|  |  | 115            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: SENSOR-/AKTORBUS (WJ)<br>SLAVE CTR.STAND(FULL), ENGINE 1/2: SENSOR/ACTUATOR BUS (WJ)         |  |   |  |   |  |   |  | 140            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC MCS   |  |   |  |  |  |  |  |
|  |  | 116            |  | NEBENFAHRSTAND(VOLL), MOTOR 1: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(FULL), ENGINE 1: EMERG.-STOP PUSHBUTTON                  |  |   |  |   |  |   |  | 141            |  | NEBENF.ST.(RED.), MOTOR 1/2: STROMVER. 24VDC RCS (VIA DIODE)<br>SL.CTR.ST.(RED.), ENG.1/2: POWER SUPPLY 24VDC RCS(VIA DIODE) |  |   |  |  |  |  |  |
|  |  | 117            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(FULL), ENGINE 2: EMERG.-STOP PUSHBUTTON                  |  |   |  |   |  |   |  | 142            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC RCS   |  |   |  |  |  |  |  |
|  |  | 118            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHL INSTRUM. (GIORGIO)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INST. (GIORGIO)  |  |   |  |   |  |   |  | 143            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC MCS   |  |   |  |  |  |  |  |
|  |  | 119            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: RESERVE<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPARE                                      |  |   |  |   |  |   |  | 144            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC MCS   |  |   |  |  |  |  |  |
|  |  | 120            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHL INSTRUM. (HORN EAD)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INST.(HORN EAD) |  |   |  |   |  |   |  | 145            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC RCS   |  |   |  |  |  |  |  |
|  |  | 121            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ANZEIGEINST./CPP (HORN EAD)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: IND.INSTR./CPP (HORN EAD) |  |   |  |   |  |   |  | 146            |  | NEBENFAHRSTAND(RED.), MOTOR 1/2: BEDIENSTATION ROS7<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: OPERATING STATION ROS7          |  |   |  |  |  |  |  |
|  |  | 122            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: ANZEIGEINSTR./WJ (HORN EAD)<br>SLAVE CTR. STAND(FULL), ENGINE 1/2: IND.INSTR./WJ (HORN EAD)  |  |   |  |   |  |   |  | 147            |  | FAHRSTAND 1-6, MOTOR 1: PRINZIPVERKABELUNG ROS9<br>CONTROL STAND 1-6, ENGINE 1: PRINCIPLE WIRING ROS9                        |  |   |  |  |  |  |  |
|  |  | 123            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHL INSTRUM. (HORN EA)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INST. (HORN EA)  |  |   |  |   |  |   |  | 148            |  | FAHRSTAND 1-6, MOTOR 1&2: PRINZIPVERKABELUNG ROS7<br>CONTROL STAND 1-6, ENGINE 1&2: PRINCIPLE WIRING ROS7                    |  |   |  |  |  |  |  |
|  |  | 124            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHL INSTRUMENTE (VDO)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INSTRUM. (VDO)    |  |   |  |   |  |   |  | 149            |  | FAHRSTAND 1-6, MOTOR 1&2&3: PRINZIPVERKABELUNG ROS11<br>CONTROL STAND 1-6, ENGINE 1&2&3: PRINCIPLE WIRING ROS11              |  |   |  |  |  |  |  |
|  |  | 125            |  | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHL INSTRUMENTE (VDO)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INSTRUM. (VDO)    |  |   |  |   |  |   |  | 150            |  | FAHRSTAND 1-6, MOTOR 3&4: PRINZIPVERKABELUNG ROS7<br>CONTROL STAND 1-6, ENGINE 3&4: PRINCIPLE WIRING ROS7                    |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |
|  |  |                |  |                                                                                                                               |  |   |  |   |  |   |  |                |  |                                                                                                                              |  |   |  |  |  |  |  |



|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|-----|----------------------------------------------------------------------------------------------------------------------------|---|---|---|---|-------|-------------------------------------------------------------------------------------------------------------------------------|---|--|---|-----|-----------------------------------------------------------------------------------------|--|---|--|---|---|---|---|---|---|---|-----|--------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
|     |                                                                                                                            | 1 |   | 2 |   | 3     |                                                                                                                               | 4 |  | 5 |     | 6                                                                                       |  | 7 |  | 8 |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| A   | B                                                                                                                          | C | D | E | F | Blatt | Benennungen                                                                                                                   |   |  |   |     |                                                                                         |  |   |  |   | A | B | C | D | E | F |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | Sheet | Titles                                                                                                                        |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 151   | NEBENFAHRSTAND(RED.), MOTOR 1: FELDBUS<br>SLAVE CONTROL STAND(ERD.), ENGINE 1: FIELD BUS                                      |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 176 | FREMD RCS, MOTOR 1/2: DREHZAHLVORGABE<br>NON-MTU RCS, ENGINE 1/2: SPEED DEMAND                                           |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 152   | NEBENFAHRSTAND(RED.), MOTOR 2: FELDBUS<br>SLAVE CONTROL STAND(RED.), ENGINE 2: FIELD BUS                                      |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 177 | FREMD RCS, MOTOR 1/2: ANALOG-SIGNALE<br>NON-MTU RCS, ENGINE 1/2: ANALOG SIGNALS                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 153   | NEBENFAHRSTAND(RED.), MOTOR 1/2/5/6: SENSOR-/AKTORBUS<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SENSOR/ACTUATOR BUS           |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 178 | FREMD RCS, MOTOR 1/2: BINAR-SIGNALE<br>NON-MTU RCS, ENGINE 1/2: BINARY SIGNALS                                           |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 154   | NEBENFAHRSTAND(RED.), MOTOR 1: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(RED.), ENGINE 1: EMERG.-STOP PUSHBUTTON                  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 179 | FREMD RCS, MOTOR 1/2: FAHRSTANDSUMSCHALTUNG<br>NON-MTU RCS, ENGINE 1/2: CONTROL STAND TRANSFER                           |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 155   | NEBENFAHRSTAND(RED.), MOTOR 2: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(RED.), ENGINE 2: EMERG.-STOP PUSHBUTTON                  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 180 | FREMD RCS, MOTOR 1/2: KUPPLUNGSSTEUERUNG<br>NON-MTU RCS, ENGINE 1/2: CLUTCH CONTROL                                      |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 156   | NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHL INSTRUM. (GIORGIO)<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INST. (GIORGIO)  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 181 | FREMD RCS, MOTOR 1/2: KUPPLUNGSRÜCKMELDUNGEN<br>NON-MTU RCS, ENGINE 1/2: CLUTCH FEEDBACKS                                |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 157   | NEBENFAHRSTAND(RED.), MOTOR 1/2: RESERVE<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPARE                                      |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 182 | FREMD MCS (PIM4), MOTOR 1/2: STROMVERSORGUNG & FELDBUS<br>NON-MTU MCS (PIM4), ENGINE 1/2: POWER SUPPLY & FIELD BUS       |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 158   | NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHL INSTRUM. (HORN EAD)<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INST.(HORN EAD) |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 183 | FREMD MCS (PIM4), MOTOR 1/2: START/STOPP/NOTSTOPP<br>NON-MTU MCS (PIM4), ENGINE 1/2: START/STOP/EMERGENCY STOP           |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 159   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ANZEIGEINST./RES.(HORN EAD)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: IND.INST./RES.(HORN EAD)  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 184 | FREMD MCS (PIM4), MOTOR 1/2: BINAR-SIGNALE (LOP_KEY_GND)<br>NON-MTU MCS (PIM4), ENGINE 1/2: BINARY SIGNALS (LOP_KEY_GND) |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 160   | NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHL INSTRUM. (HORN EA)<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INST. (HORN EA)  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 185 | FREMD MCS (PIM4), MOTOR 1/2: BINAR-SIGNALE (PIM4_KEY_GND)<br>NON-MTU MCS (PIM4), ENGINE 1/2: BIN. SIGNALS (PIM4_KEY_GND) |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 161   | NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHL INSTRUMENTE (VDO)<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INSTRUM. (VDO)    |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 186 | FREMD MCS (PIM4), MOTOR 1/2: BINAR-SIGNALE<br>NON-MTU MCS (PIM4), ENGINE 1/2: BINARY SIGNALS                             |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 162   | NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHL INSTRUMENTE (VDO)<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INSTRUM. (VDO)    |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 187 | FREMD MCS (PIM4), MOTOR 1/2: ANALOG-SIGNALE<br>NON-MTU MCS (PIM4), ENGINE 1/2: ANALOG SIGNALS                            |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 163   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUMENTE<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTRUMENTS         |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 188 | FREMD MCS (PIM4), MOTOR 1/2: SERIELLE SCHNITTSTELLE<br>NON-MTU MCS (PIM4), ENGINE 1/2: SERIAL INTERFACE                  |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 164   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST. (GIORGIO)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INST.(GIORGIO)  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 189 | FREMD MCS (PIM4), MOTOR 1/2: SCHNITTSTELLE MODBUS<br>NON-MTU MCS (PIM4), ENGINE 1/2: INTERFACE MODBUS                    |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 165   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST. (GIORGIO)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INST.(GIORGIO)  |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 190 | FREMD MCS (PIM4), MOTOR 1/2: SCHNITTSTELLE VDR<br>NON-MTU MCS (PIM4), ENGINE 1/2: INTERFACE VDR                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 166   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)   |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 191 | PRINZIPVERKABELUNG MOTOR 1/2: ZUSÄTZLICHE SCHNITTSTELLEN<br>PRINCIPLE WIRING ENGINE 1/2: ADDITIONAL INTERFACES           |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 167   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)   |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 192 | BUSTOPOLOGIE, MOTORRAUM<br>BUS TOPOLOGY, ENGINE ROOM                                                                     |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   | 168   | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTR. (VDO)     |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   | 193 | BUSTOPOLOGIE, HAUPTFAHRSTAND<br>BUS TOPOLOGY, MAIN CONTROL STAND                                                         |  |  |  |  |  |  |  |  |  |
| 169 | NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTR. (VDO)  |   |   |   |   |       |                                                                                                                               |   |  |   | 194 | BUSTOPOLOGIE, NEBENFAHRSTAND (VOLL)<br>BUS TOPOLOGY, SLAVE CONTROL STAND (FULL)         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 170 | HAUPTFAHRSTAND, MOTOR 1/2: SCHNITTSTELLE MODBUS<br>MAIN CONTROL STAND, ENGINE 1/2: INTERFACE MODBUS                        |   |   |   |   |       |                                                                                                                               |   |  |   | 195 | BUSTOPOLOGIE, NEBENFAHRSTAND (REDUZIERT)<br>BUS TOPOLOGY, SLAVE CONTROL STAND (REDUCED) |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 171 | HAUPTFAHRSTAND, MOTOR 1/2: SCHNITTSTELLE VDR<br>MAIN CONTROL STAND, ENGINE 1/2: INTERFACE VDR                              |   |   |   |   |       |                                                                                                                               |   |  |   | 196 | BUSTOPOLOGIE, FREMD MCS (PIM4)<br>BUS TOPOLOGY, NON-MTU MCS (PIM4)                      |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 172 | HAUPTFAHRSTAND, MOTOR 1/2: SCHNITTSTELLE FREMD ÜBERWACHUNG<br>MAIN CONTROL STAND, ENGINE 1/2: INTERFACE NON-MTU MONITORING |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 173 | HAUPT- ODER NEBENFAHRSTAND, MOTOR 1: SCHNITTSTELLE PARALLEL<br>MAIN OR SLAVE CONTROL STAND, ENGINE 1: INTERFACE PARALLEL   |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 174 | HAUPT- ODER NEBENFAHRSTAND, MOTOR 2: SCHNITTSTELLE PARALLEL<br>MAIN OR SLAVE CONTROL STAND, ENGINE 2: INTERFACE PARALLEL   |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
| 175 | PRINZIPVERKABELUNG MOTOR 1/2: ZUSÄTZLICHE SCHNITTSTELLEN<br>PRINCIPLE WIRING ENGINE 1/2: ADDITIONAL INTERFACES             |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |
|     |                                                                                                                            |   |   |   |   |       |                                                                                                                               |   |  |   |     |                                                                                         |  |   |  |   |   |   |   |   |   |   |     |                                                                                                                          |  |  |  |  |  |  |  |  |  |



1. KEIN SCHLÜSSELSCHALTER ANGESCHLOSSEN  
--> ANLAGENAUTOMATION IST IMMER "EIN"
2. SCHLÜSSELSCHALTER ANGESCHLOSSEN  
--> ANLAGENAUTOMATION IST "EIN", SOBALD EINER DER SCHLÜSSELSCHALTER EIN IST
3. BRÜCKEN VON -X100:154 NACH -X100:155  
UND VON -X100:354 NACH -X100:355  
--> FERNÜBERWACHUNG IST IMMER "EIN" UND LÄSST SICH ÜBER DIE SCHLÜSSELSCHALTER NICHT AUSSCHALTEN  
--> DIE SCHLÜSSELSCHALTER SCHALTEN NUR DIE VORORTÜBERWACHUNG (MASCHINENRAUM) UND DIE FERNSTEUERUNG DES ENTSPRECHENDEN MOTORS AUS.

1. NO KEYSWITCH CONNECTED:  
--> PLANT AUTOMATION "ON" CONTINUOSLY
2. KEYSWITCH CONNECTED  
--> PLANT AUTOMATION IS "ON" AS SOON AS ONE  
OF THE KEYSWITCHES IN ON
3. JUMPERS FROM -X100:154 TO -X100:155  
AND FROM -X100:354 TO -X100:355  
--> REMOTE MONITORING IS ALWAYS ON AND CANNOT  
BE SWITCHED OFF VIA THE KEYSWITCHES  
--> THE KEYSWITCHES ONLY SWITCH OFF THE LOCAL  
MONITORING (ENGINE ROOM) AND REMOTE MONITORING  
FOR THE RELEVANT ENGINE.

| CCP3                           |       |
|--------------------------------|-------|
| GERÄT/DEVICE                   | -F005 |
| 1 x PIM3                       | 0.32A |
| 1 x DIM3                       | 0.10A |
| 5 x PAN                        | 1.10A |
| 1 x SUMMER<br>1 x BUZZER       | 0.02A |
| 12 x INSTRUM.                  | 0.60A |
|                                |       |
|                                |       |
|                                |       |
|                                |       |
|                                |       |
|                                |       |
| 3 x PIM5 ERW.<br>3 x PIM5 EXT. | 0.62A |
| 1 x PAN                        | 0.22A |
| 2 x ROS                        | 1.0A  |
| SUMME:                         | 3.98A |
|                                |       |
| max. STROM<br>max. CURRENT     | 6.00A |
|                                |       |

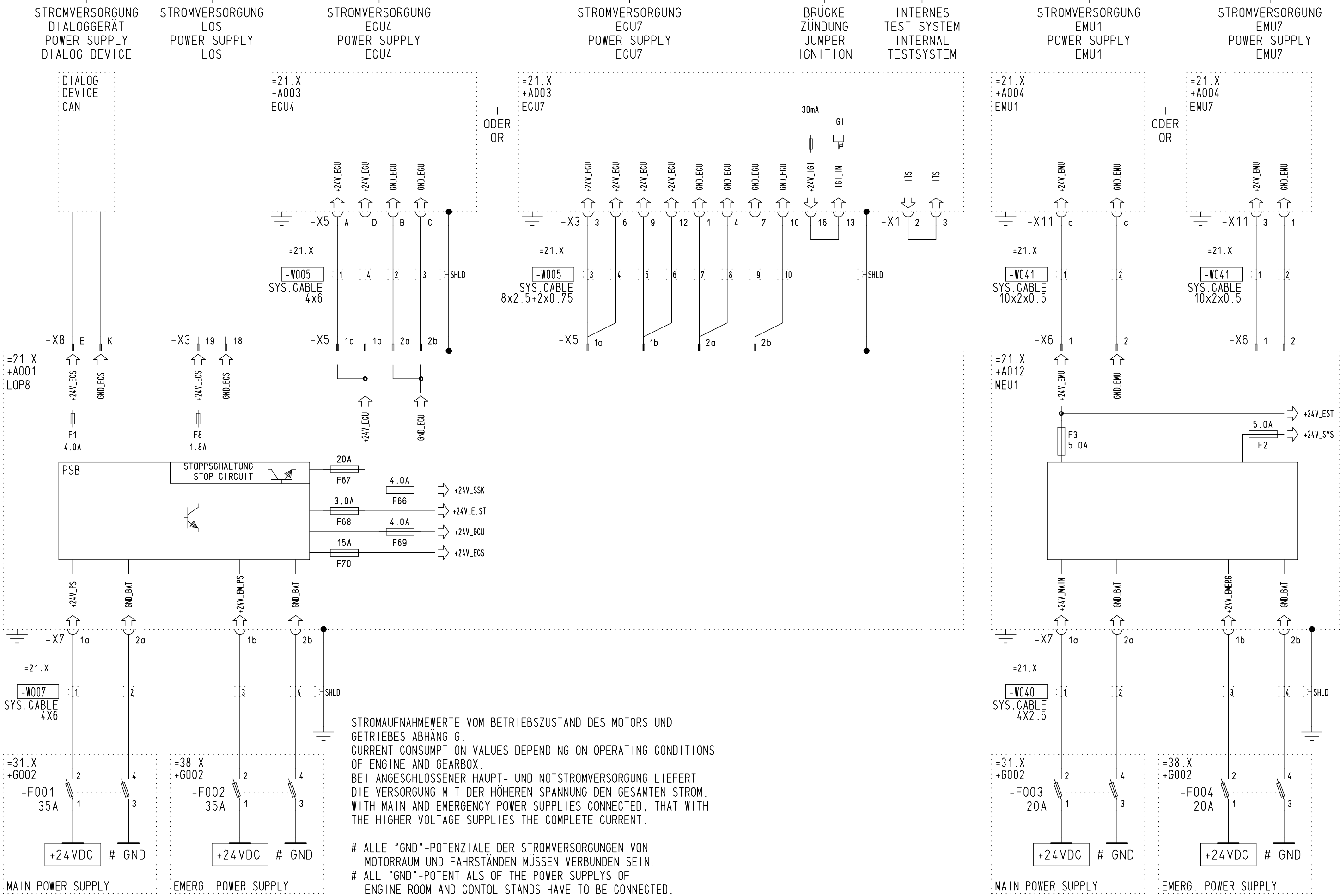
MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|                                  |                      |  |              |            |      |        |            |        |                                                                                                                |                                                                                                  |                                                  |          |                               |           |                |   |
|----------------------------------|----------------------|--|--------------|------------|------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------|----------|-------------------------------|-----------|----------------|---|
|                                  |                      |  |              |            |      | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HINWEISE: STROMVERSORGUNG & STROMBILANZ          |          |                               |           |                |   |
|                                  |                      |  |              |            |      | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | INSTRUCTIONS: POWER SUPPLY & CURRENT CONSUMPTION |          |                               |           |                |   |
|                                  |                      |  |              |            |      | Gep.   | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                  |          |                               |           |                |   |
| i                                | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                  |          |                               |           |                |   |
| Buchst.                          | Änderung / Revision  |  | Auftrags-Nr. | Datum/Date | Name | Abt.   | TDE        |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing-No. | X00031757 | Blatt<br>Sheet | 6 |
| Referenz-Nr./Reference-No. ----- |                      |  |              |            |      |        |            |        |                                                                                                                |                                                                                                  |                                                  |          |                               |           |                |   |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F



|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



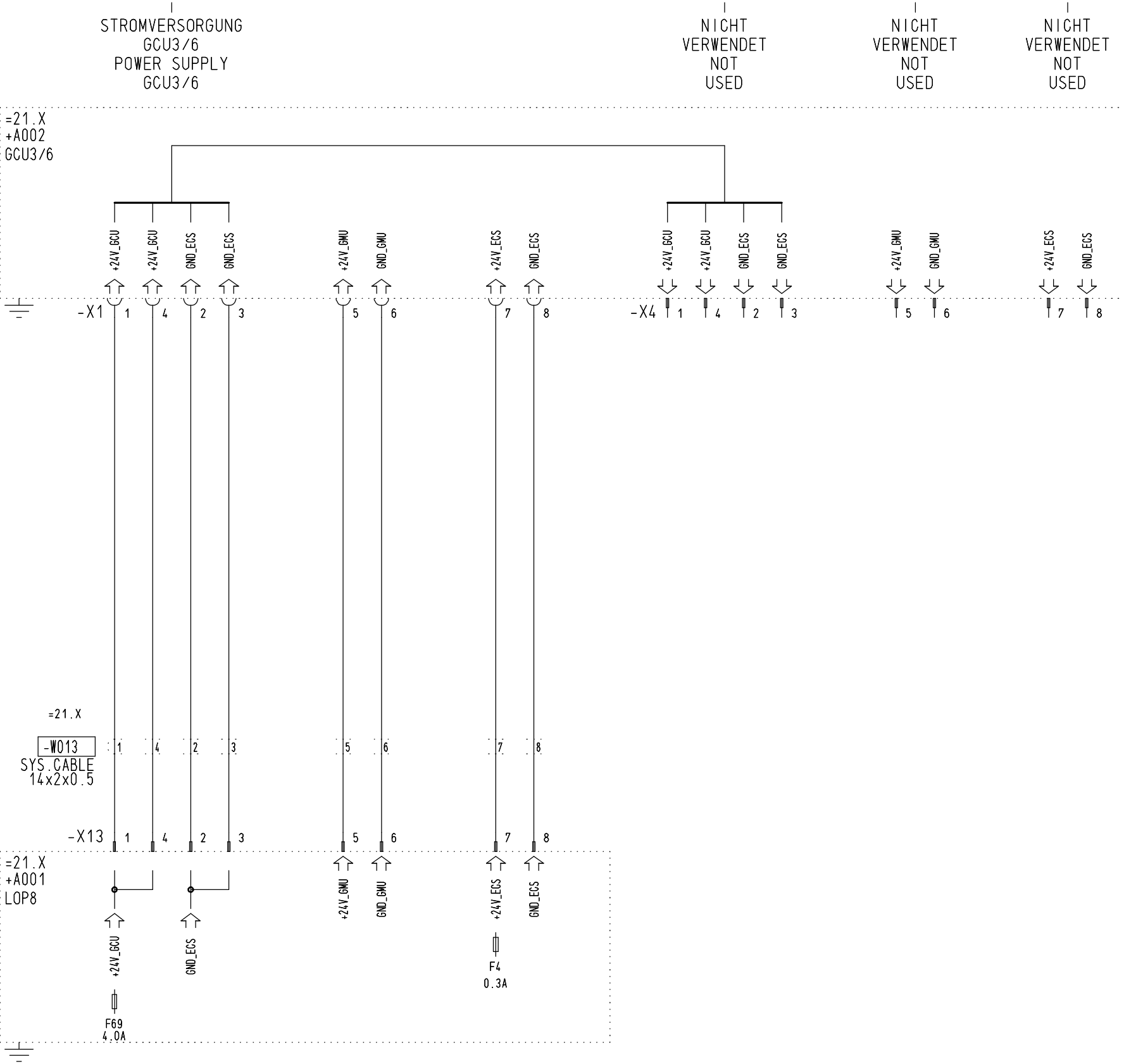
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                            |                                         |
|--------------------------------------------------------------------------------------------|-----------------------------------------|
| MOTORRAUM, MOTOR 1/2: STROMVERSORGUNG 24VDC<br>ENGINE ROOM, ENGINE 1/2: POWER SUPPLY 24VDC |                                         |
| Auftrag/Projekt<br>BLUE-VIS                                                                | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                 | Blatt<br>Sheet 7                        |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

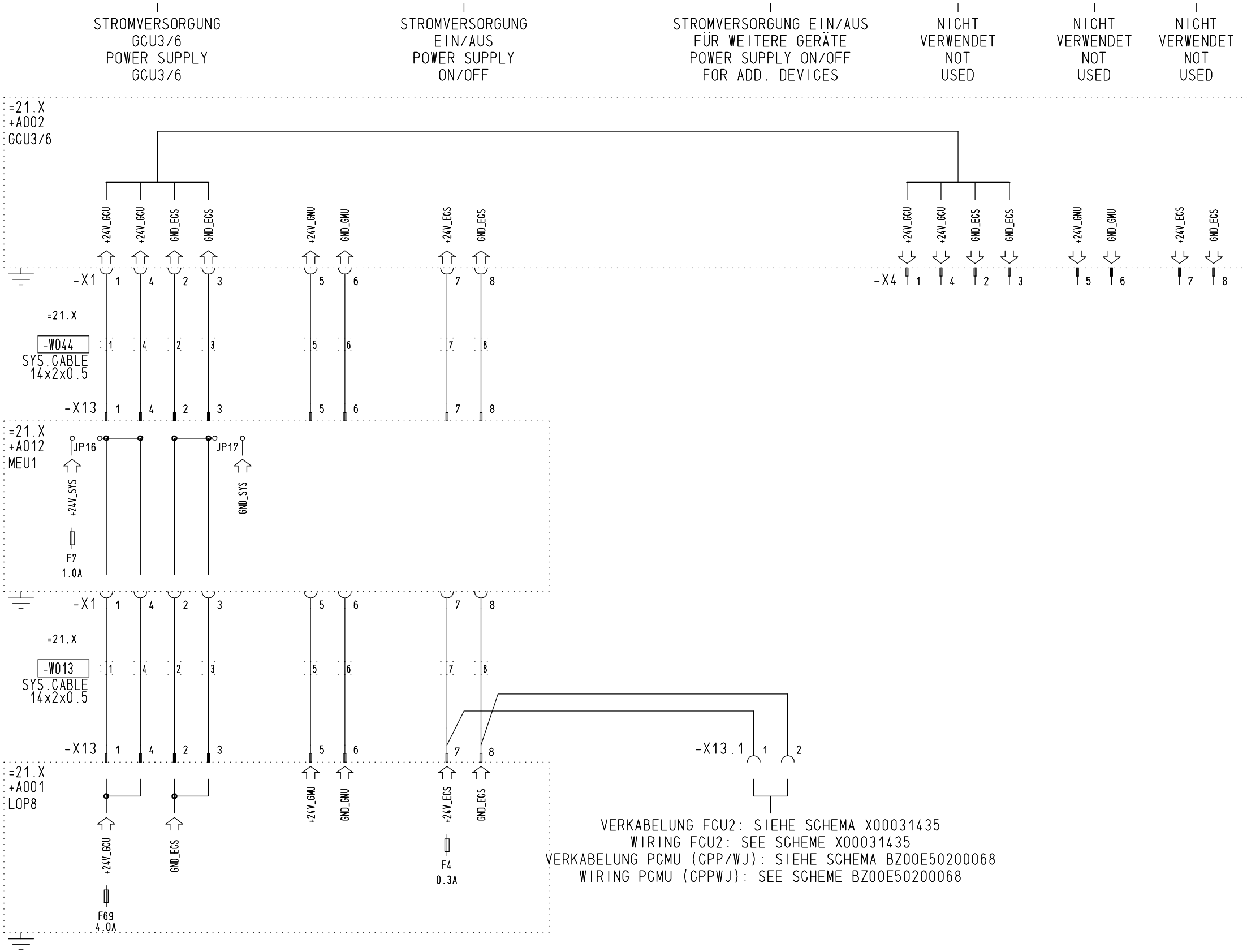
© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



|         |                      |  |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                                                                                      |                            |          |                               |           |                |   |
|---------|----------------------|--|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------|----------|-------------------------------|-----------|----------------|---|
|         |                      |  |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | MOTORRAUM, MOTOR 1/2: STROMVERSORGUNG 24VDC, GRUNDUMFANG<br>ENGINE ROOM, ENGINE 1/2: POWER SUPPLY 24VDC, BASIC SCOPE |                            |          |                               |           |                |   |
|         |                      |  |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                                      |                            |          |                               |           |                |   |
|         |                      |  |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                                                                                      |                            |          |                               |           |                |   |
|         |                      |  |              |            |      | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                                      |                            |          |                               |           |                |   |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll |          |            |        |                                                                                                                |                                                                                                  |                                                                                                                      | Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 8 |
| Buchst. | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  |                                                                                                                      | Referenz-Nr./Reference-No. | -----    |                               |           |                |   |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 1/2: STROMVERSORGUNG 24VDC, ERWEIT. UMFANG  
ENGINE ROOM, ENGINE 1/2: POWER SUPPLY 24VDC, EXTENDED SCOPE

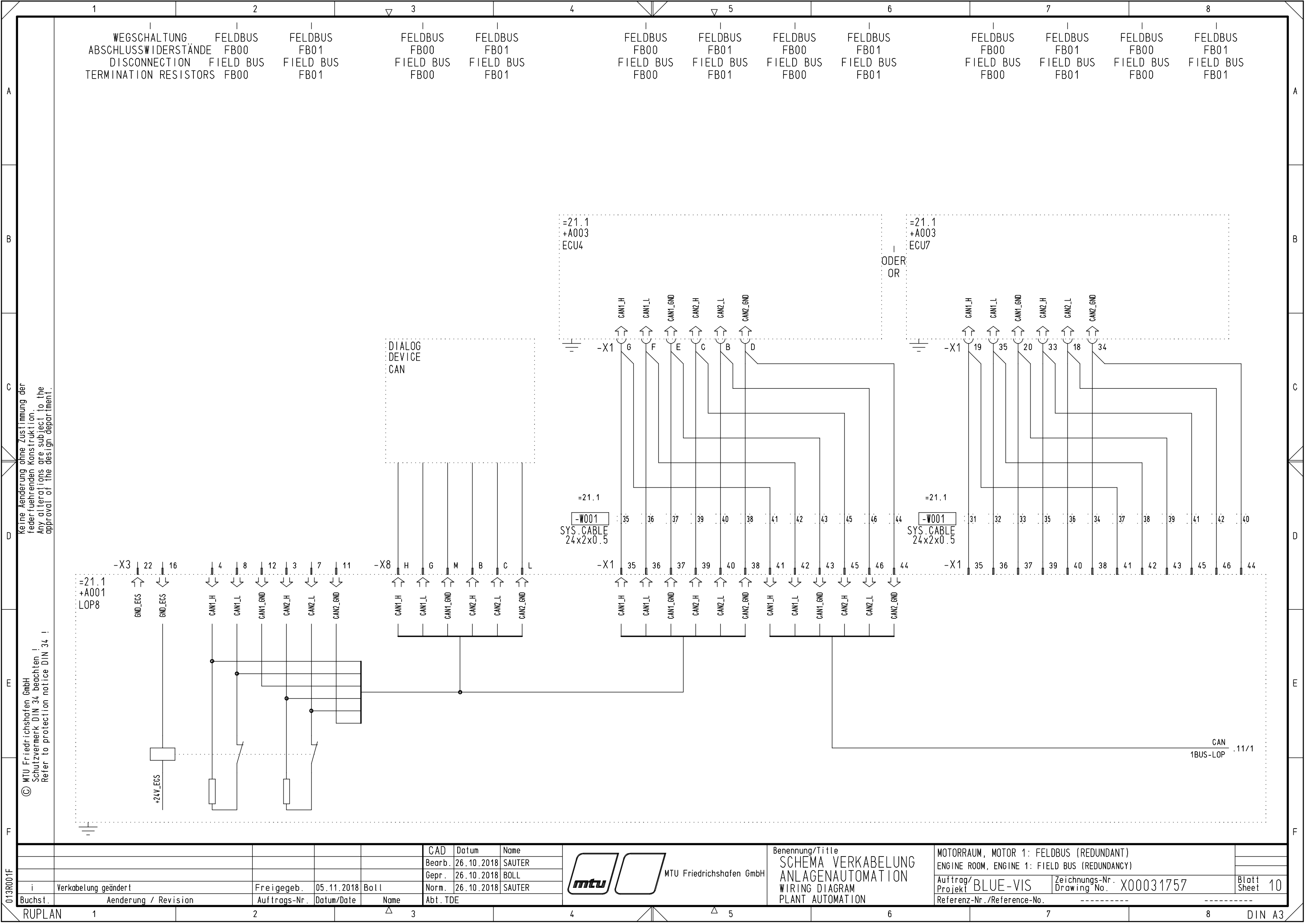
Auftrag/Projekt  
BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

Referenz-Nr./Reference-No. -----

Blatt  
Sheet 9





Keine Änderung ohne Zustimmung der federführenden Konstruktion.  
Any alterations are subject to the approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |        |
|---------|----------------------|--------------|------------|--------|
|         |                      | CAD          | Datum      | Name   |
|         |                      | Bearb.       | 26.10.2018 | SAUTER |
|         |                      | Gepr.        | 26.10.2018 | BOLL   |
|         |                      | Norm.        | 26.10.2018 | SAUTER |
|         |                      | Abt. TDE     |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll   |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 1: FELDBUS (REDUNDANT)  
ENGINE ROOM, ENGINE 1: FIELD BUS (REDUNDANCY)

Auftrag/Projekt  
BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

Referenz-Nr./Reference-No.

Blatt  
Sheet 10

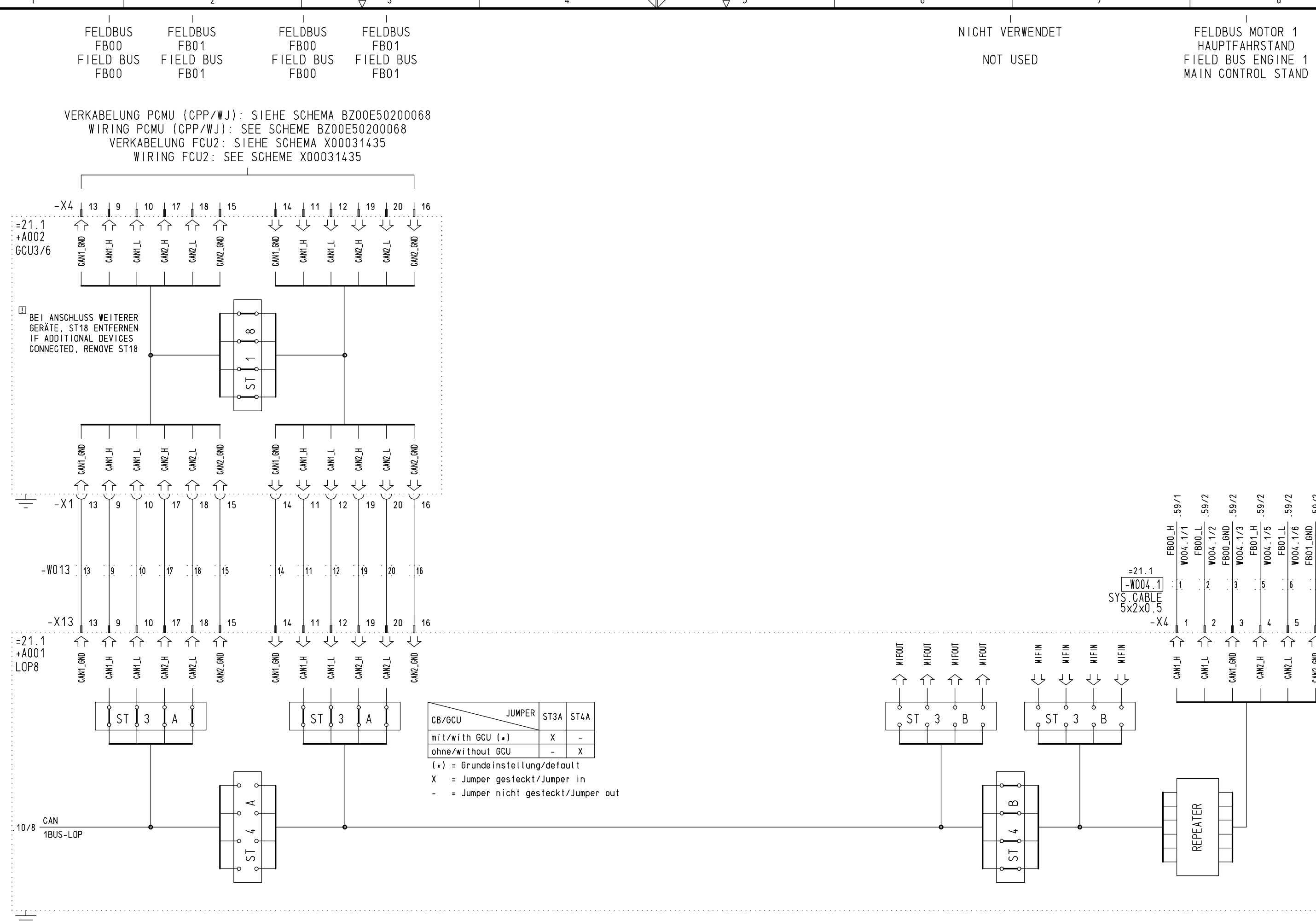
013R001F



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F



|          |  |            |        |
|----------|--|------------|--------|
| CAD      |  | Datum      | Name   |
| Bearb.   |  | 26.10.2018 | SAUTER |
| Gepr.    |  | 26.10.2018 | BOLL   |
| Norm.    |  | 26.10.2018 | SAUTER |
| Abt. TDE |  |            |        |

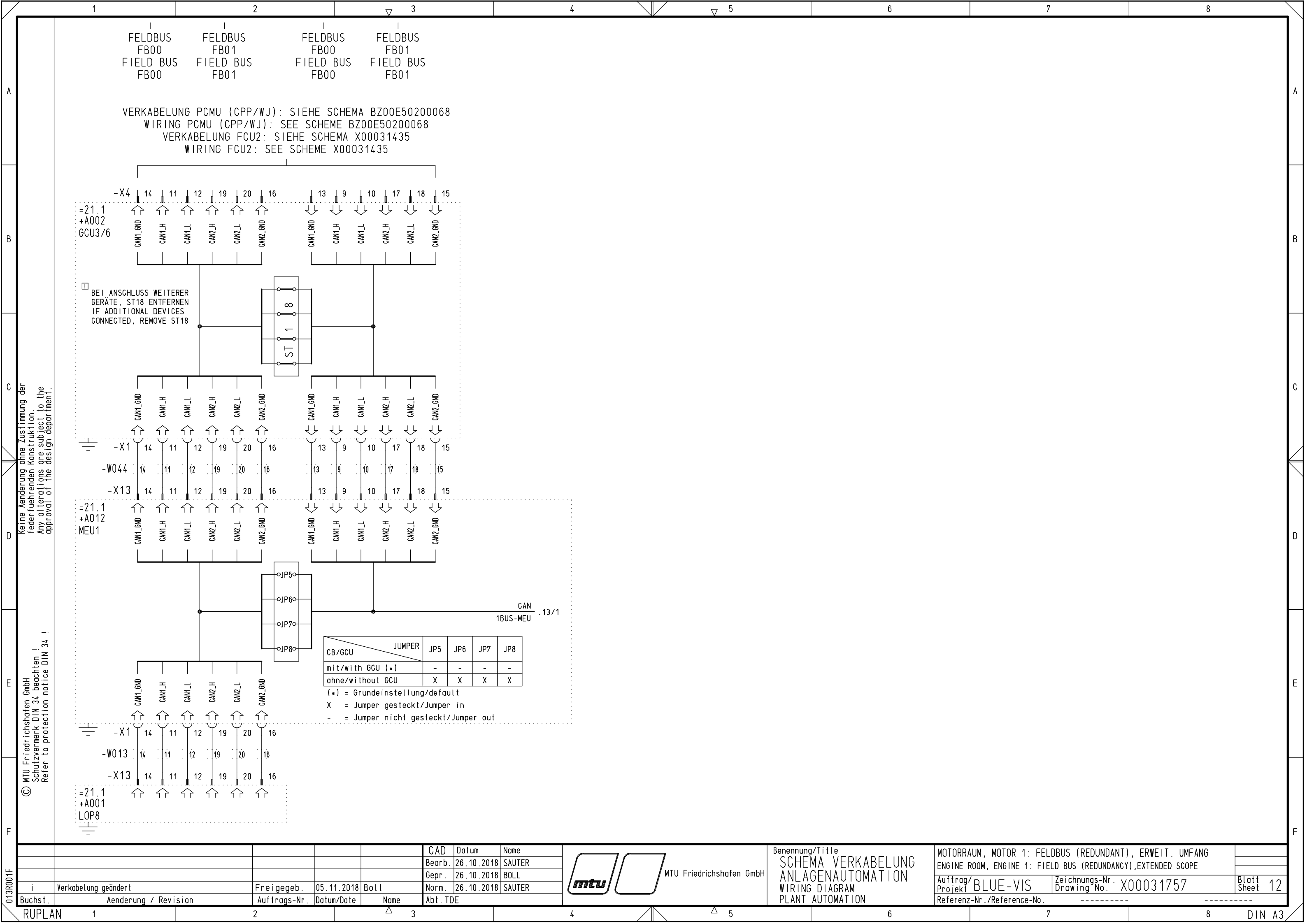
|         |                      |              |            |      |
|---------|----------------------|--------------|------------|------|
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |

MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 1: FELDBUS (REDUNDANT), GRUNDUMFANG  
ENGINE ROOM, ENGINE 1: FIELD BUS (REDUNDANCY), BASIC SCOPE

|                            |          |                               |           |                |    |
|----------------------------|----------|-------------------------------|-----------|----------------|----|
| Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 11 |
| Referenz-Nr./Reference-No. |          | -----                         |           |                |    |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

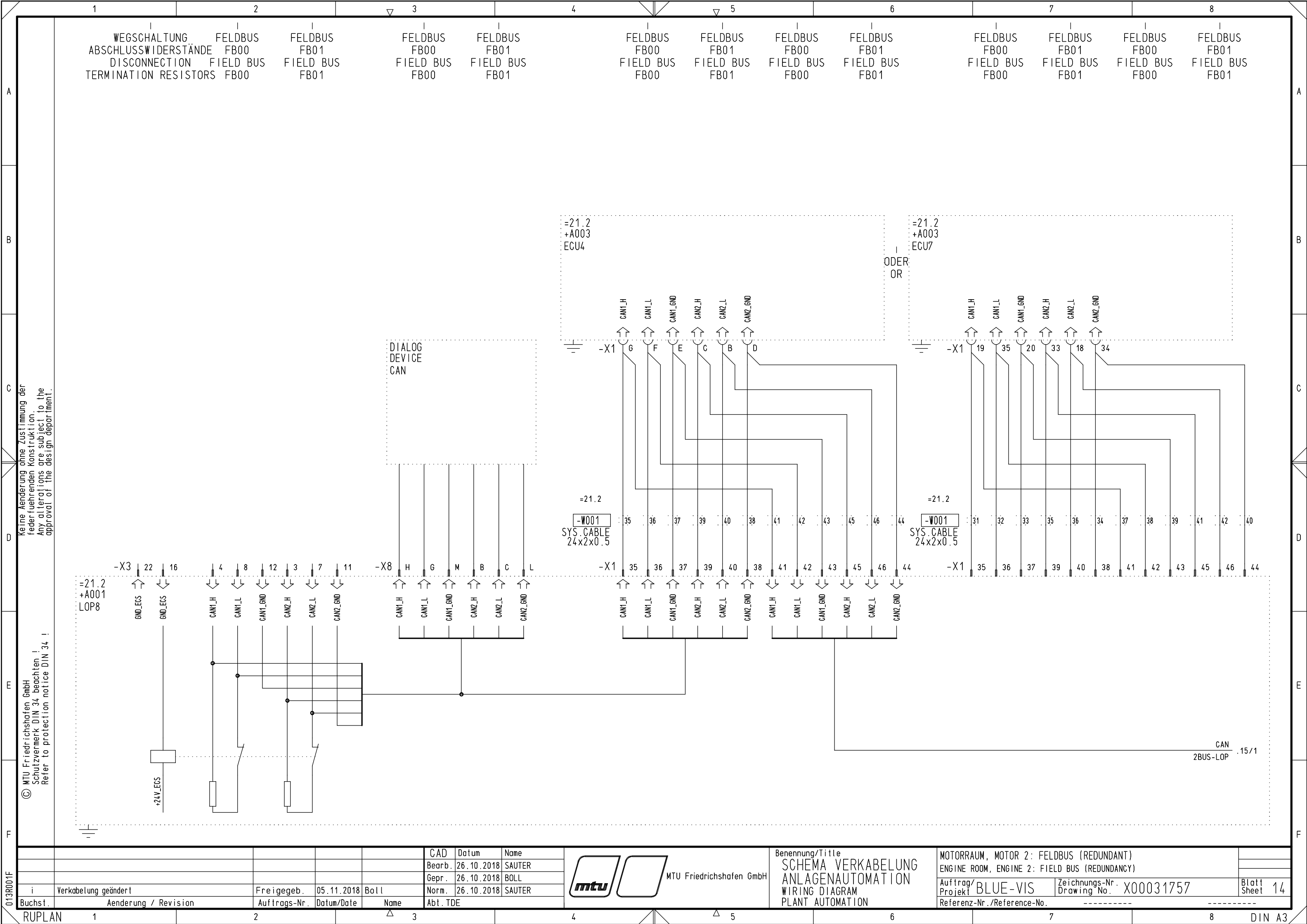
Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 1: FELDBUS (REDUNDANT), ERWEIT. UMFANG  
ENGINE ROOM, ENGINE 1: FIELD BUS (REDUNDANCY), EXTENDED SCOPE

Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. Drawing No. X00031757

Referenz-Nr./Reference-No. -----

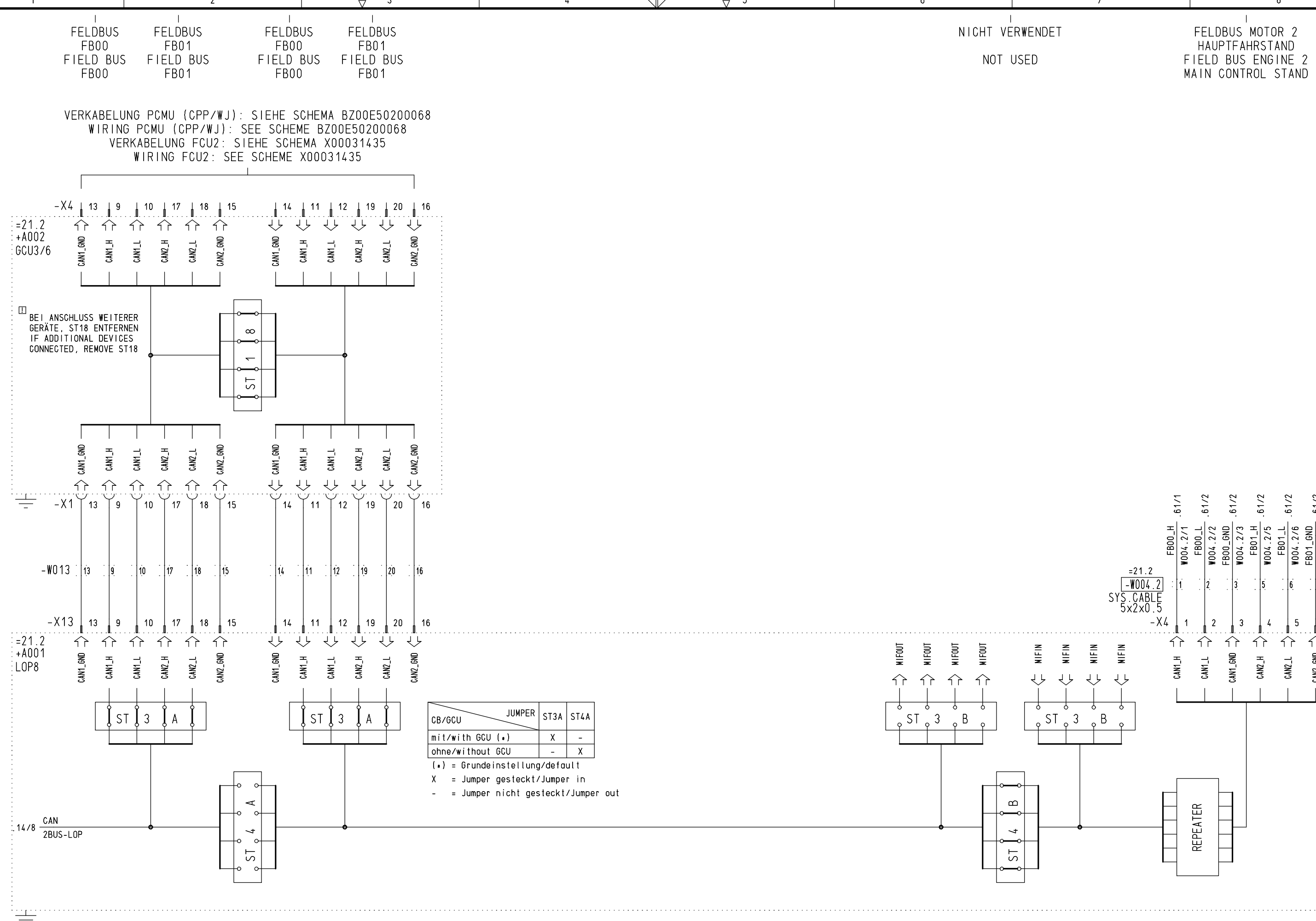
Blatt  
Sheet 13



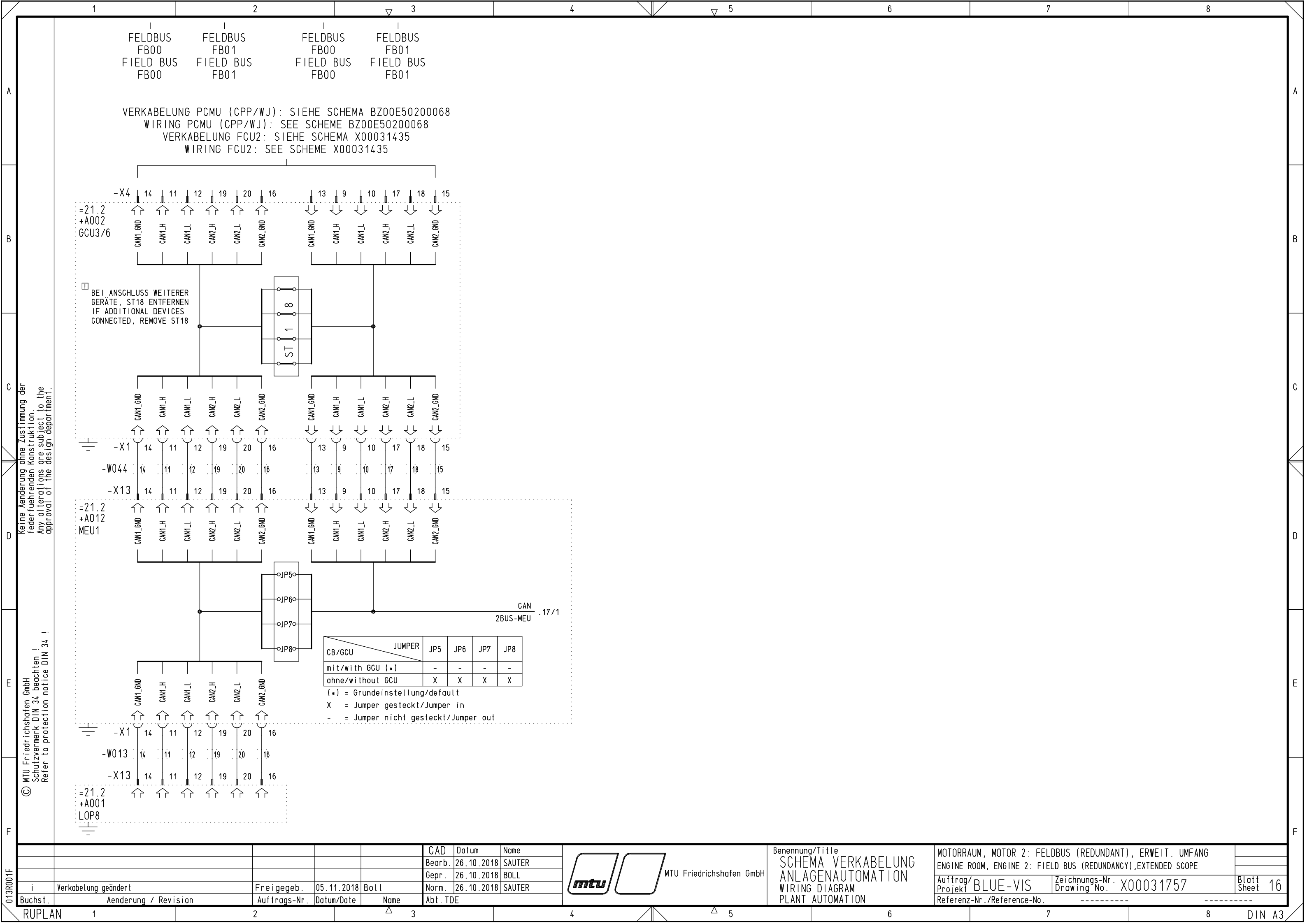
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F



|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                                                                                    |          |                               |           |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | MOTORRAUM, MOTOR 2: FELDBUS (REDUNDANT), GRUNDUMFANG<br>ENGINE ROOM, ENGINE 2: FIELD BUS (REDUNDANCY), BASIC SCOPE |          |                               |           |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                                    |          |                               |           |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                                                                                    |          |                               |           |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                                    |          |                               |           |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                                                                                | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 2: FELDBUS (REDUNDANT), ERWEIT. UMFANG  
ENGINE ROOM, ENGINE 2: FIELD BUS (REDUNDANCY), EXTENDED SCOPE

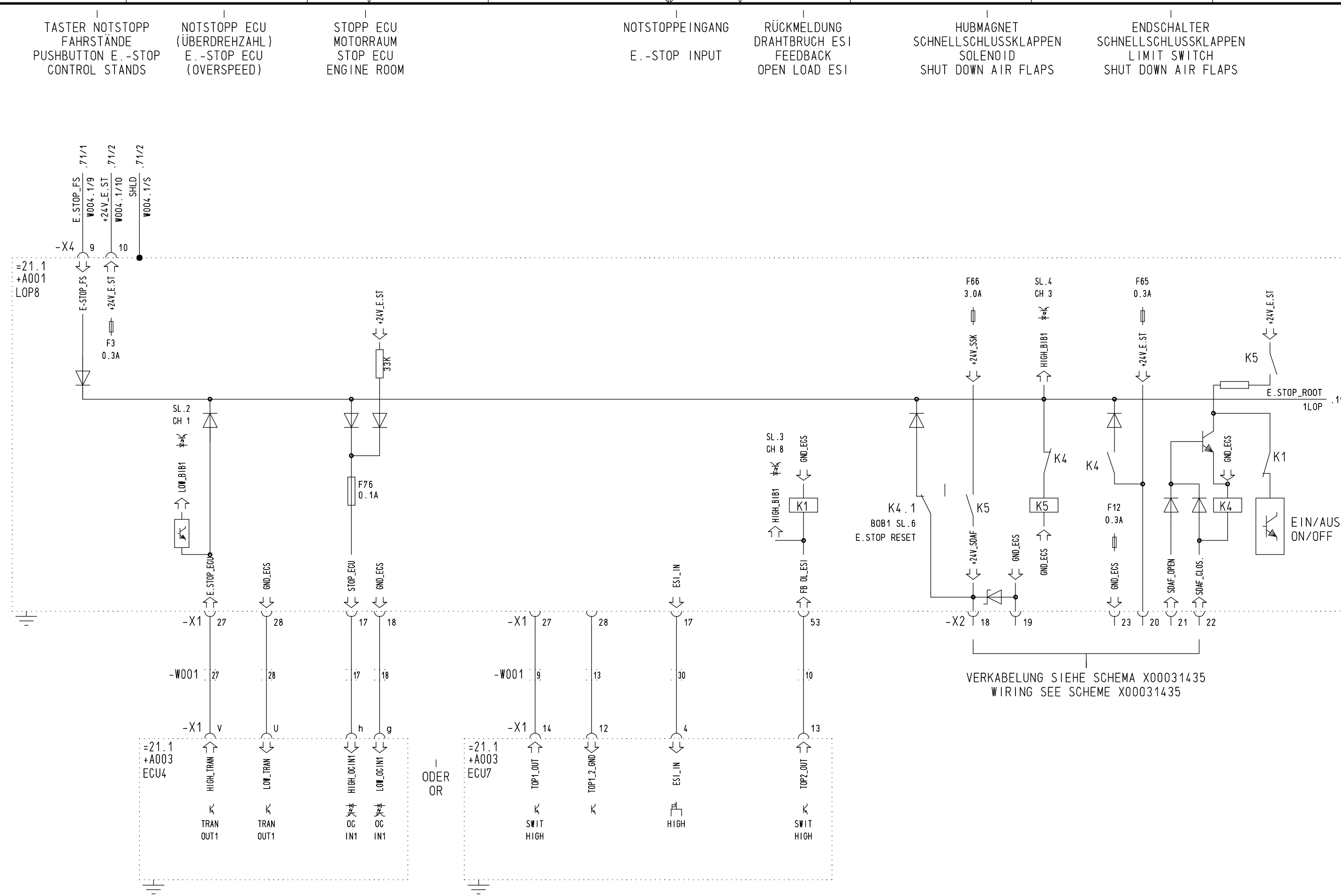
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. Drawing No. X00031757

Referenz-Nr./Reference-No. -----

Blatt  
Sheet 17

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

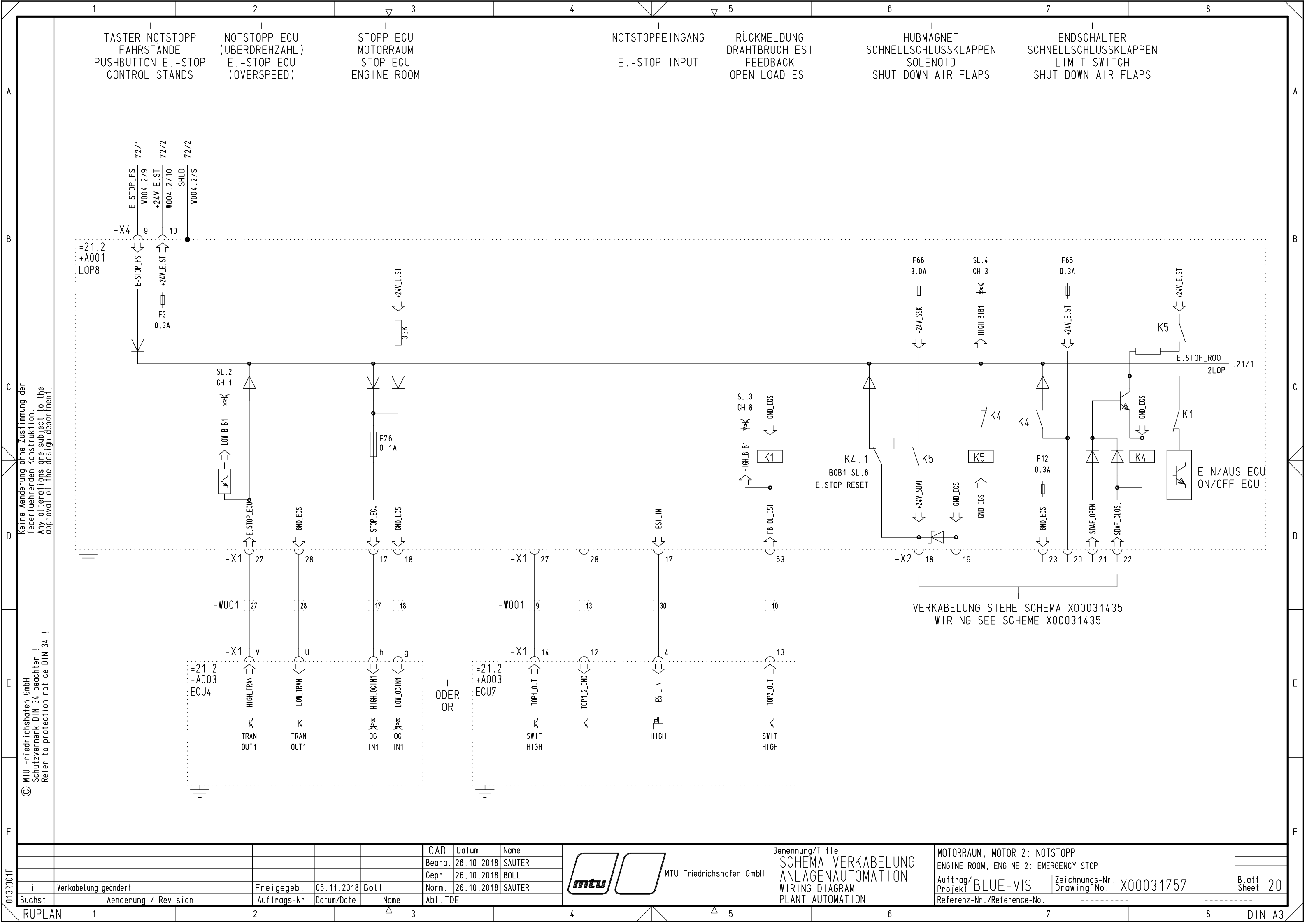


VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

|         |                      |  |              |            |        |            |        |                                                                                                                |                                                                                                  |                                                                       |          |                               |                   |  |
|---------|----------------------|--|--------------|------------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------|-------------------------------|-------------------|--|
|         |                      |  |              |            | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | MOTORRAUM, MOTOR 1: NOTSTOPP<br>ENGINE ROOM, ENGINE 1: EMERGENCY STOP |          |                               | Blatt<br>Sheet 18 |  |
|         |                      |  |              |            | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                       |          |                               |                   |  |
|         |                      |  |              |            | Gepr.  | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                                       |          |                               |                   |  |
|         |                      |  |              |            | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                       |          |                               |                   |  |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll   |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                                   | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757         |  |
| Buchst. | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE   |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                            | -----    |                               |                   |  |







Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

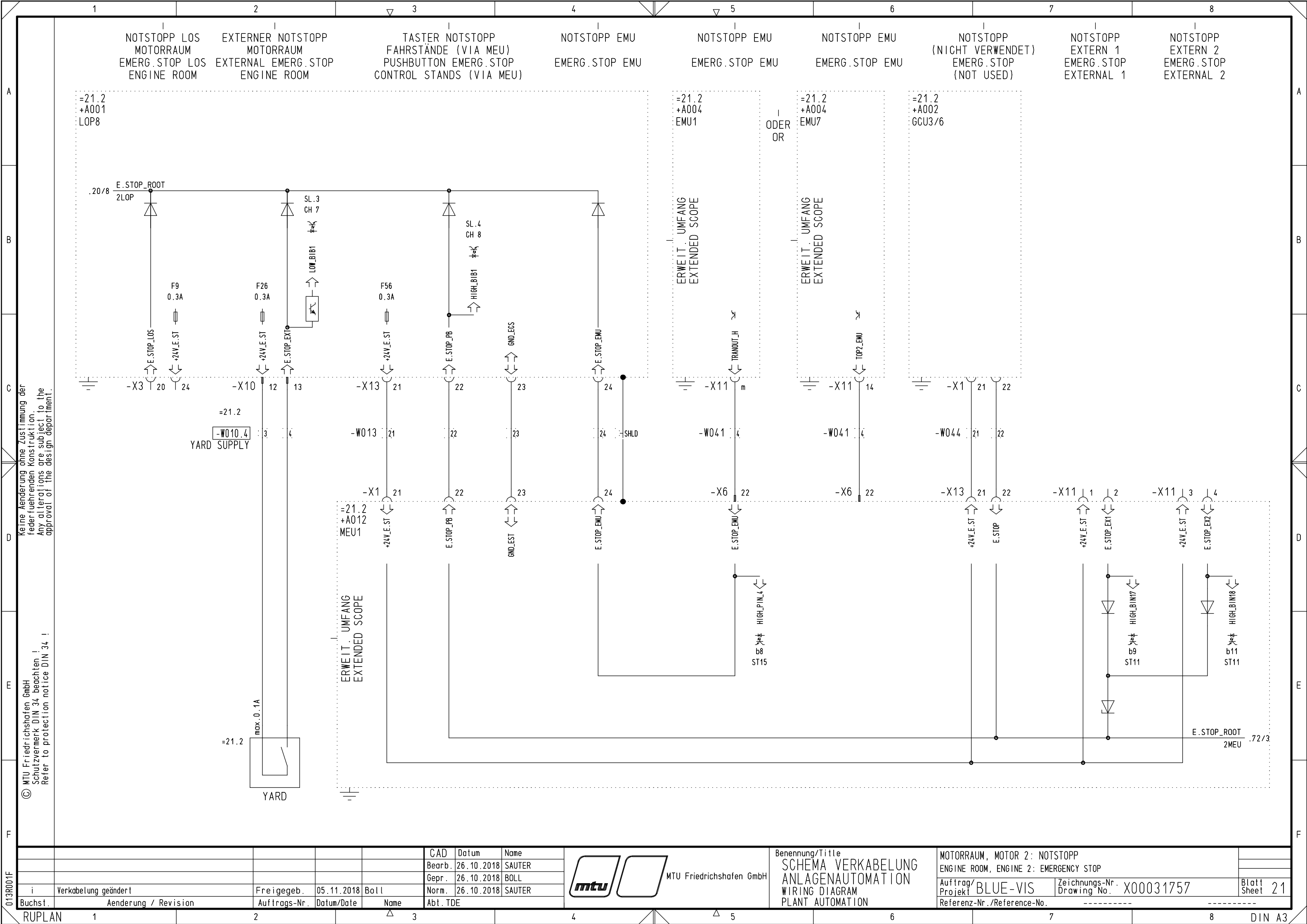
MOTORRAUM, MOTOR 2: NOTSTOPP  
ENGINE ROOM, ENGINE 2: EMERGENCY STOP

Auftrag/  
Projekt BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

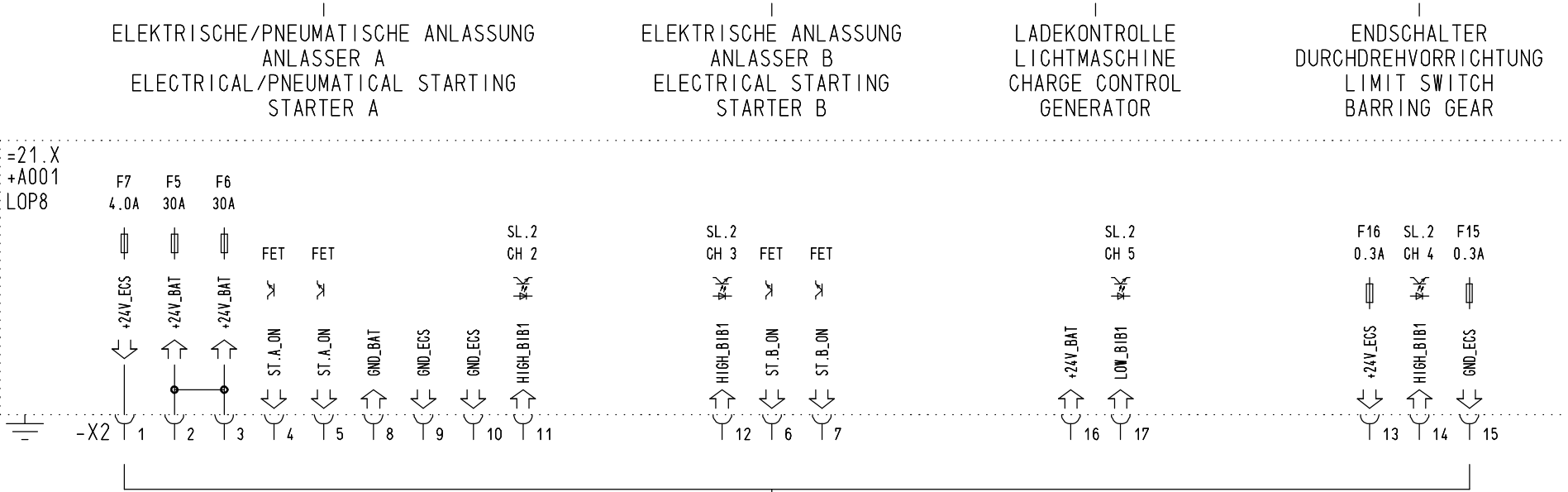
Referenz-Nr./Reference-No.

Blatt  
Sheet 20



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

|         |                      |  |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |    |
|---------|----------------------|--|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------|-------------------------------|-----------|----------------|----|
|         |                      |  |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | MOTORRAUM, MOTOR 1/2: ANLASSUNG, LICHTMASCHINE DURCHDREH.  |          |                               |           |                |    |
|         |                      |  |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | ENGINE ROOM, ENGINE 1/2: STARTING, GENERATOR, BARRING GEAR |          |                               |           |                |    |
|         |                      |  |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |    |
|         |                      |  |              |            |      | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |    |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll |          |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projek                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 22 |
| Buchst. | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                 |          | -----                         |           |                |    |



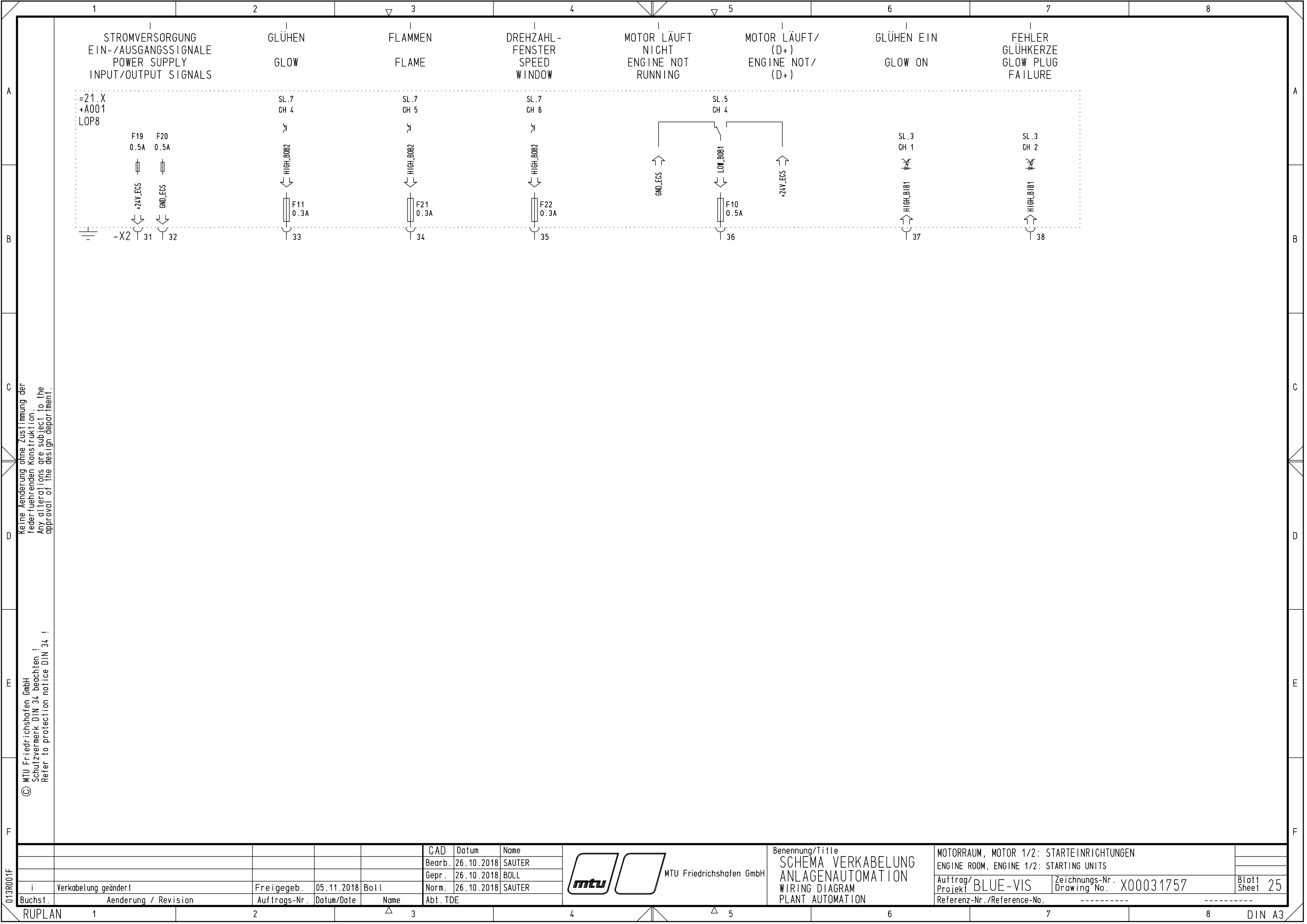
VORSCHMIERPUMPE  
1x230 VAC  
PRIMING PUMP  
1x230 VAC

⋮ L1 1x230VAC/50Hz N

Blatt  
Sheet 24

|                            |       |       |
|----------------------------|-------|-------|
| Referenz-Nr./Reference-No. | ----- | ----- |
|----------------------------|-------|-------|



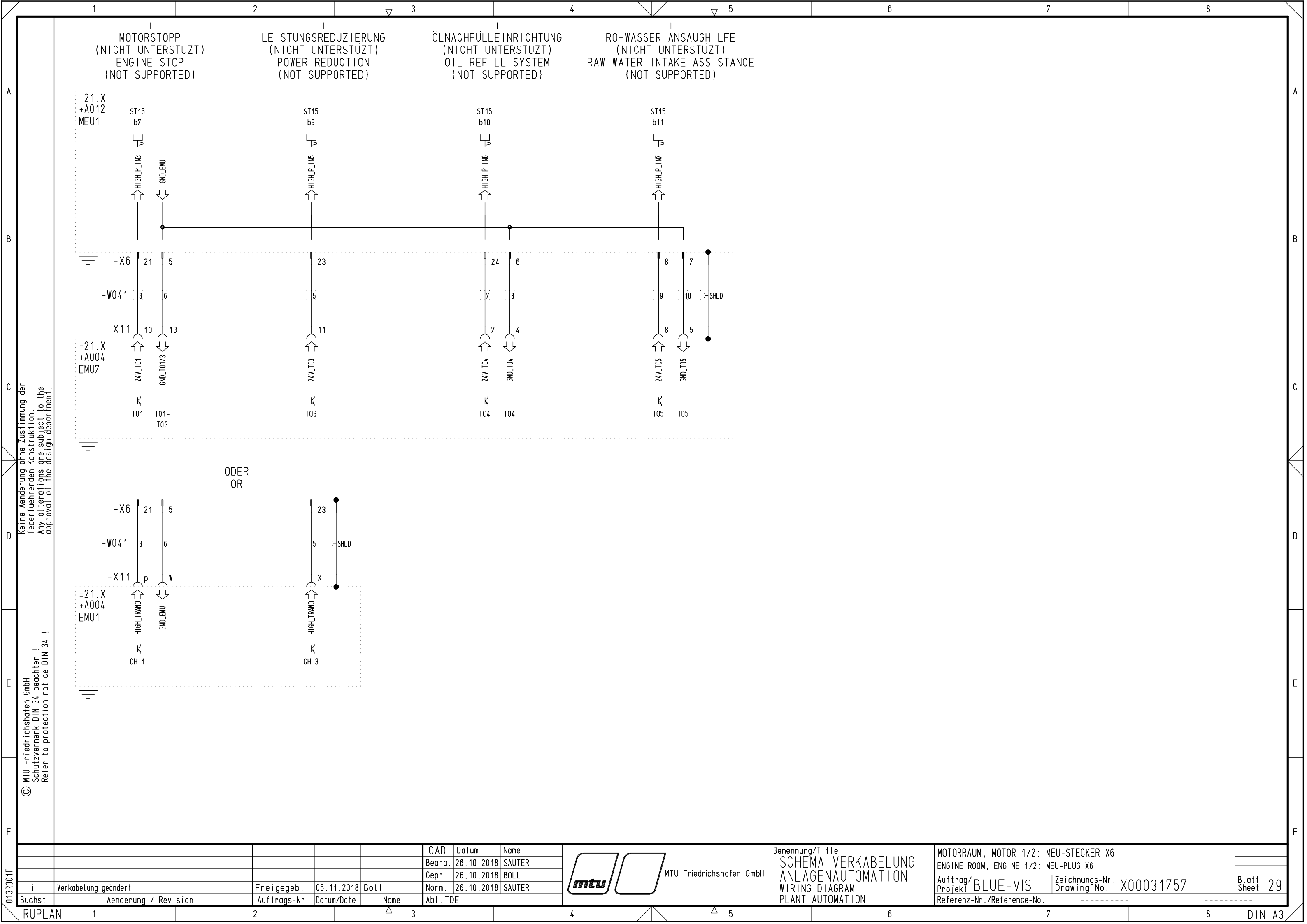












Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

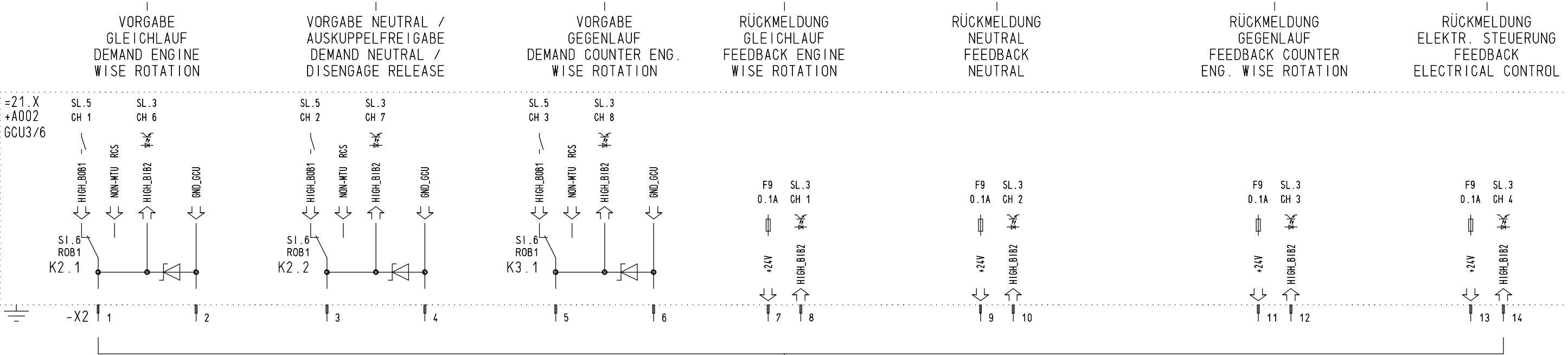
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gedr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



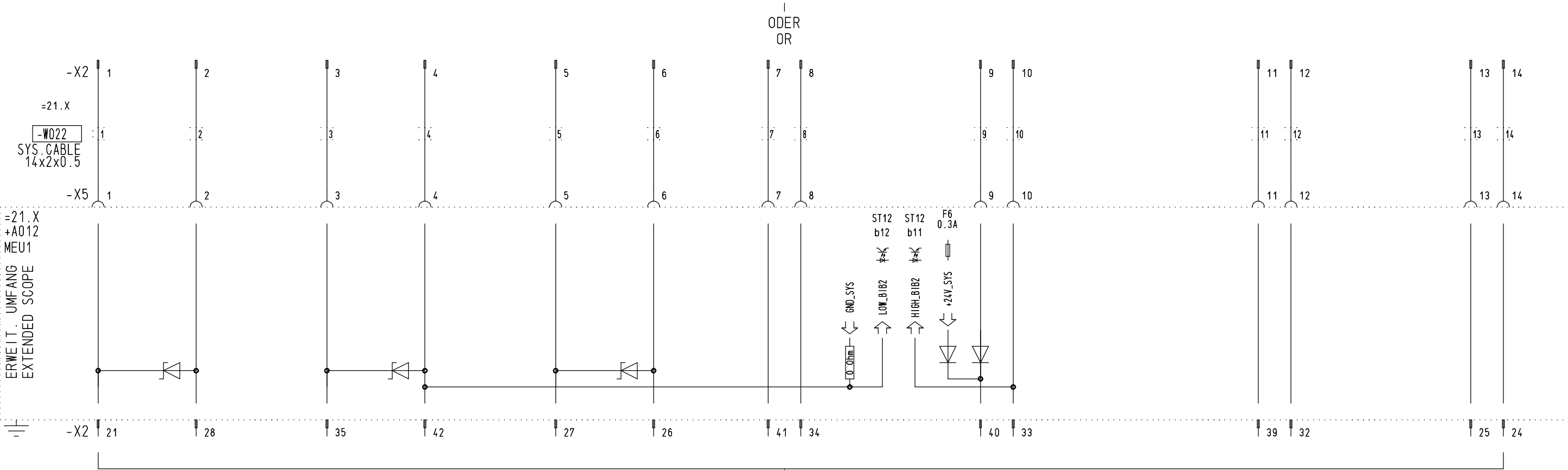
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                      |                                         |
|--------------------------------------------------------------------------------------|-----------------------------------------|
| MOTORRAUM, MOTOR 1/2: GCU/MEU-STECKER X2<br>ENGINE ROOM, ENGINE 1/2: GCU/MEU-PLUG X2 |                                         |
| Auftrag/Projekt<br>BLUE-VIS                                                          | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                           | Blatt<br>Sheet 30                       |



VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435



VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

=21.X  
+A002  
GCU3/6

P-GETRIEBE-  
STEUERÖL  
P-GEAR  
CONTROL OIL

P-GETRIEBE-  
SCHMIERÖL  
P-GEAR  
LUBE OIL

T-GETRIEBEÖL  
T-GEAR OIL

GETRIEBEÖLFILTER  
VERSCHMUTZT  
GEAR OIL FILTER  
CLOGGED

F6 SL.4  
0.1A CH 1

F7 SL.4  
0.1A CH 2

SL.4 SL.4  
CH 3 CH 3

F11 SL.1  
0.1A CH 8

+24V  
UREF\_-A1B1

+24V  
UREF\_-A1B1

UDIF+\_A1B1  
UDIF\_-A1B1

+24V  
HIGH\_LB1B1

VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

ODER  
OR

=21.X  
+A012  
MEU1

ERWEIT. UMFANG  
EXTENDED SCOPE

VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

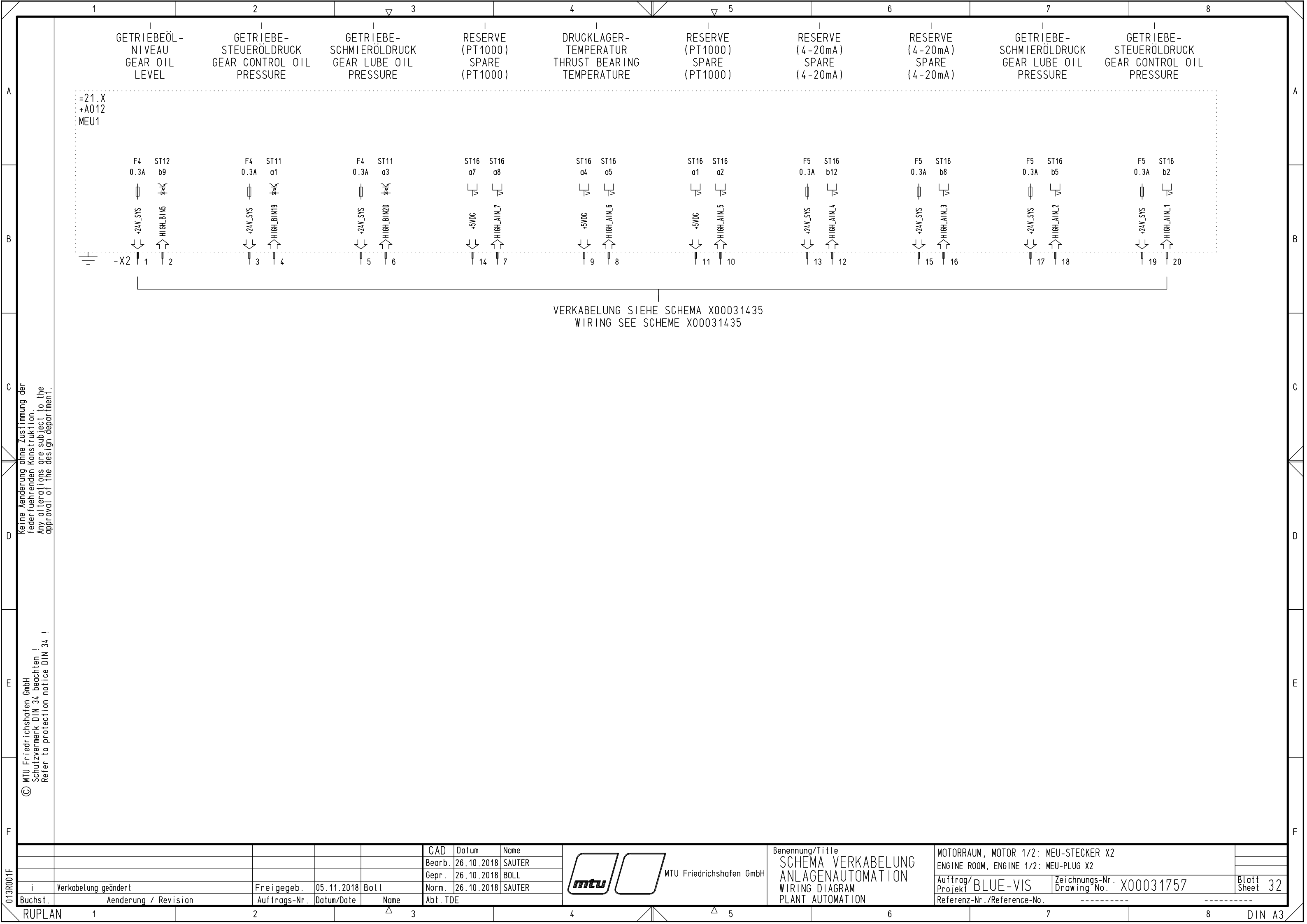
| CAD      | Datum      | Name   |
|----------|------------|--------|
| Bearb.   | 26.10.2018 | SAUTER |
| Gepr.    | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |



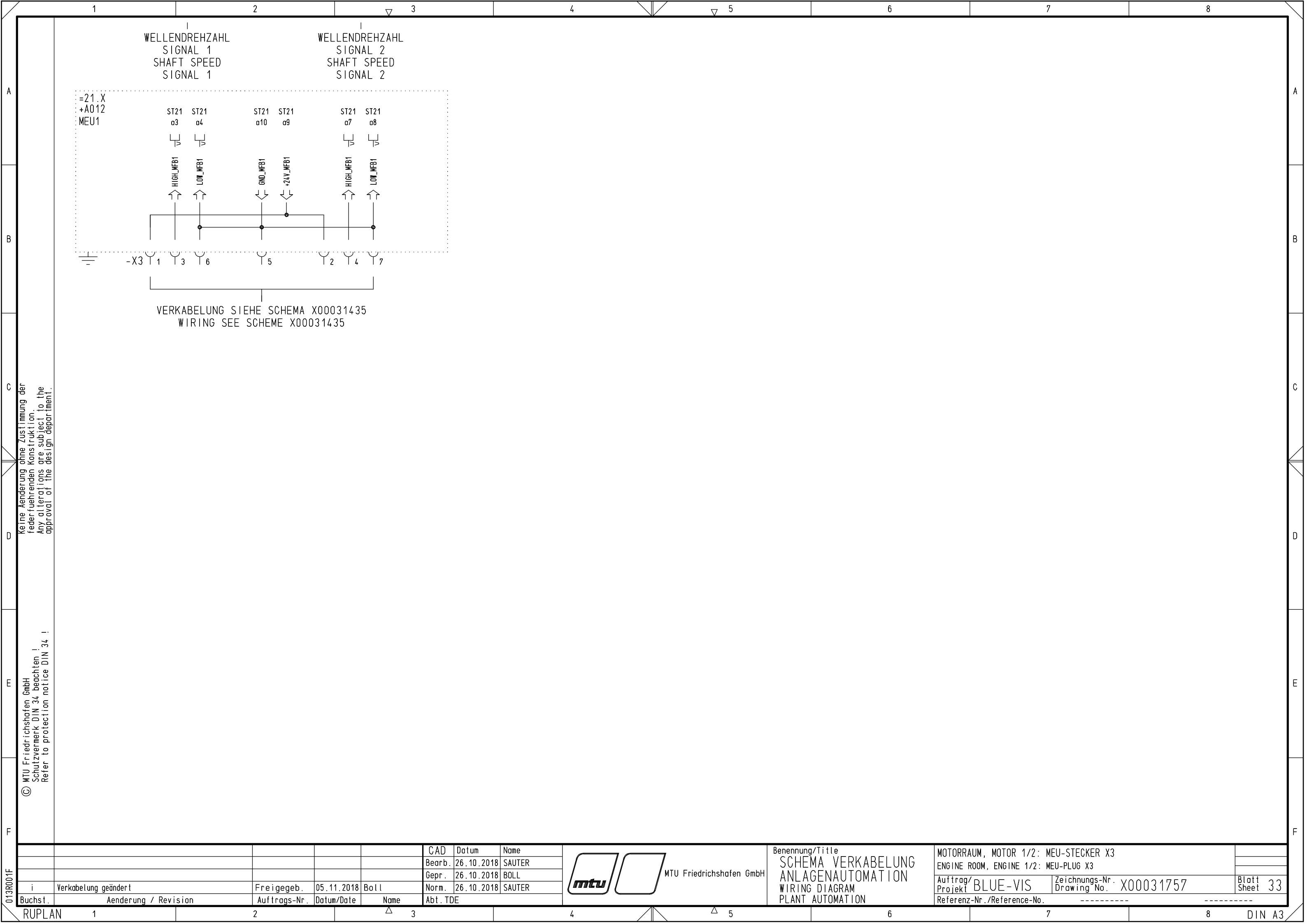
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                      |                                  |                                         |                   |
|--------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------|-------------------|
| MOTORRAUM, MOTOR 1/2: GCU/MEU-STECKER X2<br>ENGINE ROOM, ENGINE 1/2: GCU/MEU-PLUG X2 |                                  | Zeichnungs-Nr.<br>Drawing No. X00031757 | Blatt<br>Sheet 31 |
| Auftrag/<br>Projekt BLUE-VIS                                                         | Referenz-Nr./Reference-No. ----- |                                         |                   |

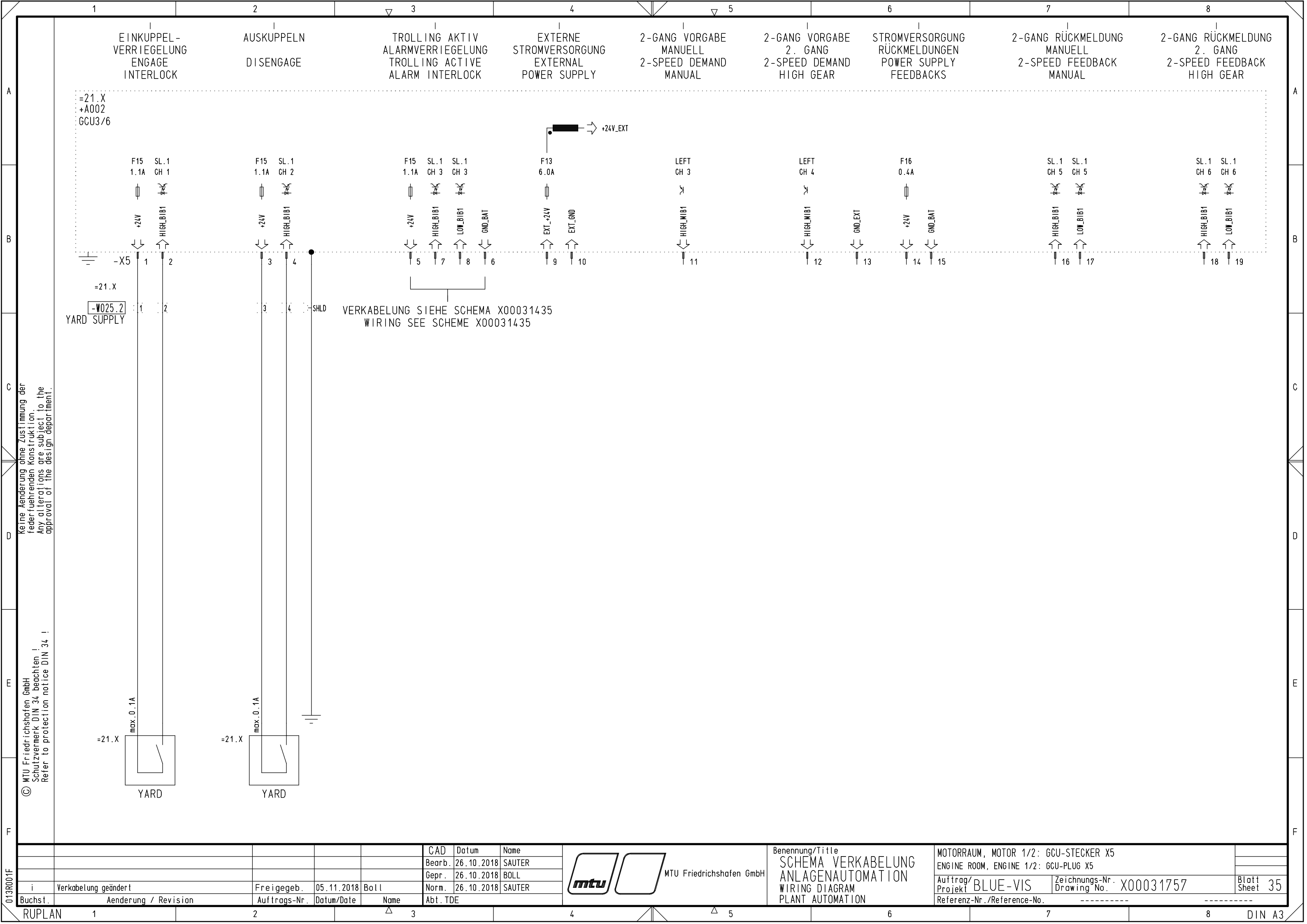












Keine Änderung ohne Zustimmung der federführenden Konstruktion.  
Any alterations are subject to the approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

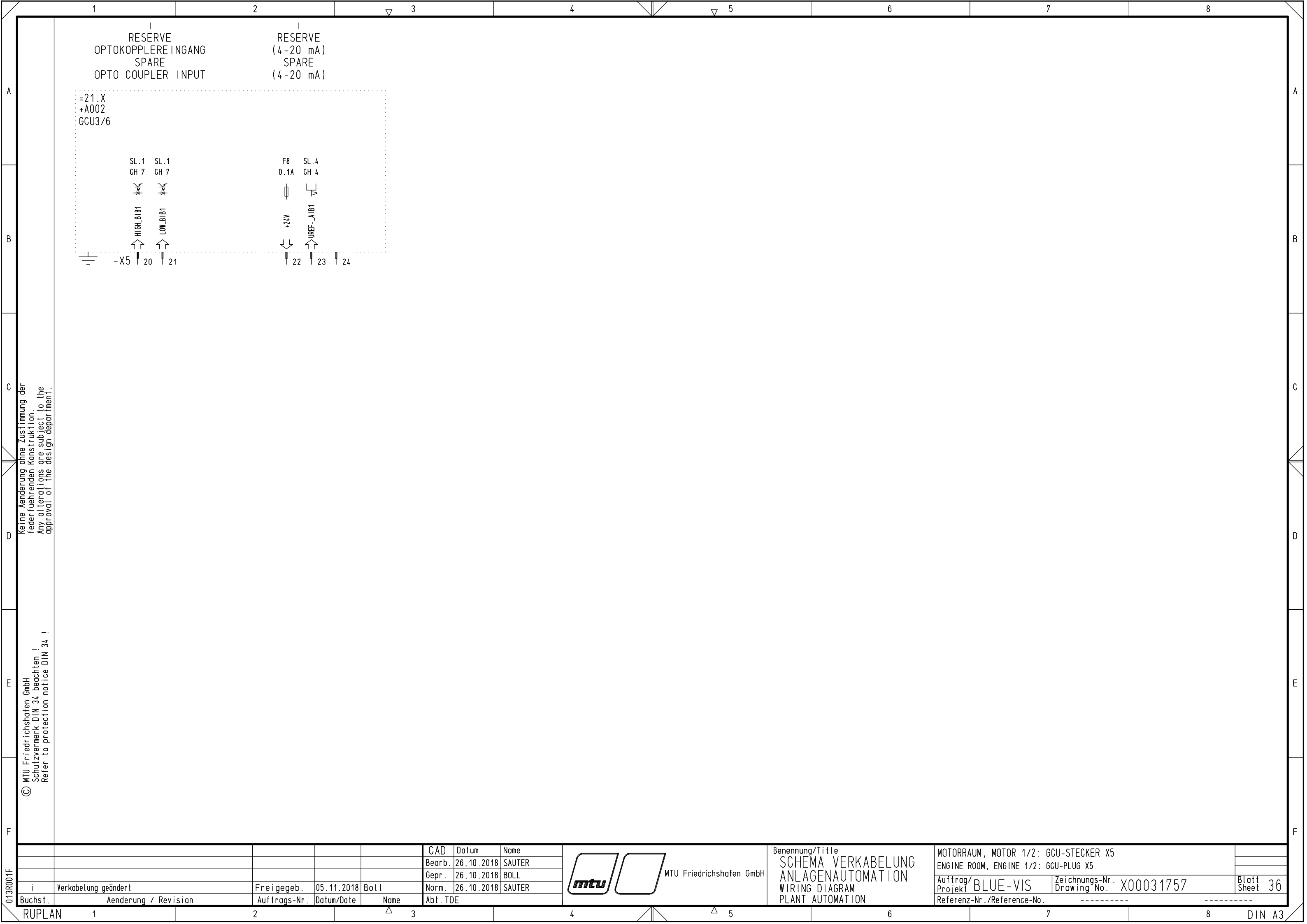
Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

MOTORRAUM, MOTOR 1/2: GCU-STECKER X5  
ENGINE ROOM, ENGINE 1/2: GCU-PLUG X5

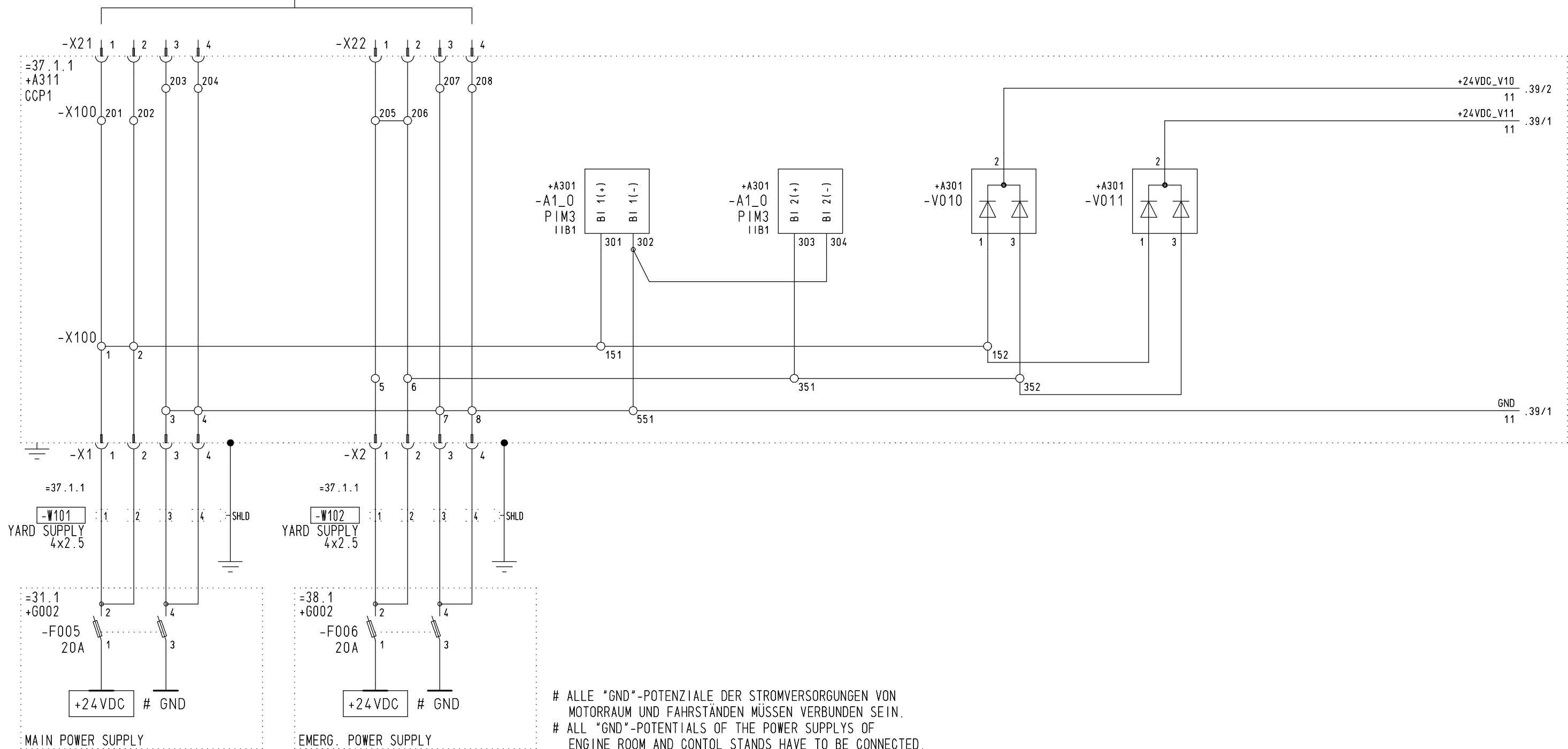
Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt/Sheet: 35



|                      |                     |                                     |                                   |                   |                   |
|----------------------|---------------------|-------------------------------------|-----------------------------------|-------------------|-------------------|
| HAUPTSTROMVERSORGUNG | NOTSTROMVERSORGUNG  | ÜBERWACHUNG<br>HAUPTSTROMVERSORGUNG | ÜBERWACHUNG<br>NOTSTROMVERSORGUNG | SPANNUNGSKOPPLUNG | SPANNUNGSKOPPLUNG |
| MAIN POWER SUPPLY    | EMERG. POWER SUPPLY | MAIN POWER SUPPLY<br>MONITORING     | EMERG. POWER SUPPLY<br>MONITORING | POWER COUPLING    | POWER COUPLING    |

SIEHE HINWEISBLATT "STROMVERSORGUNG"  
SEE INSTRUCTION SHEET "POWER SUPPLY"



# ALLE "GND"-POTENZIALE DER STROMVERSORGUNGEN VON  
MOTORRAUM UND FAHRSTÄNDEN MÜSSEN VERBUNDEN SEIN.  
# ALL "GND"-POTENTIALS OF THE POWER SUPPLYS OF  
ENGINE ROOM AND CONTOL STANDS HAVE TO BE CONNECTED.

|        |            |        |
|--------|------------|--------|
| CAD    | Datum      | Name   |
| Bearb. | 26.10.2018 | SAUTER |
| Gepr.  | 26.10.2018 | BOLL   |
| Norm.  | 26.10.2018 | SAUTER |



/ MTU Friedrichshafen GmbH

|   |                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|
| H | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |
|---|--------------------------------------------------------------------------------------------------|

|                                                              |  |
|--------------------------------------------------------------|--|
| HAUPTFAHRSTAND, MOTOR 1/2: STROMVERSG. 24VDC (RCS VIA DIODE) |  |
| MAIN CTR.STAND, ENG. 1/2: POWER SUPPLY 24VDC (RCS VIA DIODE) |  |

|                     |          |                               |           |
|---------------------|----------|-------------------------------|-----------|
| Auftrag/<br>Projekt | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
|---------------------|----------|-------------------------------|-----------|

Blatt 37  
Sheet 37

|                            |       |
|----------------------------|-------|
| Referenz-Nr./Reference-No. | ----- |
|----------------------------|-------|

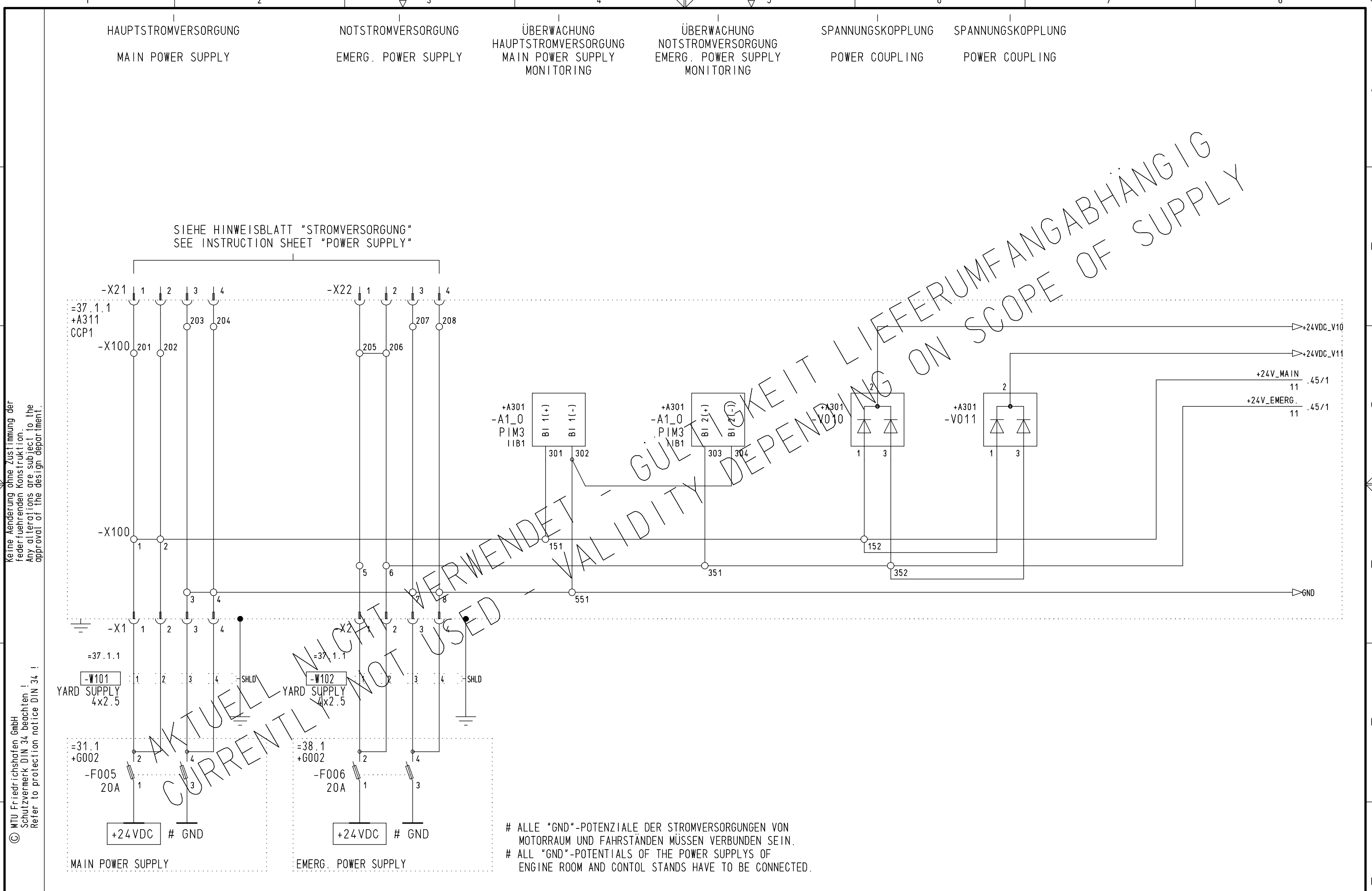
Keine Aenderung ohne Zustimmung der federfuehrenden Konstruktion.  
Any alterations are subject to the approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

RUPLAN

DIN A3

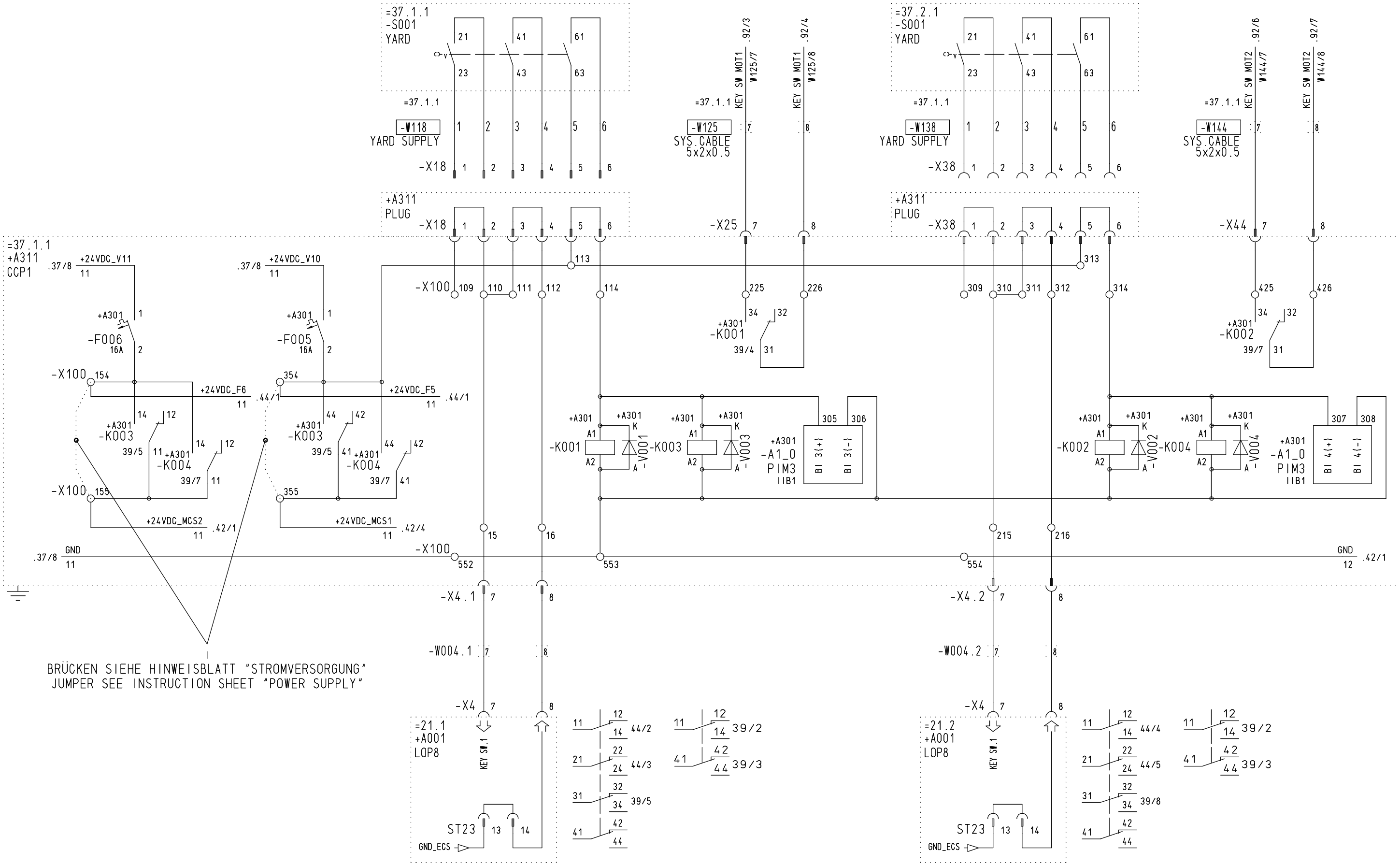


|         |                      |   |              |            |      |        |            |        |                                                                                                                |                                                                                                  |                                                    |                               |                                                                                       |                   |        |
|---------|----------------------|---|--------------|------------|------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|-------------------|--------|
|         |                      |   |              |            |      | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, MOTOR 1/2: STROMVERSORGUNG 24VDC   |                               |  | Blatt<br>Sheet 38 |        |
|         |                      |   |              |            |      | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | MAIN CONTROL STAND, ENGINE 1/2: POWER SUPPLY 24VDC |                               |                                                                                       |                   |        |
|         |                      |   |              |            |      | Gepr.  | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                    |                               |                                                                                       |                   |        |
|         |                      |   |              |            |      | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                    |                               |                                                                                       |                   |        |
| i       | Verkabelung geändert |   | Freigegeb.   | 05.11.2018 | Boll |        |            |        | Abt. TDE                                                                                                       | Auftrag/Projekt                                                                                  | BLUE-VIS                                           | Zeichnungs-Nr.<br>Drawing No. | X00031757                                                                             |                   |        |
| Buchst. | Aenderung / Revision |   | Auftrags-Nr. | Datum/Date | Name | Abt.   | TDE        |        |                                                                                                                | Referenz-Nr./Reference-No.                                                                       | -----                                              |                               | -----                                                                                 |                   |        |
| RUPLAN  |                      | 1 | 2            |            | 3    |        |            | 4      |                                                                                                                | 5                                                                                                |                                                    | 6                             | 7                                                                                     | 8                 | DIN A3 |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F



BRÜCKEN SIEHE HINWEISBLATT "STROMVERSORGUNG"  
JUMPER SEE INSTRUCTION SHEET "POWER SUPPLY"

| CAD      | Datum      | Name   |
|----------|------------|--------|
| Bearb.   | 26.10.2018 | SAUTER |
| Gedr.    | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: 24VDC EIN/AUS (RCS VIA DIODE)  
MAIN CONTROL STAND, ENGINE 1/2: 24VDC ON/OFF (RCS VIA DIODE)  
Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt  
Sheet 39



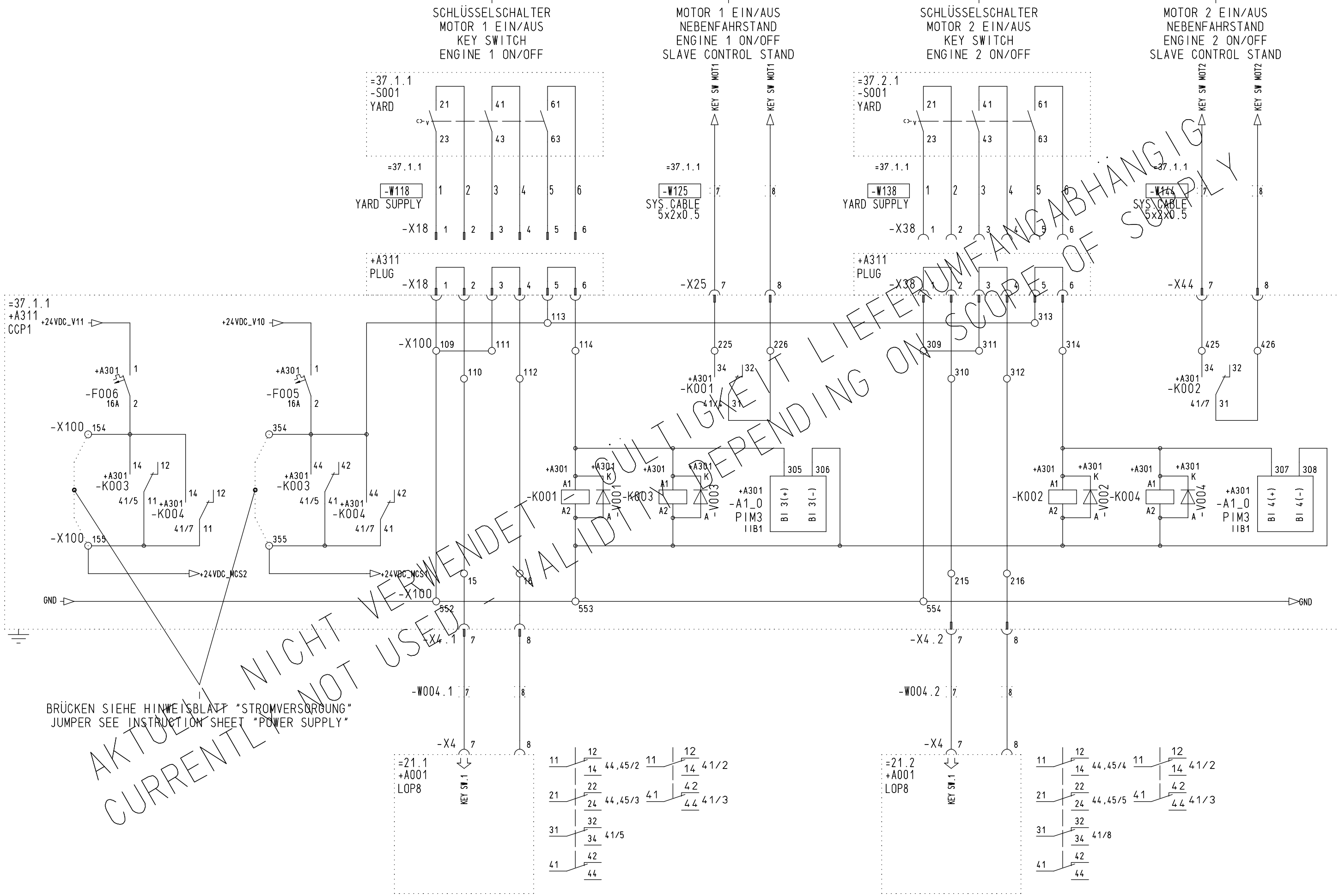


Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F

RUPLAN



BRÜCKEN SIEHE HINWEISBLATT "STROMVERSÖRGUNG"  
JUMPER SEE INSTRUCTION SHEET "POWER SUPPLY"

| CAD      | Datum      | Name   |
|----------|------------|--------|
| Bearb.   | 26.10.2018 | SAUTER |
| Gedr.    | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |



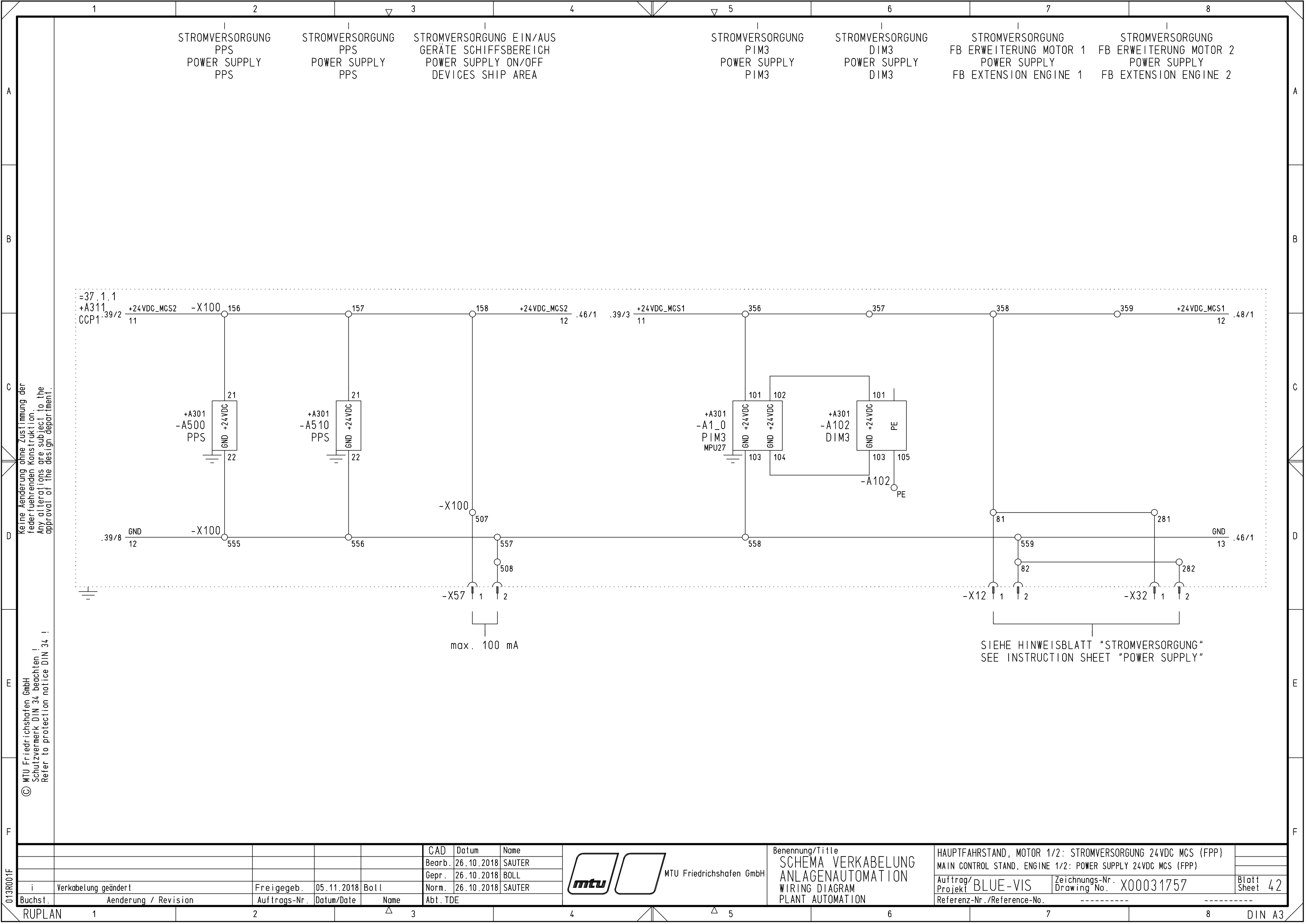
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

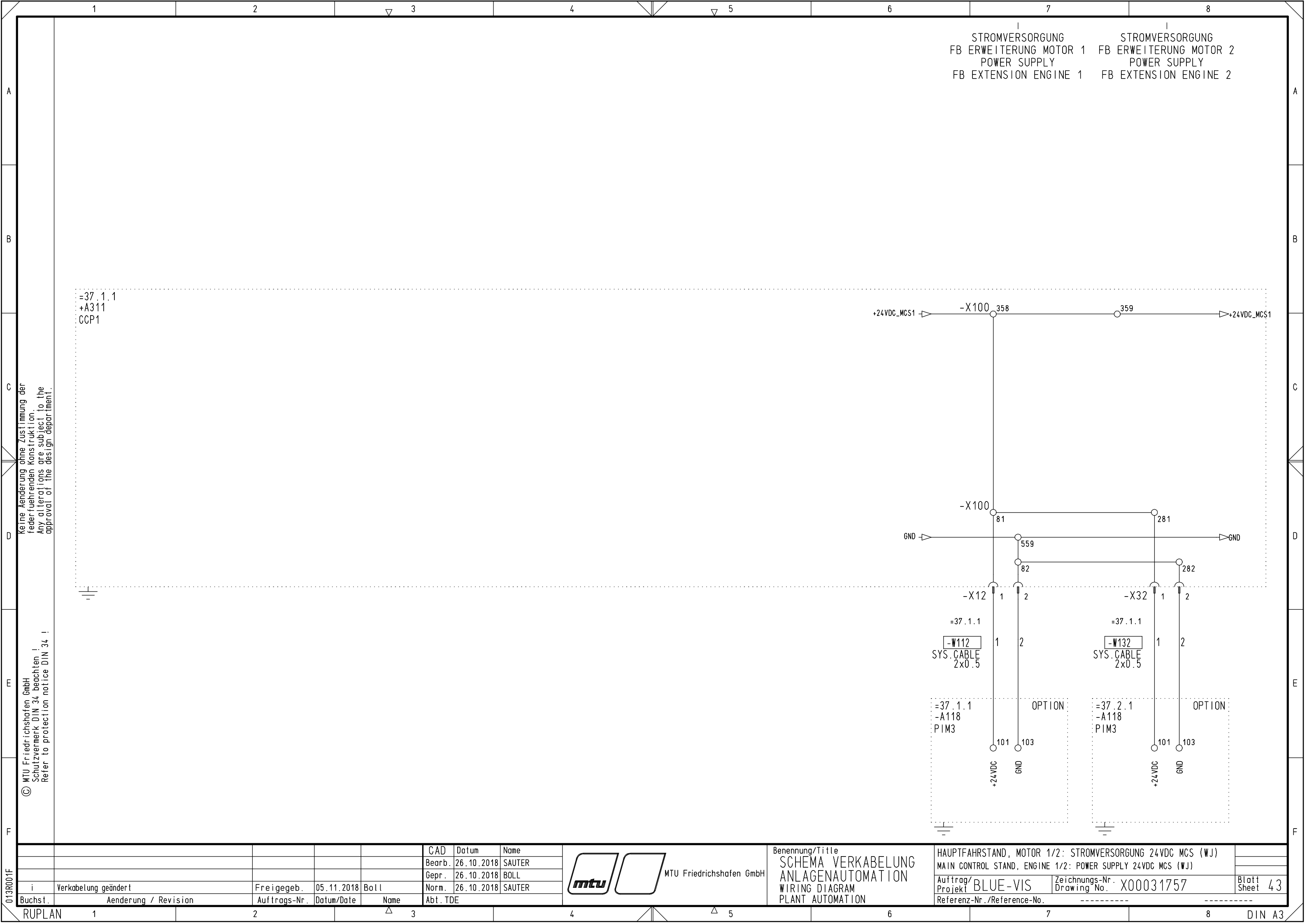
HAUPTFAHRSTAND, MOTOR 1/2: 24VDC EIN/AUS (CCP\_KEY\_GND)  
MAIN CONTROL STAND, ENGINE 1/2: 24VDC ON/OFF (CCP\_KEY\_GND)  
Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

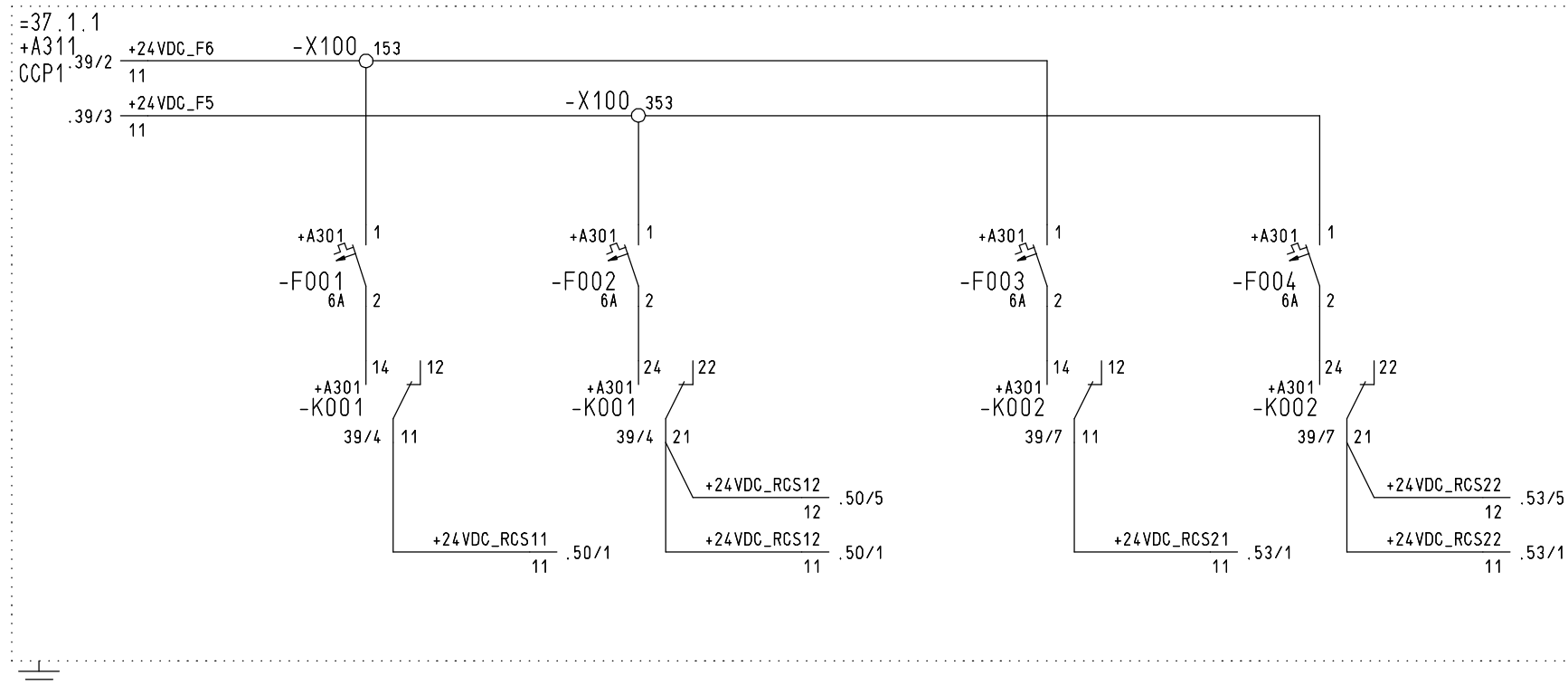
Blatt  
Sheet 41

DIN A3









Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

RUPLAN

| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |
|---------|----------------------|--------------|------------|------|----------|
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          |

| CAD    | Datum      | Name   |
|--------|------------|--------|
| Bearb. | 26.10.2018 | SAUTER |
| Gepr.  | 26.10.2018 | BOLL   |
| Norm.  | 26.10.2018 | SAUTER |

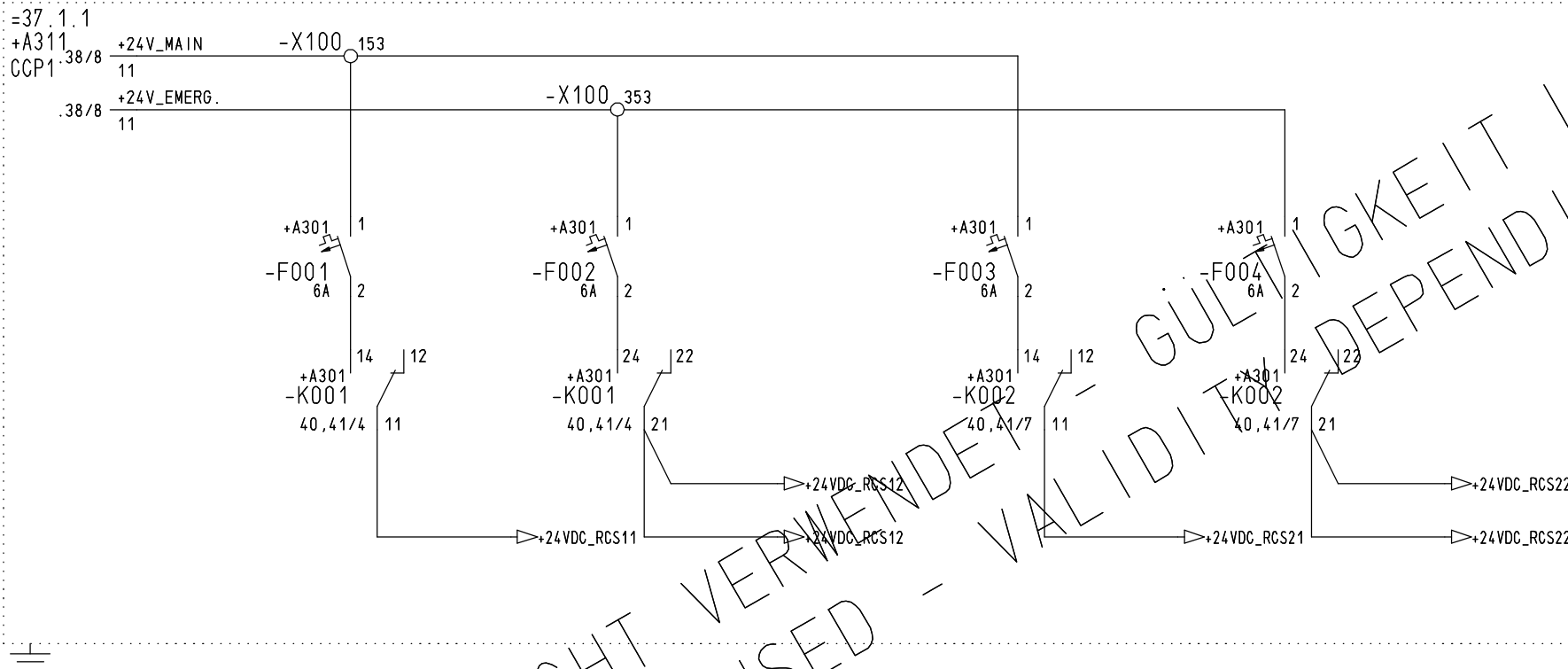


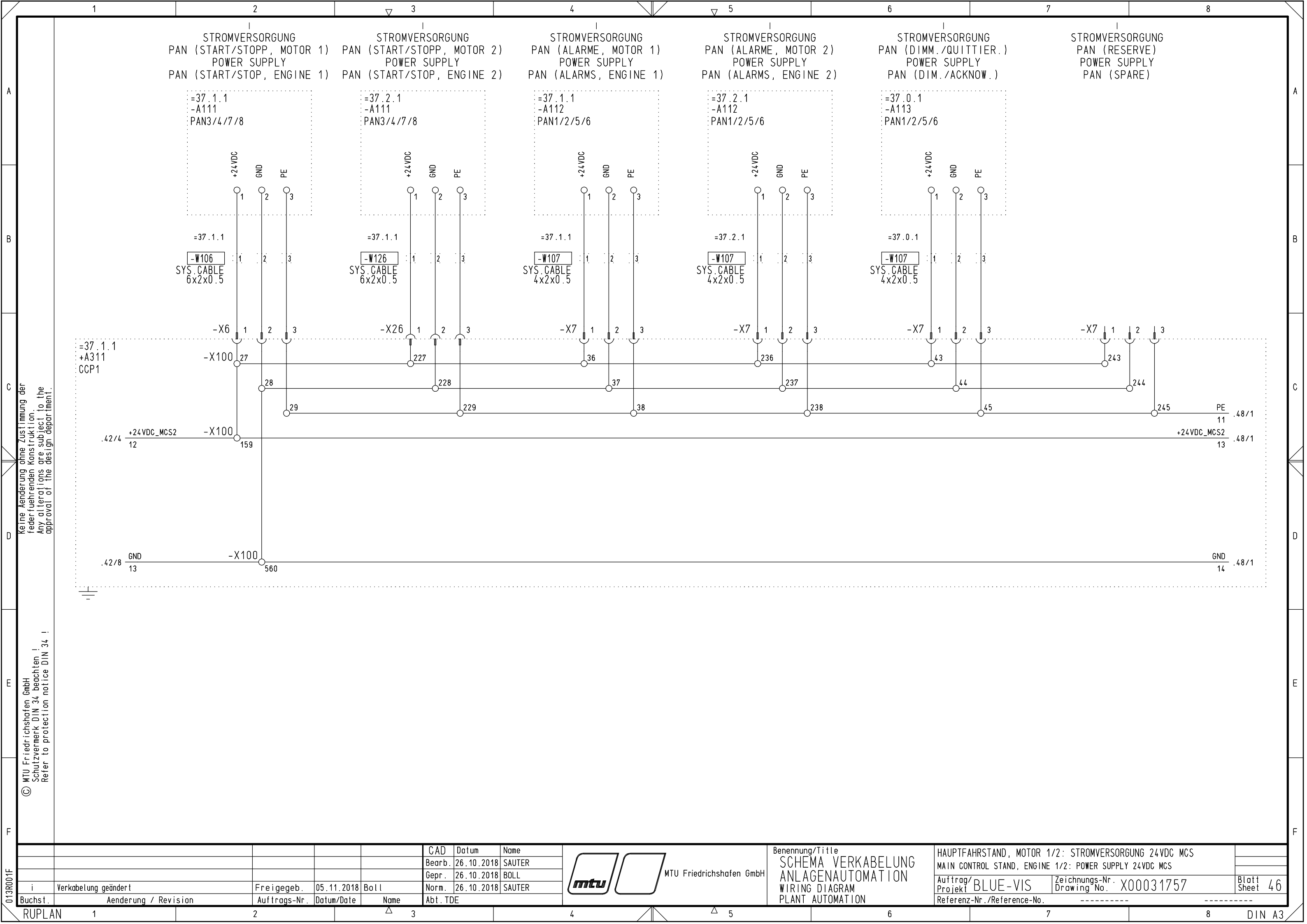
MTU Friedrichshafen GmbH

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
| Benennung/Title | SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |
|-----------------|-------------------------------------------------------------------------------|

|                                                                                                                |                              |                                         |                   |
|----------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------|-------------------|
| HAUPTFAHRSTAND, MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>MAIN CONTROL STAND, ENGINE 1/2: POWER SUPPLY 24VDC RCS | Auftrag/<br>Projekt BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 | Blatt<br>Sheet 45 |
| Referenz-Nr./Reference-No.                                                                                     | -----                        | -----                                   | -----             |

DIN A3





|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

|                            |                                                        |                                         |
|----------------------------|--------------------------------------------------------|-----------------------------------------|
| Benennung/Title            | HAUPTFAHRSTAND, MOTOR 1/2: STROMVERSORGUNG 24VDC MCS   |                                         |
|                            | MAIN CONTROL STAND, ENGINE 1/2: POWER SUPPLY 24VDC MCS |                                         |
| Auftrag/<br>Projekt        | BLUE-VIS                                               | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No. | -----                                                  | Blatt<br>Sheet 46                       |

STROMVERSORGUNG  
LEITSTANDSRECHNER 1  
POWER SUPPLY  
MANAG.COMPUTER UNIT 1

STROMVERSORGUNG  
LEITSTANDSRECHNER 2  
POWER SUPPLY  
MANAG.COMPUTER UNIT 2

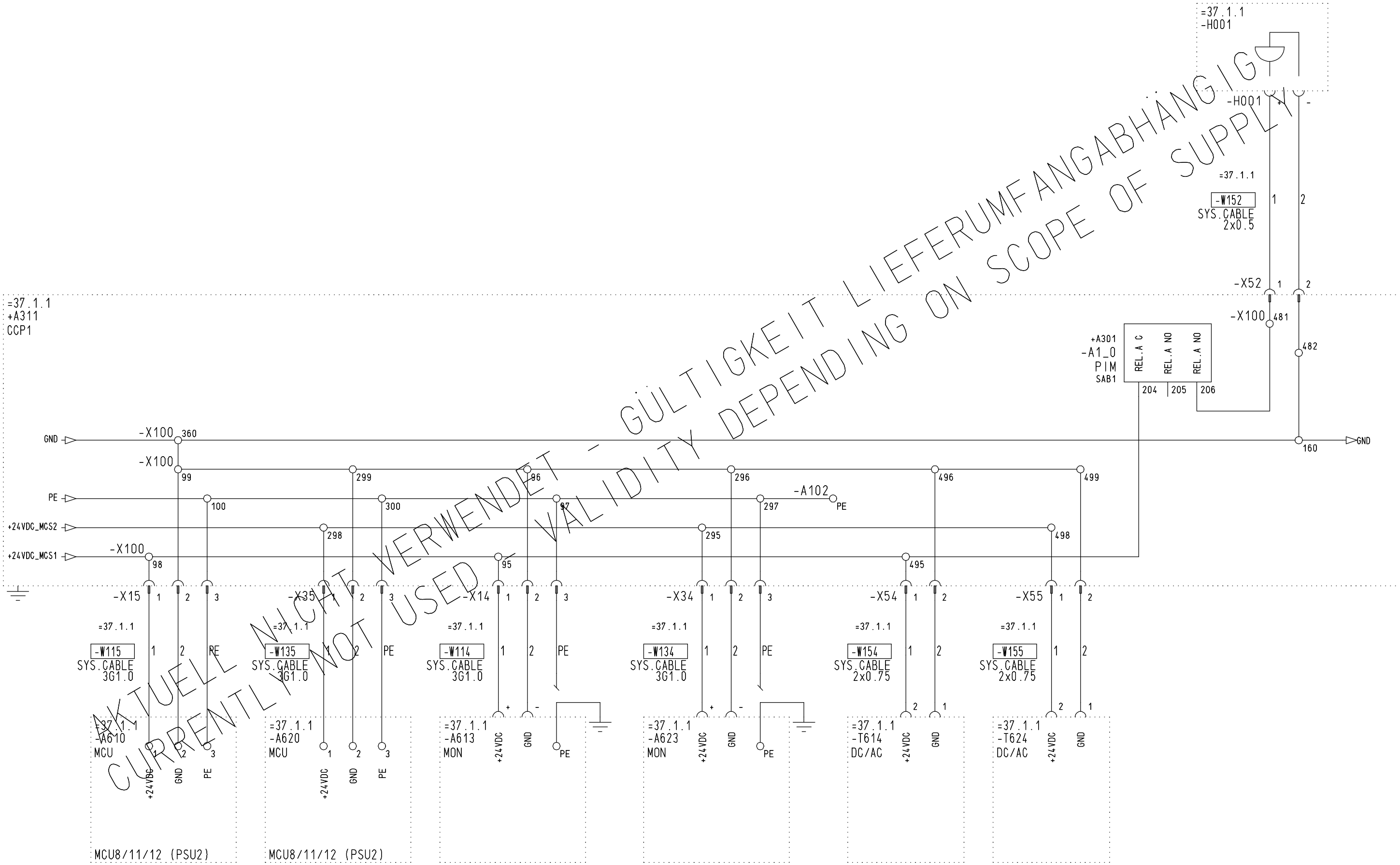
STROMVERSORGUNG  
BILDSCHIRM 1  
POWER SUPPLY  
MONITOR 1

STROMVERSORGUNG  
BILDSCHIRM 2  
POWER SUPPLY  
MONITOR 2

STROMVERSORGUNG  
DRUCKER 1  
POWER SUPPLY  
PRINTER 1

STROMVERSORGUNG  
DRUCKER 2  
POWER SUPPLY  
PRINTER 2

ALARM  
SUMMER  
ALARM  
BUZZER



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gedr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, STROMVERS. 24VDC, GRAFIKST. 1&2 (MCU8/11/12)  
MAIN CONTROL STAND, P.S.24VDC, GRAPHIC STAT.1&2 (MCU8/11/12)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 47

STROMVERSORGUNG  
LEITSTANDSRECHNER 1  
POWER SUPPLY  
MANAG.COMPUTER UNIT 1

STROMVERSORGUNG  
LEITSTANDSRECHNER 2  
POWER SUPPLY  
MANAG.COMPUTER UNIT 2

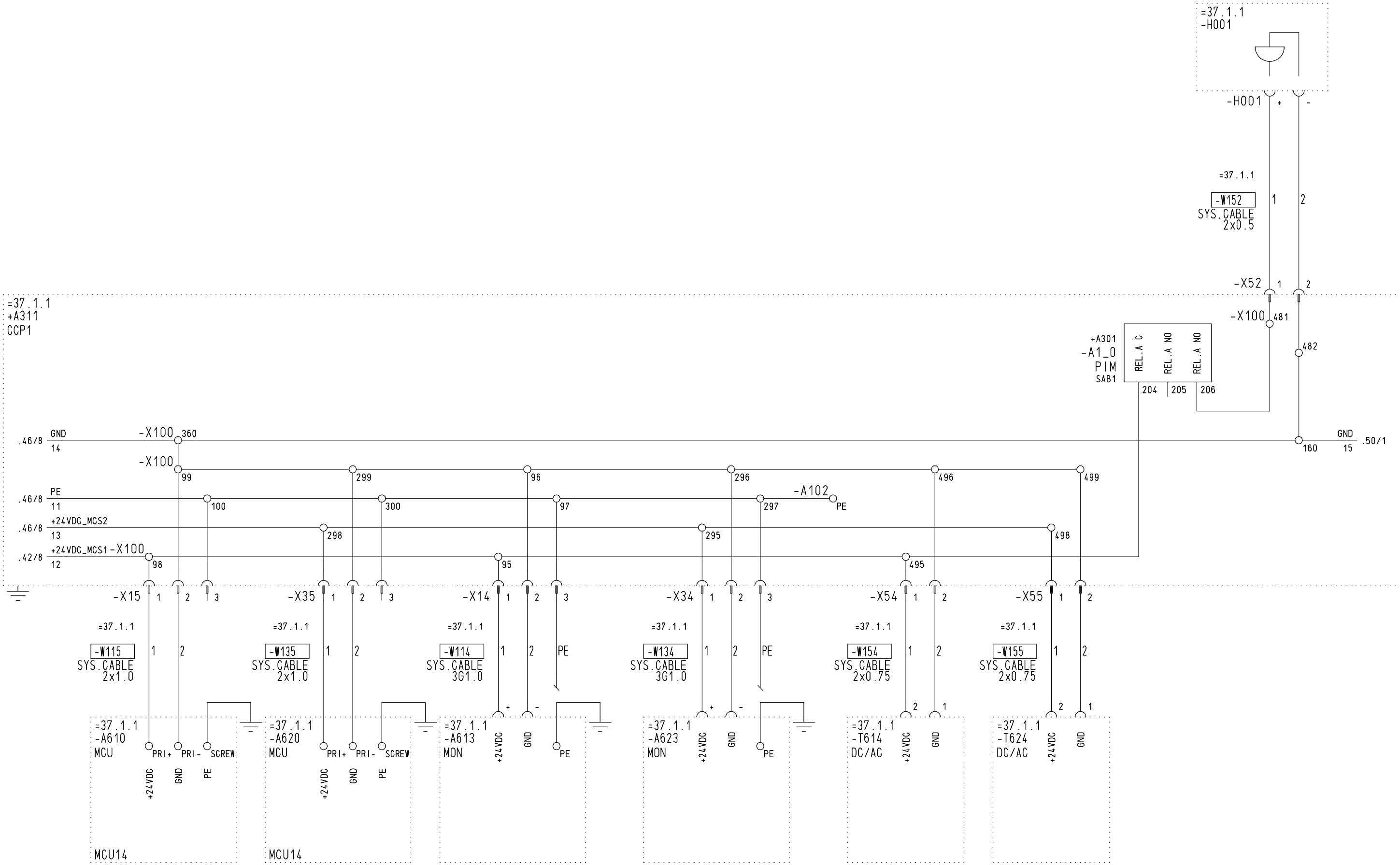
STROMVERSORGUNG  
BILDSCHIRM 1  
POWER SUPPLY  
MONITOR 1

STROMVERSORGUNG  
BILDSCHIRM 2  
POWER SUPPLY  
MONITOR 2

STROMVERSORGUNG  
DRUCKER 1  
POWER SUPPLY  
PRINTER 1

STROMVERSORGUNG  
DRUCKER 2  
POWER SUPPLY  
PRINTER 2

ALARM  
SUMMER  
ALARM  
BUZZER



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

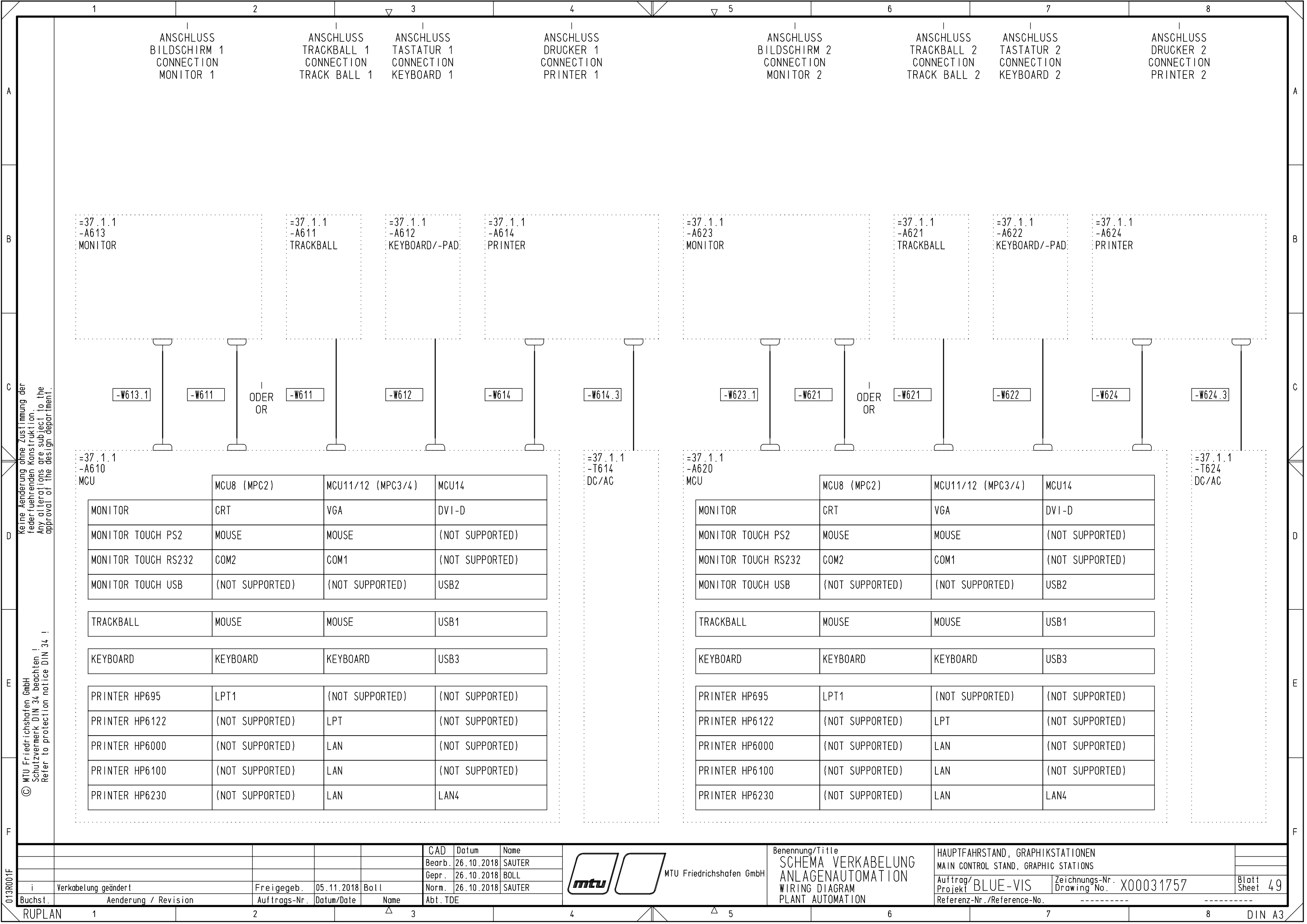
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                         |          |                            |           |                |
|---------------------------------------------------------|----------|----------------------------|-----------|----------------|
| HAUPTFAHRSTAND, STROMVERS. 24VDC, GRAFIKST. 1&2 (MCU14) |          |                            |           |                |
| MAIN CONTROL STAND, P.S.24VDC, GRAPHIC STAT.1&2 (MCU14) |          |                            |           |                |
| Auftrag/Projekt                                         | BLUE-VIS | Zeichnungs-Nr. Drawing No. | X00031757 | Blatt Sheet 48 |
| Referenz-Nr./Reference-No.                              |          |                            |           |                |





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1: STROMVERSORGUNG 24VDC RCS (FPP)  
MAIN CONTROL STAND, ENGINE 1: POWER SUPPLY 24VDC RCS (FPP)

Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. Drawing No. X00031757

Referenz-Nr./Reference-No. Blatt Sheet 50



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

|                            |  |                                                            |                              |               |
|----------------------------|--|------------------------------------------------------------|------------------------------|---------------|
| Benennung/Title            |  | HAUPTFAHRSTAND, MOTOR 1: STROMVERSORGUNG 24VDC RCS (CPP)   |                              |               |
| SCHEMA VERKABELUNG         |  | MAIN CONTROL STAND, ENGINE 1: POWER SUPPLY 24VDC RCS (CPP) |                              |               |
| ANLAGENAUTOMATION          |  |                                                            |                              |               |
| WIRING DIAGRAM             |  |                                                            |                              |               |
| PLANT AUTOMATION           |  |                                                            |                              |               |
| Auftrag/Projekt            |  | BLUE-VIS                                                   | Zeichnungs-Nr. / Drawing No. | Blatt / Sheet |
| Referenz-Nr./Reference-No. |  | X00031757                                                  |                              | 51            |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                      |          |                                         |
|----------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| HAUPTFAHRSTAND, MOTOR 1: STROMVERSORGUNG 24VDC RCS (WJ)<br>MAIN CONTROL STAND, ENGINE 1: POWER SUPPLY 24VDC RCS (WJ) |          |                                         |
| Auftrag/<br>Projekt                                                                                                  | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                                           |          | Blatt<br>Sheet 52                       |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (FPP)  
MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (FPP)

Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. Drawing No. X00031757

Referenz-Nr./Reference-No. -----

Blatt Sheet 53

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                            |          |                               |                |
|------------------------------------------------------------|----------|-------------------------------|----------------|
| HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (CPP)   |          |                               |                |
| MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (CPP) |          |                               |                |
| Auftrag/<br>Projekt                                        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | Blatt<br>Sheet |
| Referenz-Nr./Reference-No.                                 |          | X00031757                     | 54             |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |

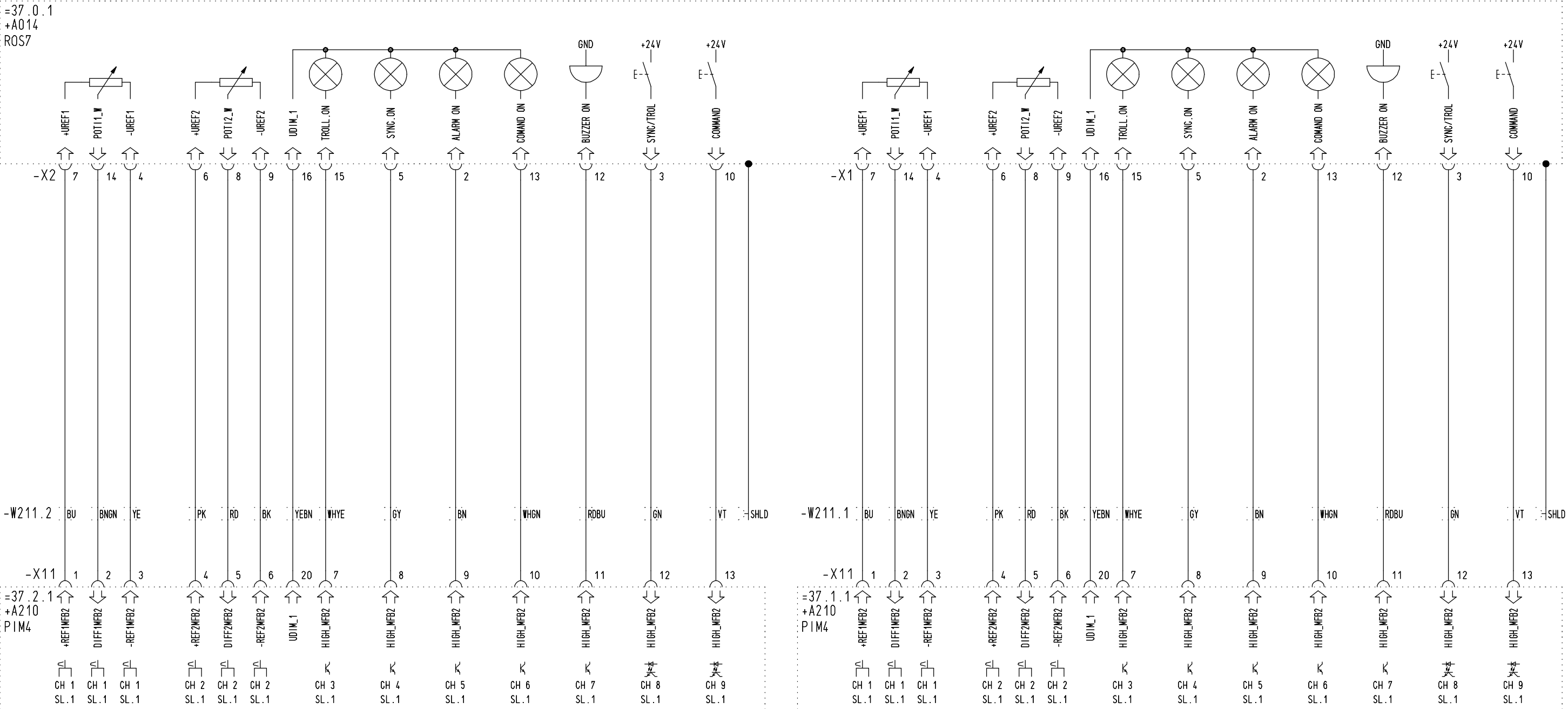


MTU Friedrichshafen GmbH

|                    |  |                                                           |  |                          |
|--------------------|--|-----------------------------------------------------------|--|--------------------------|
| Benennung/Title    |  | HAUPTFAHRSTAND, MOTOR 2: STROMVERSORGUNG 24VDC RCS (WJ)   |  |                          |
| SCHEMA VERKABELUNG |  | MAIN CONTROL STAND, ENGINE 2: POWER SUPPLY 24VDC RCS (WJ) |  |                          |
| ANLAGENAUTOMATION  |  |                                                           |  |                          |
| WIRING DIAGRAM     |  | Auftrag/Projekt BLUE-VIS                                  |  | Zeichnungs-Nr. X00031757 |
| PLANT AUTOMATION   |  | Referenz-Nr./Reference-No.                                |  | Blatt Sheet 55           |

DREHZAHLVORGABE  
MOTOR 2  
SPEED DEMAND  
ENGINE 2

DREHZAHLVORGABE  
MOTOR 1  
SPEED DEMAND  
ENGINE 1



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

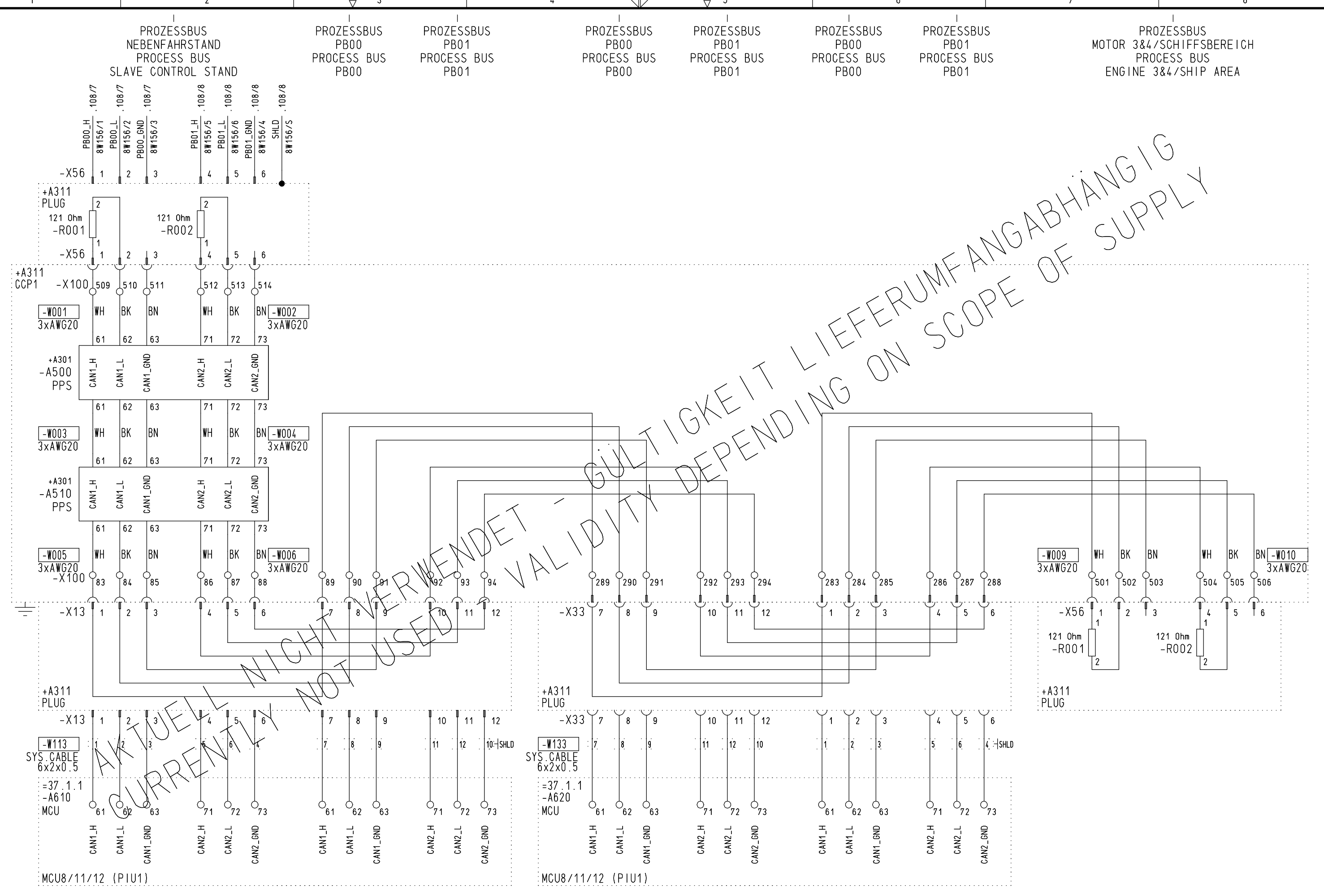
|                                                                                                         |          |                               |           |
|---------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| HAUPTFAHRSTAND, MOTOR 1/2: BEDIENSTATION ROS7<br>MAIN CONTROL STAND, ENGINE 1/2: OPERATING STATION ROS7 |          |                               |           |
| Auftrag/<br>Projekt                                                                                     | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                              |          | -----                         |           |
| RUPLAN                                                                                                  |          | DIN A3                        |           |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F



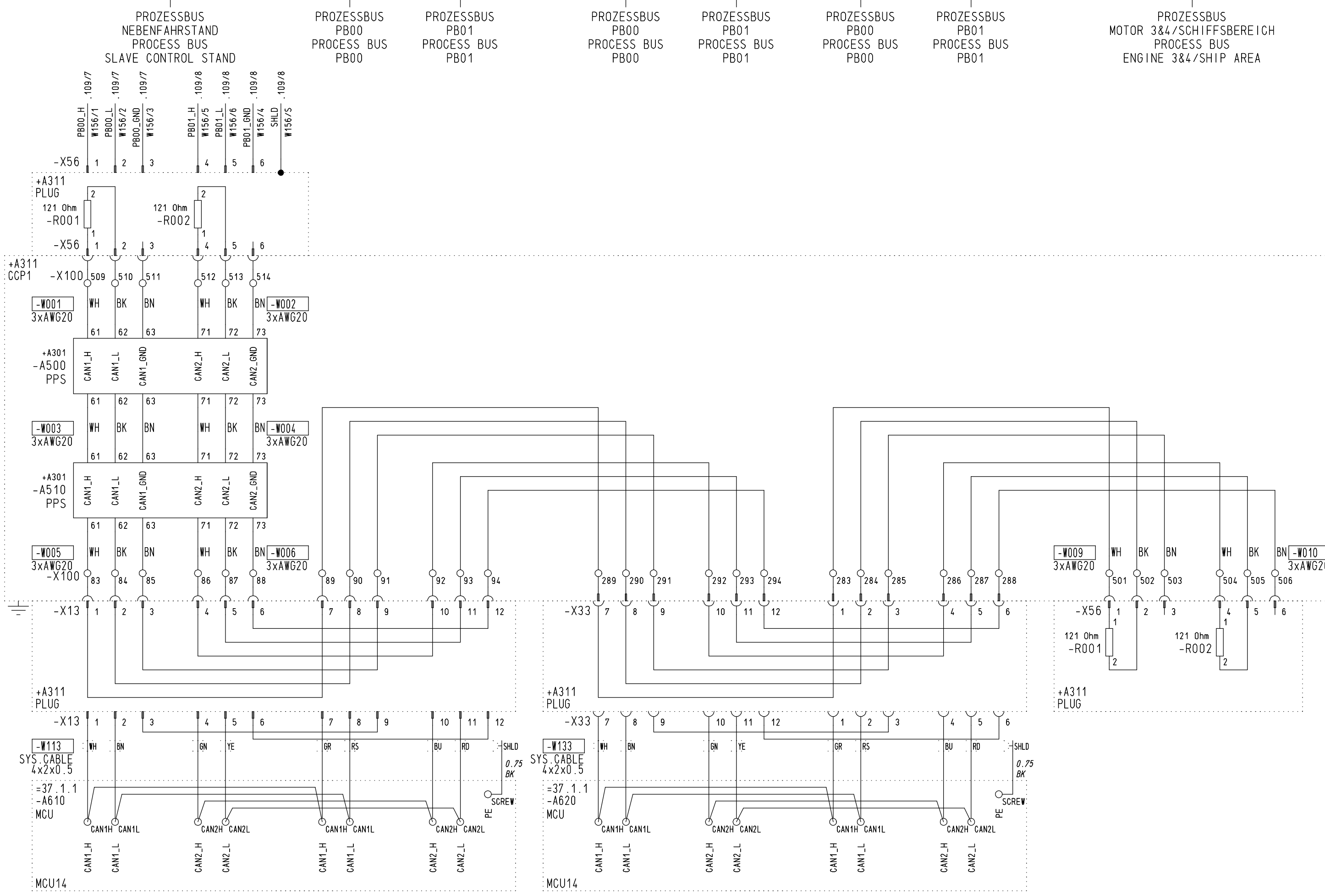
|         |                      |  |              |            |        |            |        |                                                                                                                |                                                                                                  |                                              |  |          |                               |           |                |    |
|---------|----------------------|--|--------------|------------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------|--|----------|-------------------------------|-----------|----------------|----|
|         |                      |  |              |            | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, PROZESSBUS (MCU8/11/12)      |  |          |                               |           |                |    |
|         |                      |  |              |            | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | MAIN CONTROL STAND, PROCESS BUS (MCU8/11/12) |  |          |                               |           |                |    |
|         |                      |  |              |            | Gedr.  | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                              |  |          |                               |           |                |    |
|         |                      |  |              |            | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                              |  |          |                               |           |                |    |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll   |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                          |  | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 57 |
| Buchst. | Änderung / Revision  |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE   |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                   |  |          |                               |           |                |    |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

RUPLAN



| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |
|---------|----------------------|--------------|------------|------|----------|
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          |

| CAD    | Datum      | Name   |
|--------|------------|--------|
| Bearb. | 26.10.2018 | SAUTER |
| Gedr.  | 26.10.2018 | BOLL   |
| Norm.  | 26.10.2018 | SAUTER |

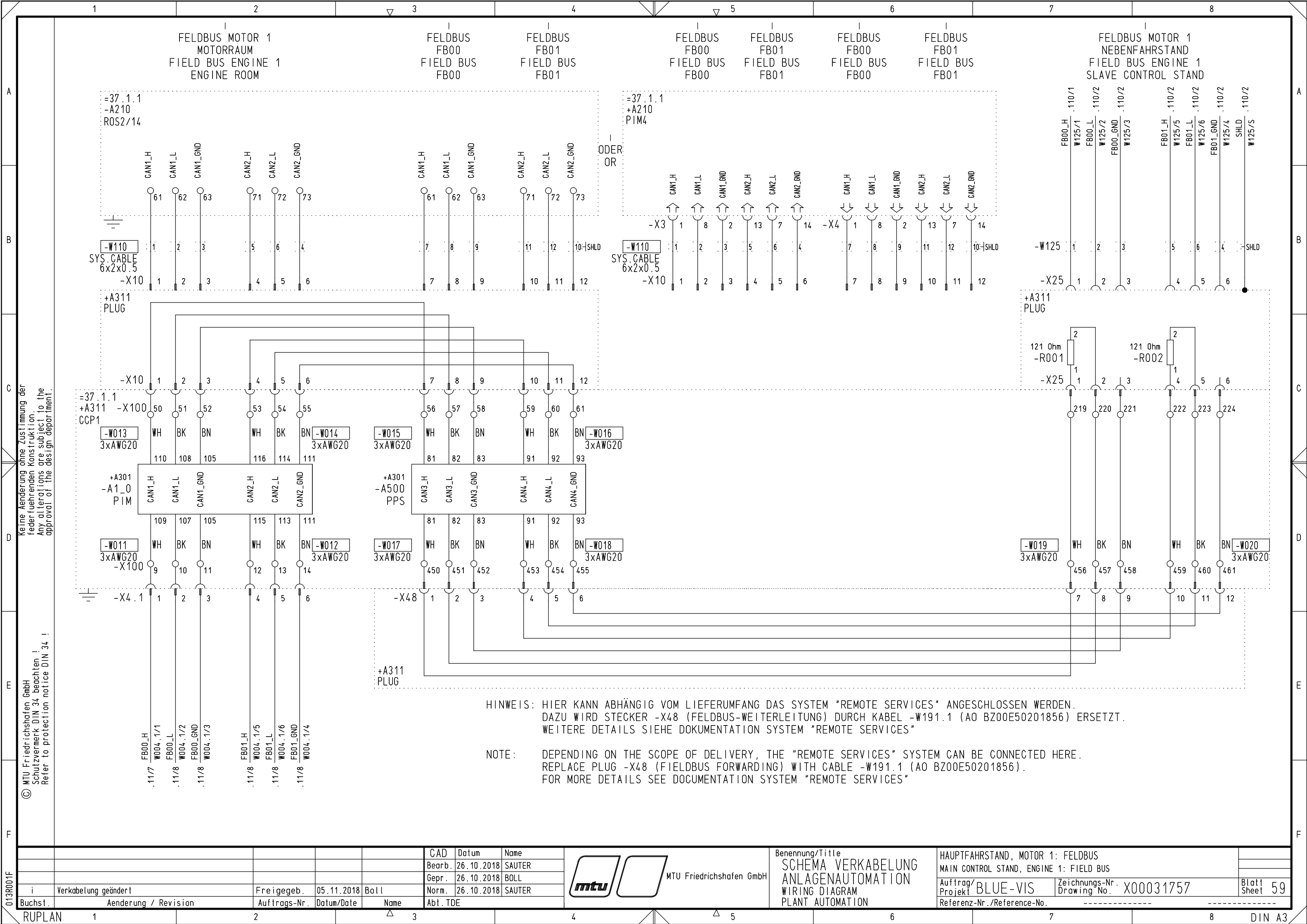


MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                               |                                         |                   |
|-------------------------------------------------------------------------------|-----------------------------------------|-------------------|
| HAUPTFAHRSTAND, PROZESSBUS (MCU14)<br>MAIN CONTROL STAND, PROCESS BUS (MCU14) |                                         | Blatt<br>Sheet 58 |
| Auftrag/<br>Projekt BLUE-VIS                                                  | Zeichnungs-Nr.<br>Drawing No. X00031757 |                   |
| Referenz-Nr./Reference-No.                                                    |                                         |                   |







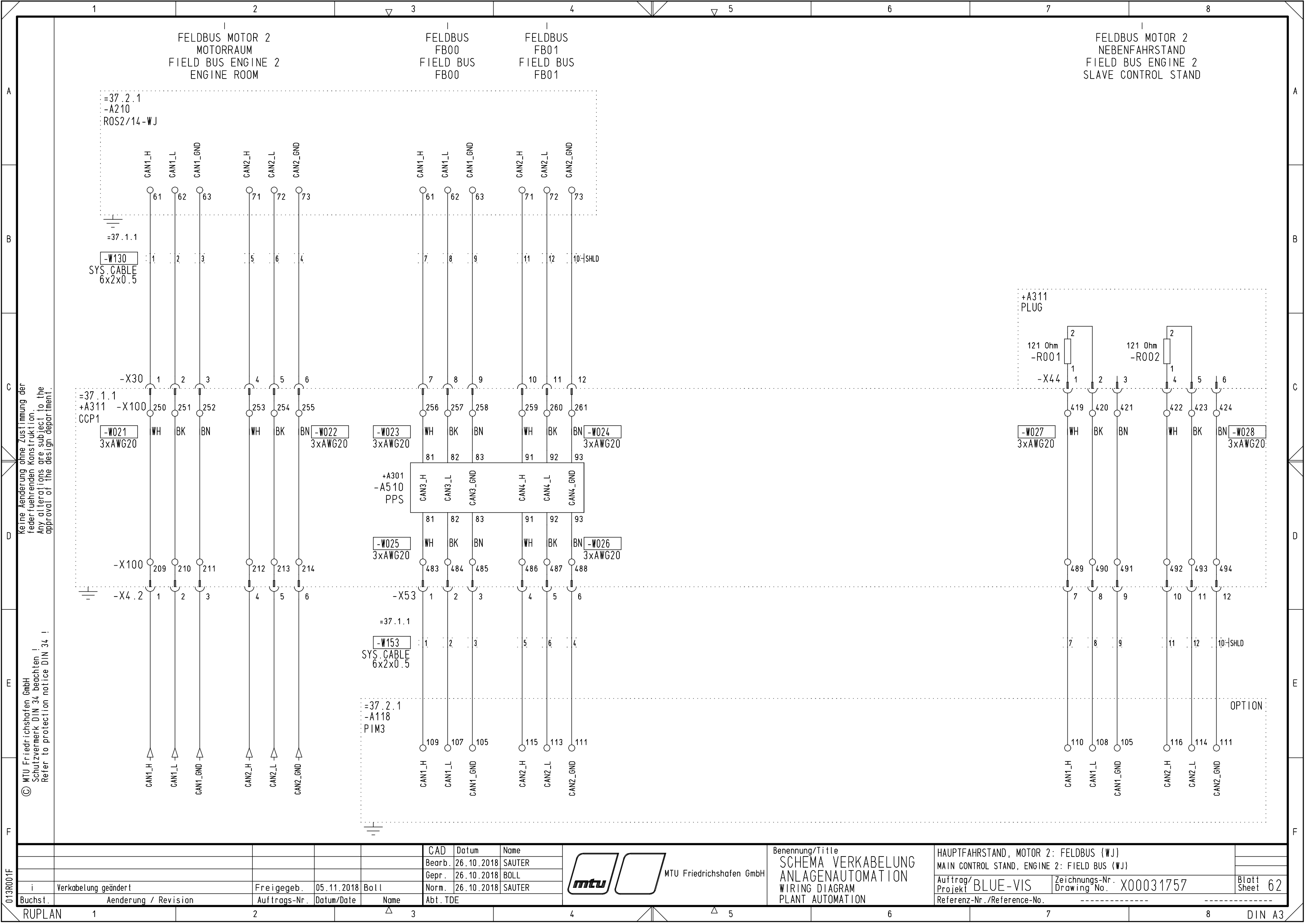
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



|                                                                                                  |          |                                                                             |           |                |
|--------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------|-----------|----------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | HAUPTFAHRSTAND, MOTOR 2: FELDBUS<br>MAIN CONTROL STAND, ENGINE 2: FIELD BUS |           |                |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                               | X00031757 | Blatt<br>Sheet |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                       |           |                |



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 2: FELDBUS (WJ)  
MAIN CONTROL STAND, ENGINE 2: FIELD BUS (WJ)

Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt Sheet: 62

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: KOORDINATIONSBUS RCS-5FPP/CPP/WJ  
MAIN CTR.STAND, ENGINE 1/2: COORDINATION BUS RCS-5FPP/CPP/WJ  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 63

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: ANSCHLUSS HANDSTEUERGERÄT ROS8  
MAIN CTR. STAND, ENGINE 1/2:CONNECTING PORT.REMOTE CTR. ROS8

Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr./Drawing No. X00031757

Referenz-Nr./Reference-No. Blatt Sheet 64



FELDBUS  
CB00  
FIELD BUS  
CB00

STROMVERSORGUNG  
NICHT VERWENDET  
POWER SUPPLY  
NOT USED

FELDBUS  
CB00  
FIELD BUS  
CB00

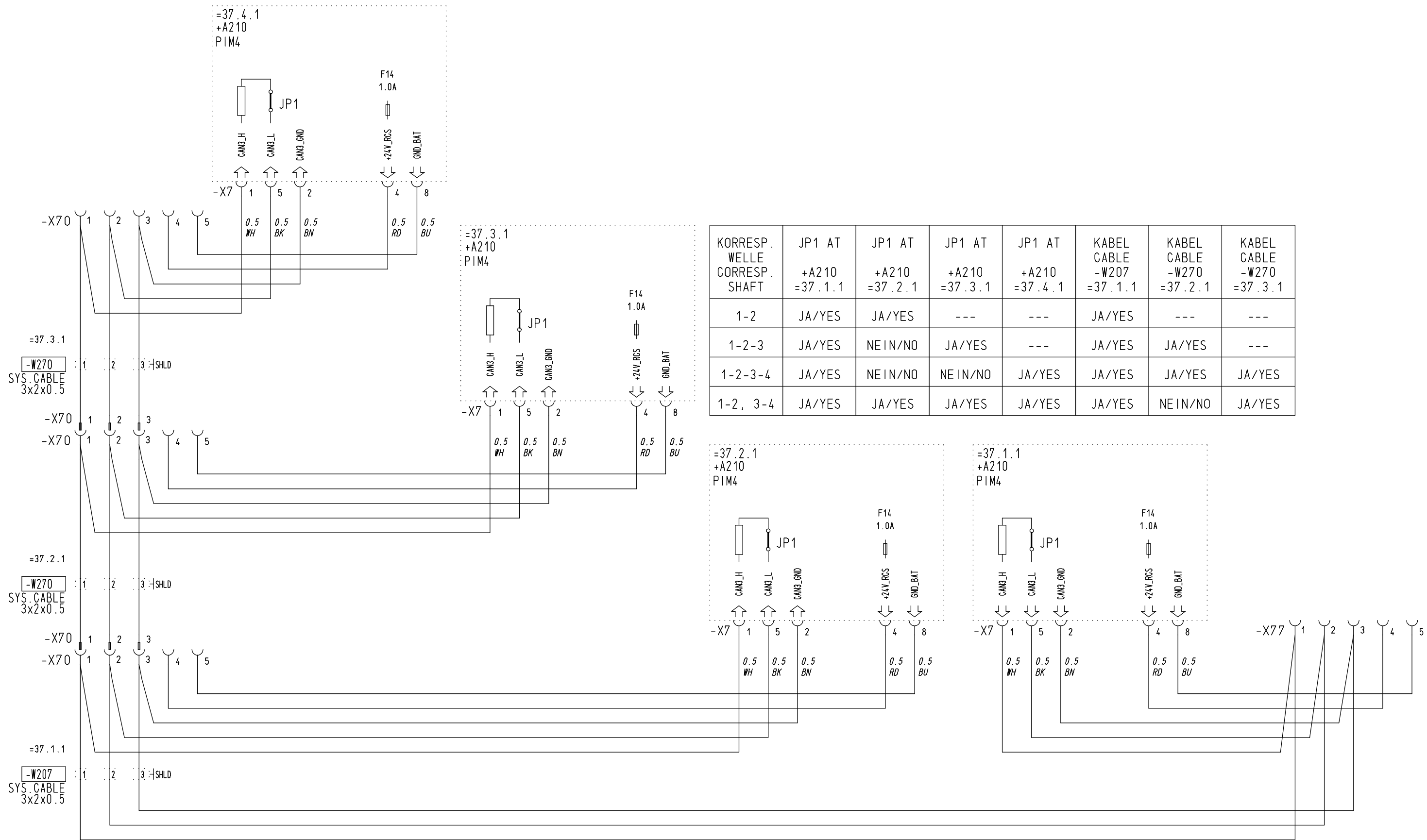
STROMVERSORGUNG  
NICHT VERWENDET  
POWER SUPPLY  
NOT USED

FELDBUS  
CB00  
FIELD BUS  
CB00

STROMVERSORGUNG  
NICHT VERWENDET  
POWER SUPPLY  
NOT USED

FELDBUS  
CB00  
FIELD BUS  
CB00

STROMVERSORGUNG  
HANDSTEUERGERÄT  
POWER SUPPLY  
PORTABLE REMOTE CTR.

[illegible]

|         |                      |              |            |      |        |            |        |
|---------|----------------------|--------------|------------|------|--------|------------|--------|
|         |                      |              |            |      | CAD    | Datum      | Name   |
|         |                      |              |            |      | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.  | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt.   | TDE        |        |



MTU Friedrichshafen GmbH

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
| Benennung/Title | SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |
|-----------------|-------------------------------------------------------------------------------|

|                                                            |
|------------------------------------------------------------|
| HAUPTFAHRSTAND, MOTOR 1-4: KOORDINATIONSBUS RCS-5DUO/TRIO  |
| MAIN CTR. STAND, ENGINE 1-4:COORDINATION BUS RCS-5DUO/TRIO |

|                     |          |                               |           |
|---------------------|----------|-------------------------------|-----------|
| Auftrag/<br>Projekt | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
|---------------------|----------|-------------------------------|-----------|

|       |    |
|-------|----|
| Blatt | 65 |
| Sheet |    |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                  |          |                               |                   |
|------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-------------------|
| HAUPTFAHRSTAND, HILFSFAHRSTÄNDE 1-4, KOORDINATIONSBUS<br>MAIN CTR. STAND, AUX. CTR. STANDS 1-4, COORDINATION BUS |          |                               |                   |
| Auftrag/<br>Projekt                                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757         |
| Referenz-Nr./Reference-No.                                                                                       |          | -----                         |                   |
|                                                                                                                  |          |                               | Blatt<br>Sheet 66 |



STROMVERSORGUNG  
E/A-MODULE  
POWER SUPPLY  
I/O MODULES

ANKER  
AB  
DROP  
ANCHOR

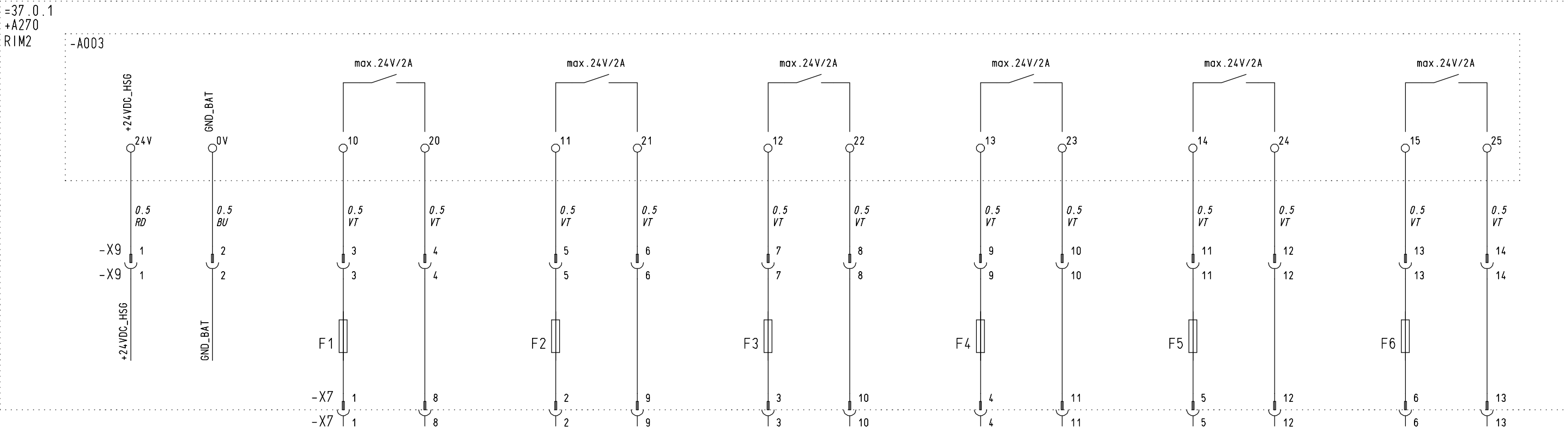
ANKER  
LICHTEN  
RAISE  
ANCHOR

BUGSTRAHLRUDER  
BACKBORD  
BOW THRUSTER  
PORT DIRECTION

BUGSTRAHLRUDER  
STEUERBORD  
BOW THRUSTER  
STARBOARD DIRECTION

HECKSTRAHLRUDER  
BACKBORD  
STERN THRUSTER  
PORT DIRECTION

HECKSTRAHLRUDER  
STEUERBORD  
STERN THRUSTER  
STARBOARD DIRECTION



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

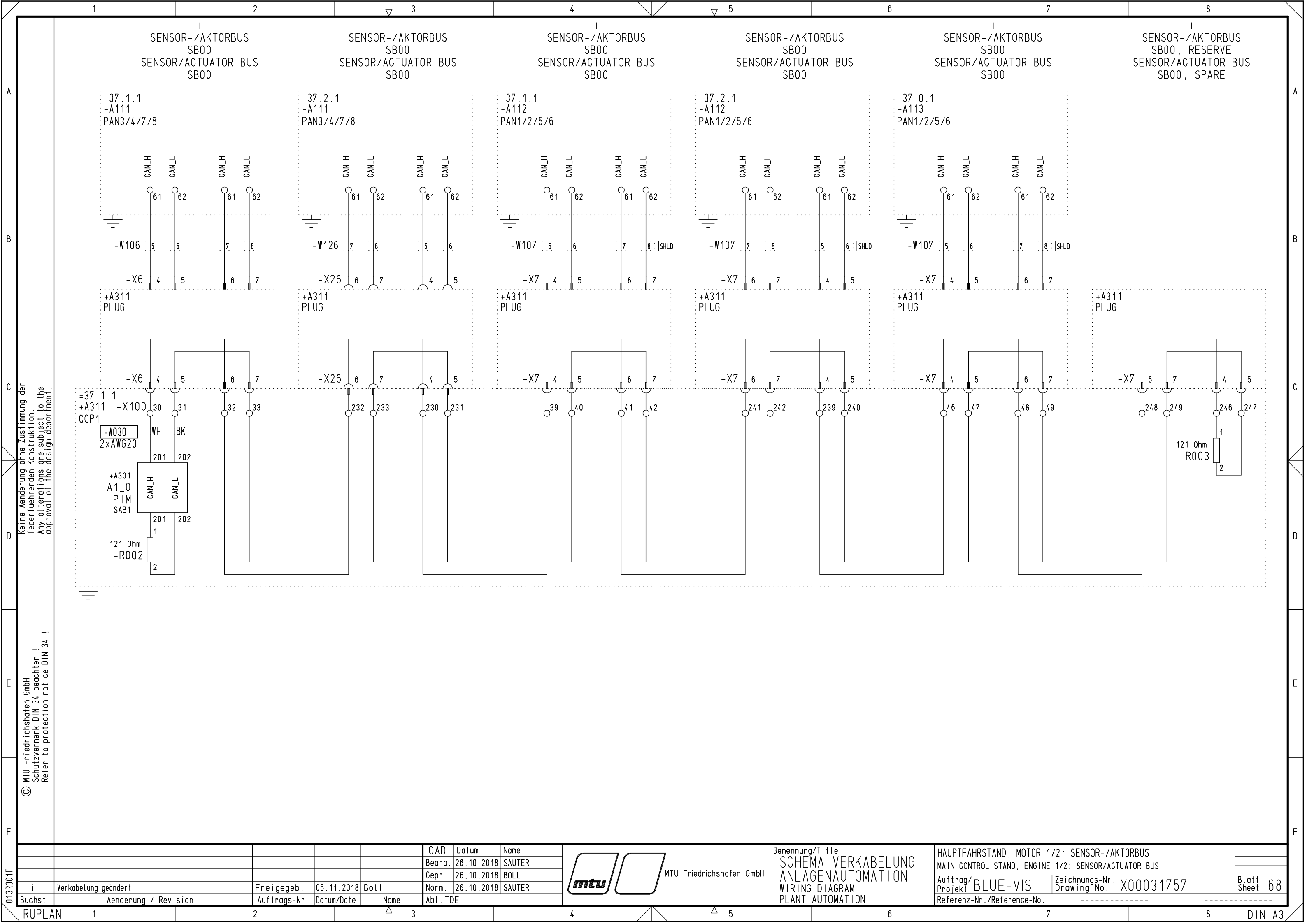
|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |

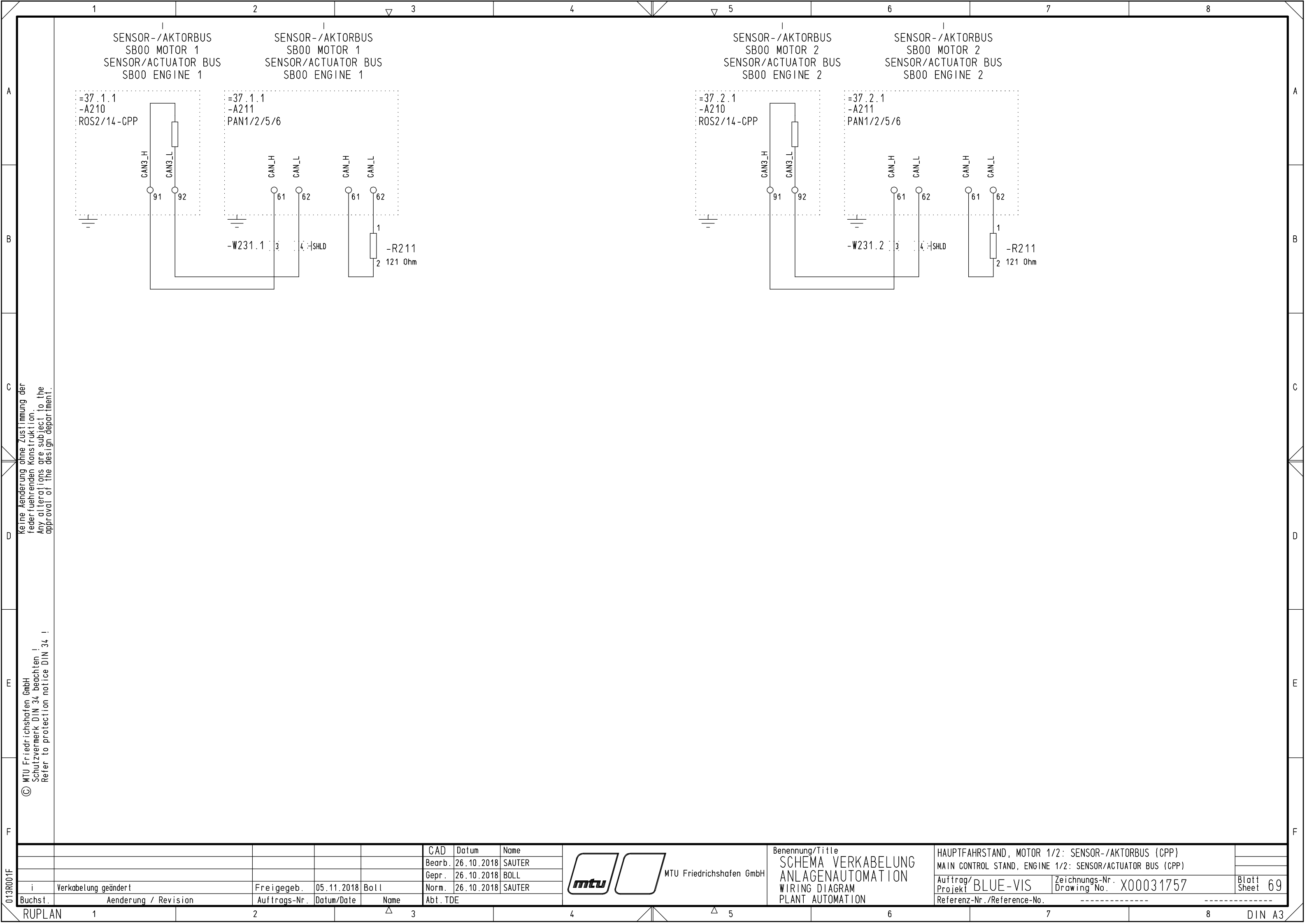


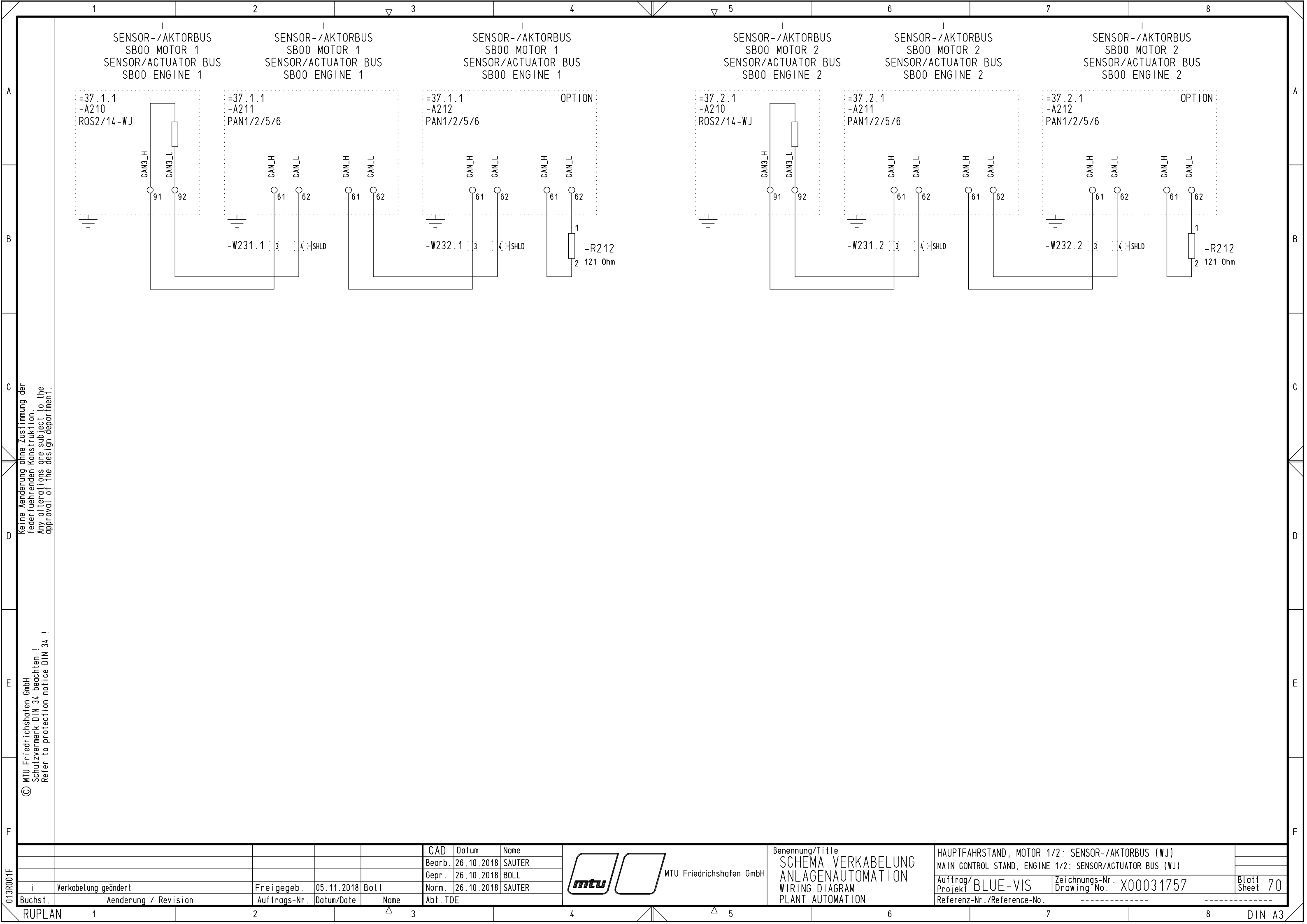
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                               |          |                               |           |
|---------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| HAUPTFAHRSTAND, ANSCHLUSSSCHNITTSTELLE (ANKER, ...)<br>MAIN CONTROL STAND, CONNECTION INTERFACE (ANCHOR, ...) |          |                               |           |
| Auftrag/<br>Projekt                                                                                           | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                    |          | -----                         |           |
| RUPLAN                                                                                                        |          | DIN A3                        |           |



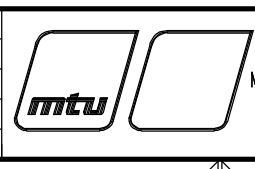




Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: SENSOR-/AKTORBUS (WJ)  
MAIN CONTROL STAND, ENGINE 1/2: SENSOR/ACTUATOR BUS (WJ)

Auftrag/  
Projekt

BLUE-VIS

Zeichnungs-Nr.  
Drawing No.

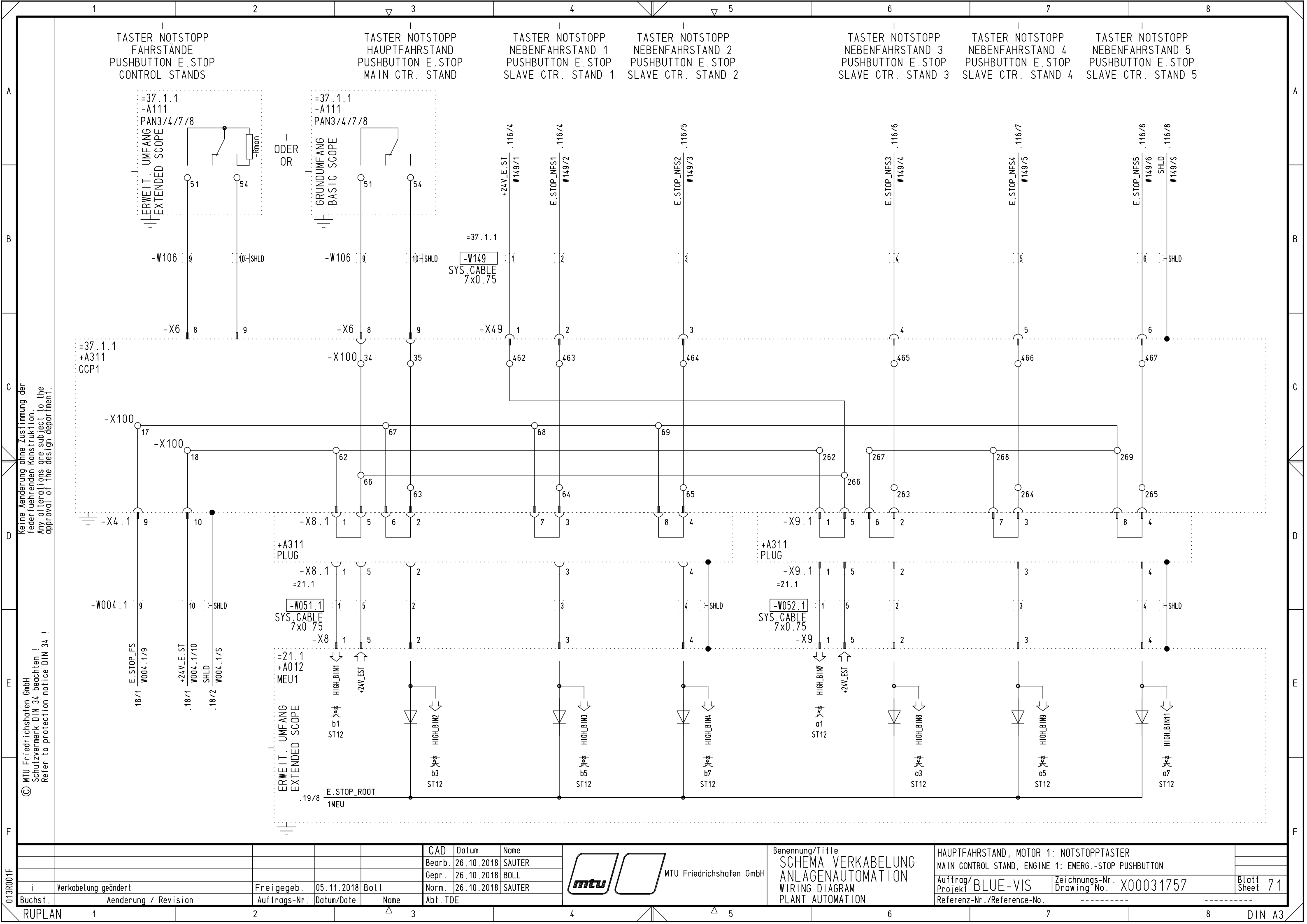
X00031757

Referenz-Nr./Reference-No.

-----

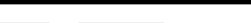
Blatt  
Sheet

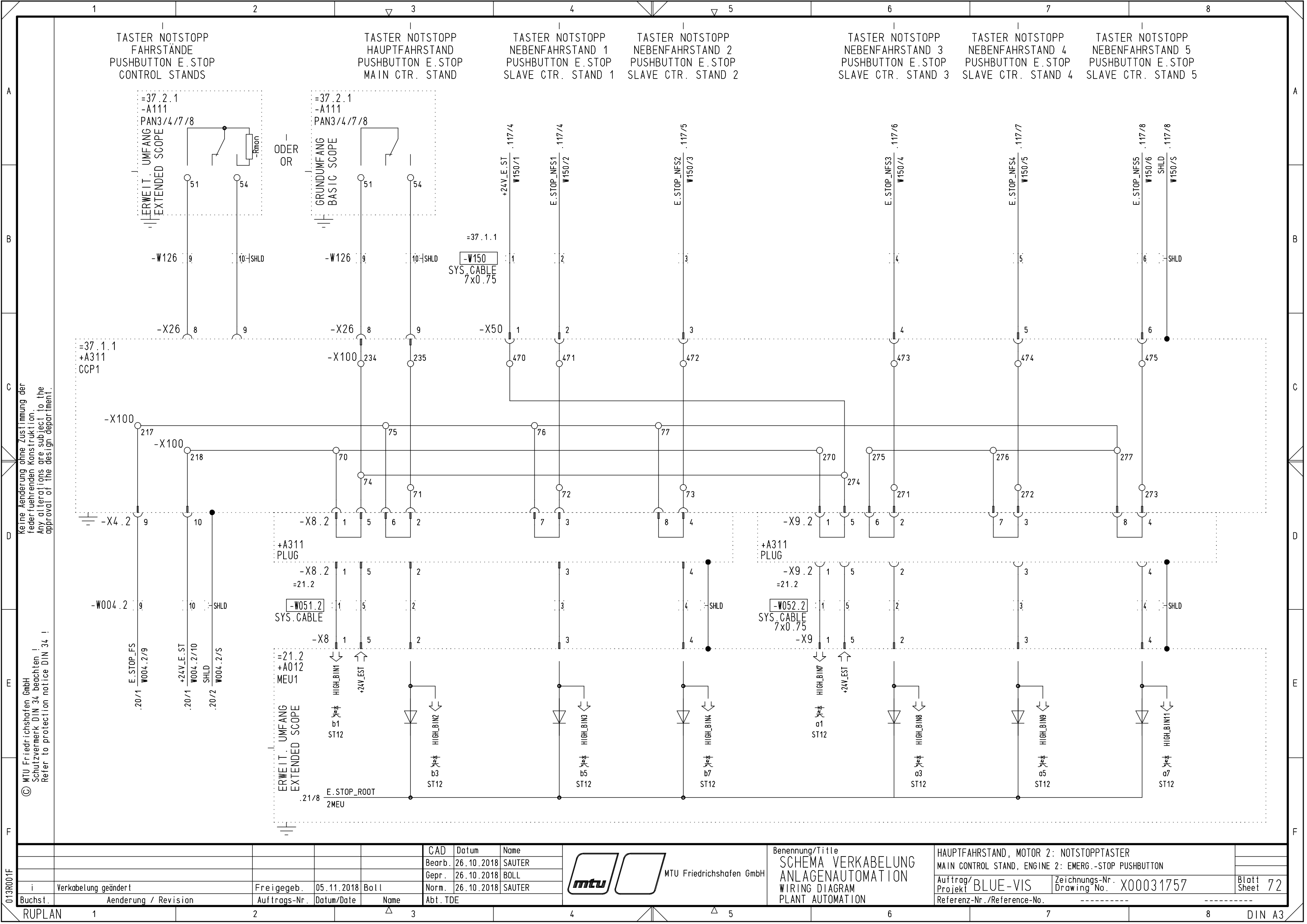
70



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                                                                 |       |  |                   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------|--|-------------------|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, MOTOR 1: NOTSTOPPTASTER<br>MAIN CONTROL STAND, ENGINE 1: EMERG.-STOP PUSHBUTTON |       |  |                   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                 |       |  |                   |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                                                                 |       |  |                   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                 |       |  |                   |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                | Referenz-Nr./Reference-No.                                                                       |                                                                                                 | ----- |  | Blatt<br>Sheet 71 |

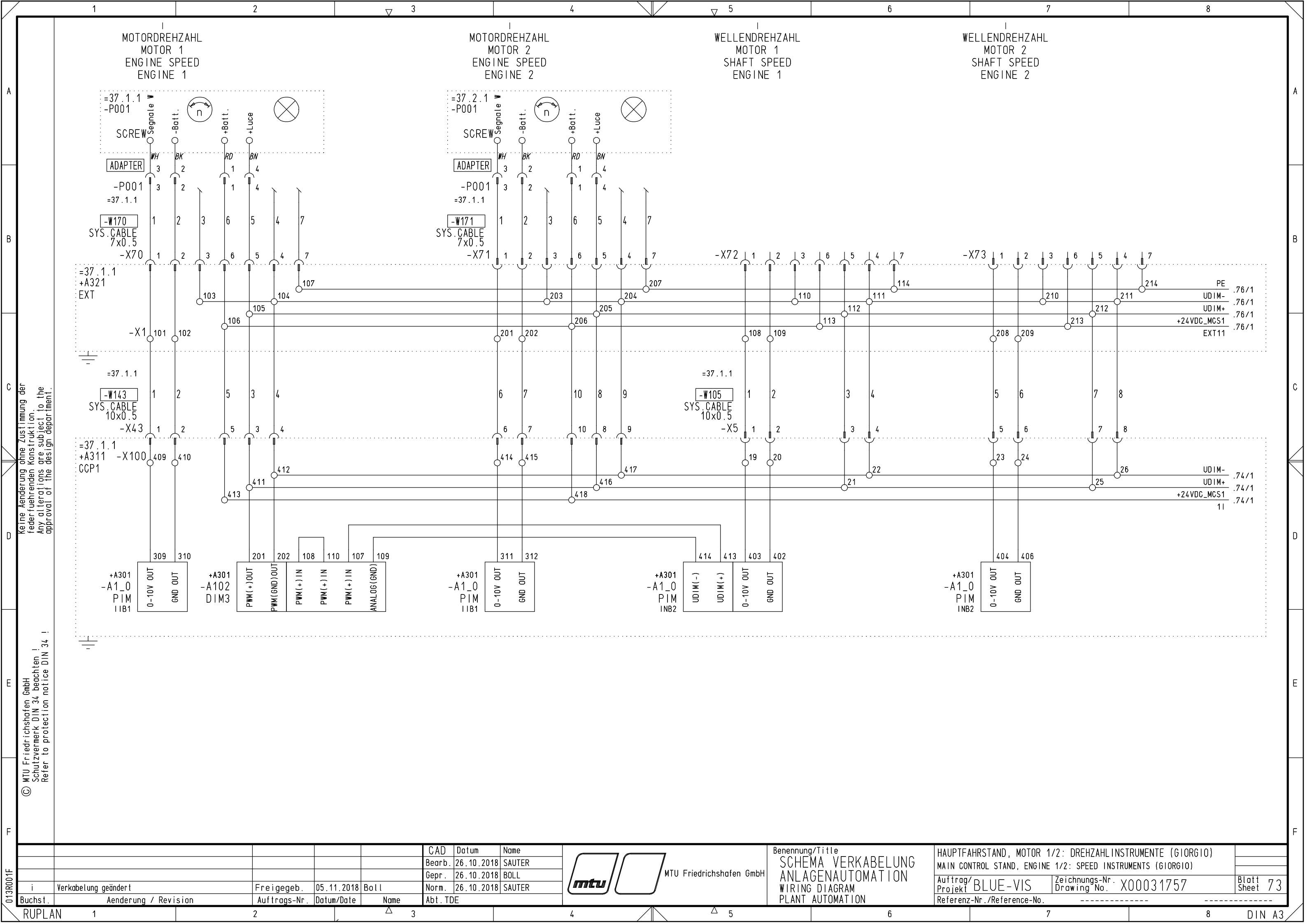


Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                                                                 |          |                               |           |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, MOTOR 2: NOTSTOPPTASTER<br>MAIN CONTROL STAND, ENGINE 2: EMERG.-STOP PUSHBUTTON |          |                               |           |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                                                                 |          |                               |           |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                                                                 |          |                               |           |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                                                             | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                                                      |          | -----                         |           |





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHLINSTRUMENTE (GIORGIO)  
MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (GIORGIO)

Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt/Sheet: 73

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

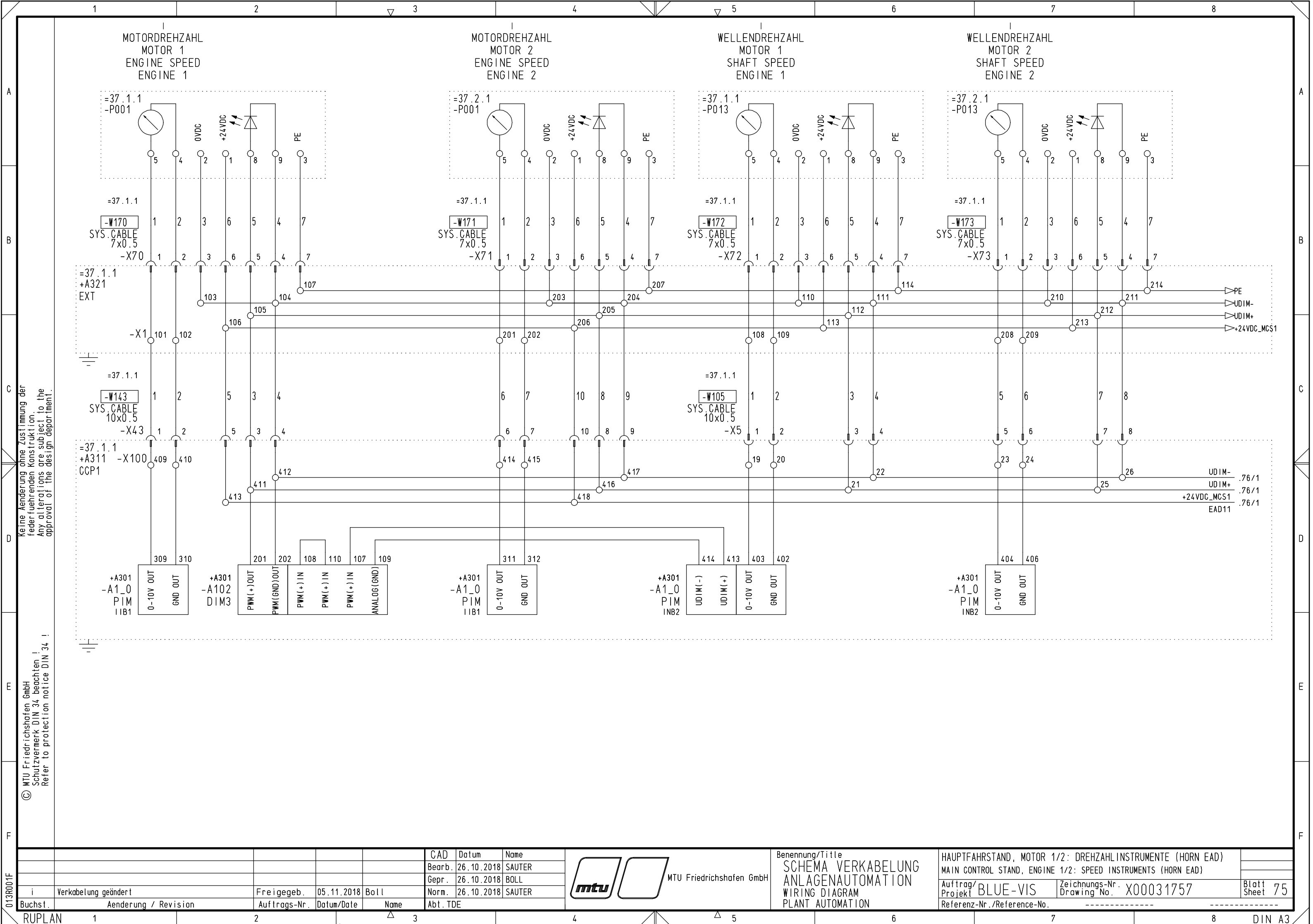
|         |                      |              |            |      |          |        |            |        |
|---------|----------------------|--------------|------------|------|----------|--------|------------|--------|
|         |                      |              |            |      |          | CAD    | Datum      | Name   |
|         |                      |              |            |      |          | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      |          | Gepr.  | 26.10.2018 | BOLL   |
|         |                      |              |            |      |          | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |        |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                             |          |                               |           |
|-----------------------------------------------------------------------------|----------|-------------------------------|-----------|
| HAUPTFAHRSTAND, MOTOR 1/2: RESERVE<br>MAIN CONTROL STAND, ENGINE 1/2: SPARE |          |                               |           |
| Auftrag/<br>Projekt                                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                  |          | -----                         |           |





| CAD    | Datum      | Name   |
|--------|------------|--------|
| Bearb. | 26.10.2018 | SAUTER |
| Gepr.  | 26.10.2018 | BOLL   |
| Norm.  | 26.10.2018 | SAUTER |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHLINSTRUMENTE (HORN EAD)  
MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (HORN EAD)

Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757

Referenz-Nr./Reference-No.:  
Blatt/Sheet: 75

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



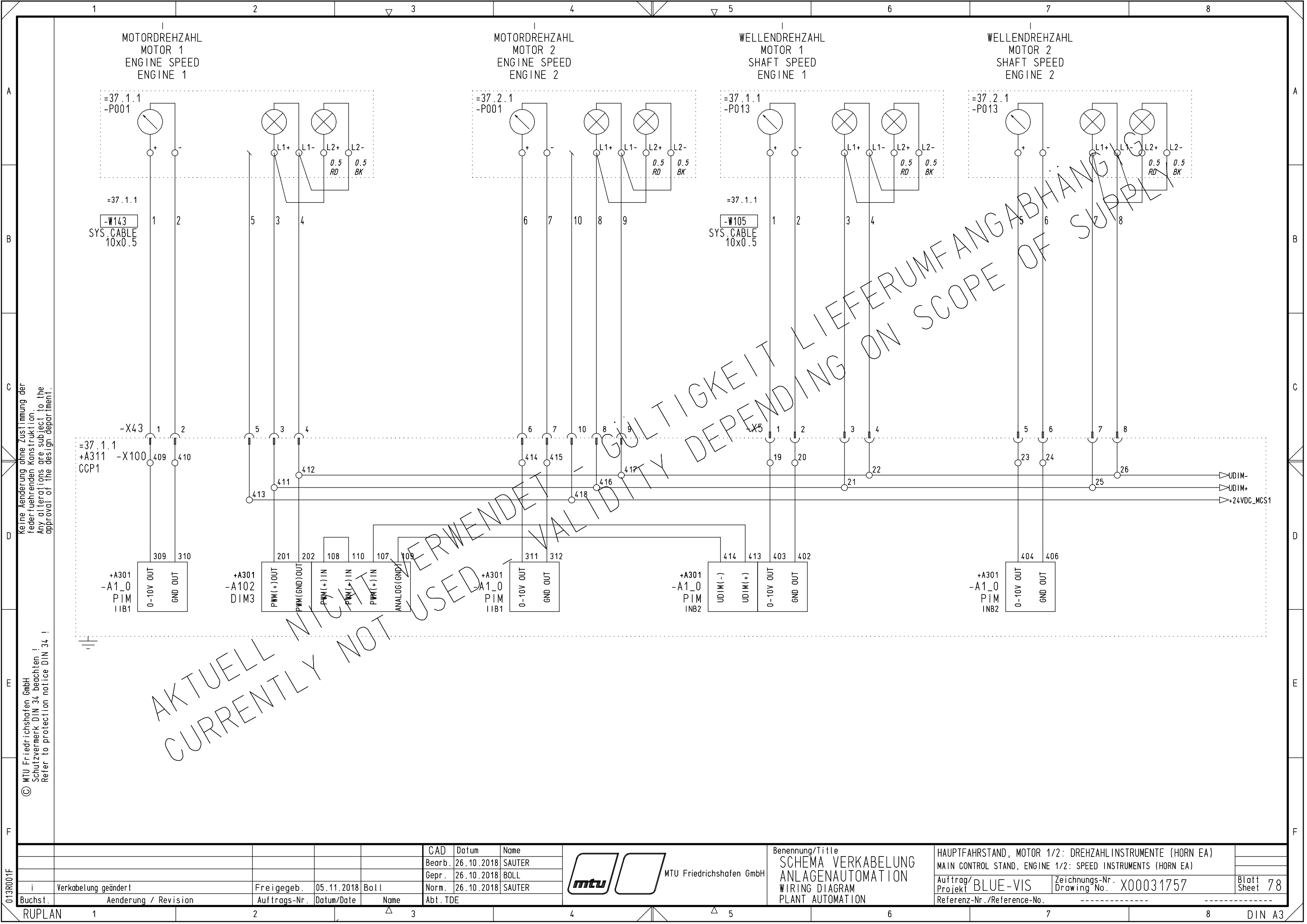
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

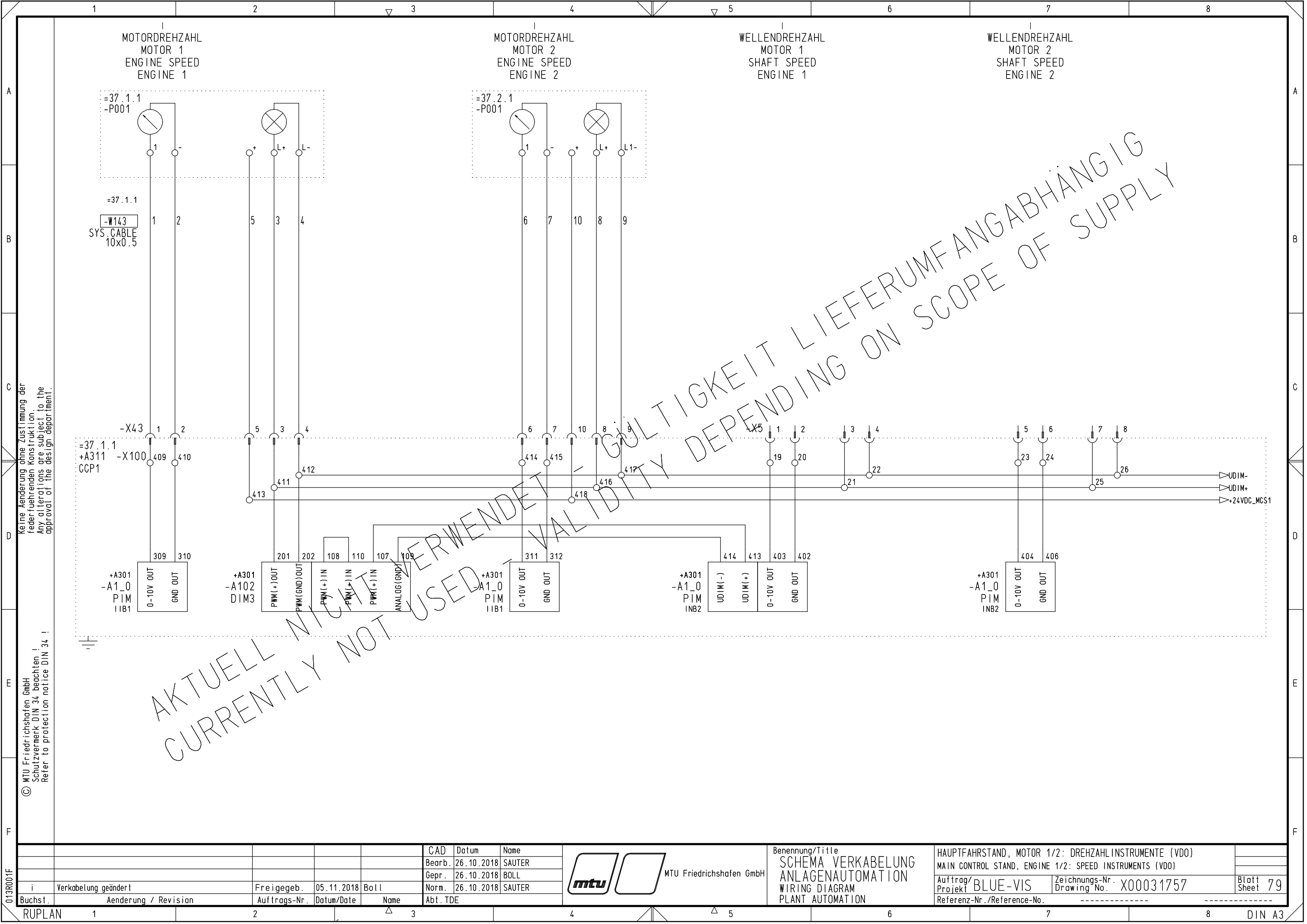
HAUPTFAHRSTAND, MOTOR 1/2: ANZEIGEINSTRUMENTE/CPP (HORN EAD)  
MAIN CONTROL STAND, ENGINE 1/2: IND. INSTR./CPP (HORN EAD)

|                            |          |                               |           |                |    |
|----------------------------|----------|-------------------------------|-----------|----------------|----|
| Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 76 |
| Referenz-Nr./Reference-No. | -----    |                               |           |                |    |





|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                             |          |                               |           |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------|-------------------------------|-----------|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHLINSTRUMENTE (HORN EA)    |          | Blatt<br>Sheet 78             |           |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (HORN EA) |          |                               |           |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                             |          |                               |           |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                             |          |                               |           |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                  |          | -----                         |           |



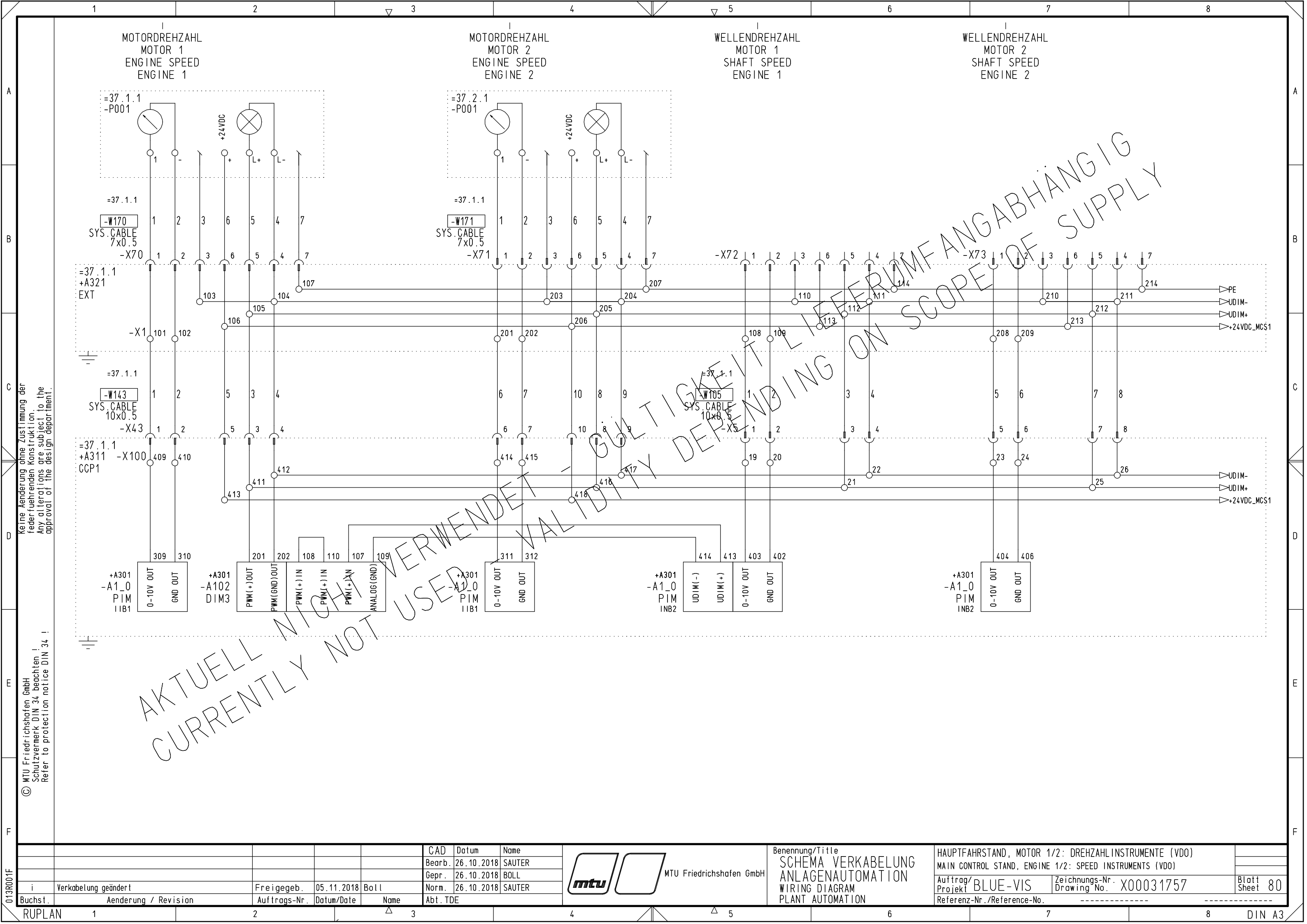
Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |

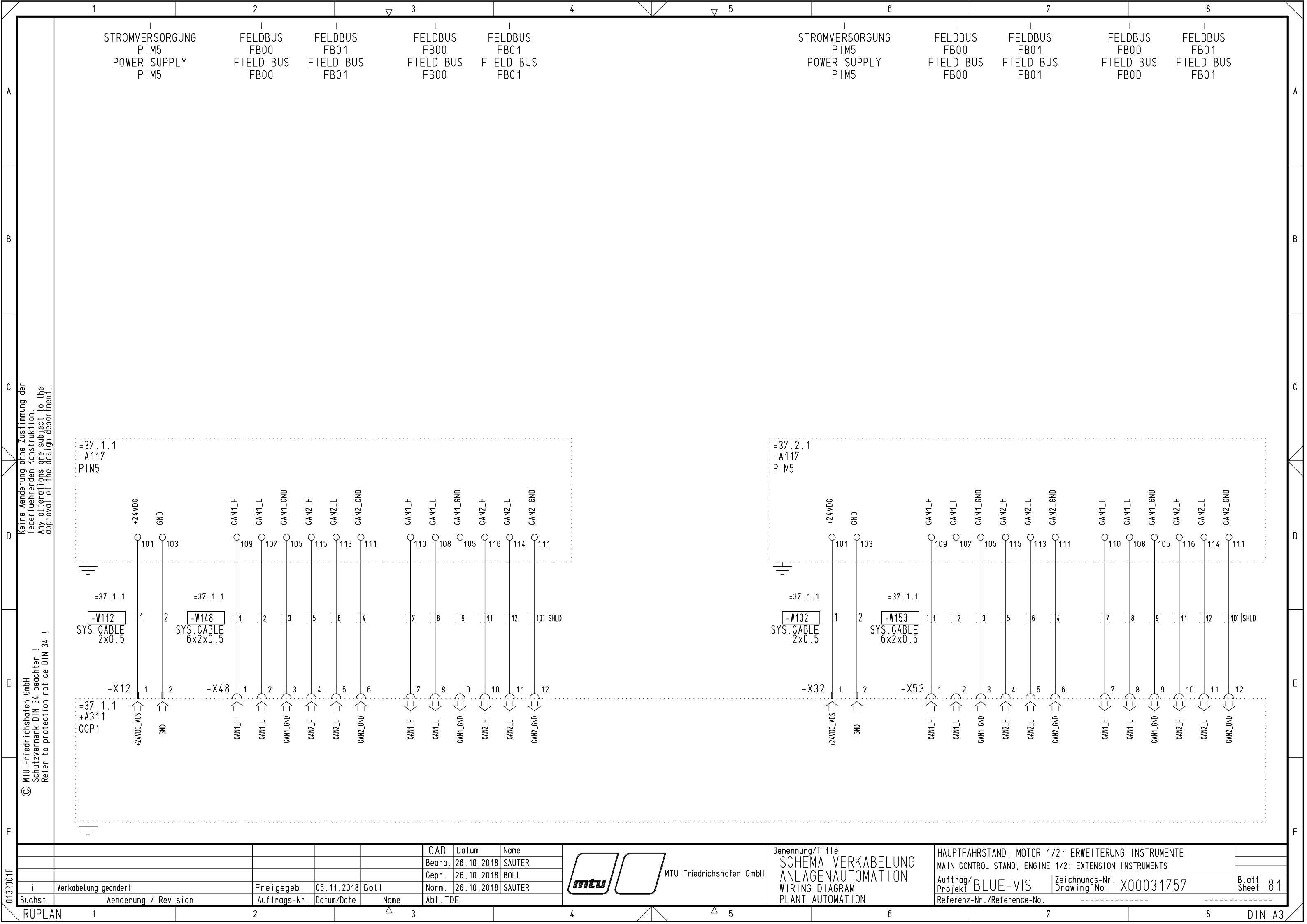


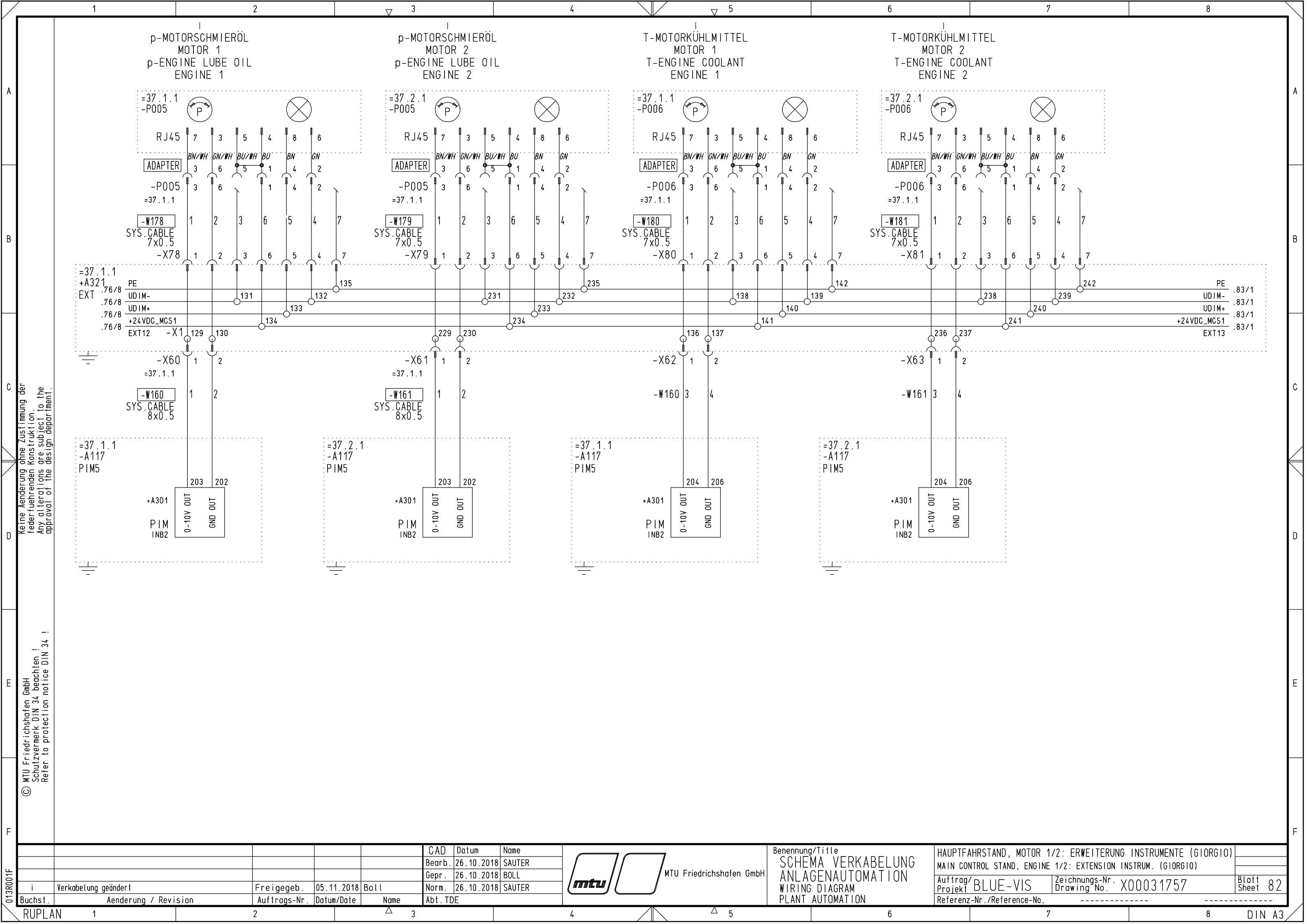
|                                                                                                  |          |                                                                                                                 |           |                |
|--------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|-----------|----------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHLINSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (VDO) |           |                |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                   | X00031757 | Blatt<br>Sheet |
| Referenz-Nr./Reference-No.                                                                       |          |                                                                                                                 |           | 79             |



|                                |                      |  |  |              |            |        |                                                                                                                |                                                                                                  |                                                         |          |                               |           |
|--------------------------------|----------------------|--|--|--------------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------|-------------------------------|-----------|
|                                |                      |  |  | CAD          | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | HAUPTFAHRSTAND, MOTOR 1/2: DREHZAHLINSTRUMENTE (VDO)    |          | Blatt<br>Sheet 80             |           |
|                                |                      |  |  | Bearb.       | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | MAIN CONTROL STAND, ENGINE 1/2: SPEED INSTRUMENTS (VDO) |          |                               |           |
|                                |                      |  |  | Gepr.        | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                         |          |                               |           |
|                                |                      |  |  | Norm.        | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                         |          |                               |           |
| i                              | Verkabelung geändert |  |  | Freigegeb.   | 05.11.2018 | Boll   |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                     | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Buchst.                        | Änderung / Revision  |  |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE                                                                                                       |                                                                                                  | Referenz-Nr./Reference-No.                              |          | -----                         |           |
| DIPDI AN 1 2 3 4 5 6 7 8 DIN A |                      |  |  |              |            |        |                                                                                                                |                                                                                                  |                                                         |          |                               |           |











Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUM. (HORN EAD)  
MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM.(HORN EAD)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 84

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

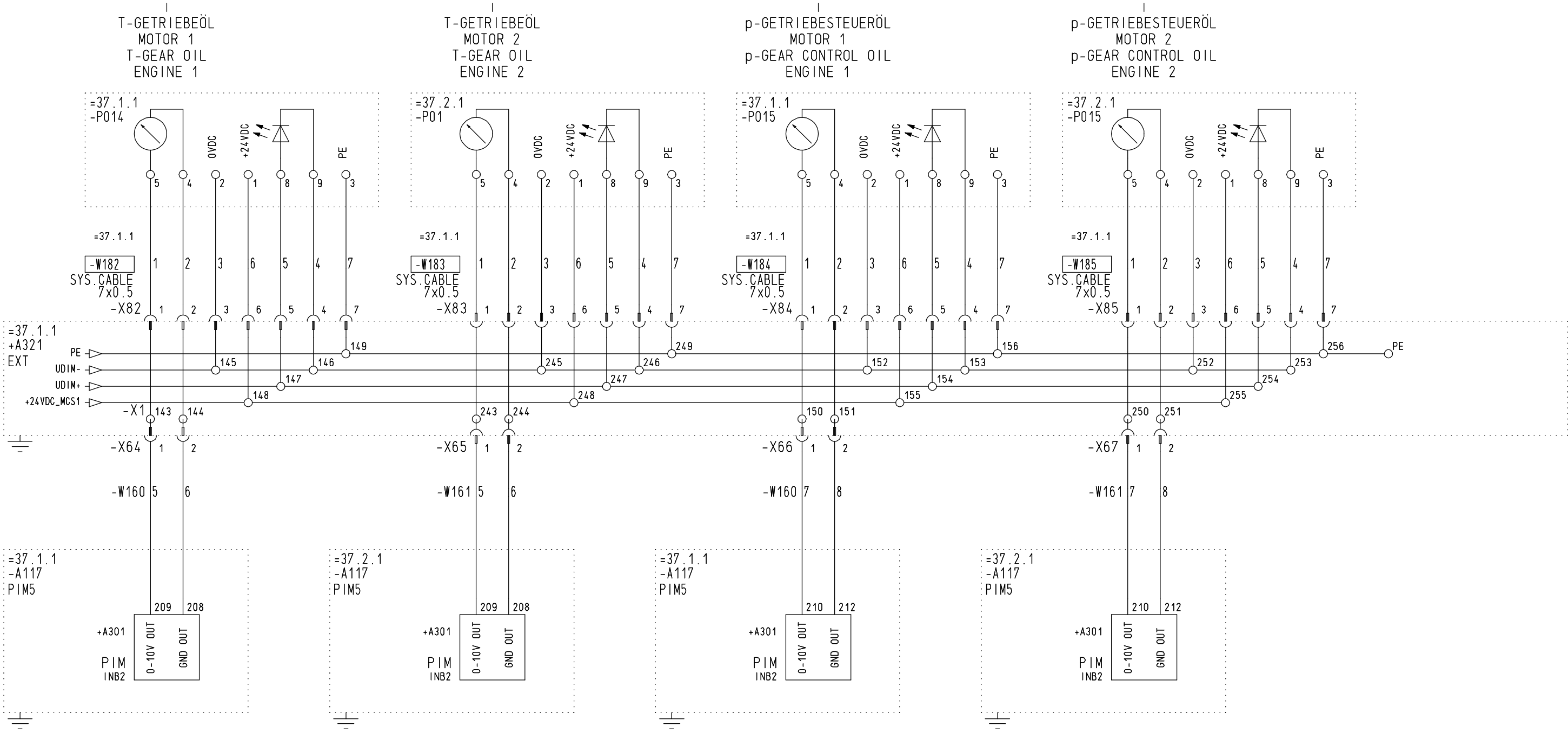
© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

|                                                                                                  |          |                                                                                                                            |           |
|--------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------|-----------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUM. (HORN EAD)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUM.(HORN EAD) |           |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                              | X00031757 |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                                                                      |           |
| RUPLAN                                                                                           |          | DIN A3                                                                                                                     |           |



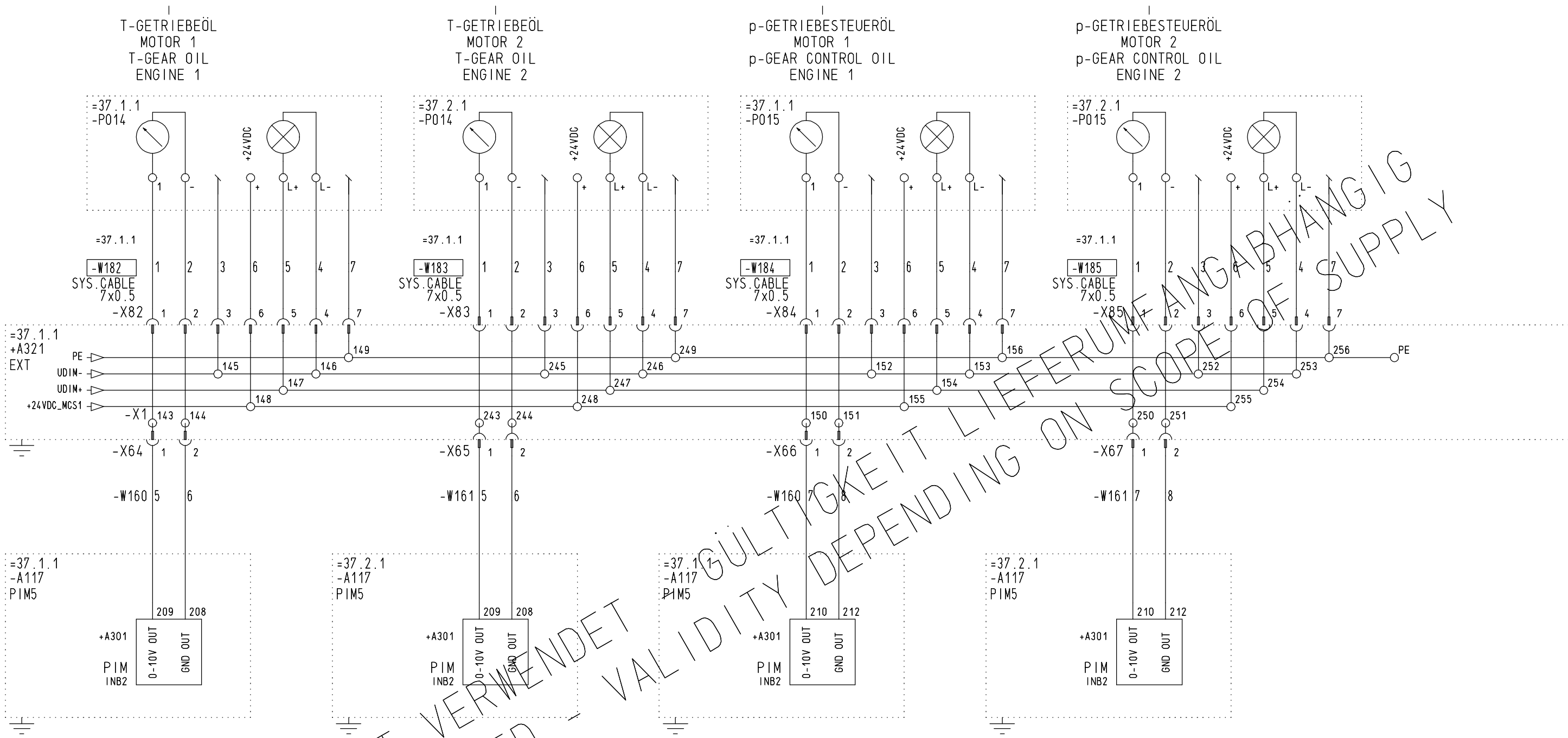


Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F

RUPLAN



013R001F

RUPLAN

|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |

 MTU Friedrichshafen GmbH

|                                                                                                  |          |                                                                                                                         |           |
|--------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|-----------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | HAUPTFAHRSTAND, MOTOR 1/2: ERWEITERUNG INSTRUMENTE (VDO)<br>MAIN CONTROL STAND, ENGINE 1/2: EXTENSION INSTRUMENTS (VDO) |           |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                           | X00031757 |
| Referenz-Nr./Reference-No.                                                                       |          |                                                                                                                         |           |
|                                                                                                  |          | Blatt<br>Sheet                                                                                                          | 87        |

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



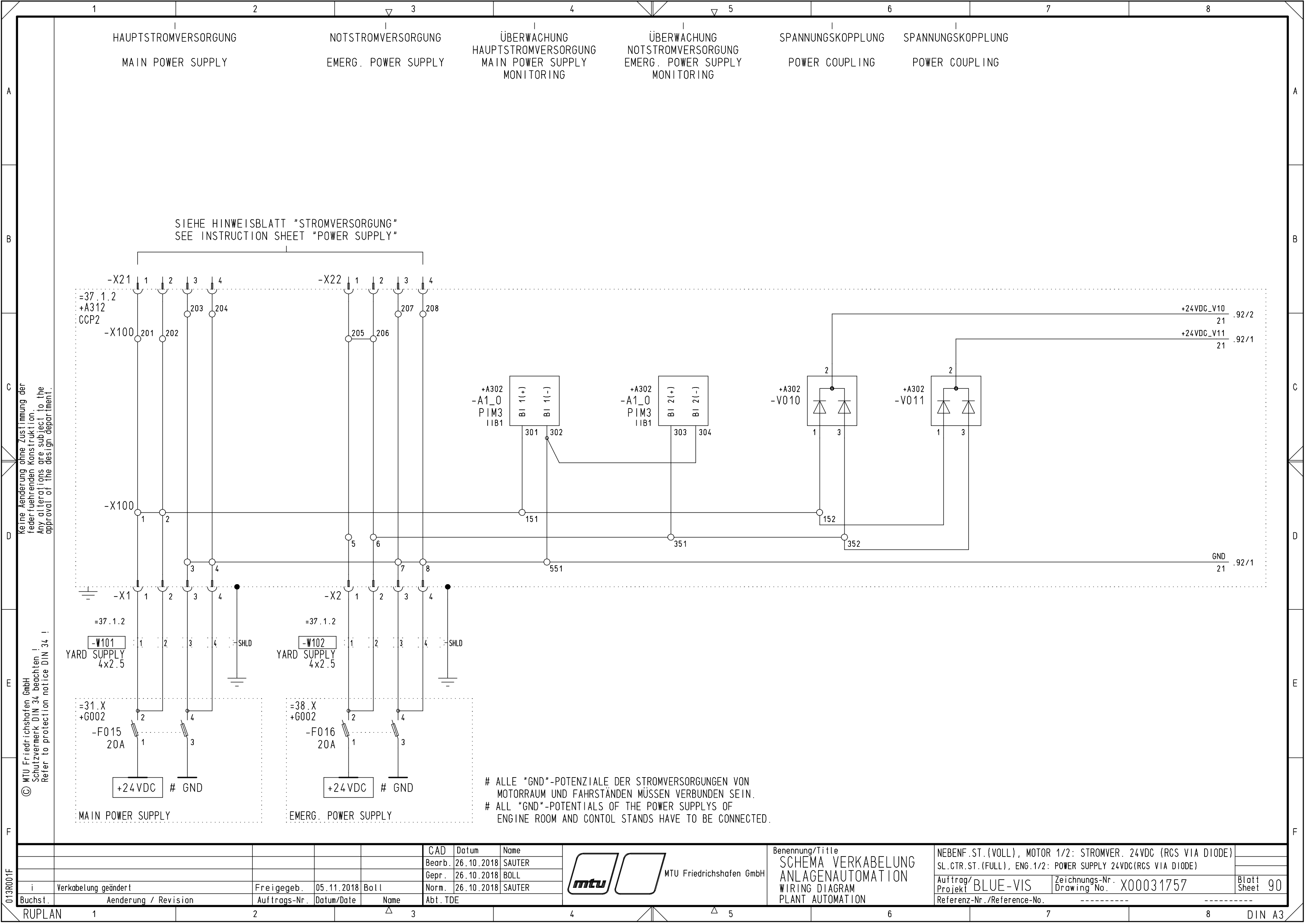
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                  |          |                                         |
|--------------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| HAUPTFAHRSTAND, MOTOR 1/2: STEUERRAD (WJ)<br>MAIN CONTROL STAND, ENGINE 1/2: STEERING WHEEL (WJ) |          |                                         |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                       |          | Blatt<br>Sheet 88                       |











Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gedr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRST.(VOLL), MOTOR 1/2: 24VDC EIN/AUS (RCS VIA DIODE)  
SL.CTR.STAND(FULL), ENGINE 1/2: 24VDC ON/OFF (RCS VIA DIODE)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 92

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

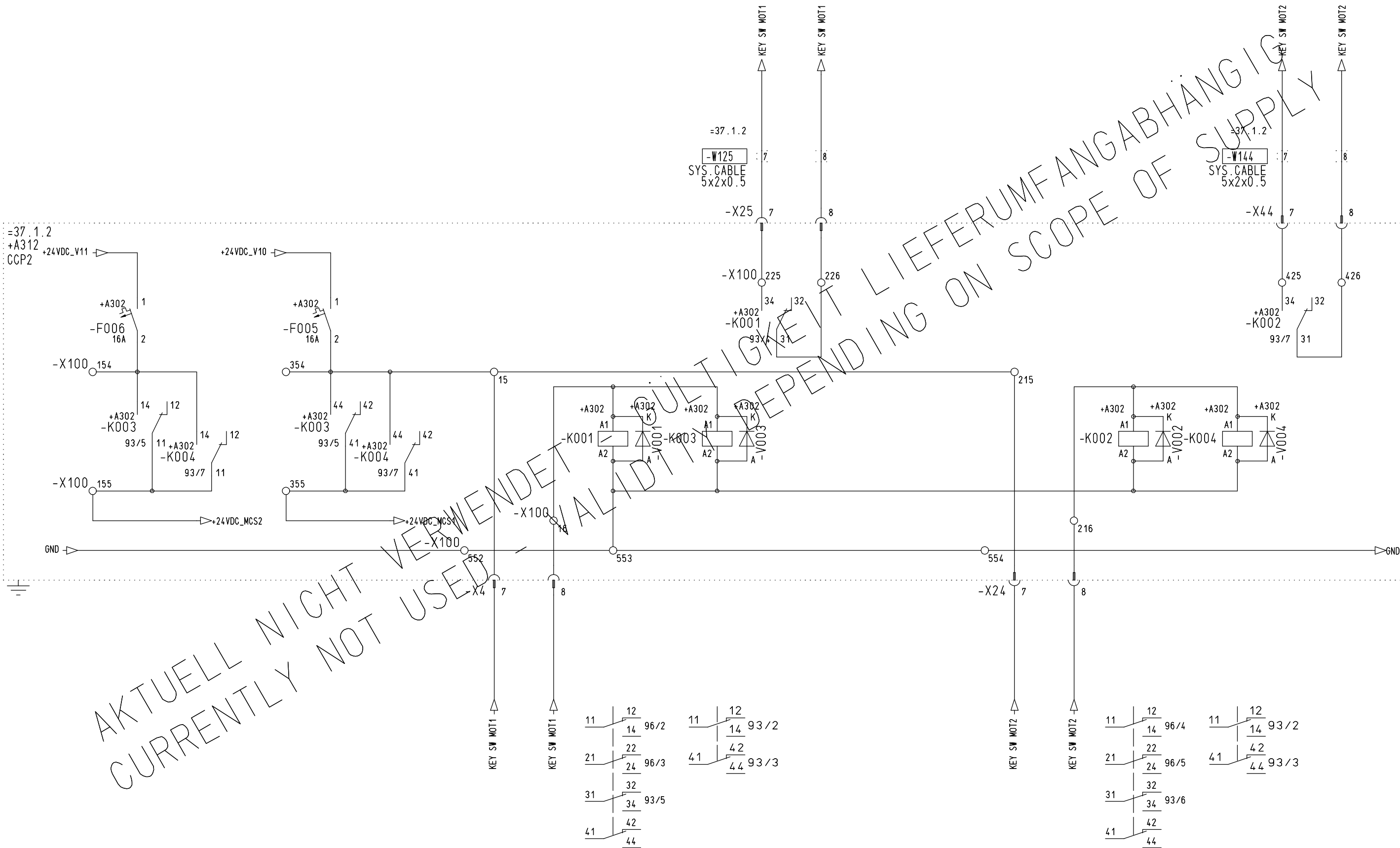
RUPLAN

SCHLÜSSELSCHALTER  
MOTOR 1 EIN/AUS  
KEY SWITCH  
ENGINE 1 ON/OFF

MOTOR 1 EIN/AUS  
NEBENFAHRSTAND  
ENGINE 1 ON/OFF  
SLAVE CONTROL STAND

SCHLÜSSELSCHALTER  
MOTOR 2 EIN/AUS  
KEY SWITCH  
ENGINE 2 ON/OFF

MOTOR 2 EIN/AUS  
NEBENFAHRSTAND  
ENGINE 2 ON/OFF  
SLAVE CONTROL STAND



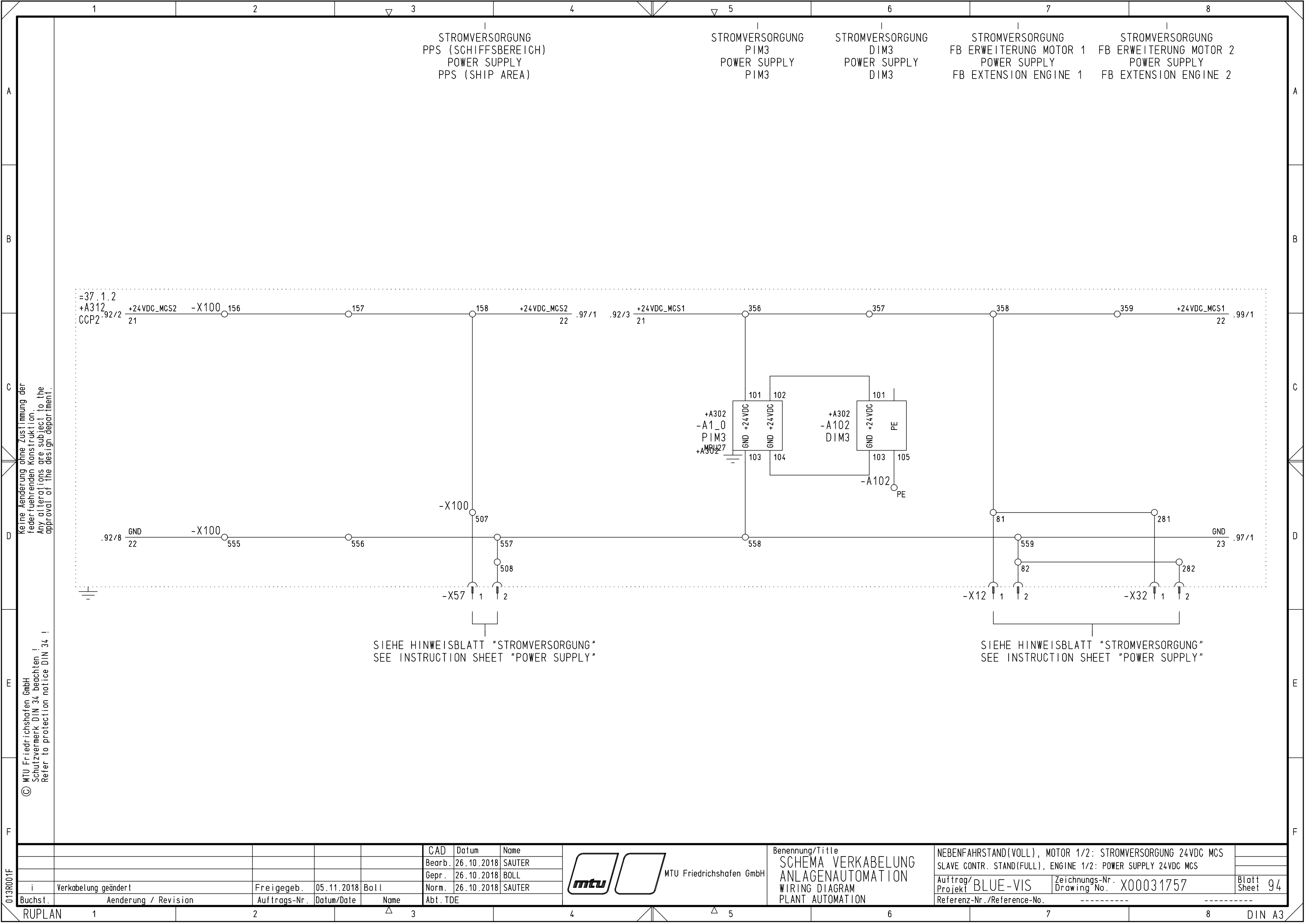
| CAD      | Datum      | Name   |
|----------|------------|--------|
| Bearb.   | 26.10.2018 | SAUTER |
| Gepr.    | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |

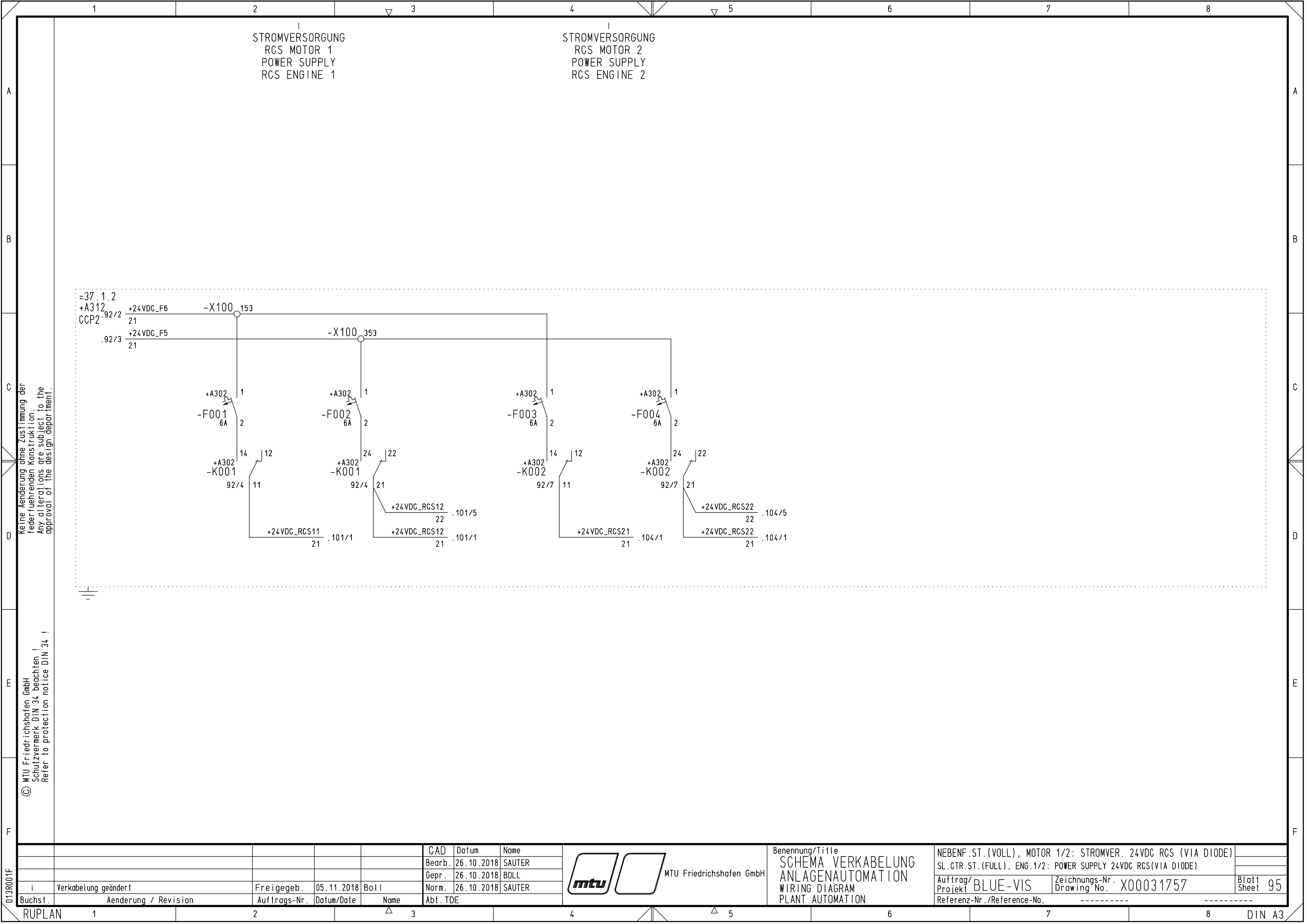


MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                       |          |                                         |
|-------------------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(VOLL), MOTOR 1/2: 24VDC EIN/AUS<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: 24VDC ON/OFF |          |                                         |
| Auftrag/<br>Projekt                                                                                   | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                            | -----    | Blatt<br>Sheet 93                       |

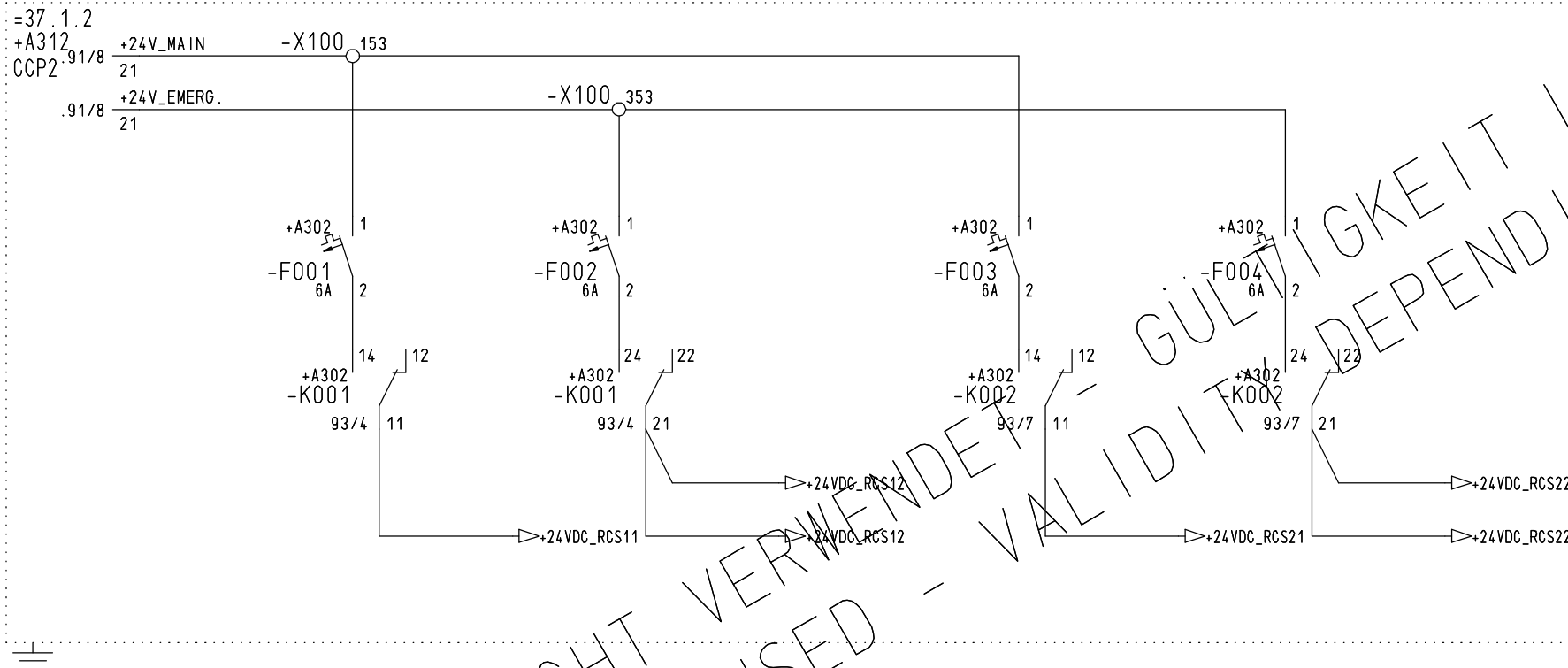




Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |                                                                                                                |                                                                                                  |                                                                                                                            |                                         |                   |
|---------|----------------------|--------------|------------|------|----------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------|
|         |                      |              |            |      |          |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(VOLL), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: POWER SUPPLY 24VDC RCS |                                         | Blatt<br>Sheet 96 |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt BLUE-VIS                                                                                               | Zeichnungs-Nr.<br>Drawing No. X00031757 |                   |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                                                                                 | -----                                   |                   |
| RUPLAN  |                      |              |            |      |          |                                                                                                                |                                                                                                  |                                                                                                                            |                                         |                   |







STROMVERSORGUNG  
LEITSTANDSRECHNER 1  
POWER SUPPLY  
MANAG.COMPUTER UNIT 1

STROMVERSORGUNG  
LEITSTANDSRECHNER 2  
POWER SUPPLY  
MANAG.COMPUTER UNIT 2

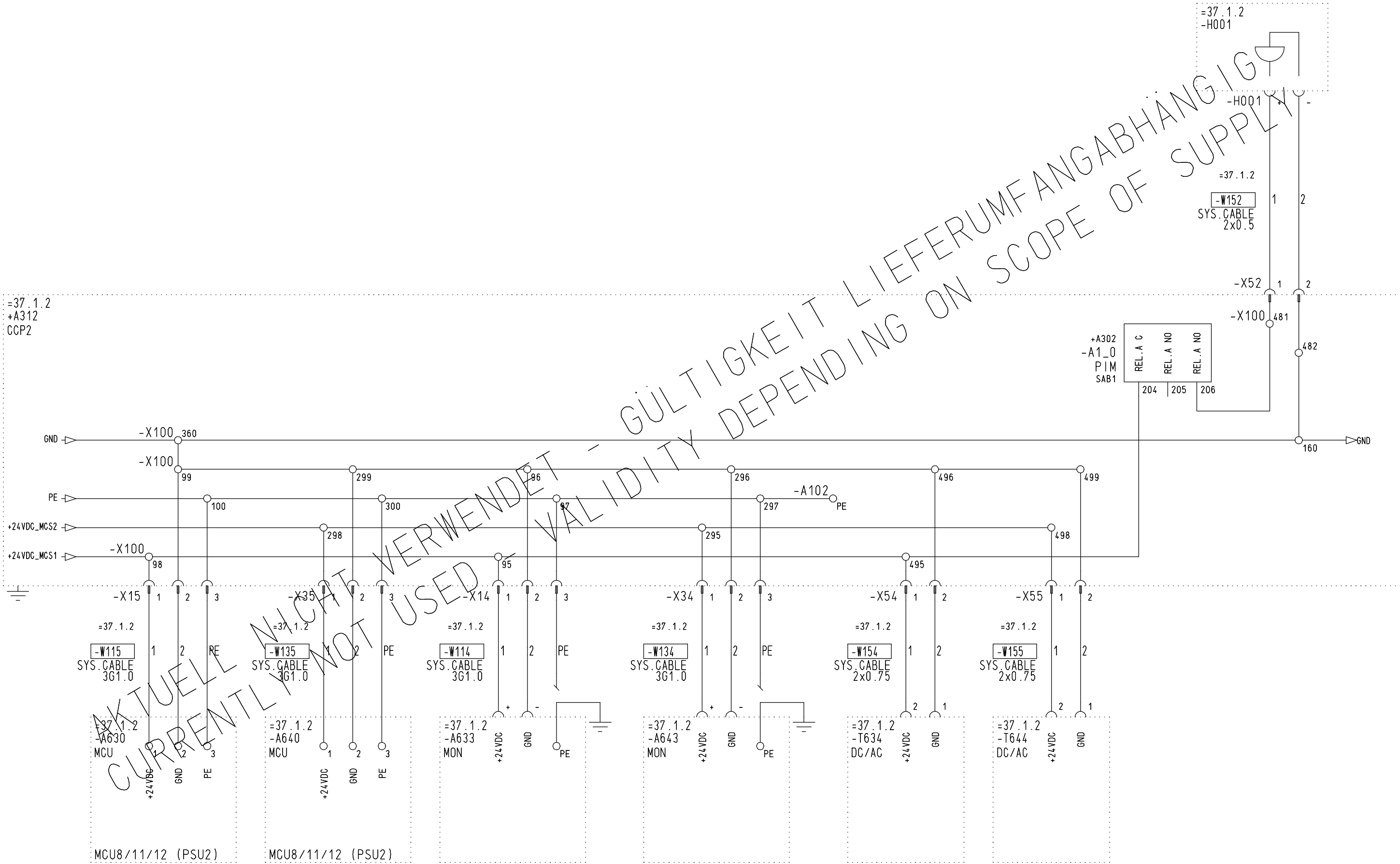
STROMVERSORGUNG  
BILDSCHIRM 1  
POWER SUPPLY  
MONITOR 1

STROMVERSORGUNG  
BILDSCHIRM 2  
POWER SUPPLY  
MONITOR 2

STROMVERSORGUNG  
DRUCKER 1  
POWER SUPPLY  
PRINTER 1

STROMVERSORGUNG  
DRUCKER 2  
POWER SUPPLY  
PRINTER 2

ALARM  
SUMMER  
ALARM  
BUZZER



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |



|                                                                                                  |          |                                                                                                                                        |           |
|--------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | NEBENFAHRST. (VOLL), STROMVERS. 24VDC, GRAFIKST. 1&2 (MCU8/11/12)<br>SLAVE CTR. ST. (FULL), P.S. 24VDC, GRAPHIC STAT. 1&2 (MCU8/11/12) |           |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                                          | X00031757 |
| Referenz-Nr./Reference-No.                                                                       |          |                                                                                                                                        |           |



STROMVERSORGUNG  
LEITSTANDSRECHNER 1  
POWER SUPPLY  
MANAG.COMPUTER UNIT 1

STROMVERSORGUNG  
LEITSTANDSRECHNER 2  
POWER SUPPLY  
MANAG.COMPUTER UNIT 2

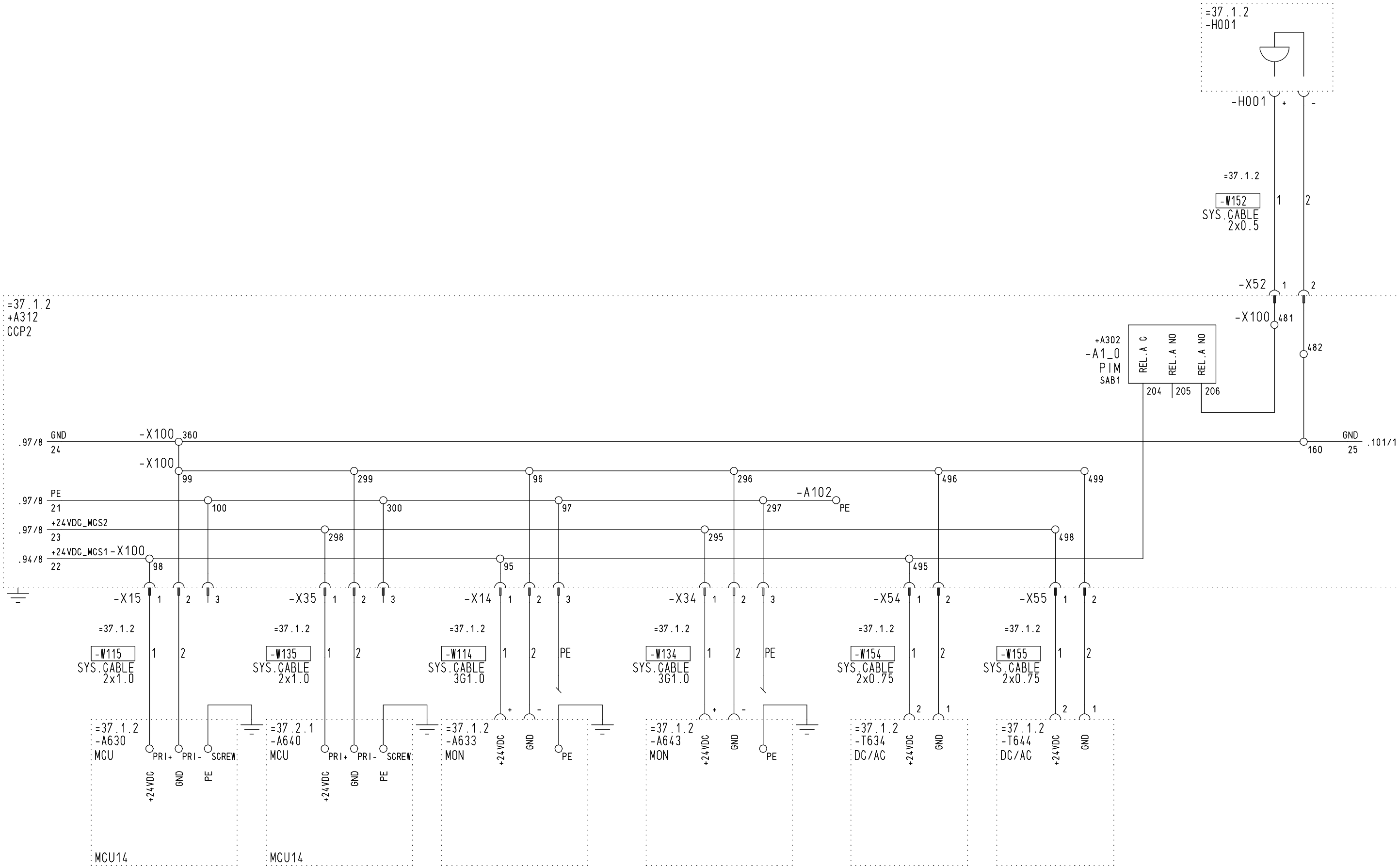
STROMVERSORGUNG  
BILDSCHIRM 1  
POWER SUPPLY  
MONITOR 1

STROMVERSORGUNG  
BILDSCHIRM 2  
POWER SUPPLY  
MONITOR 2

STROMVERSORGUNG  
DRUCKER 1  
POWER SUPPLY  
PRINTER 1

STROMVERSORGUNG  
DRUCKER 2  
POWER SUPPLY  
PRINTER 2

ALARM  
SUMMER  
ALARM  
BUZZER



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

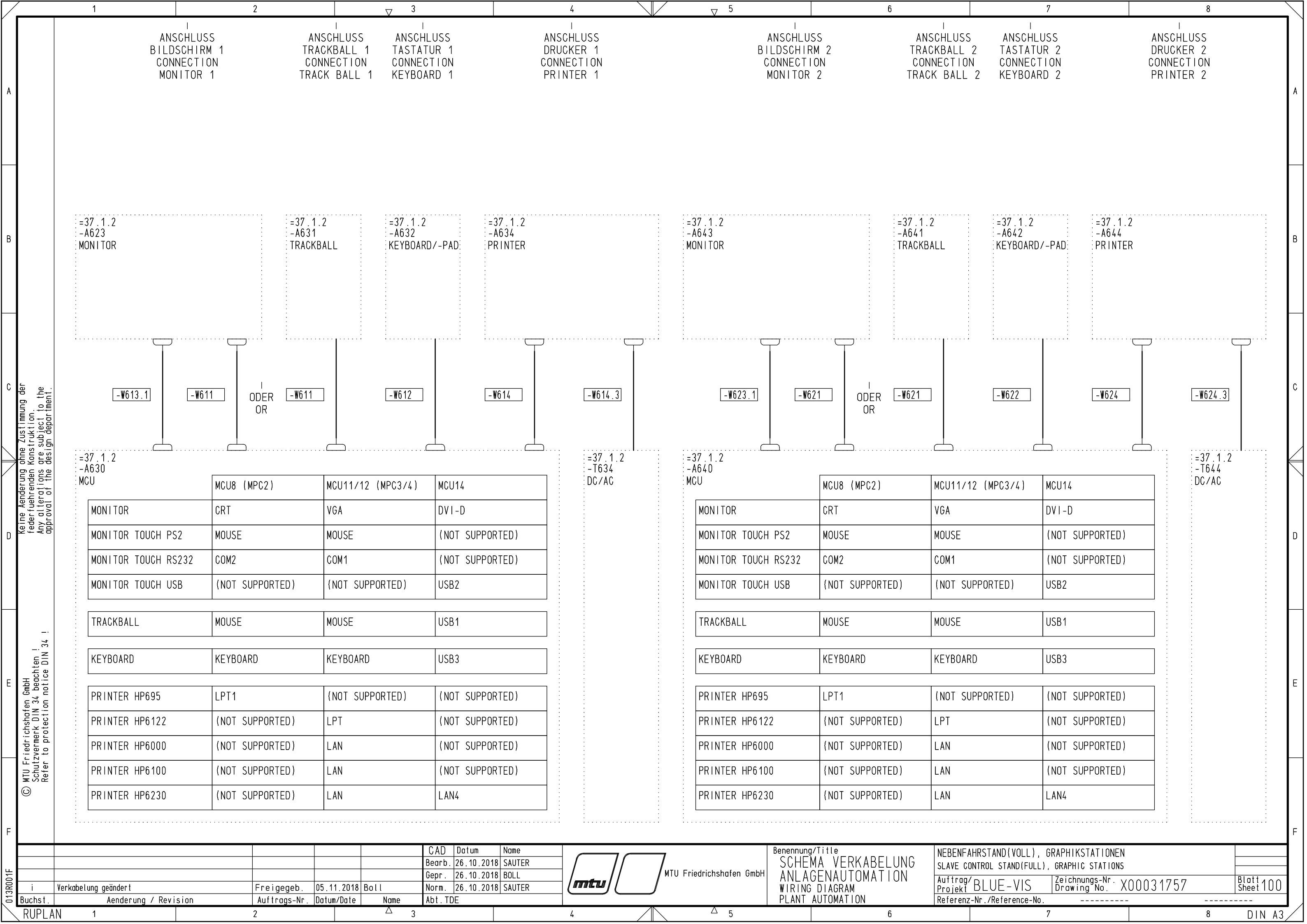
© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRST.(VOLL),STROMVERS.24VDC, GRAFIKST.1&2(MCU14)  
SLAVE CTR.ST.(FULL), P.S.24VDC, GRAPHIC STAT.1&2(MCU14)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. X00031757  
Referenz-Nr./Reference-No. -----



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1: STROMVERSORG. 24VDC RCS (FPP)  
SLAVE CONTR. STAND(FULL), ENGINE 1: POW.SUP. 24VDC RCS (FPP)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr. / Reference-No. -----

Blatt 101  
Sheet



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

|                    |  |                                                             |  |                          |
|--------------------|--|-------------------------------------------------------------|--|--------------------------|
| Benennung/Title    |  | NEBENFAHRSTAND(VOLL), MOTOR 1: STROMVERSORG. 24VDC RCS (WJ) |  |                          |
| SCHEMA VERKABELUNG |  | SLAVE CONTR. STAND(FULL), ENGINE 1: POW.SUP. 24VDC RCS (WJ) |  |                          |
| ANLAGENAUTOMATION  |  |                                                             |  |                          |
| WIRING DIAGRAM     |  | Auftrag/Projekt BLUE-VIS                                    |  | Zeichnungs-Nr. X00031757 |
| PLANT AUTOMATION   |  | Referenz-Nr./Reference-No.                                  |  | Blatt 103                |
|                    |  | -----                                                       |  | Sheet                    |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



|                                                                                                  |          |                                                                                                                              |           |
|--------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------|-----------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (FPP)<br>SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (FPP) |           |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                                | X00031757 |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                                                                        |           |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (CPP)  
SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (CPP)

|                            |          |                               |           |                |     |
|----------------------------|----------|-------------------------------|-----------|----------------|-----|
| Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 105 |
| Referenz-Nr./Reference-No. |          | -----                         |           |                |     |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

|                            |  |                                                             |                              |                   |
|----------------------------|--|-------------------------------------------------------------|------------------------------|-------------------|
| Benennung/Title            |  | NEBENFAHRSTAND(VOLL), MOTOR 2: STROMVERSORG. 24VDC RCS (WJ) |                              |                   |
| SCHEMA VERKABELUNG         |  | SLAVE CONTR. STAND(FULL), ENGINE 2: POW.SUP. 24VDC RCS (WJ) |                              |                   |
| ANLAGENAUTOMATION          |  |                                                             |                              |                   |
| WIRING DIAGRAM             |  |                                                             |                              |                   |
| PLANT AUTOMATION           |  |                                                             |                              |                   |
| Auftrag/Projekt            |  | BLUE-VIS                                                    | Zeichnungs-Nr. / Drawing No. | X00031757         |
| Referenz-Nr./Reference-No. |  | -----                                                       |                              | Blatt 106 / Sheet |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

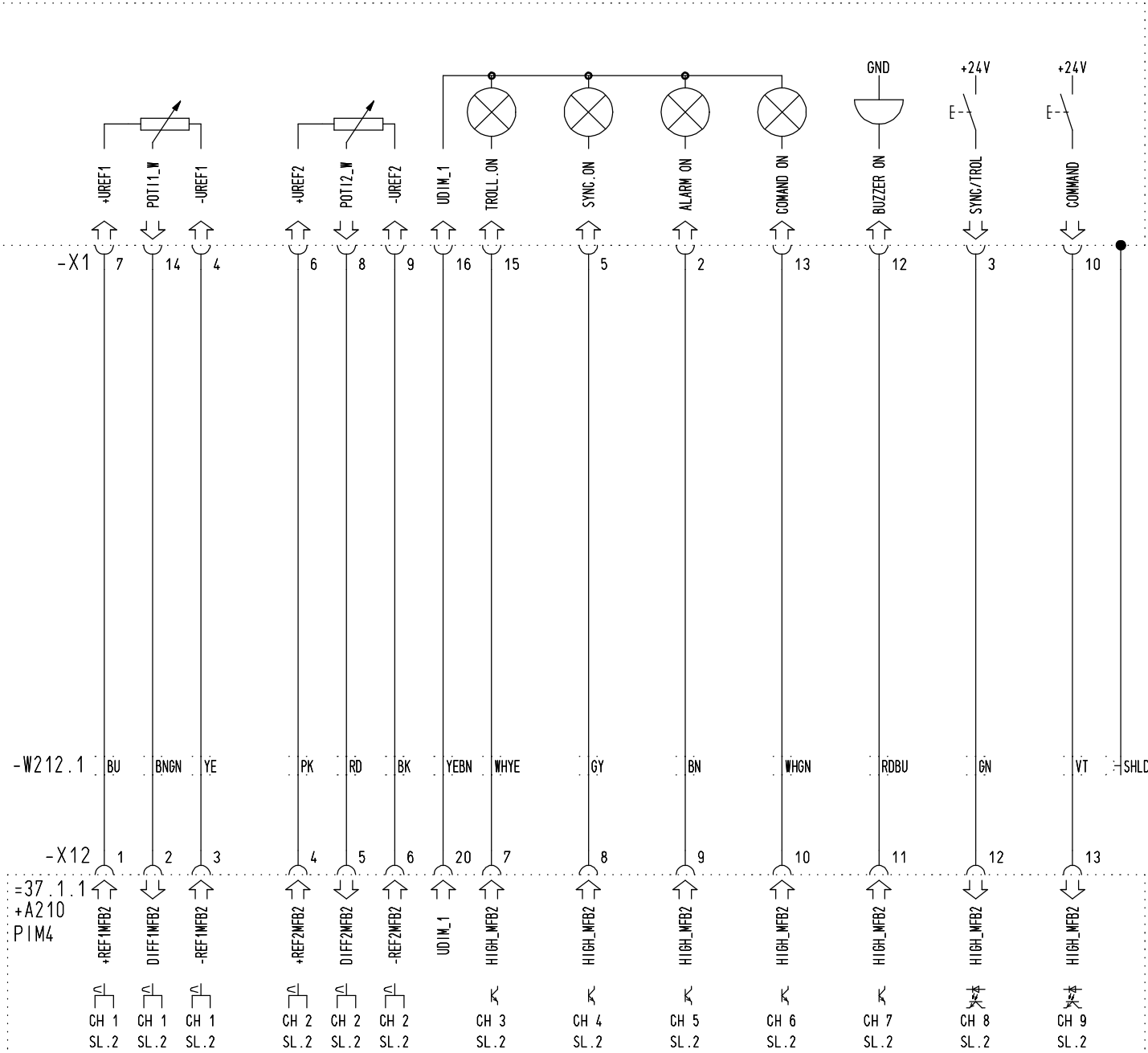
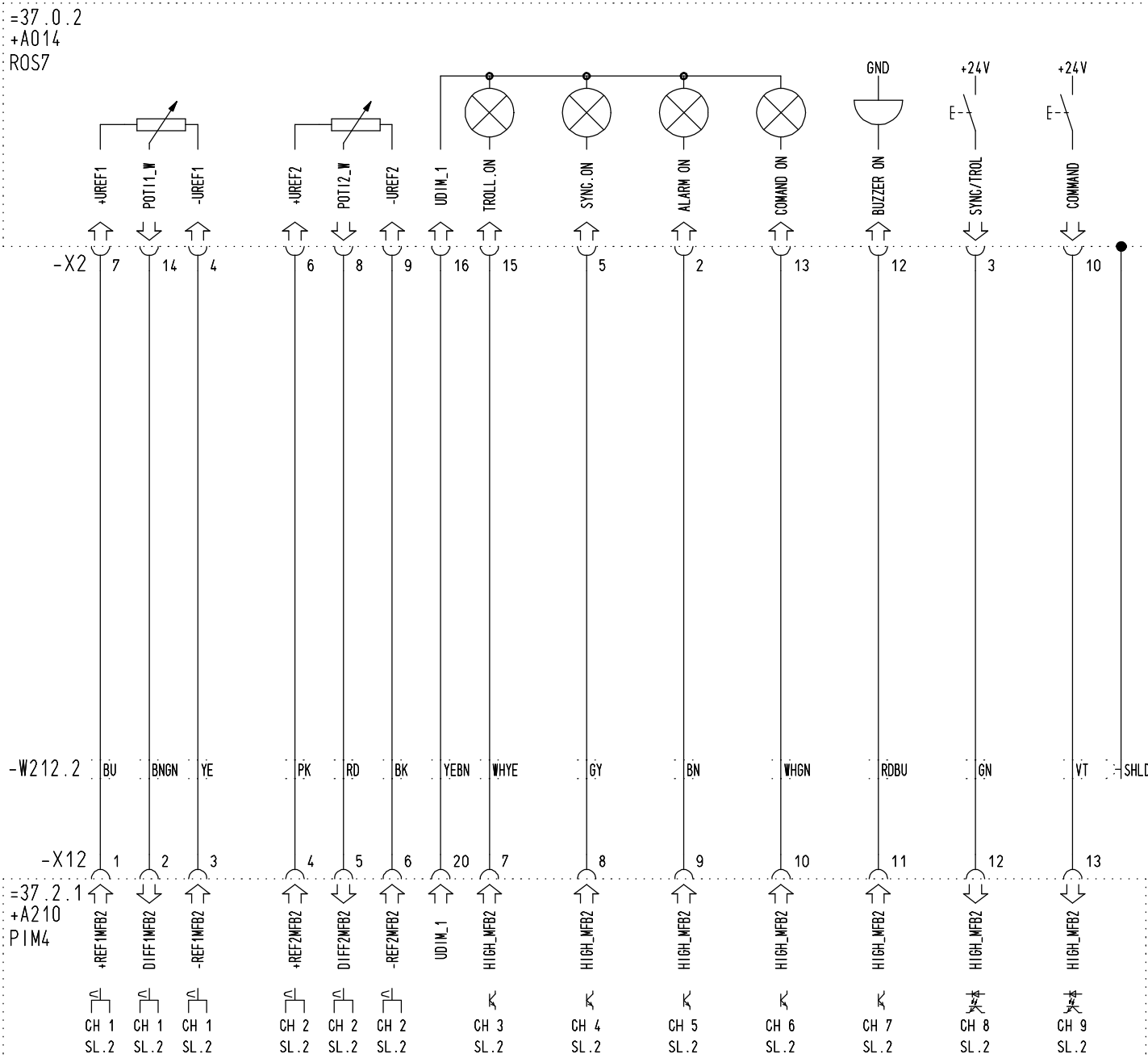
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

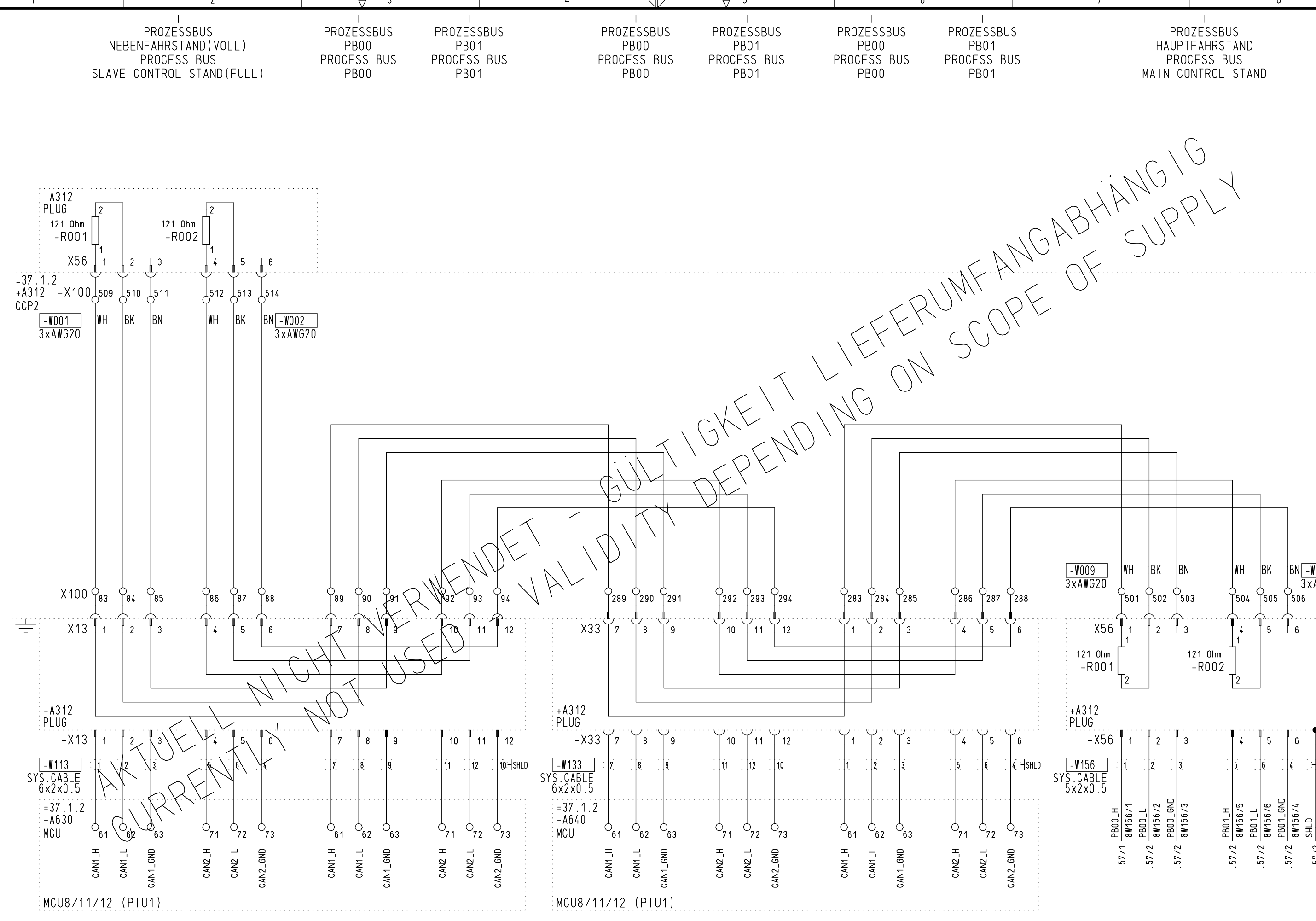
|                                                                                                                     |          |                               |                    |
|---------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|--------------------|
| NEBENFAHRSTAND(VOLL), MOTOR 1/2: BEDIENSTATION ROS7<br>SLAVE CONTR. STAND(FULL), ENGINE 1/2: OPERATING STATION ROS7 |          |                               |                    |
| Auftrag/<br>Projekt                                                                                                 | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757          |
| Referenz-Nr./Reference-No.                                                                                          |          | -----                         |                    |
|                                                                                                                     |          |                               | Blatt 107<br>Sheet |



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F



|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                     |          |                               |           |                |     |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------|-------------------------------|-----------|----------------|-----|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(VOLL), PROZESSBUS (MCU8/11/12)       |          |                               |           |                |     |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | SLAVE CONTROL STAND(FULL), PROCESS BUS (MCU8/11/12) |          |                               |           |                |     |
|         |                      |              |            |      | Gep.     | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                     |          |                               |           |                |     |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                     |          |                               |           |                |     |
|         |                      |              |            |      | Abt. TDE |            |        |                                                                                                                |                                                                                                  |                                                     |          |                               |           |                |     |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                 | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 108 |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                          |          |                               |           |                |     |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



|                                                                                                  |          |                                                                                          |           |                |
|--------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------|-----------|----------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | NEBENFAHRSTAND(VOLL), MOTOR 1: FELDBUS<br>SLAVE CONTROL STAND(FULL), ENGINE 1: FIELD BUS |           |                |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                            | X00031757 | Blatt<br>Sheet |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                                    |           |                |

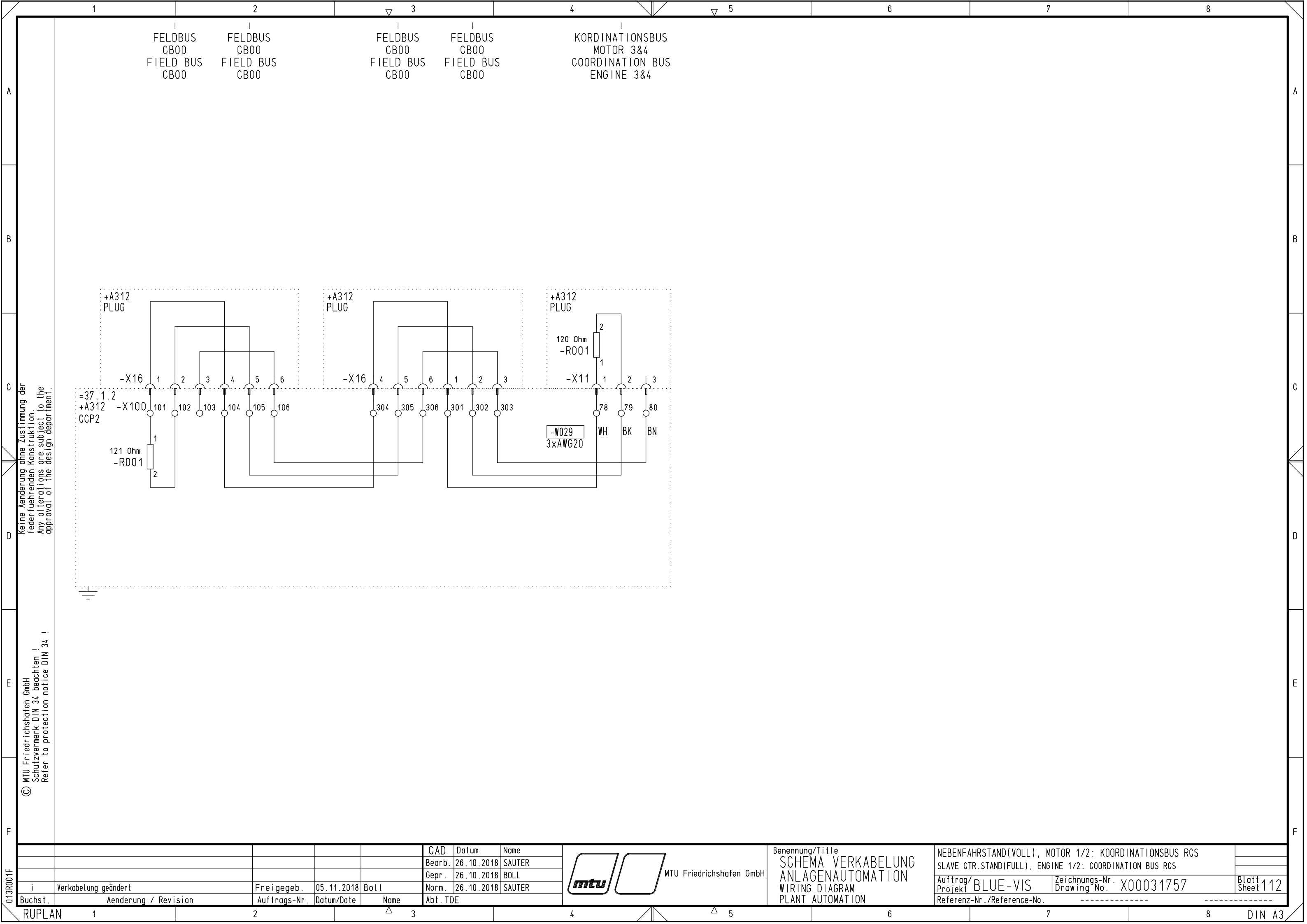
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

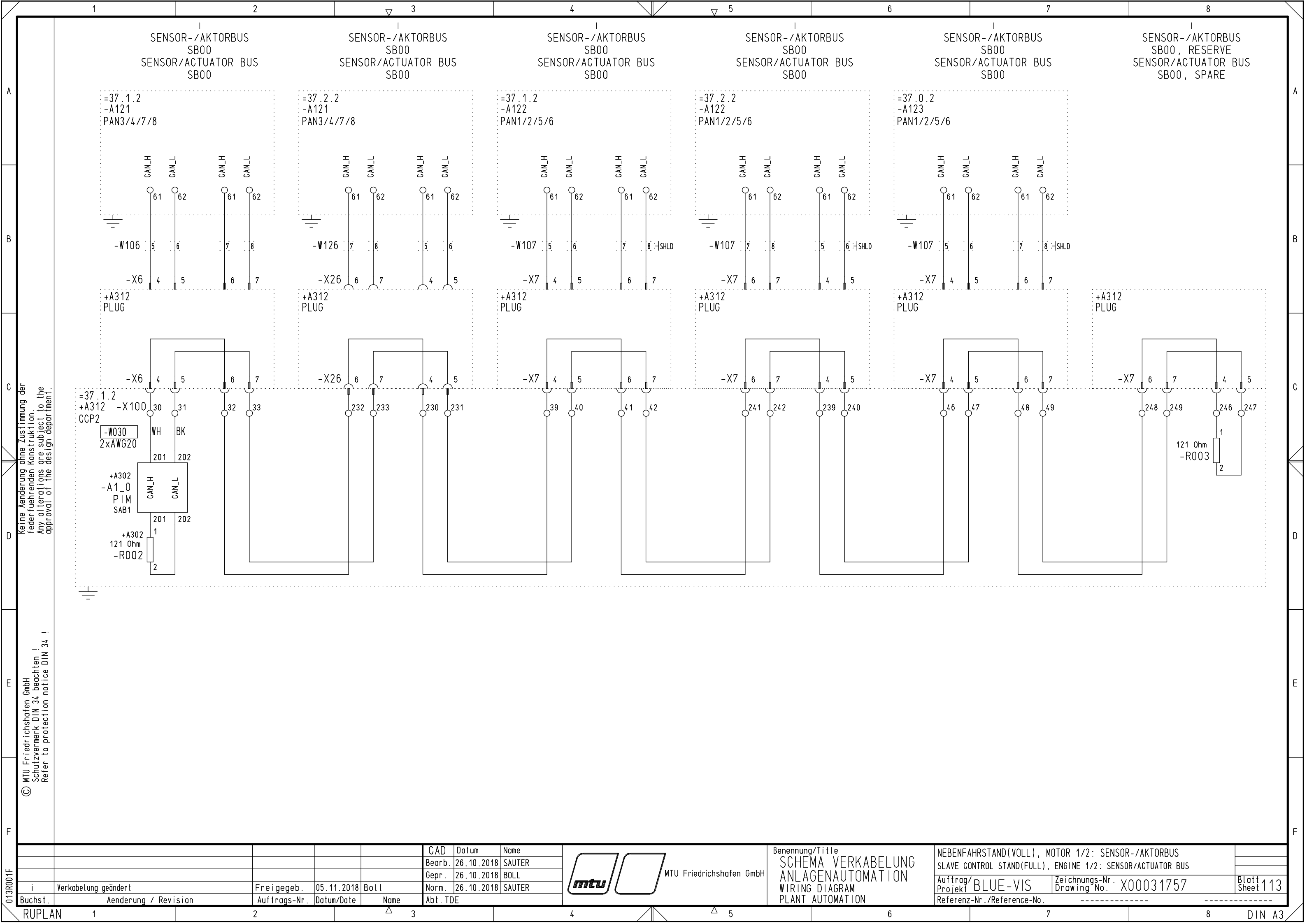
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



|                                                                                                  |          |                                                                                          |           |                    |
|--------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------|-----------|--------------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | NEBENFAHRSTAND(VOLL), MOTOR 2: FELDBUS<br>SLAVE CONTROL STAND(FULL), ENGINE 2: FIELD BUS |           |                    |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                            | X00031757 | Blatt<br>Sheet 111 |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                                    |           |                    |



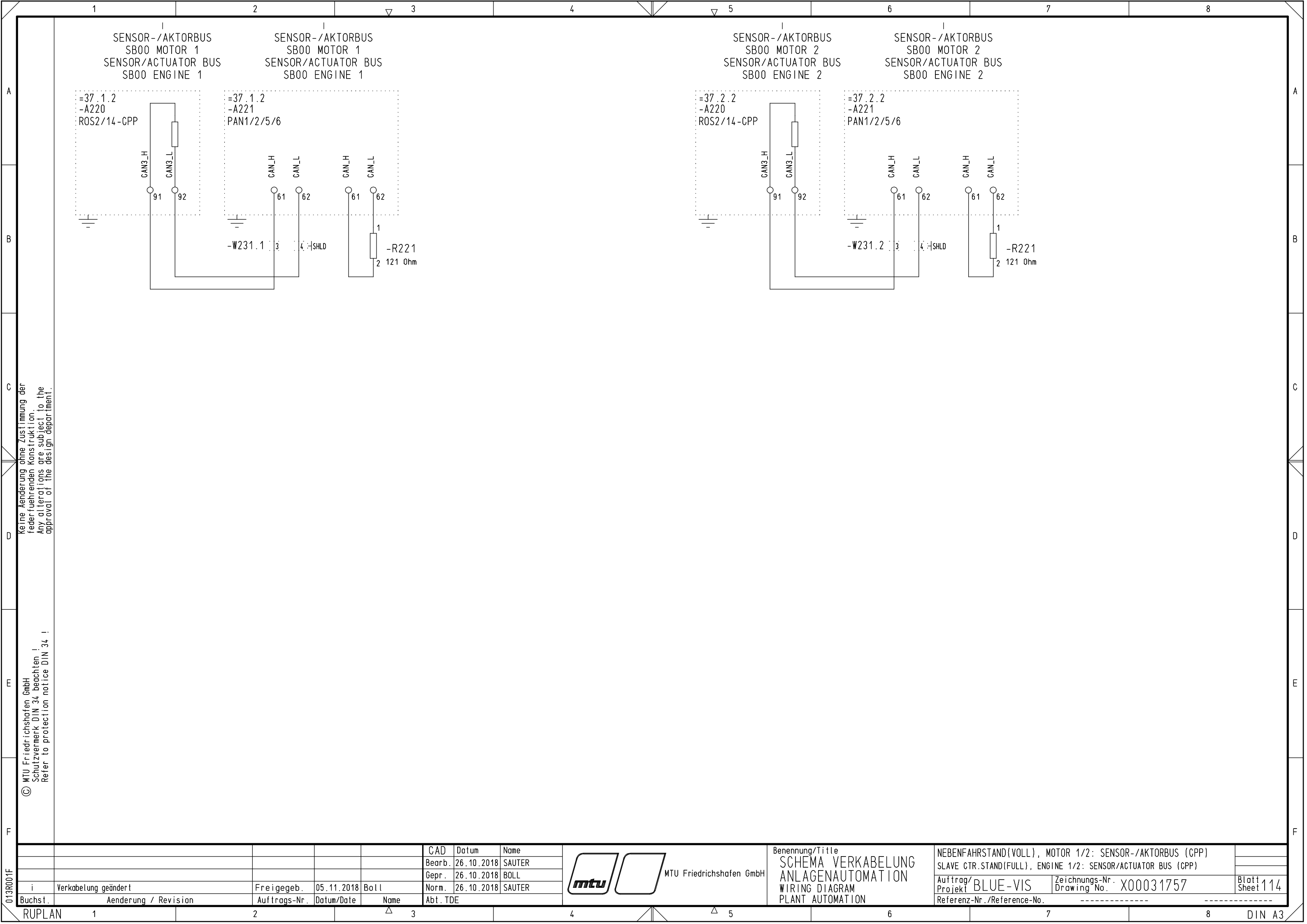




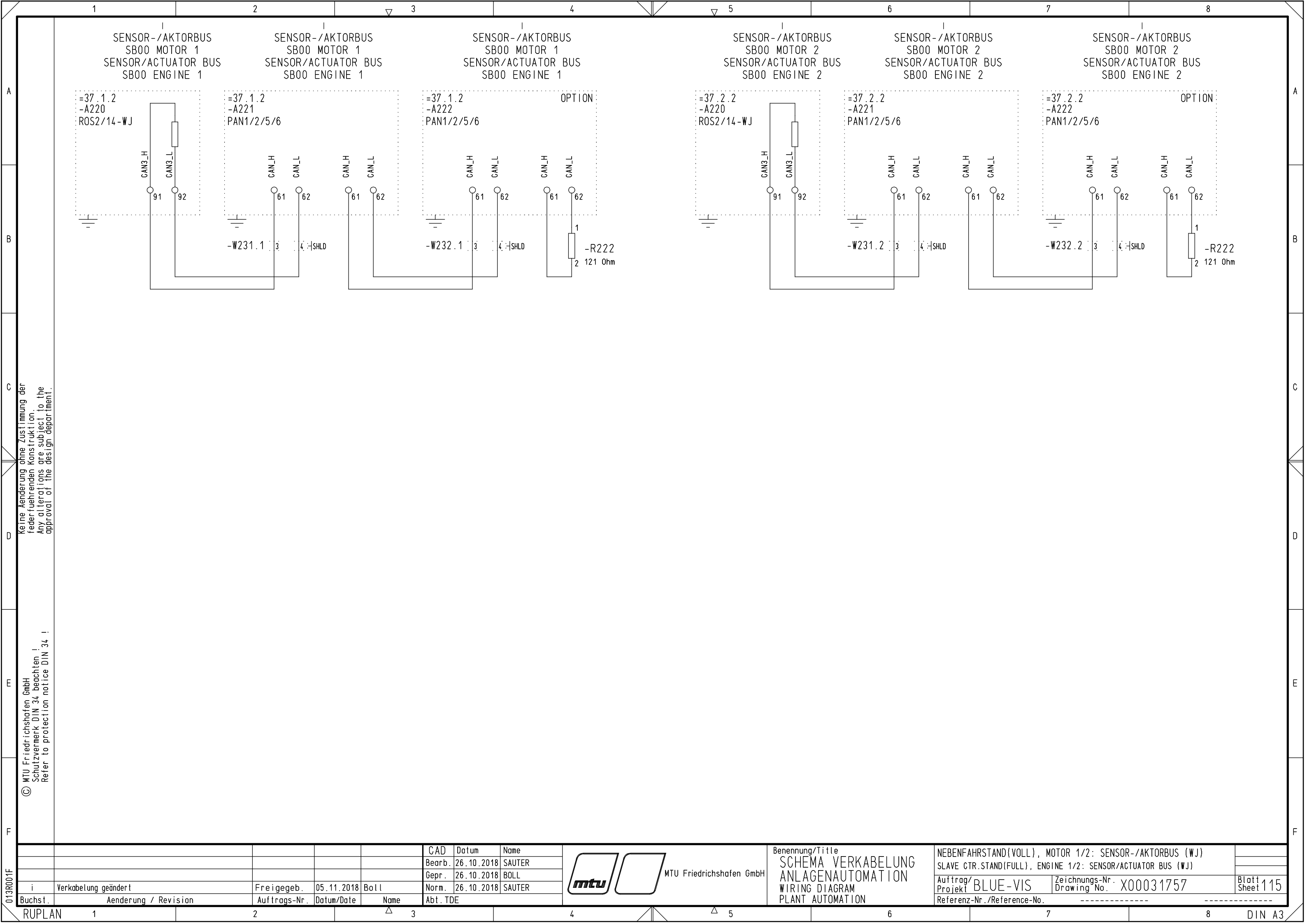
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |  |              |            |        |            |        |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |     |
|---------|----------------------|--|--------------|------------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------|-------------------------------|-----------|----------------|-----|
|         |                      |  |              |            | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(VOLL), MOTOR 1/2: SENSOR-/AKTORBUS          |          |                               |           |                |     |
|         |                      |  |              |            | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | SLAVE CONTROL STAND(FULL), ENGINE 1/2: SENSOR/ACTUATOR BUS |          |                               |           |                |     |
|         |                      |  |              |            | Gepr.  | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |     |
|         |                      |  |              |            | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                            |          |                               |           |                |     |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll   |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 113 |
| Buchst. | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE   |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No. -----                           |          |                               |           |                |     |







Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                             |          |                                         |
|-------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(VOLL), MOTOR 1/2: SENSOR-/AKTORBUS (WJ)      |          |                                         |
| SLAVE CTR.STAND(FULL), ENGINE 1/2: SENSOR/ACTUATOR BUS (WJ) |          |                                         |
| Auftrag/<br>Projekt                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                  |          | Blatt<br>Sheet 115                      |

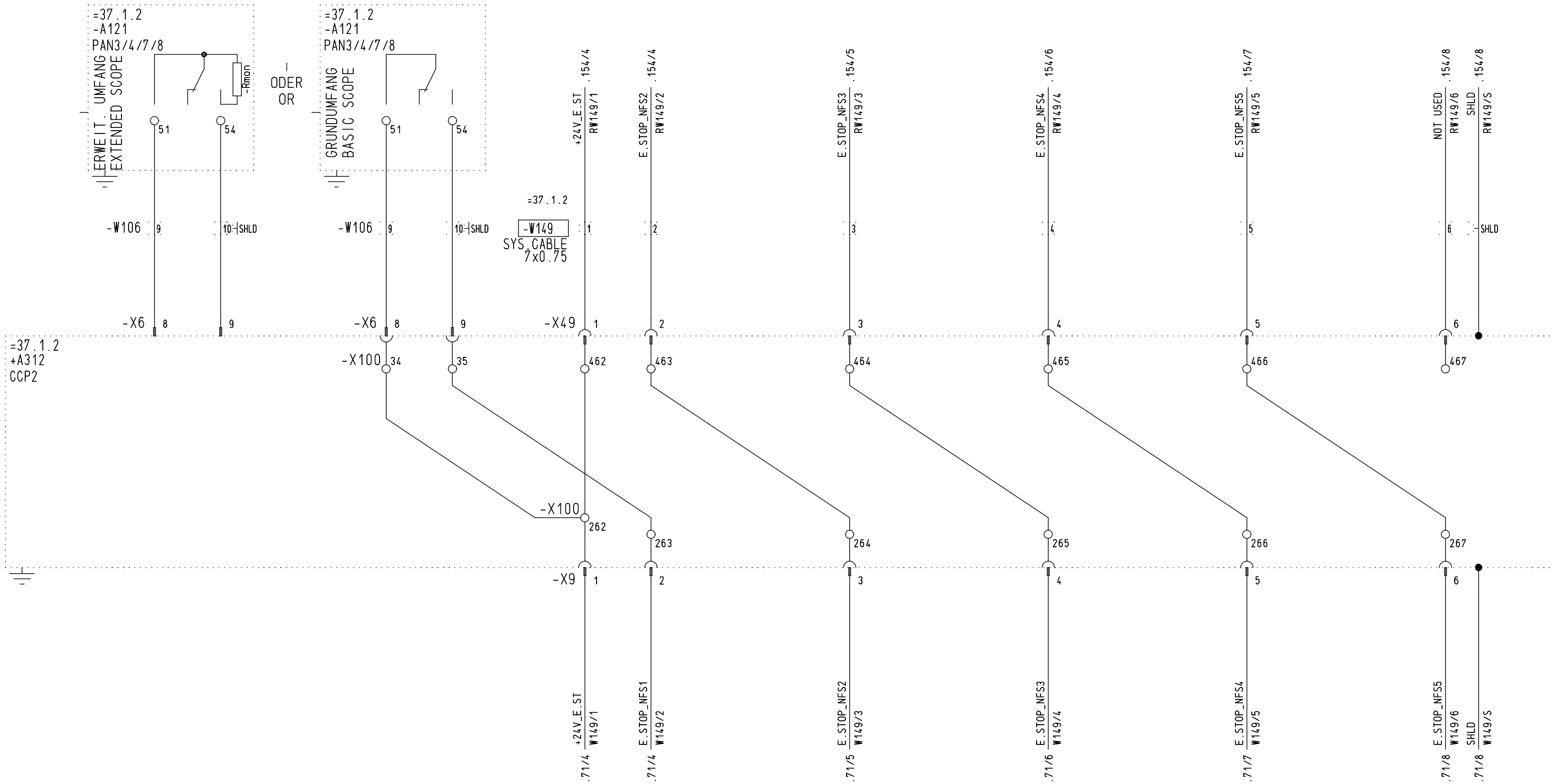
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



|                                                                                                  |          |                                                                                                              |           |                    |
|--------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------|-----------|--------------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |          | NEBENFAHRSTAND(VOLL), MOTOR 1: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(FULL), ENGINE 1: EMERG.-STOP PUSHBUTTON |           |                    |
| Auftrag/<br>Projekt                                                                              | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No.                                                                                | X00031757 | Blatt<br>Sheet 116 |
| Referenz-Nr./Reference-No.                                                                       |          | -----                                                                                                        |           |                    |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |

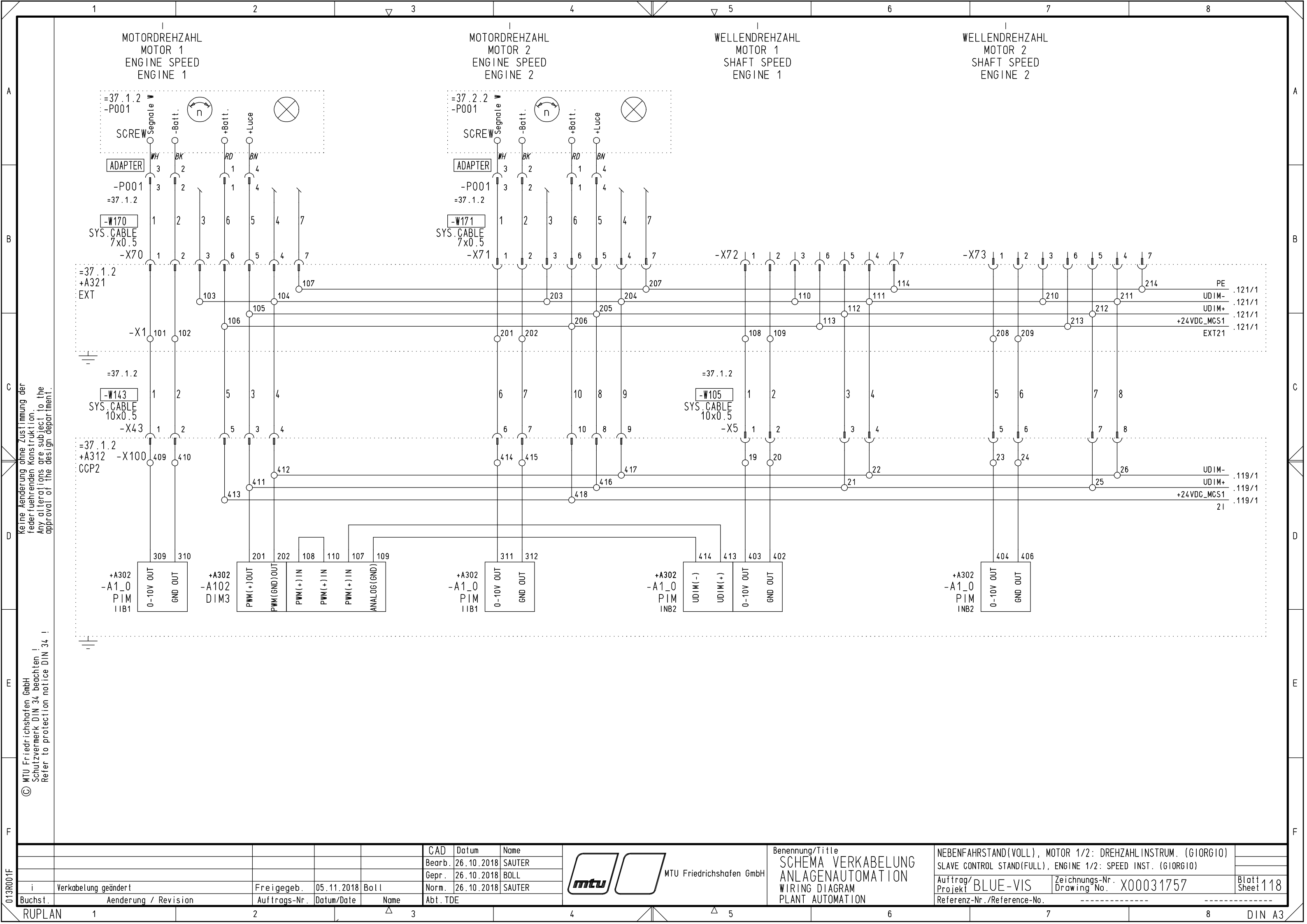


MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 2: NOTSTOPPTASTER  
SLAVE CONTROL STAND(FULL), ENGINE 2: EMERG.-STOP PUSHBUTTON  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt 117  
Sheet



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



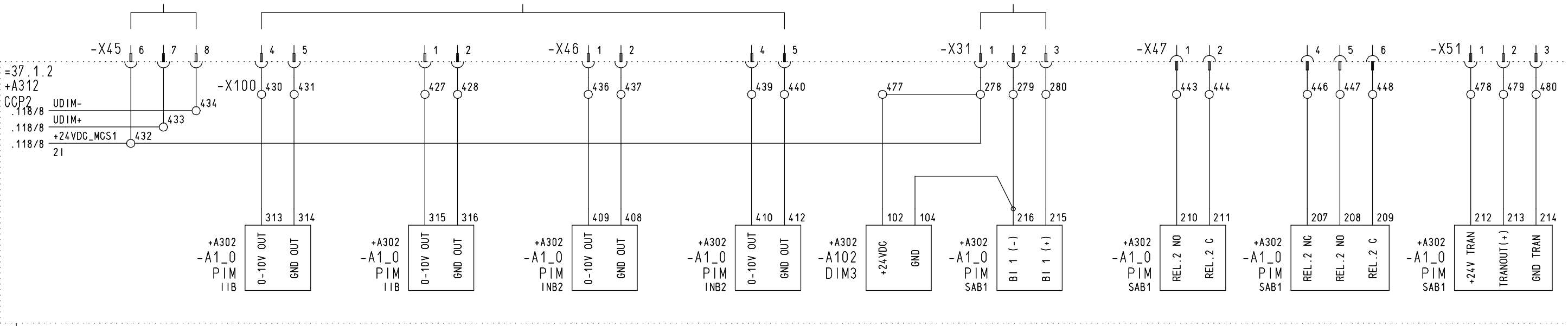
Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

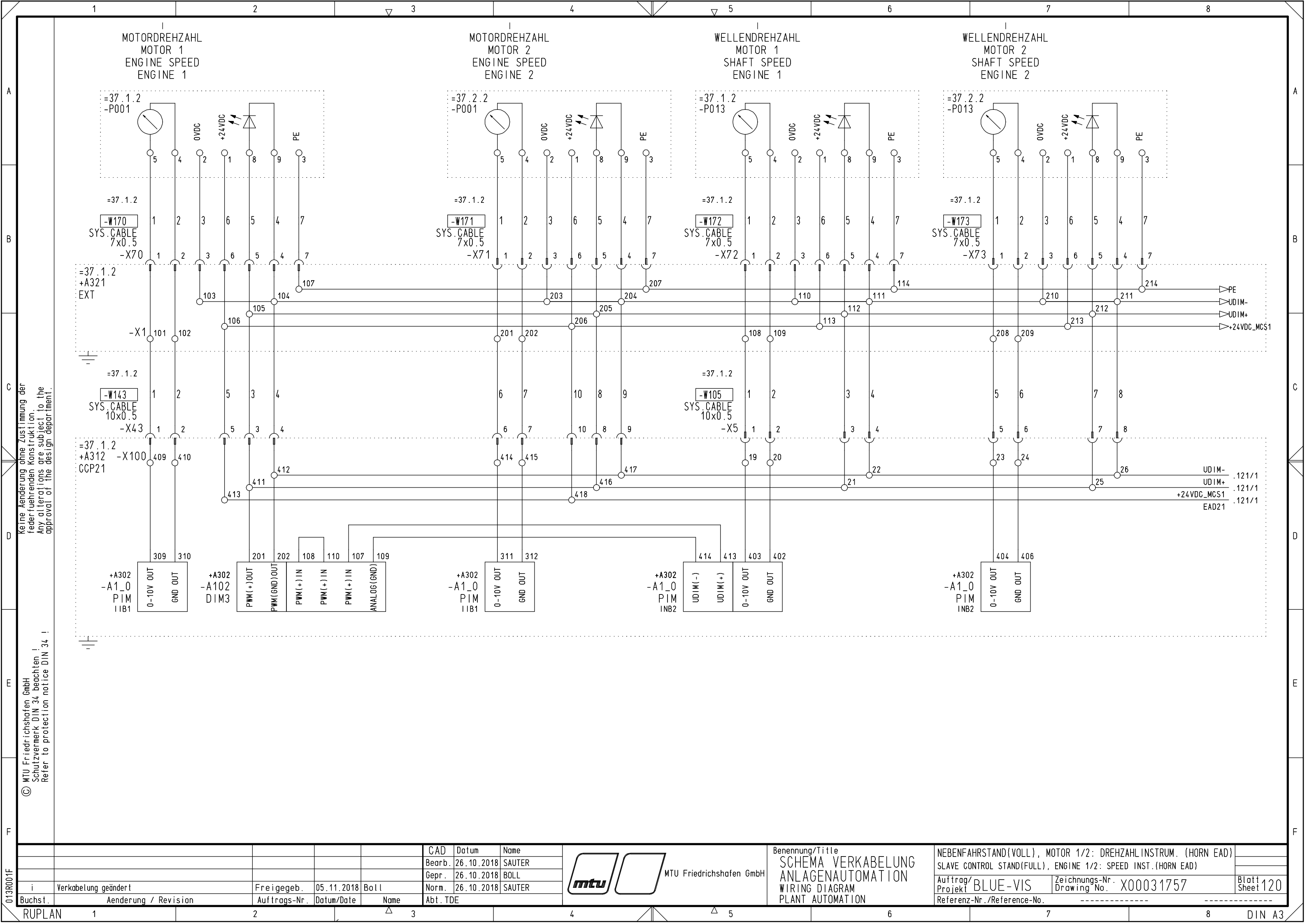
|                                                                                          |          |                                         |
|------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(VOLL), MOTOR 1/2: RESERVE<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPARE |          |                                         |
| Auftrag/<br>Projekt                                                                      | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                               |          | Blatt<br>Sheet 119                      |

SIEHE HINWEISBLATT "STROMVERSORGUNG"  
SEE INSTRUCTION SHEET "POWER SUPPLY"

SIEHE SEITEN "ANZEIGEINSTRUMENTE (CPP/WJ)"  
SEE SHEETS "IND. INSTRUMENTS (CPP/WJ)"

SIEHE SEITE "FREMD RCS: FAHRSTANDSUMSCHALTUNG"  
SEE SHEET "NON-MTU RCS: CONTROL STAND TRANSFER"







Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |

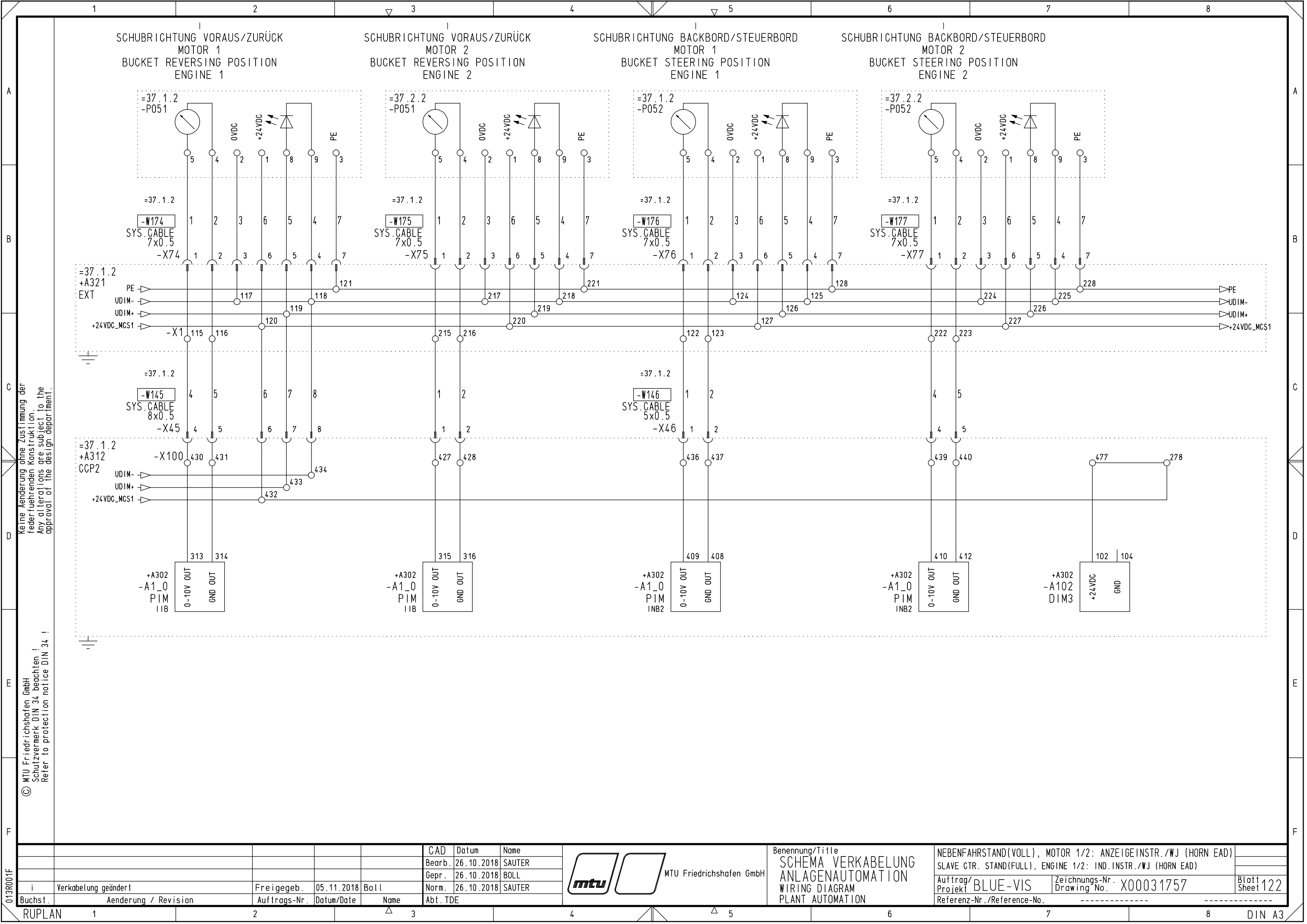


MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ANZEIGEINST./CPP (HORN EAD)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: IND.INST./CPP (HORN EAD)

|                            |          |                               |           |                |     |
|----------------------------|----------|-------------------------------|-----------|----------------|-----|
| Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 121 |
| Referenz-Nr./Reference-No. | -----    |                               |           |                |     |



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

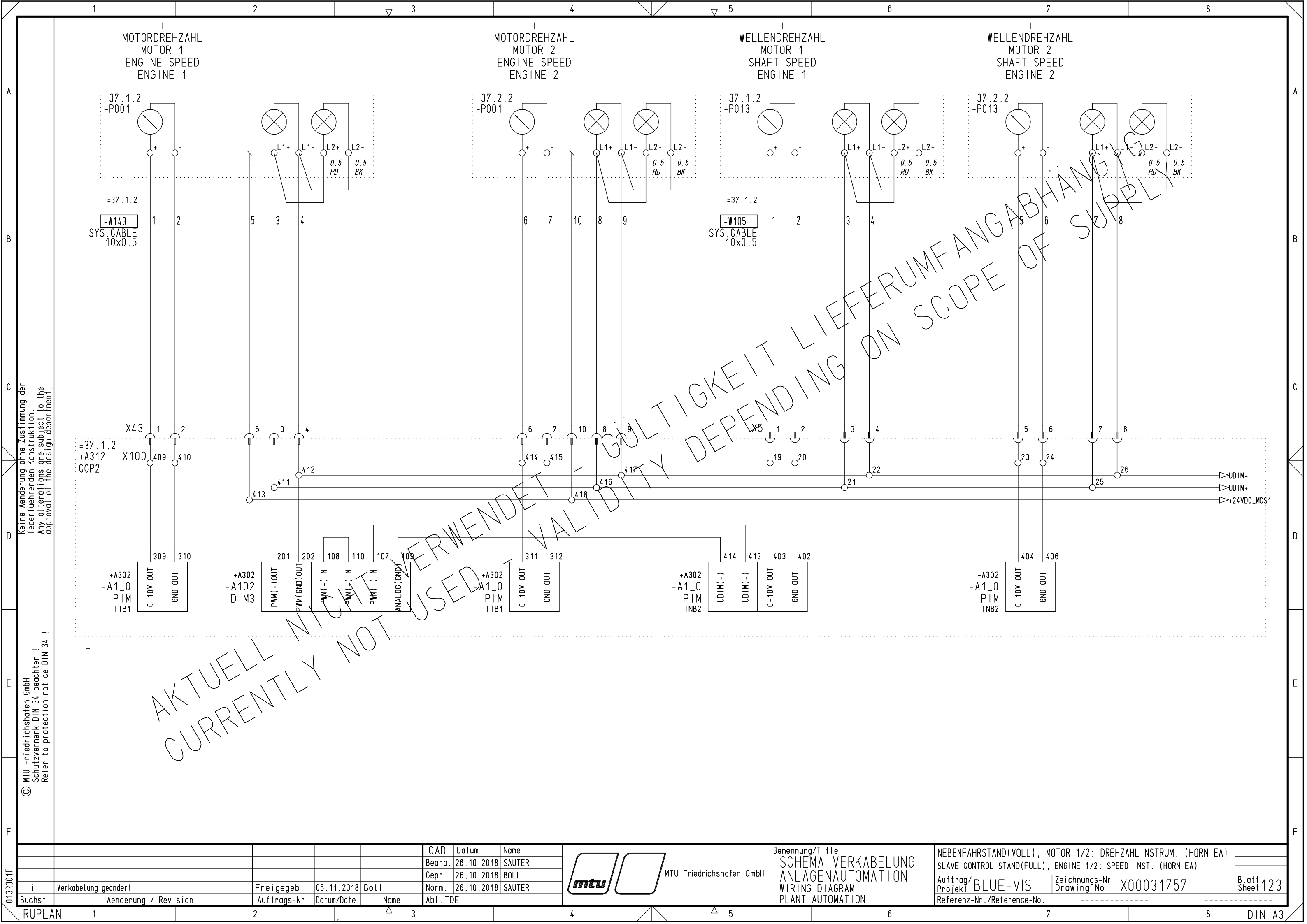
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ANZEIGEINSTR./WJ (HORN EAD)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: IND.INSTR./WJ (HORN EAD)

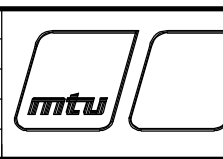
Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt/Sheet: 122



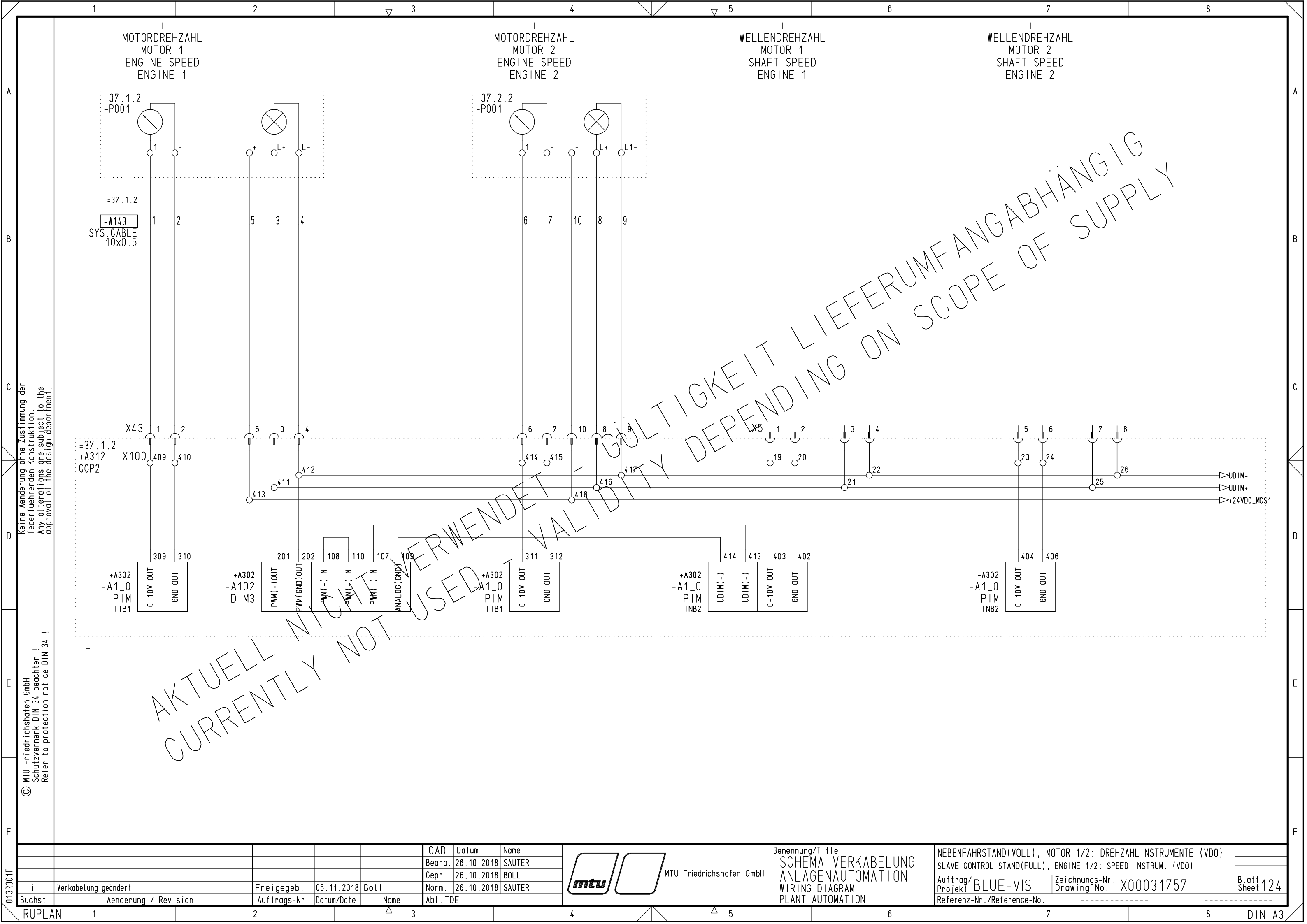


|          |                      |              |            |
|----------|----------------------|--------------|------------|
| CAD      |                      | Datum        | Name       |
| Bearb.   |                      | 26.10.2018   | SAUTER     |
| Gepr.    |                      | 26.10.2018   | BOLL       |
| Norm.    |                      | 26.10.2018   | SAUTER     |
| Abt. TDE |                      |              |            |
| Buchst.  | Aenderung / Revision | Auftrags-Nr. | Datum/Date |
| i        | Verkabelung geändert | Freigegeb.   | 05.11.2018 |
|          |                      | Name         | Boll       |



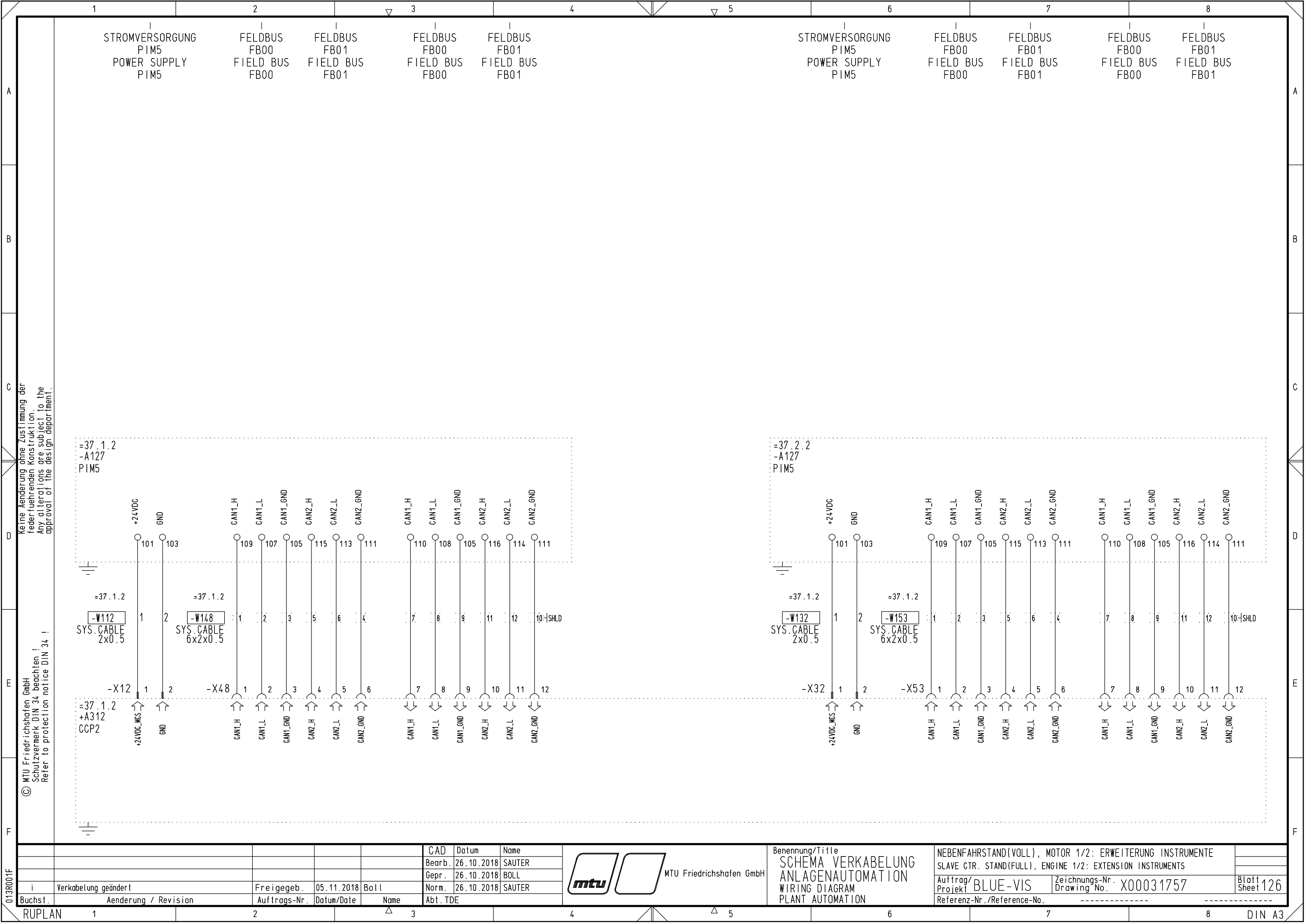
MTU Friedrichshafen GmbH

|                            |          |                                                              |           |
|----------------------------|----------|--------------------------------------------------------------|-----------|
| Benennung/Title            |          | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHLINSTRUM. (HORN EA)  |           |
| SCHEMA VERKABELUNG         |          | SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INST. (HORN EA) |           |
| ANLAGENAUTOMATION          |          |                                                              |           |
| WIRING DIAGRAM             |          |                                                              |           |
| PLANT AUTOMATION           |          |                                                              |           |
| Auftrag/Projekt            | BLUE-VIS | Zeichnungs-Nr. Drawing No.                                   | X00031757 |
| Referenz-Nr./Reference-No. |          |                                                              |           |
|                            |          | Blatt 123 Sheet 123                                          |           |



|                               |                      |  |              |            |        |            |        |                                                                                                                |                                                                                                  |                                                             |          |                               |           |  |
|-------------------------------|----------------------|--|--------------|------------|--------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------|-------------------------------|-----------|--|
|                               |                      |  |              |            | CAD    | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(VOLL), MOTOR 1/2: DREHZAHLINSTRUMENTE (VDO)  |          | Blatt<br>Sheet 124            |           |  |
|                               |                      |  |              |            | Bearb. | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | SLAVE CONTROL STAND(FULL), ENGINE 1/2: SPEED INSTRUM. (VDO) |          |                               |           |  |
|                               |                      |  |              |            | Gepr.  | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                             |          |                               |           |  |
|                               |                      |  |              |            | Norm.  | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                             |          |                               |           |  |
| i                             | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll   |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |  |
| Buchst.                       | Änderung / Revision  |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE   |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                  |          | -----                         |           |  |
| RUPLAN 1 2 3 4 5 6 7 8 DIN A3 |                      |  |              |            |        |            |        |                                                                                                                |                                                                                                  |                                                             |          |                               |           |  |





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INSTRUMENTE  
SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INSTRUMENTS

Auftrag/Projekt  
BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

Referenz-Nr./Reference-No.

Blatt  
Sheet 126

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST. (GIORGIO)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INST.(GIORGIO)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 127





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt 129  
Sheet

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |        |            |        |
|---------|----------------------|--------------|------------|------|----------|--------|------------|--------|
|         |                      |              |            |      |          | CAD    | Datum      | Name   |
|         |                      |              |            |      |          | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      |          | Gepr.  | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |        |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 130



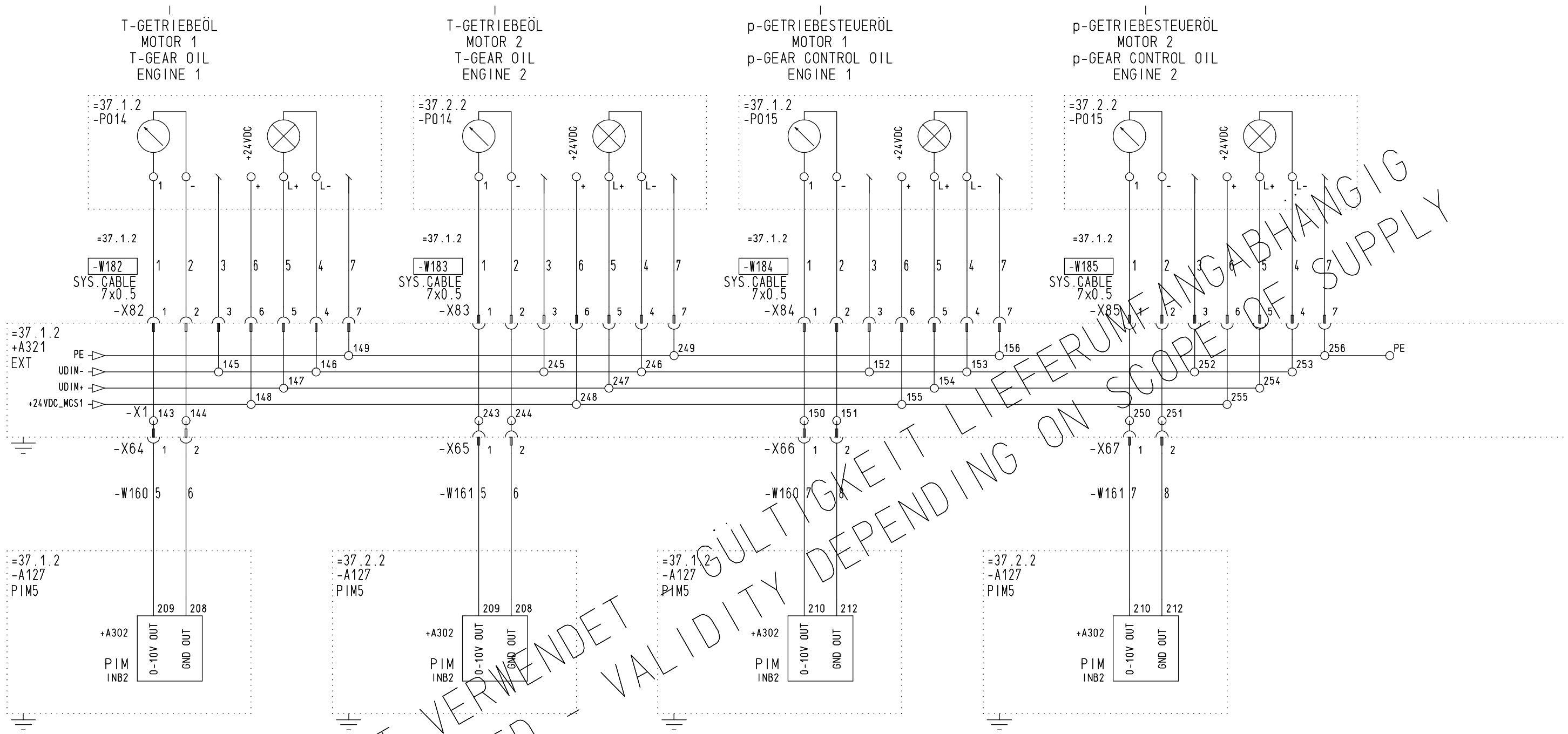


Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

RUPLAN



|  | CAD    | Datum      | Name   |
|--|--------|------------|--------|
|  | Bearb. | 26.10.2018 | SAUTER |
|  | Gepr.  | 26.10.2018 | BOLL   |
|  | Norm.  | 26.10.2018 | SAUTER |

|  | Abt. TDE |
|--|----------|
|  |          |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)  
SLAVE CTR. STAND(FULL), ENGINE 1/2: EXTENSION INSTR. (VDO)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr. / Reference-No. -----

Blatt 132  
Sheet

1 2 3 4 5 6 7 8 DIN A3

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                               |          |                                         |
|---------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(VOLL), MOTOR 1/2: STEUERRAD (WJ)<br>SLAVE CONTROL STAND(FULL), ENGINE 1/2: STEERING WHEEL (WJ) |          |                                         |
| Auftrag/<br>Projekt                                                                                           | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                                    |          | Blatt 133<br>Sheet                      |

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL) 1, MOTOR 1/2: JOYSTICK (WJ)  
SLAVE CONTROL STAND(FULL) 1, ENGINE 1/2: JOYSTICK (WJ)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt 134  
Sheet

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

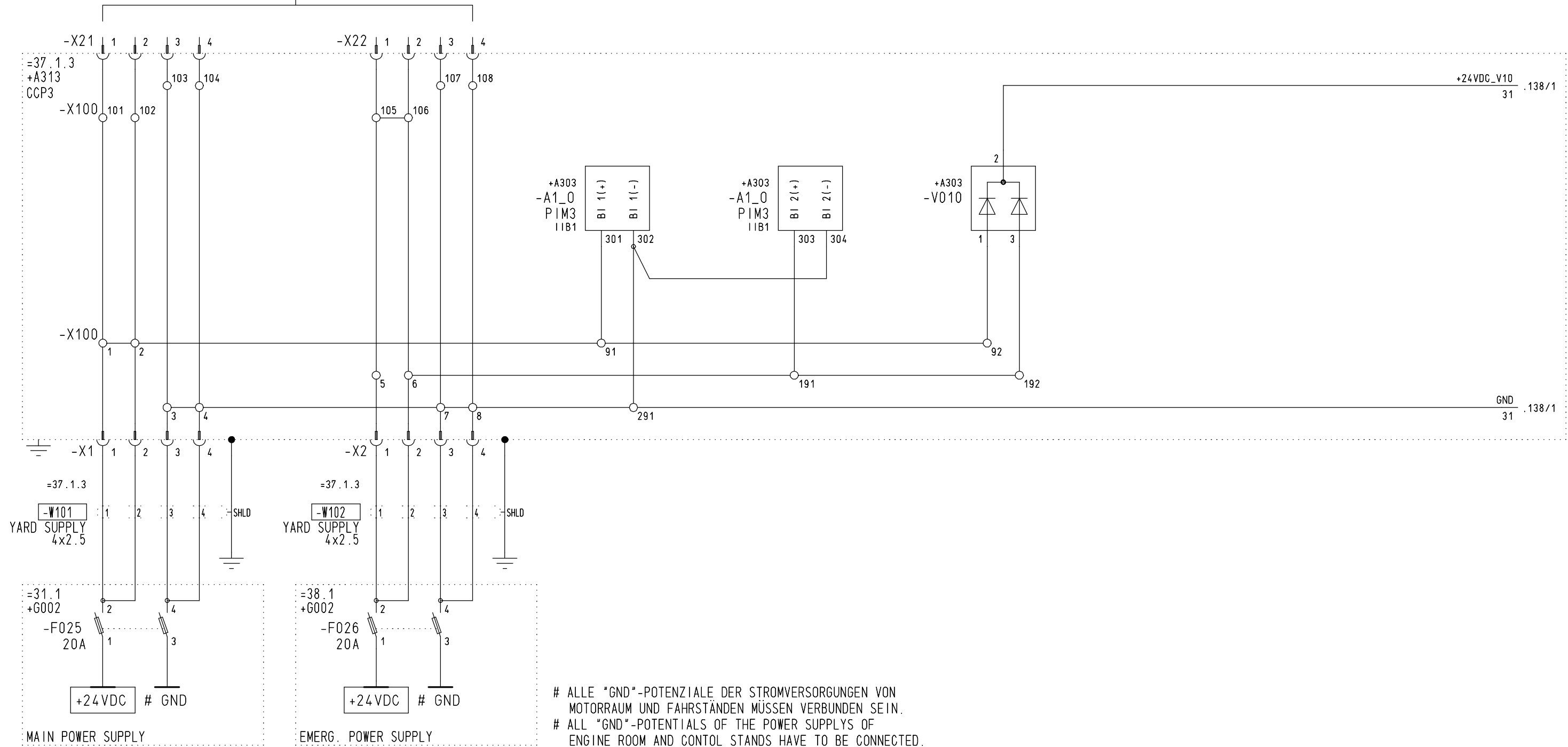
Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(VOLL) 2, MOTOR 1/2: JOYSTICK (WJ)  
SLAVE CONTROL STAND(FULL) 2, ENGINE 1/2: JOYSTICK (WJ)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt 135  
Sheet

| HAUPTSTROMVERSORGUNG | NOTSTROMVERSORGUNG  | HAUPTSTROMVERSORGUNG<br>ÜBERWACHUNG | NOTSTROMVERSORGUNG<br>ÜBERWACHUNG | SPANNUNGSKOPPLUNG |
|----------------------|---------------------|-------------------------------------|-----------------------------------|-------------------|
| MAIN POWER SUPPLY    | EMERG. POWER SUPPLY | MAIN POWER SUPPLY<br>MONITORING     | EMERG. POWER SUPPLY<br>MONITORING | POWER COUPLING    |

SIEHE HINWEISBLATT "STROMVERSORGUNG"  
SEE INSTRUCTION SHEET "POWER SUPPLY"



# ALLE "GND"-POTENZIALE DER STROMVERSORGUNGEN VON  
MOTORRAUM UND FAHRSTÄNDEN MÜSSEN VERBUNDEN SEIN.  
# ALL "GND"-POTENTIALS OF THE POWER SUPPLYS OF  
ENGINE ROOM AND CONTOL STANDS HAVE TO BE CONNECTED.

|        |            |        |
|--------|------------|--------|
| CAD    | Datum      | Name   |
| Bearb. | 26.10.2018 | SAUTER |
| Gepr.  | 26.10.2018 | BOLL   |
| Norm.  | 26.10.2018 | SAUTER |



/ MTU Friedrichshafen GmbH

|   |                                                                                                  |
|---|--------------------------------------------------------------------------------------------------|
| H | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |
|---|--------------------------------------------------------------------------------------------------|

|                                                              |  |
|--------------------------------------------------------------|--|
| NEBENF.ST.(RED.), MOTOR 1/2: STROMVER. 24VDC (RCS VIA DIODE) |  |
| SL.CTR.ST.(RED.), ENG.1/2: POWER SUPPLY 24VDC(RCS VIA DIODE) |  |

|                     |          |                               |           |
|---------------------|----------|-------------------------------|-----------|
| Auftrag/<br>Projekt | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
|---------------------|----------|-------------------------------|-----------|

Blatt 136  
Sheet 136

|                            |       |
|----------------------------|-------|
| Referenz-Nr./Reference-No. | ----- |
|----------------------------|-------|

---



Keine Änderung ohne Zustimmung der federführenden Konstruktion.  
Any alterations are subject to the approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F

HAUPTSTROMVERSORGUNG  
MAIN POWER SUPPLY

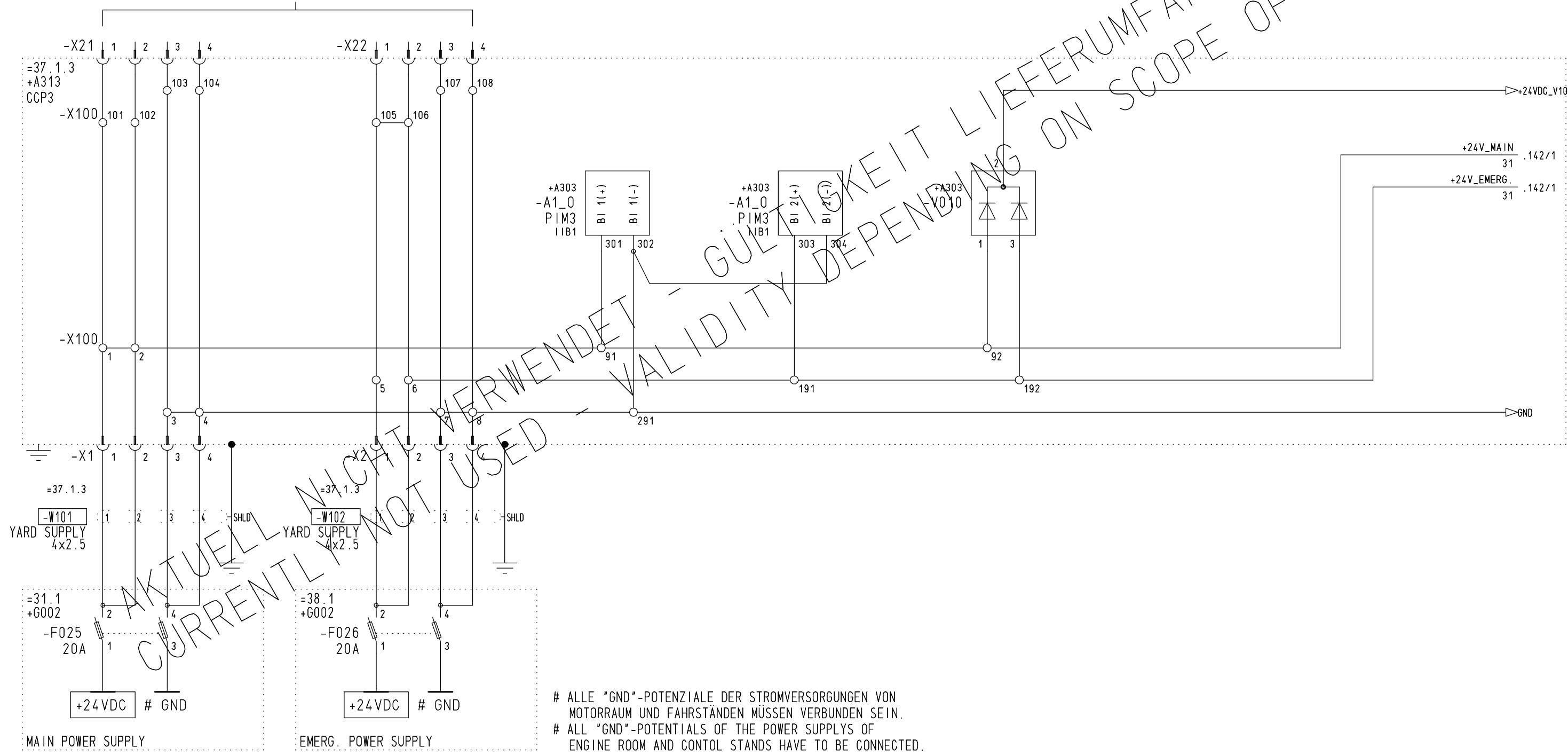
NOTSTROMVERSORGUNG  
EMERG. POWER SUPPLY

HAUPTSTROMVERSORGUNG  
ÜBERWACHUNG  
MAIN POWER SUPPLY  
MONITORING

NOTSTROMVERSORGUNG  
ÜBERWACHUNG  
EMERG. POWER SUPPLY  
MONITORING

SPANNUNGSKOPPLUNG  
POWER COUPLING

SIEHE HINWEISBLATT "STROMVERSORGUNG"  
SEE INSTRUCTION SHEET "POWER SUPPLY"



# ALLE "GND"-POTENZIALE DER STROMVERSORGUNGEN VON  
MOTORRAUM UND FAHRSTÄNDEN MÜSSEN VERBUNDEN SEIN.  
# ALL "GND"-POTENTIALS OF THE POWER SUPPLYS OF  
ENGINE ROOM AND CONTOL STANDS HAVE TO BE CONNECTED.

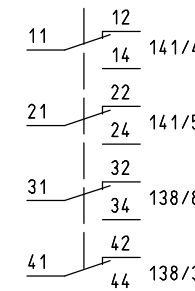
|         |                      | CAD          | Datum      | Name   |
|---------|----------------------|--------------|------------|--------|
|         |                      | Bearb.       | 26.10.2018 | SAUTER |
|         |                      | Gepr.        | 26.10.2018 | BOLL   |
|         |                      | Norm.        | 26.10.2018 | SAUTER |
|         |                      | Abt. TDE     |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll   |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                     |                                         |                    |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC |                                         |                    |
| Auftrag/Projekt<br>BLUE-VIS                                                                                         | Zeichnungs-Nr.<br>Drawing No. X00031757 | Blatt<br>Sheet 137 |
| Referenz-Nr./Reference-No. -----                                                                                    |                                         |                    |

MOTOR 2 EIN/AUS  
NEBENFAHRSTAND  
ENGINE 2 ON/OFF  
SLAVE CONTROL STAND





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |  |  |  |  |
|---------|----------------------|--------------|------------|------|--|--|--|--|--|
|         |                      |              |            |      |  |  |  |  |  |
|         |                      |              |            |      |  |  |  |  |  |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  |  |  |  |  |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  |  |  |  |  |

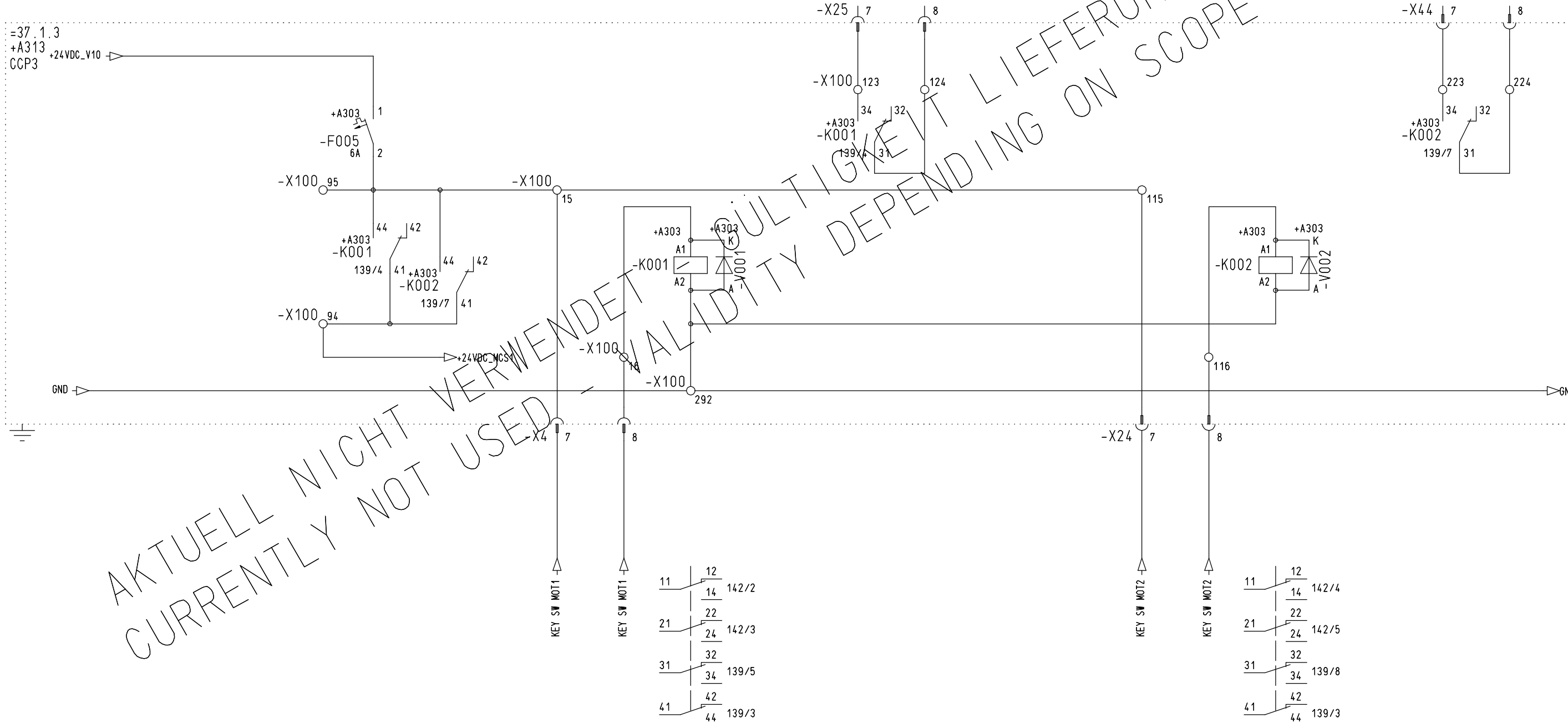
|          |            |        |
|----------|------------|--------|
| CAD      | Datum      | Name   |
| Bearb.   | 26.10.2018 | SAUTER |
| Gepr.    | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |

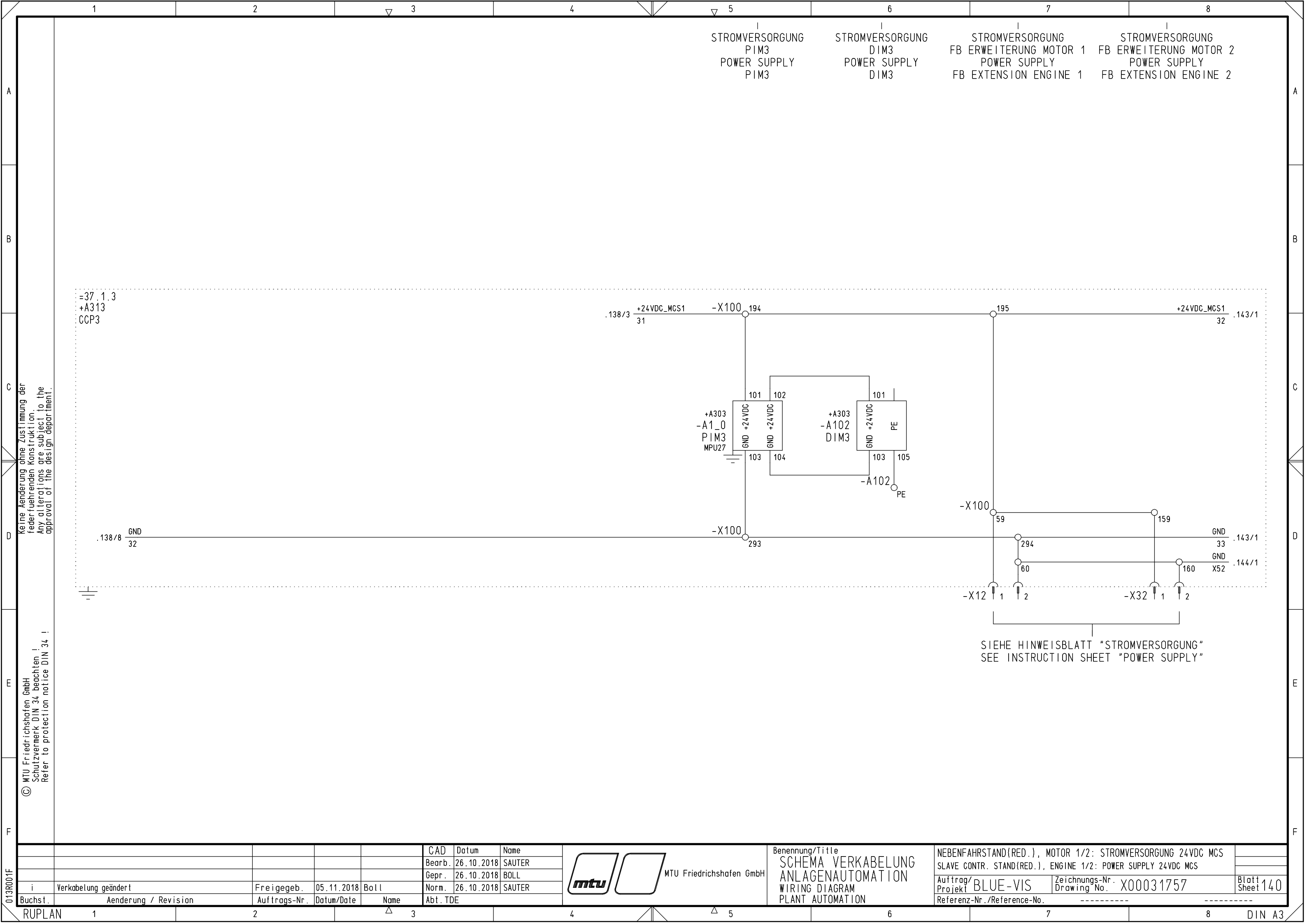


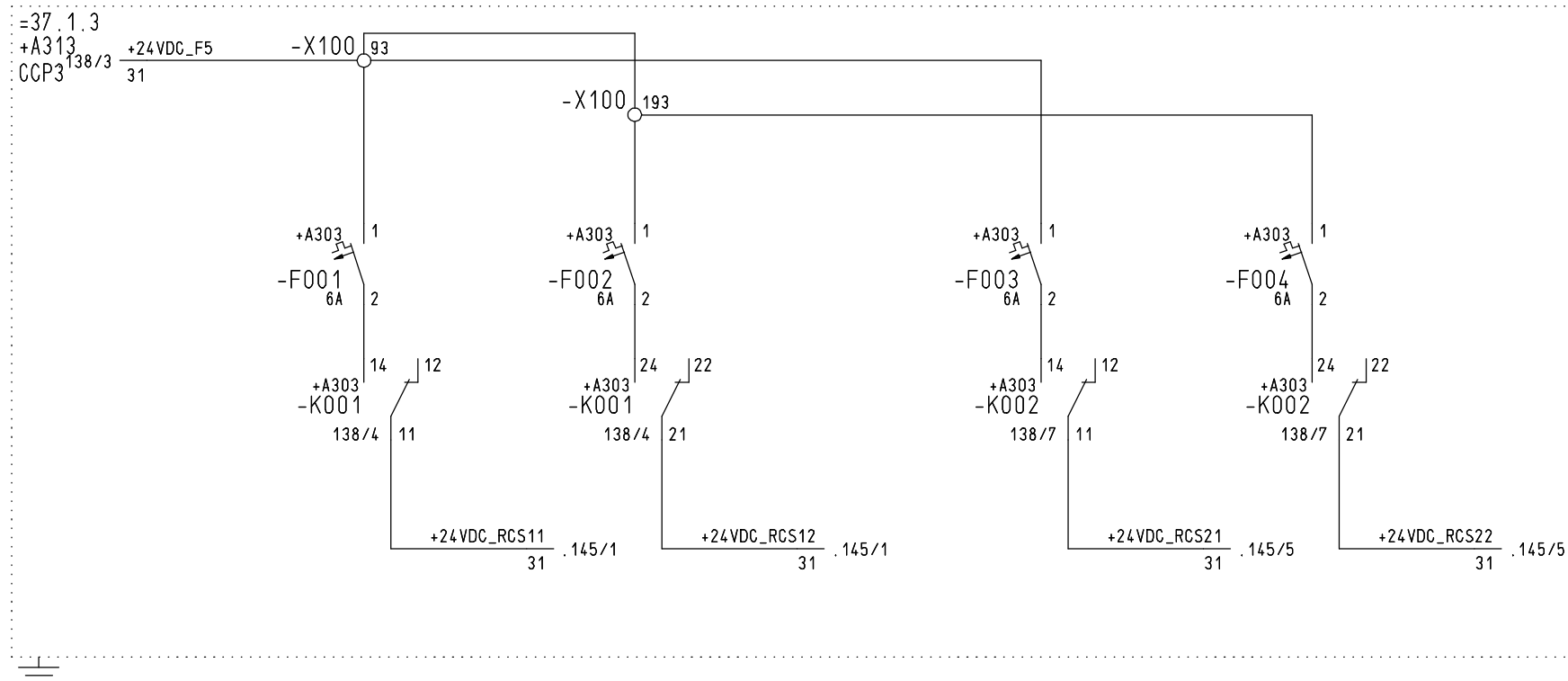
MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                       |          |                                         |
|-------------------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: 24VDC EIN/AUS<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: 24VDC ON/OFF |          |                                         |
| Auftrag/<br>Projekt                                                                                   | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                                            |          | Blatt<br>Sheet 139                      |







Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |
|---------|----------------------|--------------|------------|------|
|         |                      |              |            |      |
|         |                      |              |            |      |
|         |                      |              |            |      |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS  
SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC RCS  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. / Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

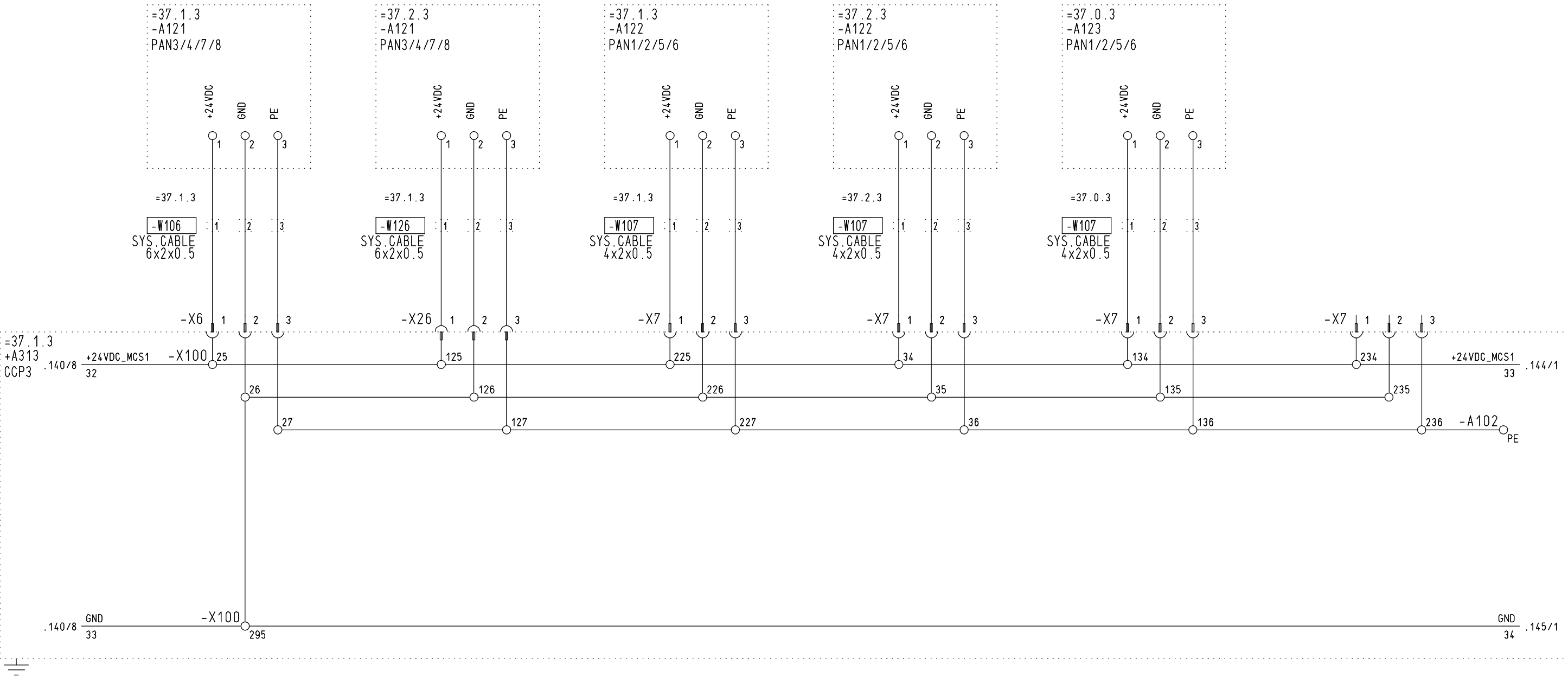
Blatt 142  
Sheet

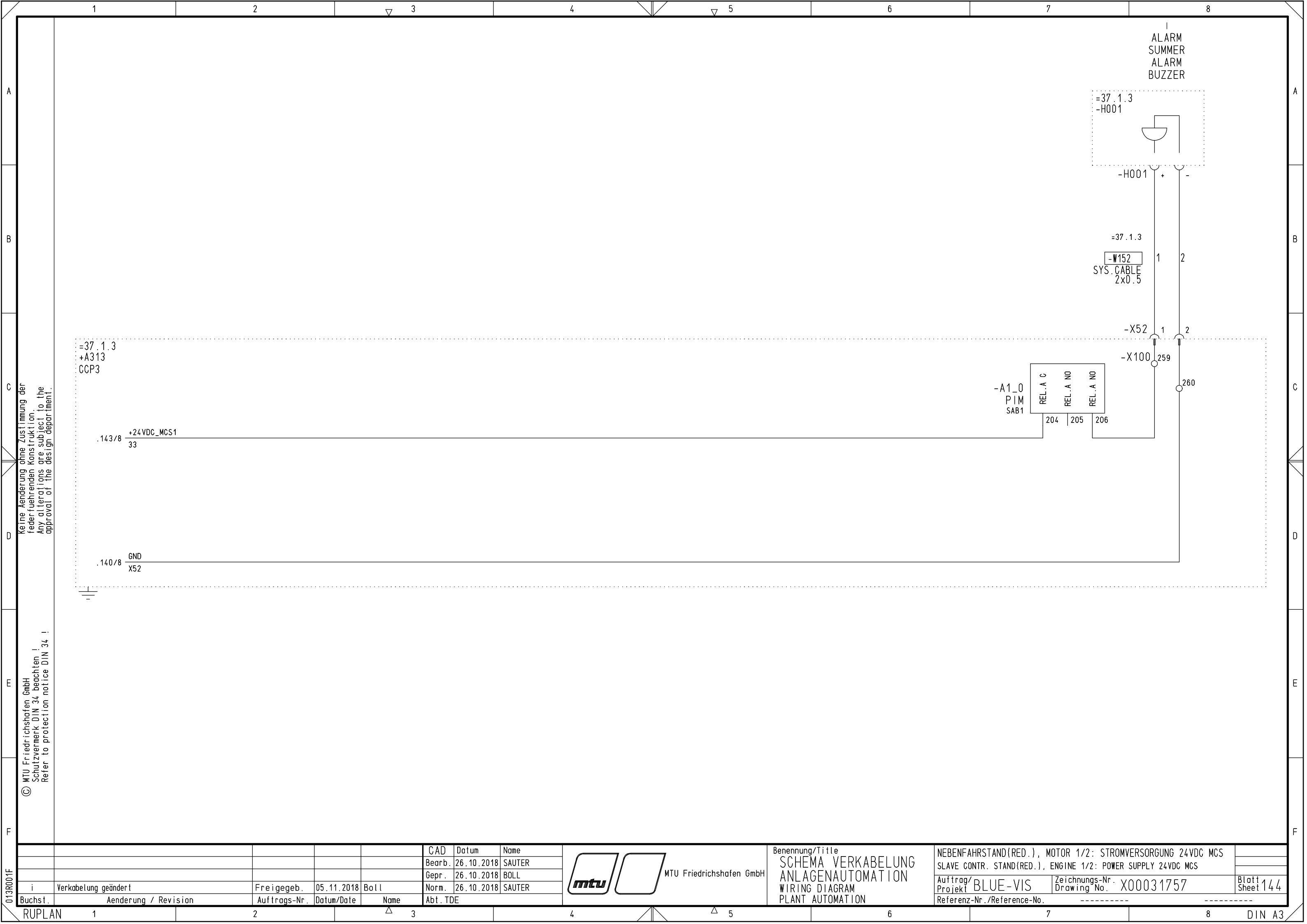
Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

|         |                      |  |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                              |          |                               |           |                |     |
|---------|----------------------|--|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------|-------------------------------|-----------|----------------|-----|
|         |                      |  |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC MCS   |          |                               |           |                |     |
|         |                      |  |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC MCS |          |                               |           |                |     |
|         |                      |  |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                              |          |                               |           |                |     |
|         |                      |  |              |            |      | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                              |          |                               |           |                |     |
| i       | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll |          |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                          | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 143 |
| Buchst. | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No. -----                             |          |                               |           |                |     |







Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |        |            |        |
|---------|----------------------|--------------|------------|------|----------|--------|------------|--------|
|         |                      |              |            |      |          | CAD    | Datum      | Name   |
|         |                      |              |            |      |          | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      |          | Gepr.  | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |        |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                            |          |                               |           |
|----------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: STROMVERSORGUNG 24VDC RCS<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: POWER SUPPLY 24VDC RCS |          |                               |           |
| Auftrag/<br>Projekt                                                                                                        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                                 |          | -----                         |           |
|                                                                                                                            |          | Blatt 145<br>Sheet            |           |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

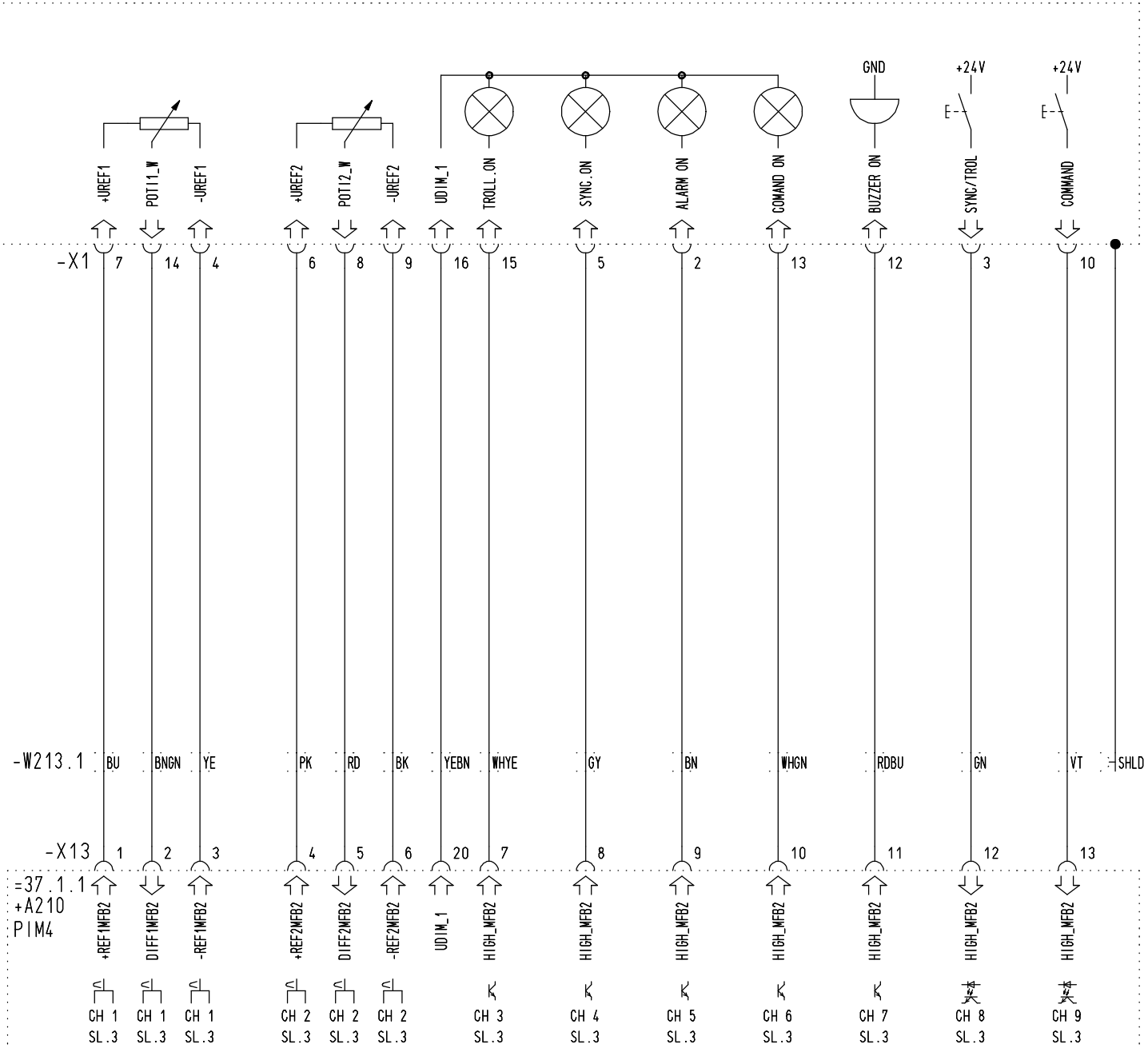
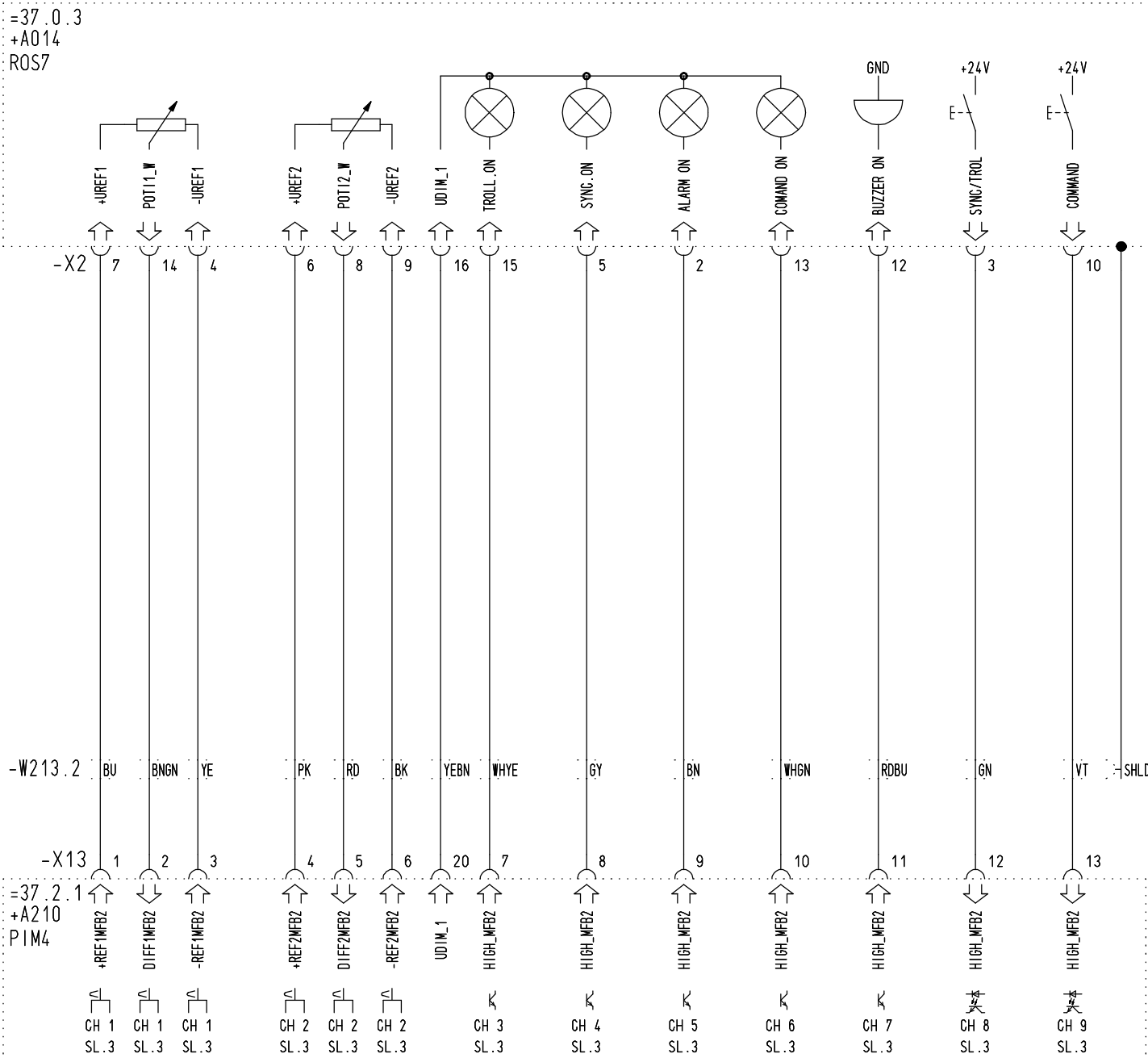
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                     |          |                               |                    |
|---------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|--------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: BEDIENSTATION ROS7<br>SLAVE CONTR. STAND(RED.), ENGINE 1/2: OPERATING STATION ROS7 |          |                               |                    |
| Auftrag/<br>Projekt                                                                                                 | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757          |
| Referenz-Nr./Reference-No.                                                                                          |          | -----                         |                    |
|                                                                                                                     |          |                               | Blatt<br>Sheet 146 |







Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

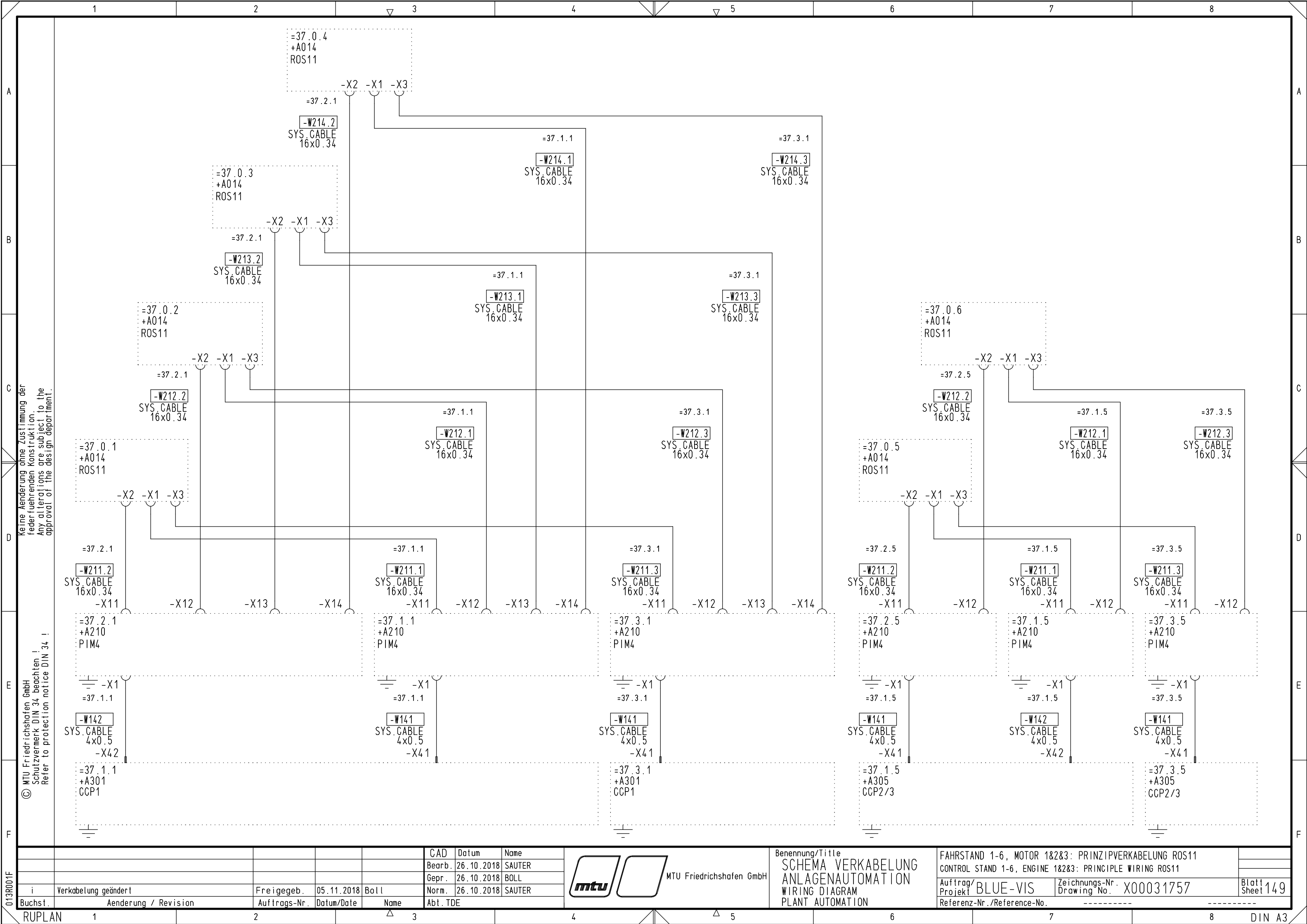
FAHRSTAND 1-6, MOTOR 1&2: PRINZIPVERKABELUNG ROS7  
CONTROL STAND 1-6, ENGINE 1&2: PRINCIPLE WIRING ROS7

Auftrag/  
Projekt BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

Blatt  
Sheet 148

Referenz-Nr./Reference-No. -----





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

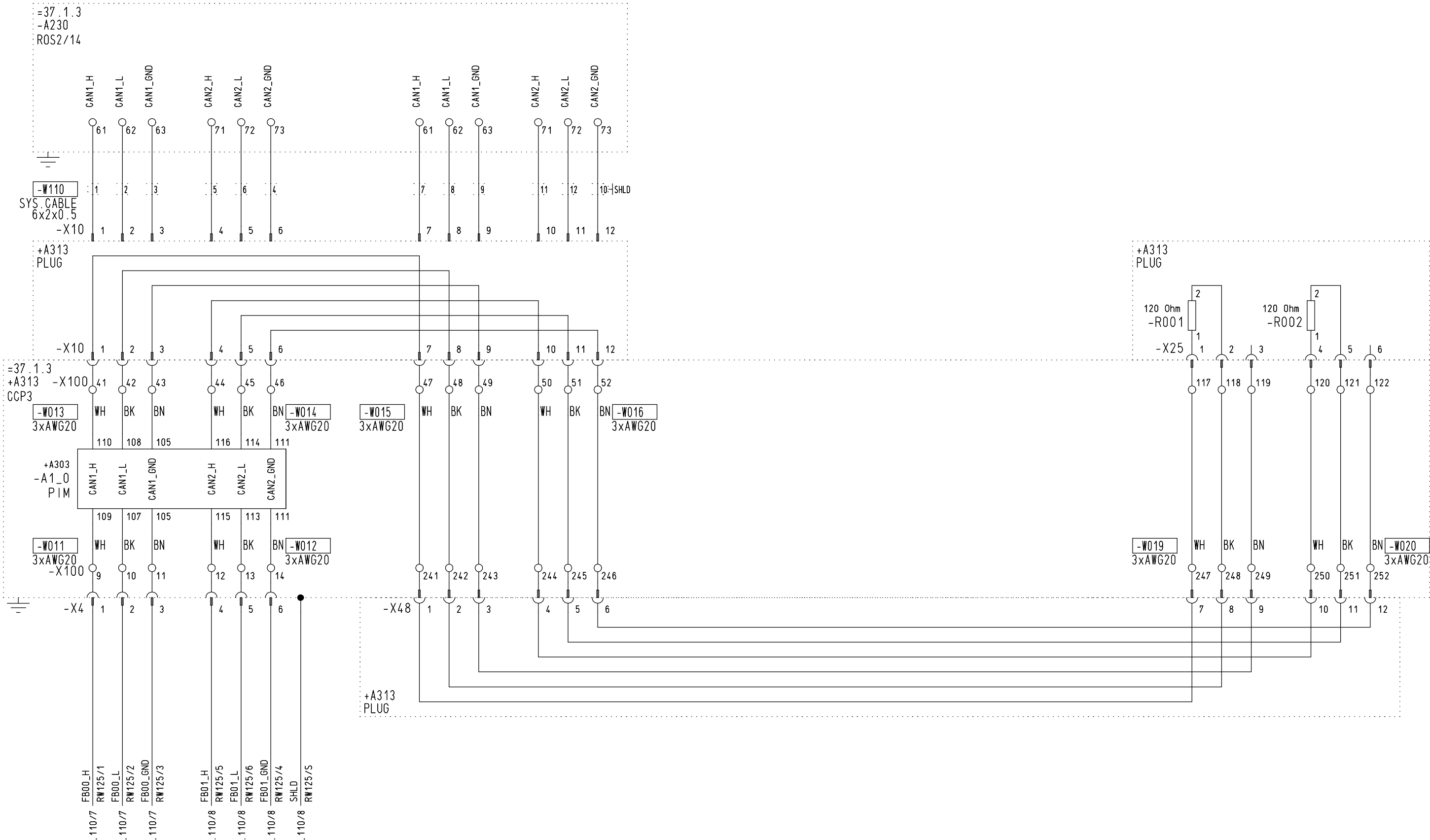
RUPLAN

FELDBUS MOTOR 1  
NEBENFAHRSTAND  
FIELD BUS ENGINE 1  
SLAVE CONTROL STAND

FELDBUS  
FB00  
FIELD BUS  
FB00

FELDBUS  
FB01  
FIELD BUS  
FB01

FELDBUS MOTOR 1  
NEBENFAHRSTAND  
FIELD BUS ENGINE 1  
SLAVE CONTROL STAND



|         |                      |              | CAD        | Datum      | Name   |
|---------|----------------------|--------------|------------|------------|--------|
|         |                      |              | Bearb.     | 26.10.2018 | SAUTER |
|         |                      |              | Gepr.      | 26.10.2018 | BOLL   |
|         |                      |              | Norm.      | 26.10.2018 | SAUTER |
|         |                      |              | Abt. TDE   |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name       |        |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll       |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                          |          |                                         |
|------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1: FELDBUS<br>SLAVE CONTROL STAND(ERD.), ENGINE 1: FIELD BUS |          |                                         |
| Auftrag/<br>Projekt                                                                      | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                               |          | Blatt<br>Sheet 151                      |

DIN A3





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |        |            |        |
|---------|----------------------|--------------|------------|------|----------|--------|------------|--------|
|         |                      |              |            |      |          | CAD    | Datum      | Name   |
|         |                      |              |            |      |          | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      |          | Gepr.  | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |        |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2/5/6: SENSOR-/AKTORBUS  
SLAVE CONTROL STAND(RED.), ENGINE 1/2: SENSOR/ACTUATOR BUS

Auftrag/Projekt  
BLUE-VIS

Zeichnungs-Nr.  
Drawing No. X00031757

Blatt  
Sheet 153

Referenz-Nr./Reference-No. -----

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                              |          |                               |           |
|--------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| NEBENFAHRSTAND(RED.), MOTOR 1: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(RED.), ENGINE 1: EMERG.-STOP PUSHBUTTON |          |                               |           |
| Auftrag/<br>Projekt                                                                                          | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                   |          | -----                         |           |



Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

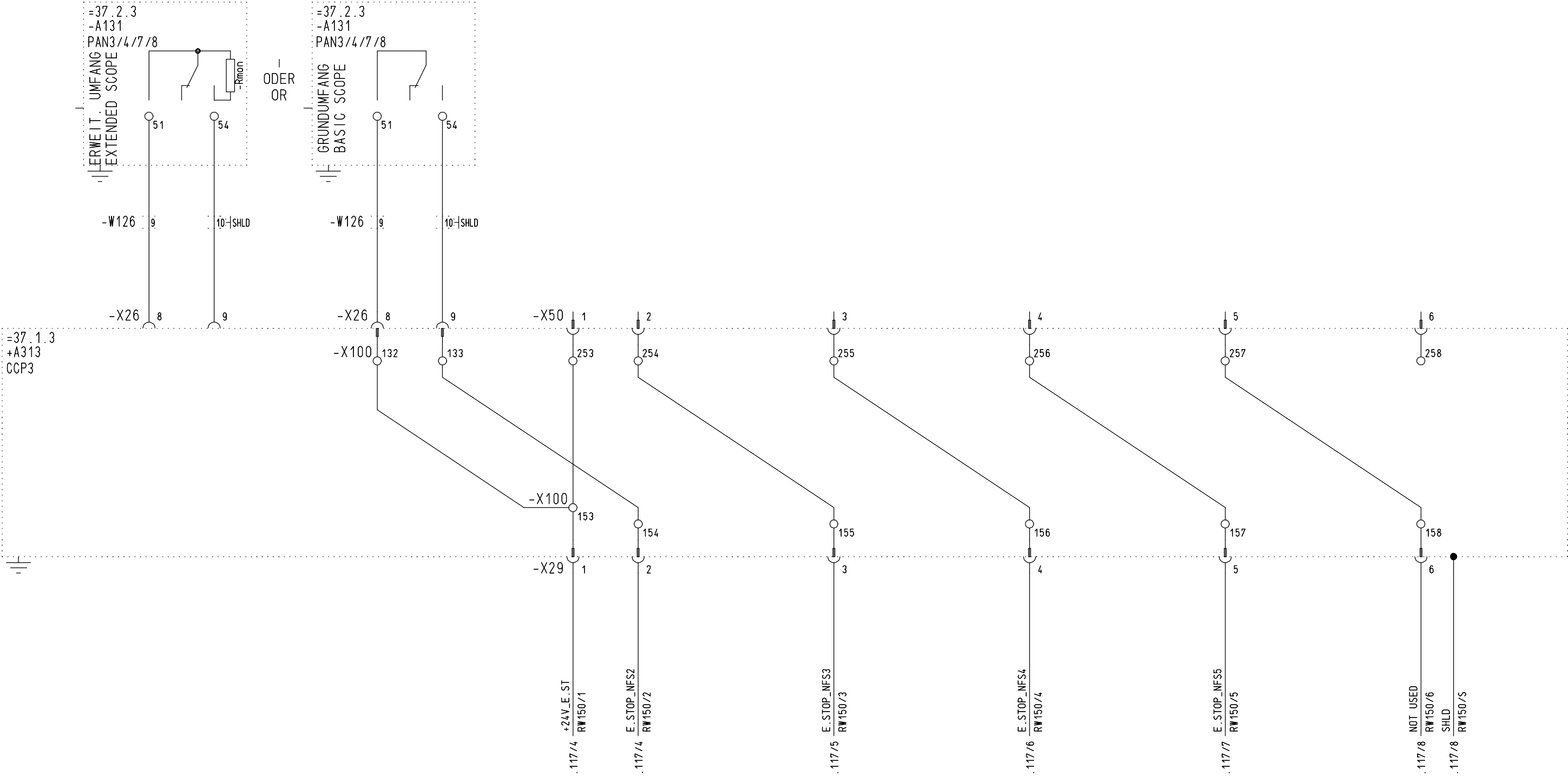
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                              |          |                               |           |
|--------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| NEBENFAHRSTAND(RED.), MOTOR 2: NOTSTOPPTASTER<br>SLAVE CONTROL STAND(RED.), ENGINE 2: EMERG.-STOP PUSHBUTTON |          |                               |           |
| Auftrag/<br>Projekt                                                                                          | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                   |          | -----                         |           |
| RUPLAN                                                                                                       |          | DIN A3                        |           |





1

2

3

4

5

6

7

8

RESERVE  
UDIM  
SPARE  
UDIM

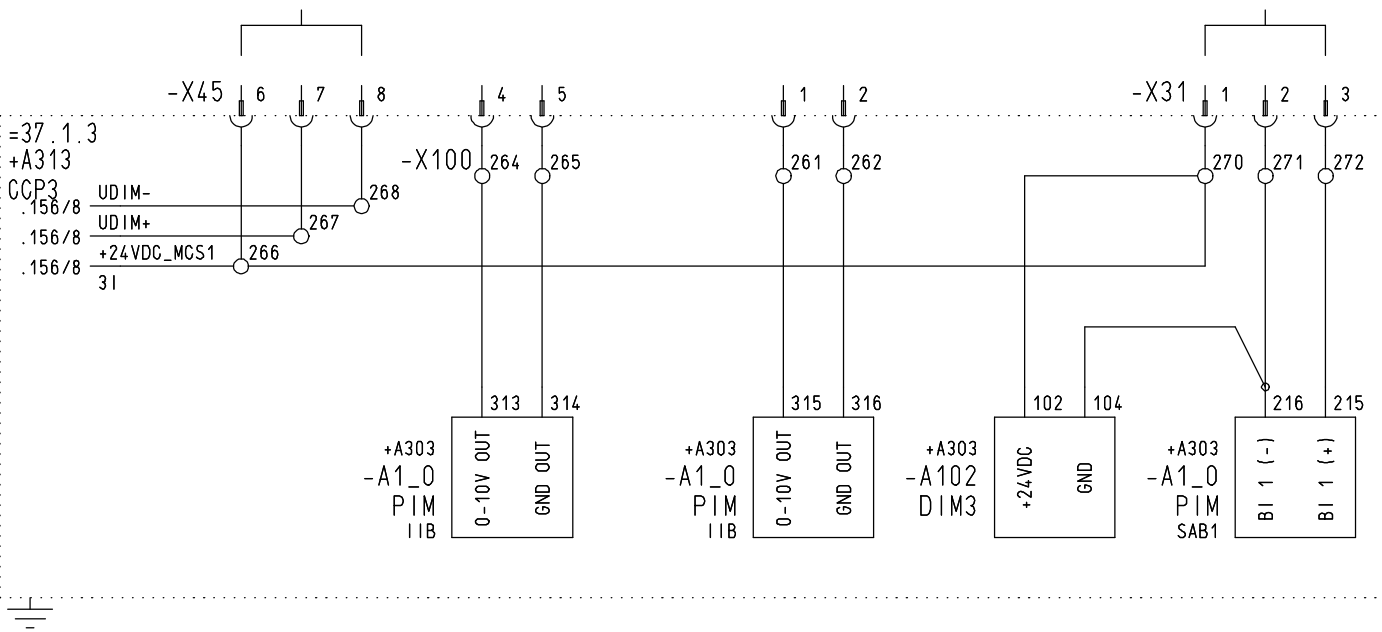
RESERVE  
0-10VDC,max.5mA  
SPARE  
0-10VDC,max.5mA

RESERVE  
0-10VDC,max.5mA  
SPARE  
0-10VDC,max.5mA

FREMD RCS  
AKTIV  
NON-MTU RCS  
ACTIVE

SIEHE HINWEISBLATT "STROMVERSORGUNG"  
SEE INSTRUCTION SHEET "POWER SUPPLY"

SIEHE SEITE "FREMD RCS: FAHRSTANDSUMSCHALTUNG"  
SEE SHEET "NON-MTU RCS: CONTROL STAND TRANSFER"



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

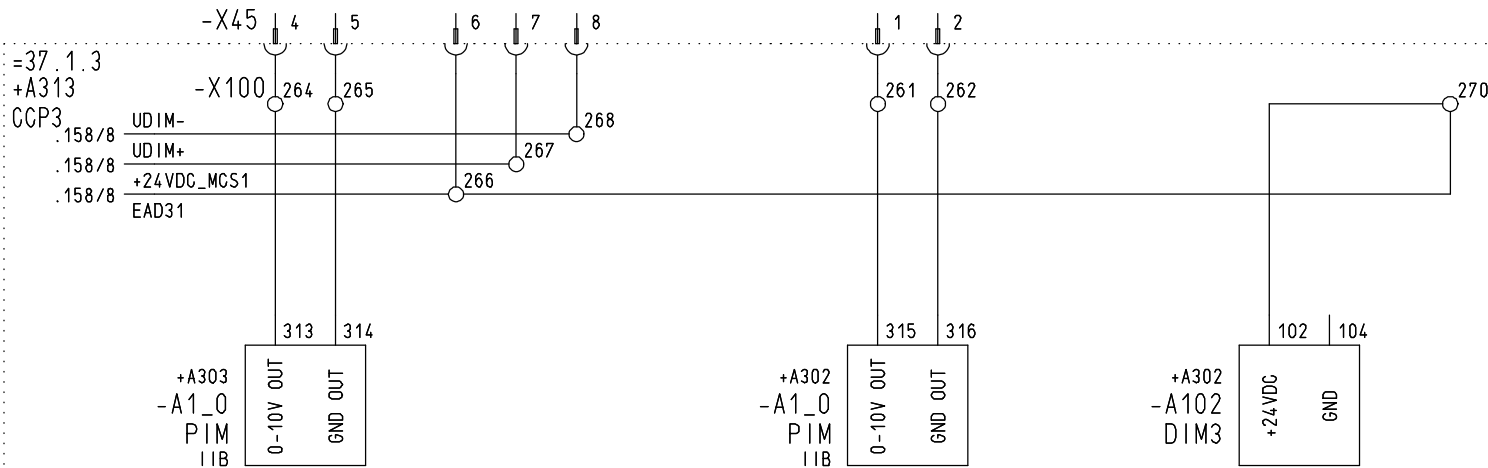
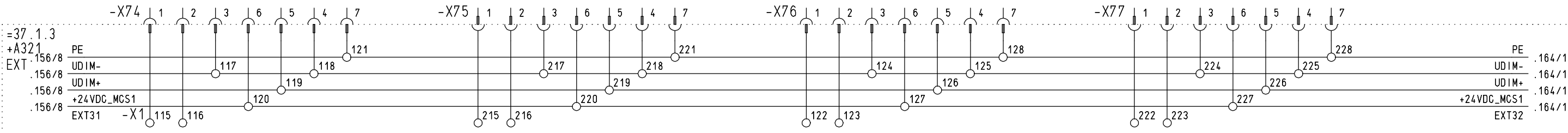
Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                          |          |                                         |
|------------------------------------------------------------------------------------------|----------|-----------------------------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: RESERVE<br>SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPARE |          |                                         |
| Auftrag/<br>Projekt                                                                      | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No.                                                               |          | Blatt<br>Sheet 157                      |

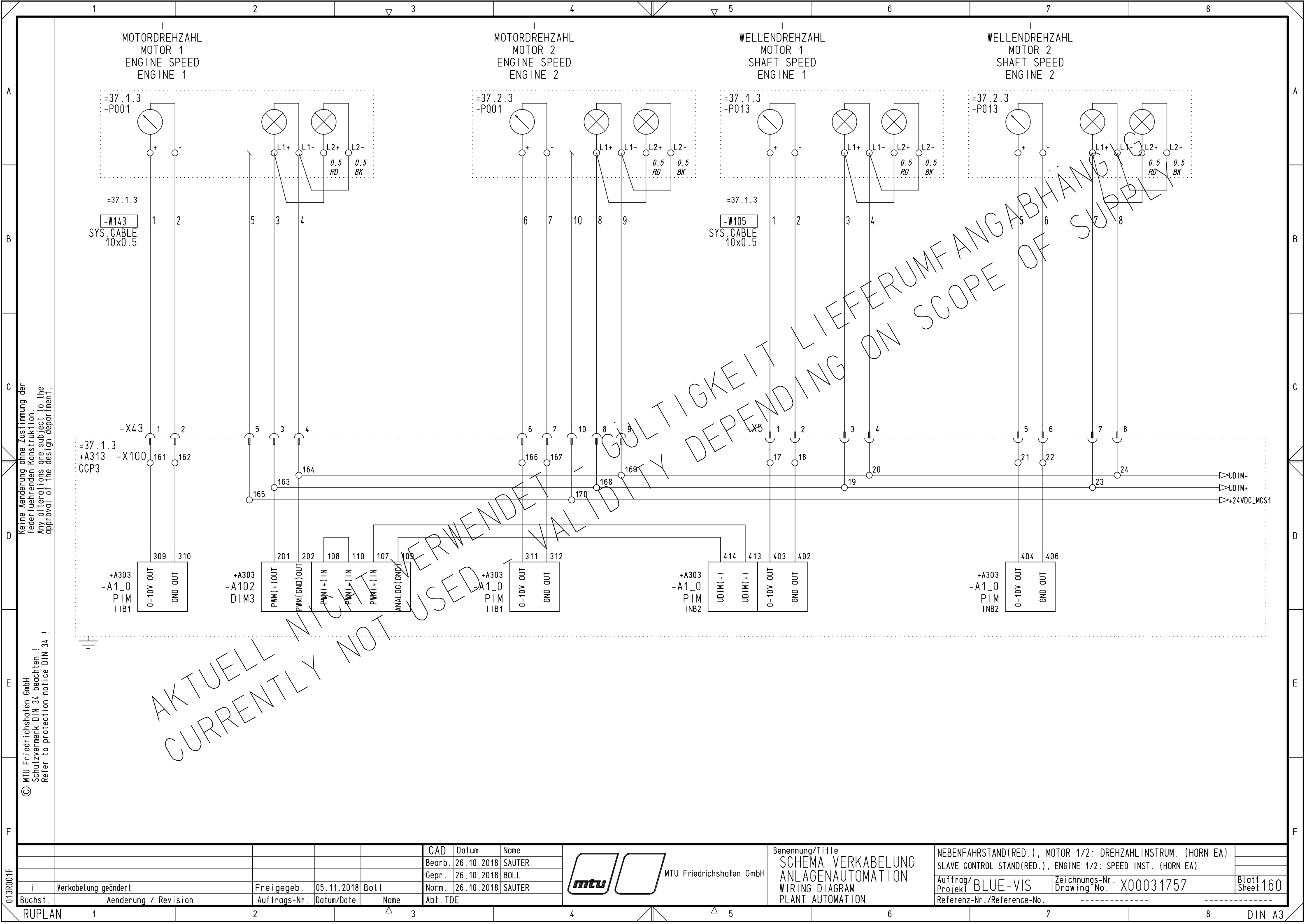


Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



|         |                      |              |            |      |  |          |            |        |                                                                                                                |                                                                                                  |                                                              |          |                                         |                    |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------|-----------------------------------------|--------------------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | NEBENFAHRSTAND(RED.), MOTOR 1/2: ANZEIGEINST./RES.(HORN EAD) |          |                                         |                    |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | SLAVE CTR. STAND(RED.), ENGINE 1/2: IND.INST./RES.(HORN EAD) |          |                                         |                    |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                              |          |                                         |                    |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                              |          |                                         |                    |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  |          |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                                          | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 | Blatt<br>Sheet 159 |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |  | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                                   |          |                                         |                    |



Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title

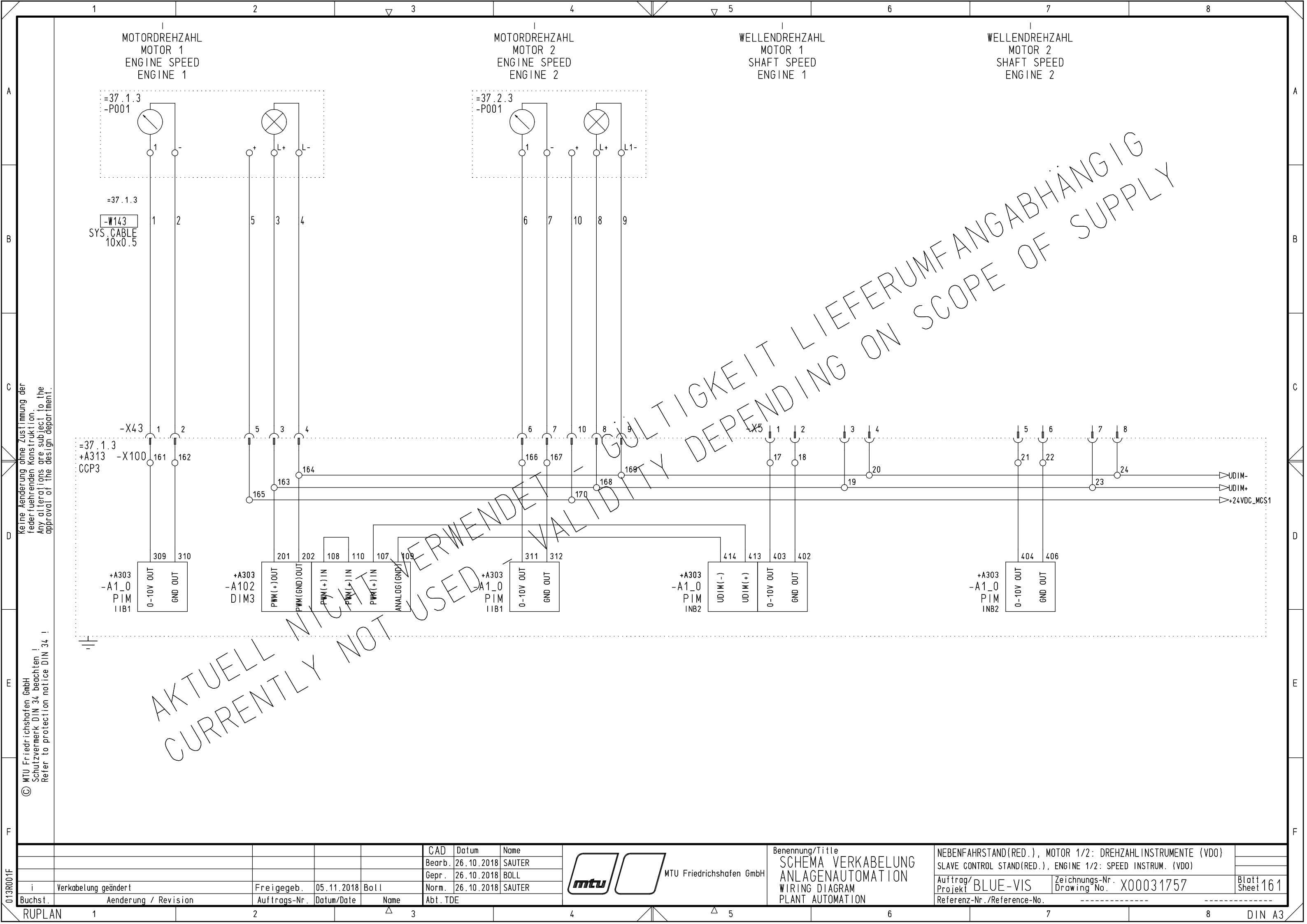
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2: DREHZAHLINSTRUM. (HORN EA)  
SLAVE CONTROL STAND(RED.), ENGINE 1/2: SPEED INST. (HORN EA)

Auftrag/Projekt: BLUE-VIS  
Zeichnungs-Nr./Drawing No.: X00031757  
Referenz-Nr./Reference-No.: -----

Blatt/Sheet: 160





AKTUELL NICHT VERWENDET  
CURRENTLY NOT USED



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND (RED.), MOTOR 1/2: DREHZAHL INSTRUMENTE (VDO)  
SLAVE CONTROL STAND (RED.), ENGINE 1/2: SPEED INSTRUM. (VDO)

|                            |          |                               |           |
|----------------------------|----------|-------------------------------|-----------|
| Auftrag/<br>Projekt        | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No. |          | -----                         |           |

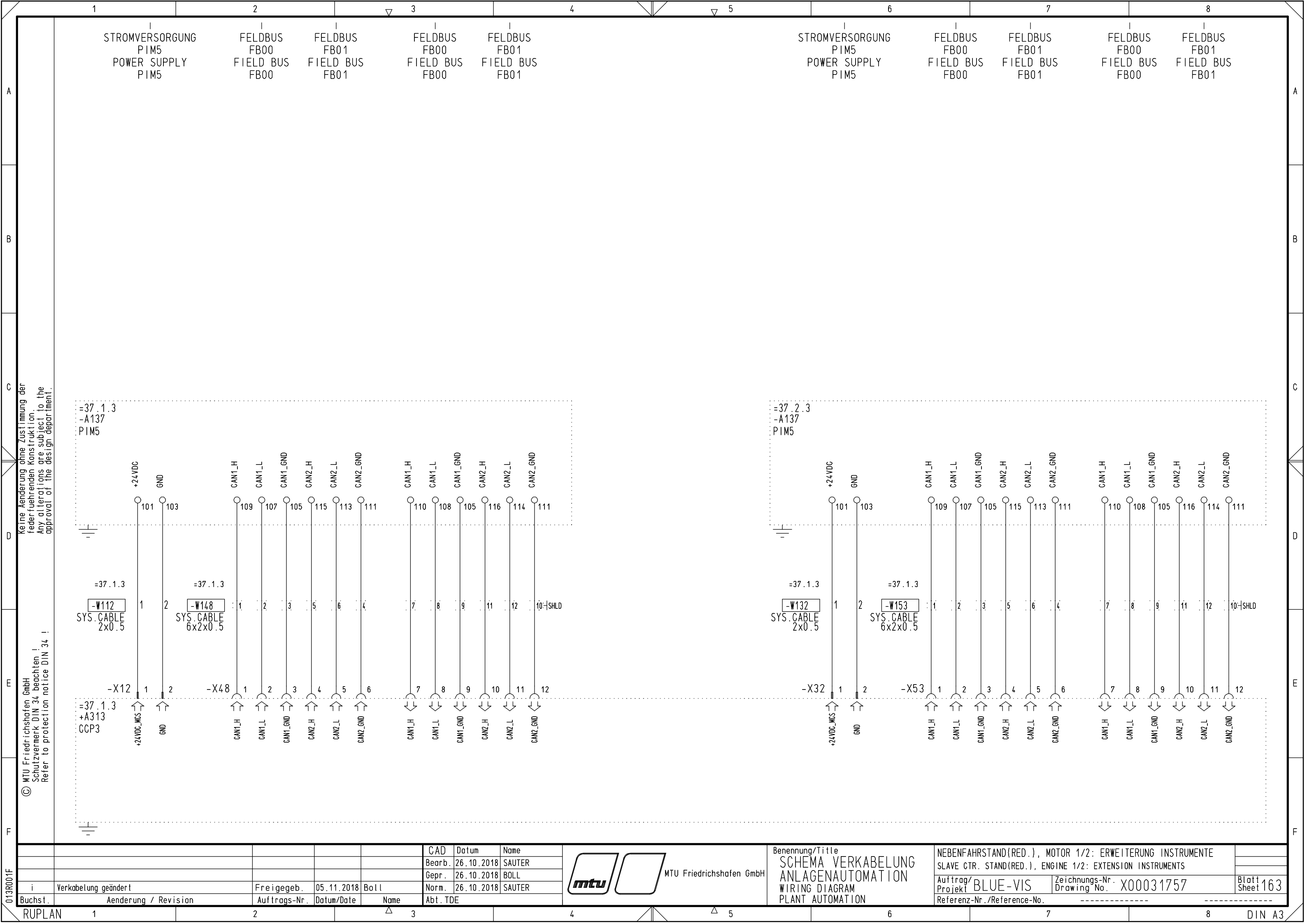
Blatt  
Sheet

161

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |







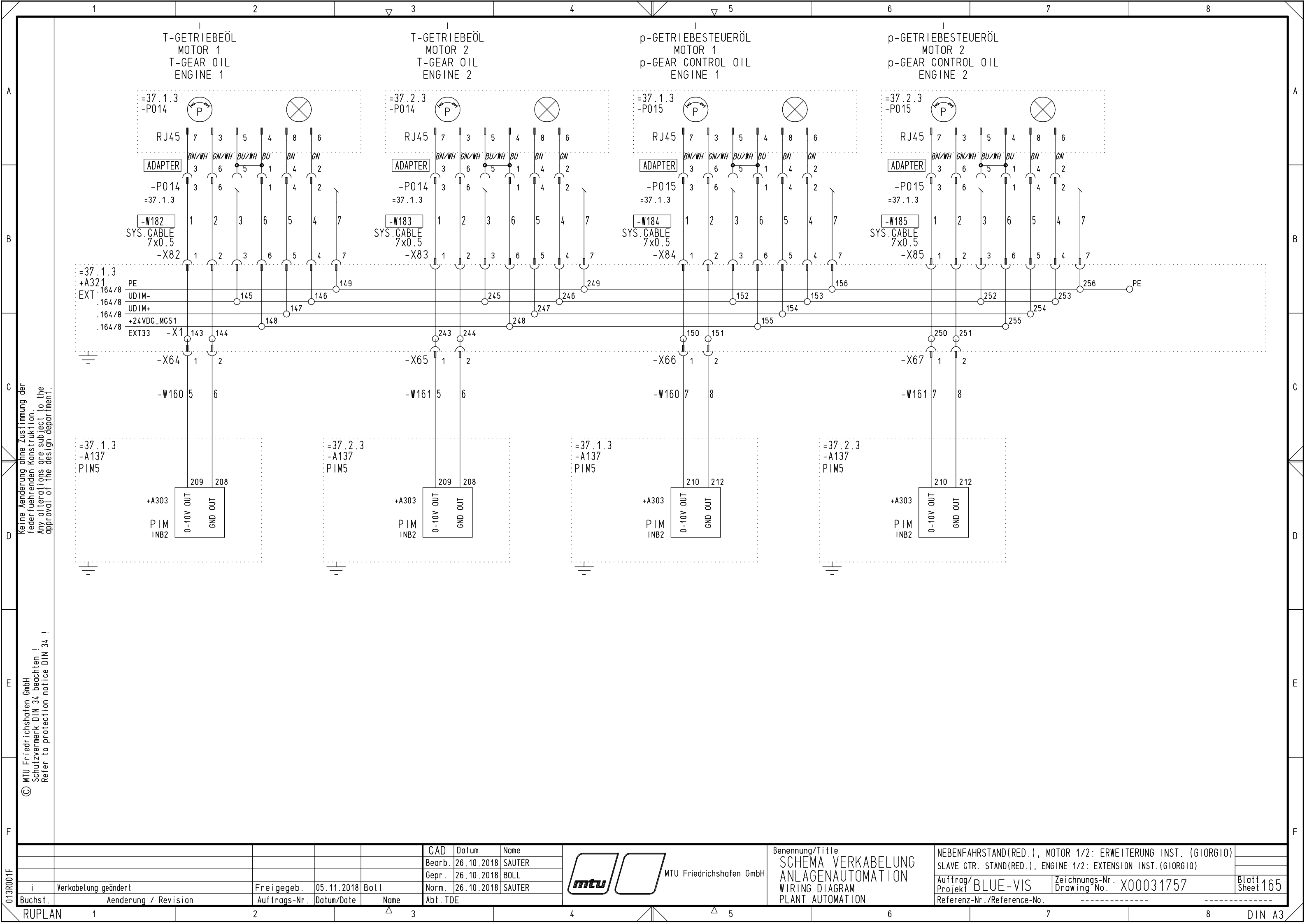
|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                       |          |                               |           |
|-----------------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUMENTE<br>SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTRUMENTS |          |                               |           |
| Auftrag/<br>Projekt                                                                                                   | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                            |          | -----                         |           |





Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |  |          |            |        |
|---------|----------------------|--------------|------------|------|--|----------|------------|--------|
|         |                      |              |            |      |  | CAD      | Datum      | Name   |
|         |                      |              |            |      |  | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      |  | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      |  | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |  | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |  |          |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)  
SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  
Auftrag/  
Projekt BLUE-VIS  
Zeichnungs-Nr.  
Drawing No. X00031757  
Referenz-Nr./Reference-No. -----

Blatt  
Sheet 166

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |        |            |        |
|---------|----------------------|--------------|------------|------|----------|--------|------------|--------|
|         |                      |              |            |      |          | CAD    | Datum      | Name   |
|         |                      |              |            |      |          | Bearb. | 26.10.2018 | SAUTER |
|         |                      |              |            |      |          | Gepr.  | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll |          | Norm.  | 26.10.2018 | SAUTER |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |        |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INST.(HORN EAD)  
SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTEN. INSTR.(HORN EAD)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. X00031757  
Referenz-Nr./Reference-No. -----

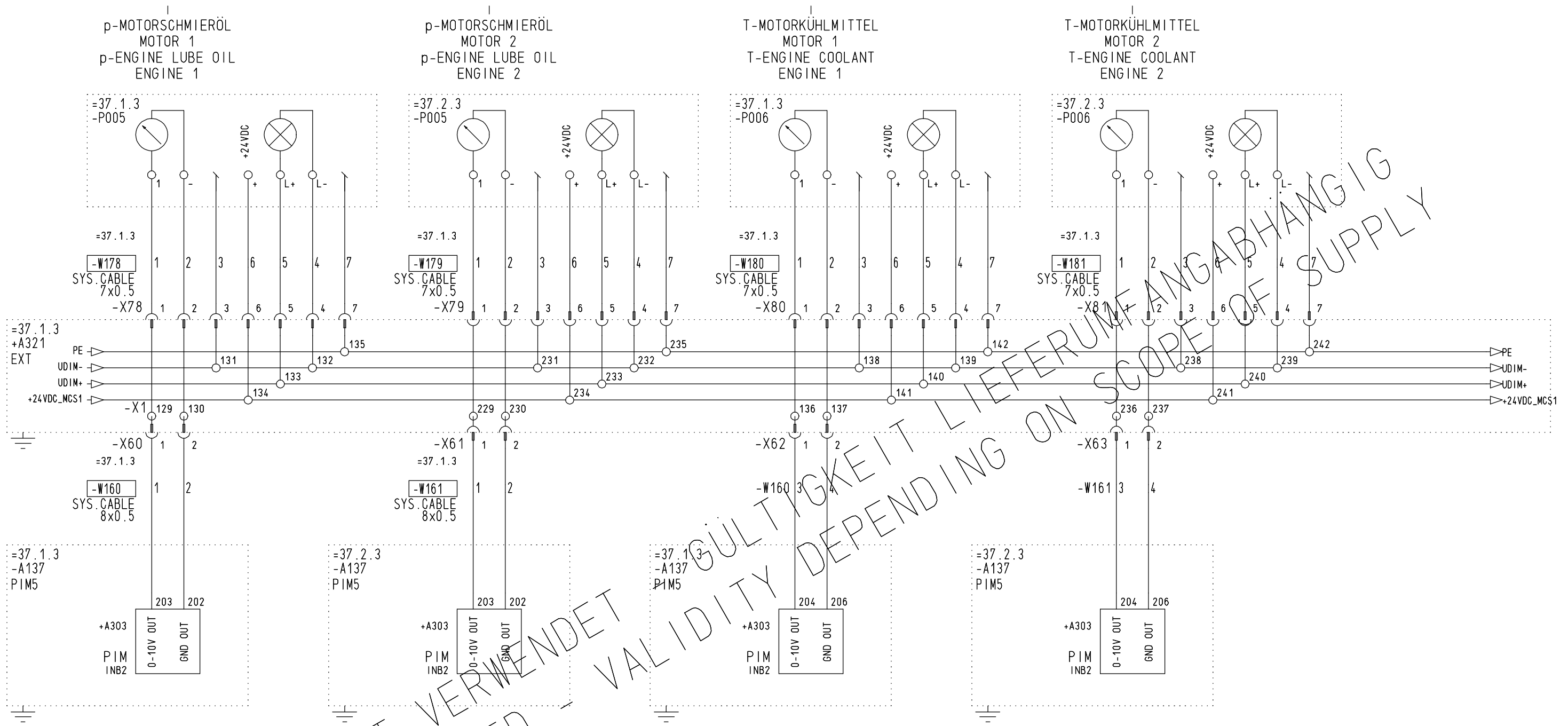
Blatt 167  
Sheet

Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F

RUPLAN



013R001F

RUPLAN

| CAD      | Datum      | Name   |
|----------|------------|--------|
| Bearb.   | 26.10.2018 | SAUTER |
| Gep.     | 26.10.2018 | BOLL   |
| Norm.    | 26.10.2018 | SAUTER |
| Abt. TDE |            |        |



MTU Friedrichshafen GmbH

Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO)  
SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTR. (VDO)  
Auftrag/Projekt BLUE-VIS  
Zeichnungs-Nr. X00031757  
Referenz-Nr./Reference-No. -----

Blatt 168  
Sheet 168

1 2 3 4 5 6 7 8 DIN A3

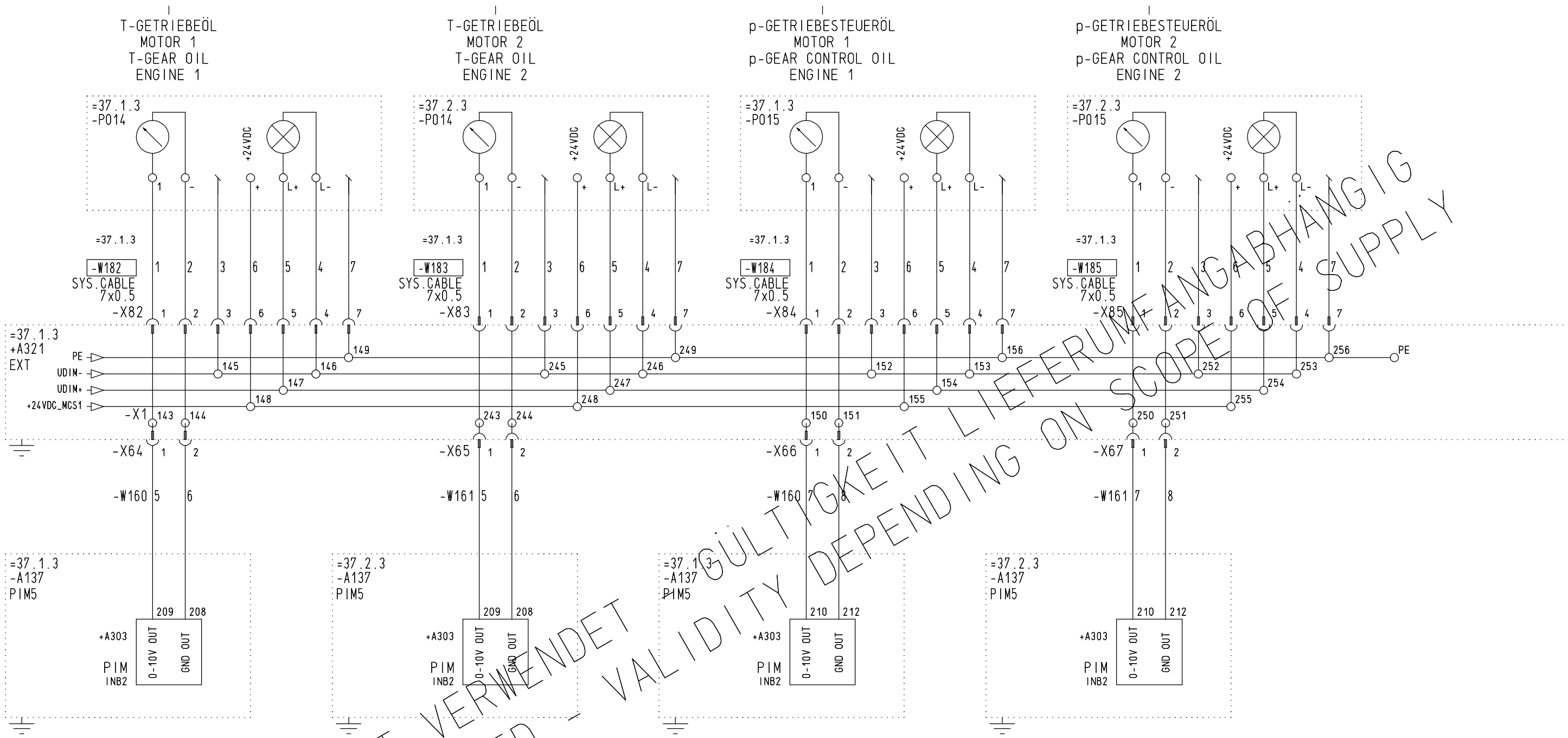


Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten!  
Refer to protection notice DIN 34!

013R001F

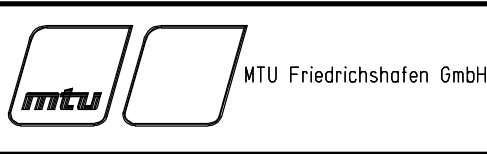
RUPLAN



013R001F

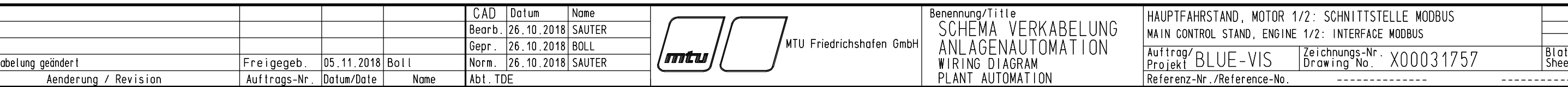
RUPLAN

|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |

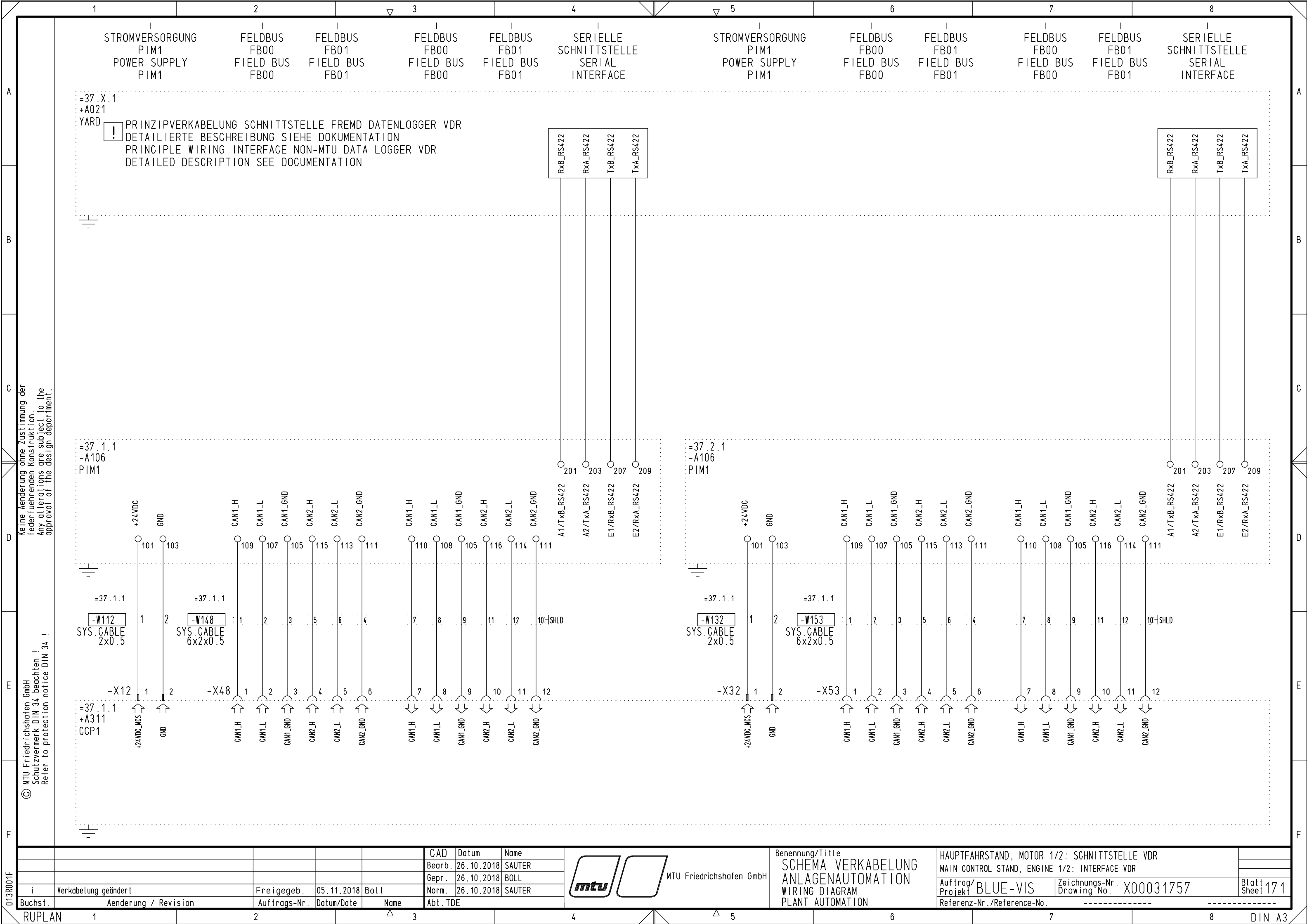


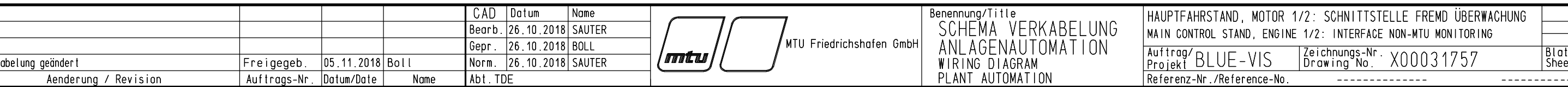
Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

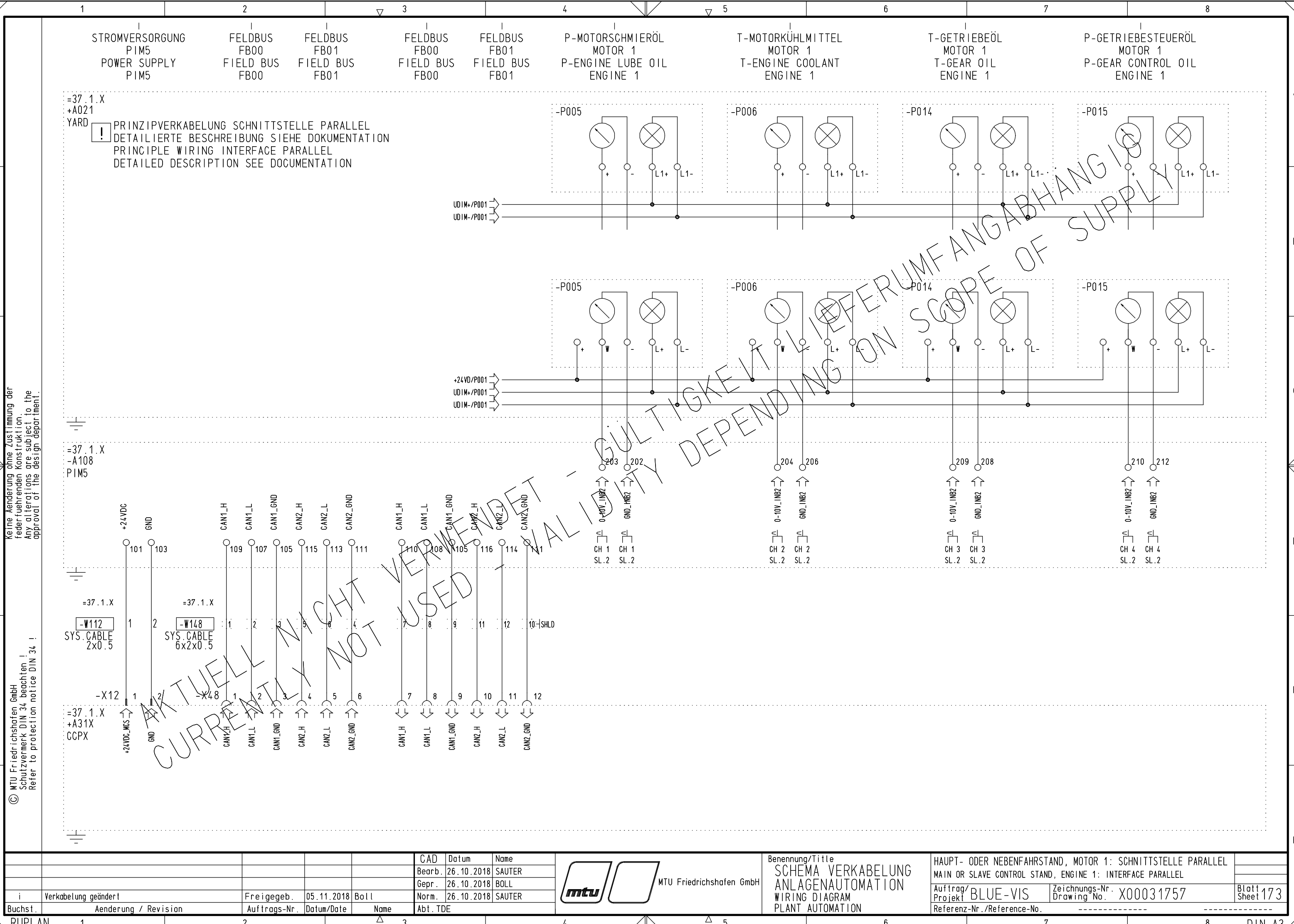
|                                                             |                              |
|-------------------------------------------------------------|------------------------------|
| NEBENFAHRSTAND(RED.), MOTOR 1/2: ERWEITERUNG INSTRUM. (VDO) |                              |
| SLAVE CTR. STAND(RED.), ENGINE 1/2: EXTENSION INSTR. (VDO)  |                              |
| Auftrag/Projekt                                             | Zeichnungs-Nr. / Drawing No. |
| BLUE-VIS                                                    | X00031757                    |
| Referenz-Nr. / Reference-No.                                |                              |





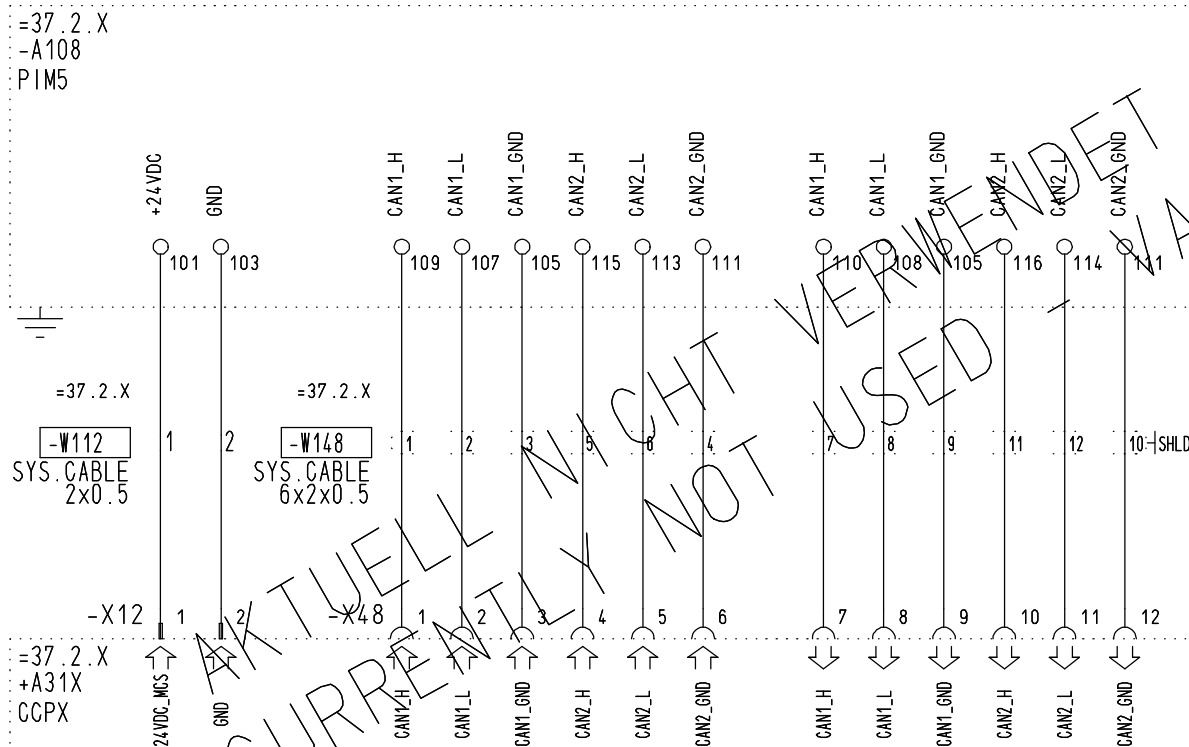
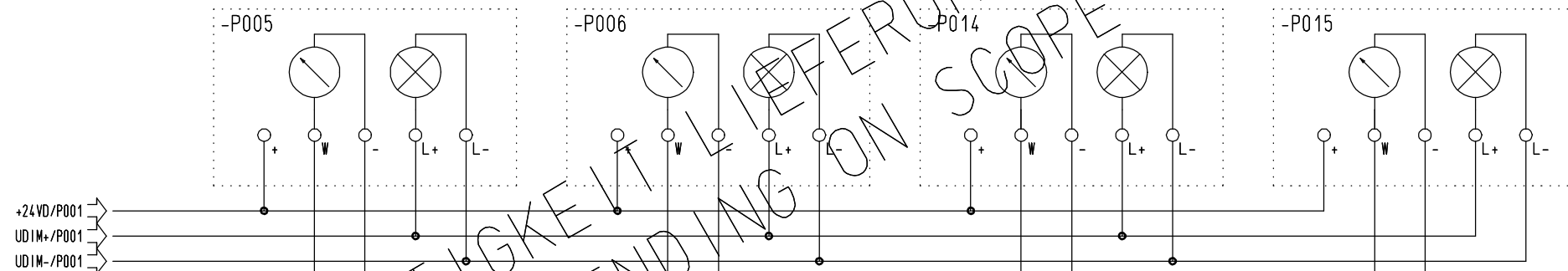
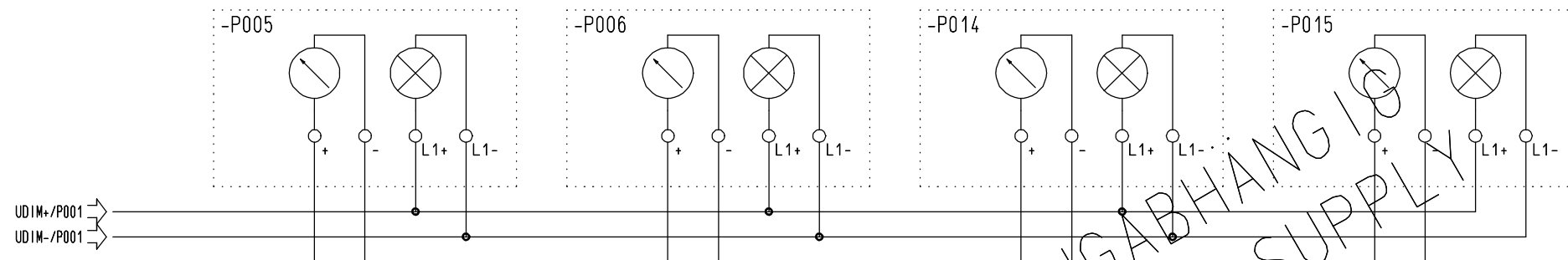






|                 |           |           |           |           |                   |                   |              |                    |
|-----------------|-----------|-----------|-----------|-----------|-------------------|-------------------|--------------|--------------------|
| STROMVERSORGUNG | FELDBUS   | FELDBUS   | FELDBUS   | FELDBUS   | P-MOTORSCHMIERÖL  | T-MOTORKÜHLMITTEL | T-GETRIEBEÖL | P-GETRIEBESTEUERÖL |
| PIM5            | FB00      | FB01      | FB00      | FB01      | MOTOR 2           | MOTOR 2           | MOTOR 2      | MOTOR 2            |
| POWER SUPPLY    | FIELD BUS | FIELD BUS | FIELD BUS | FIELD BUS | P-ENGINE LUBE OIL | T-ENGINE COOLANT  | T-GEAR OIL   | P-GEAR CONTROL OIL |
| PIM5            | FB00      | FB01      | FB00      | FB01      | ENGINE 2          | ENGINE 2          | ENGINE 2     | ENGINE 2           |

=37.2.X  
+A021  
YARD  PRINZIPVERKABELUNG SCHNITTSTELLE PARALLEL  
DETAILLIERTE BESCHREIBUNG SIEHE DOKUMENTATION  
PRINCIPLE WIRING INTERFACE PARALLEL  
DETAILED DESCRIPTION SEE DOCUMENTATION



Keine Änderung ohne Zustimmung der federführenden Konstruktion.  
Any alterations are subject to the approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |



MTU Friedrichshafen GmbH

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
| Benennung/Title | SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |
|-----------------|-------------------------------------------------------------------------------|

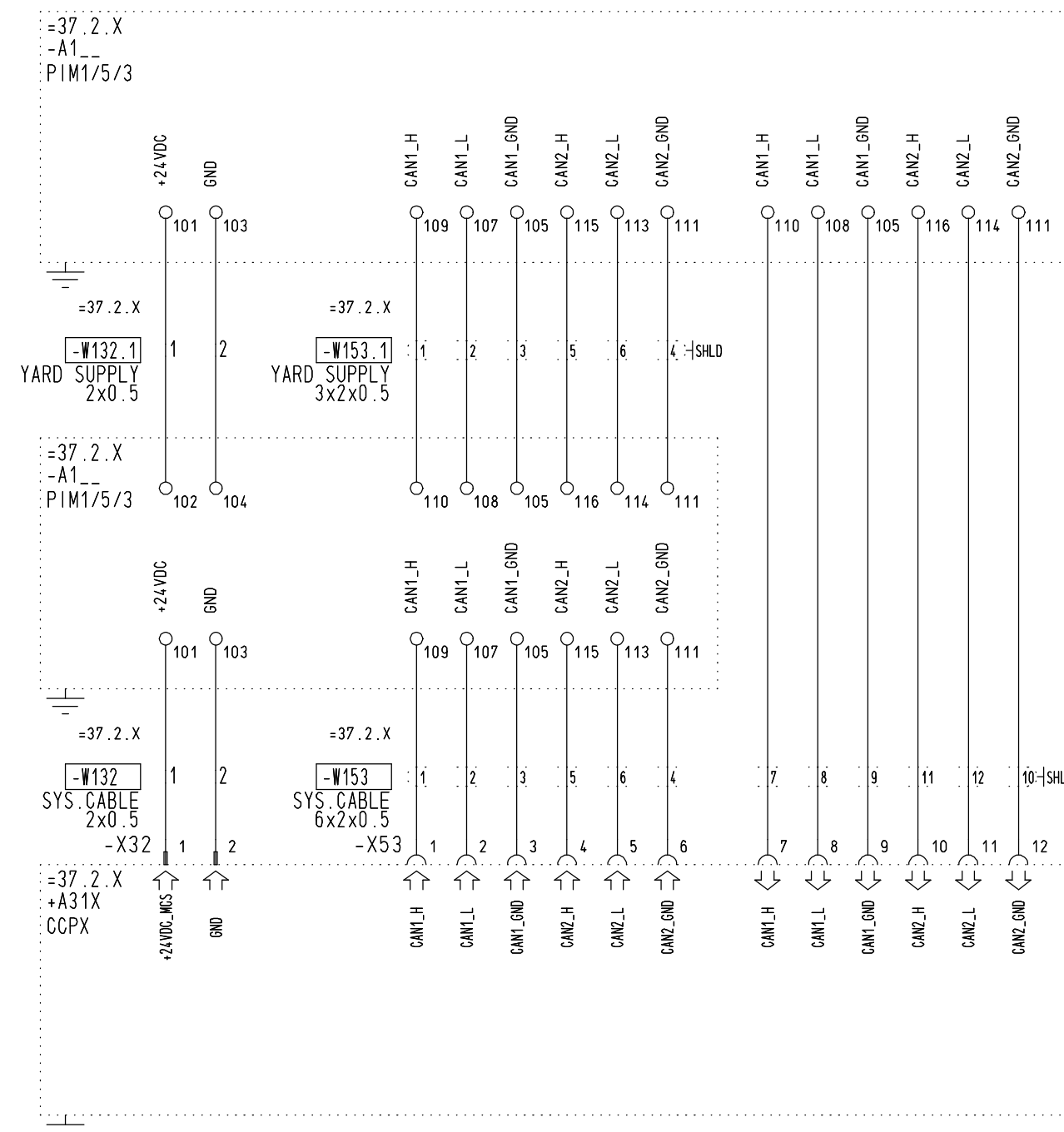
|                                                             |          |                                         |
|-------------------------------------------------------------|----------|-----------------------------------------|
| HAUPT- ODER NEBENFAHRSTAND, MOTOR 2: SCHNITTSTELLE PARALLEL |          |                                         |
| MAIN OR SLAVE CONTROL STAND, ENGINE 2: INTERFACE PARALLEL   |          |                                         |
| Auftrag/<br>Projekt                                         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 |
| Referenz-Nr./Reference-No. -----                            |          |                                         |

Blatt 174  
Sheet 174

HINWEIS: HIER KANN ABHÄNGIG VOM LIEFERUMFANG DAS SYSTEM "REMOTE SERVICES" ANGESCHLOSSEN WERDEN. DAZU WIRD AM LETZTEN PIM<sub>x</sub> UND KABEL -W153 (ÜBER STÜTZPUNKTLEMMEN) DAS KABEL -W191.2 (AO BZ00E50201867) ANGESCHLOSSEN. WEITERE DETAILS SIEHE DOKUMENTATION SYSTEM "REMOTE SERVICES"

NOTE: DEPENDING ON THE SCOPE OF DELIVERY, THE "REMOTE SERVICES" SYSTEM CAN BE CONNECTED HERE. CONNECT THE CABLE -W191.2 (AO BZ00E50201867) TO THE LAST PIMx AND CABLE -W153 (VIA TERMINALS).  
FOR MORE DETAILS SEE DOCUMENTATION SYSTEM "REMOTE SERVICES"

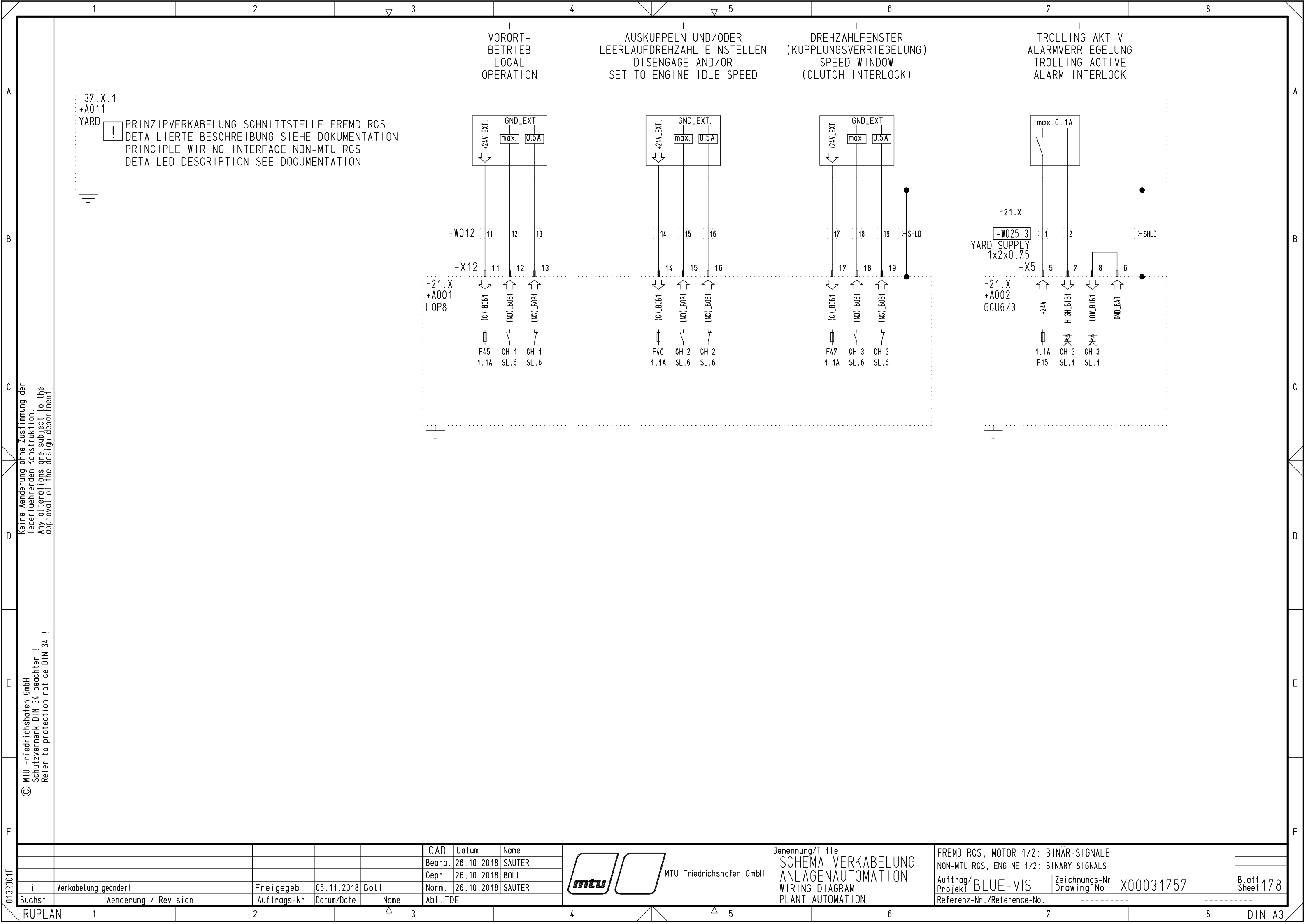
WEITERE SCHNITTSTELLENMODULE (PIM1/5/3)  
SIND SINNGEMÄSS ANZUSCHLIESSEN  
ADDITIONAL INTERFACE MODULES  
MUST BE CONNECTED ANALOGOUSLY







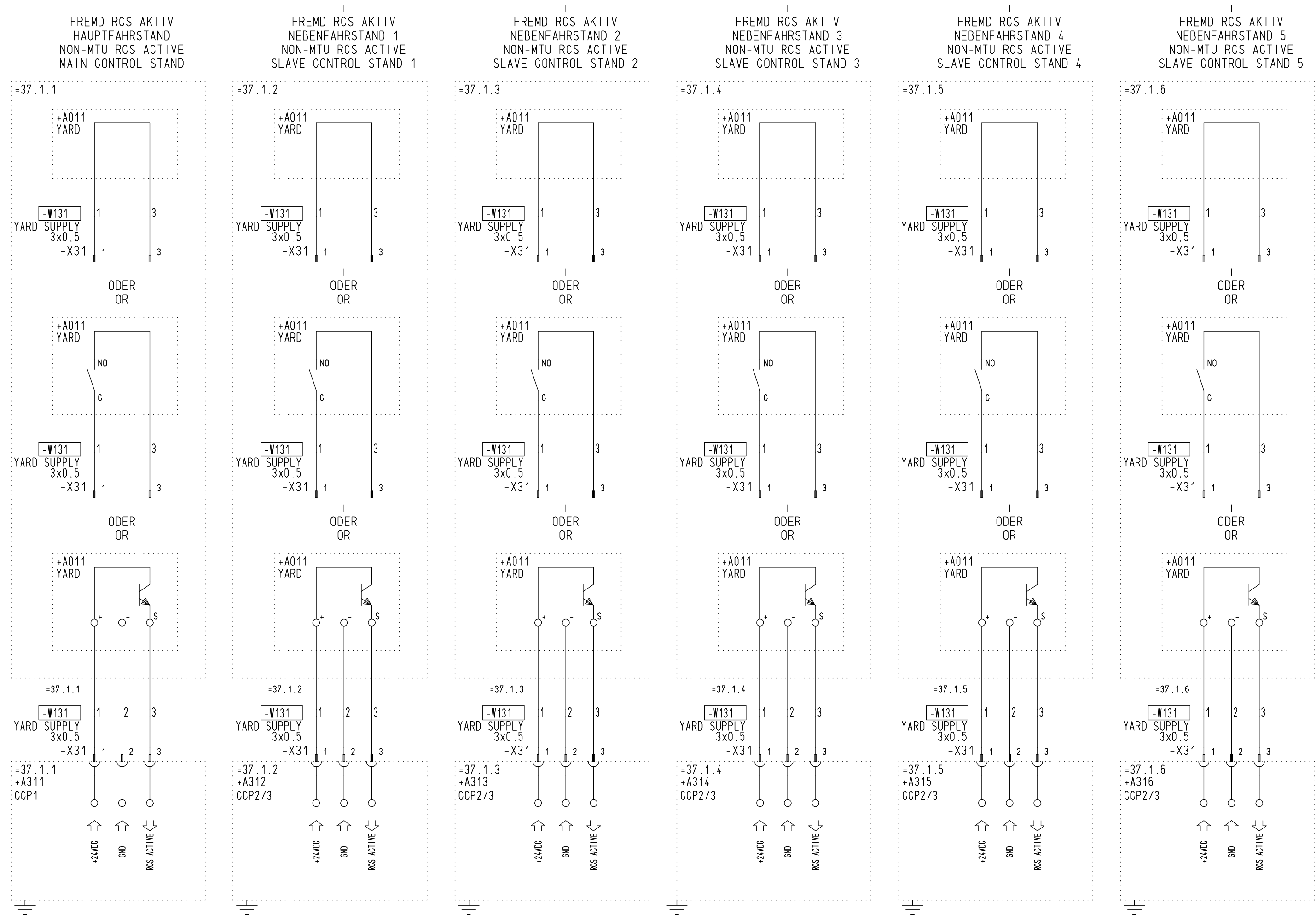






Keine Änderung ohne Zustimmung der  
Federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  |                                                 |          |                                         |                    |
|---------|----------------------|--------------|------------|------|----------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------|----------|-----------------------------------------|--------------------|
|         |                      |              |            |      | CAD      | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | FREMD RCS, MOTOR 1/2: FAHRSTANDSUMSCHALTUNG     |          |                                         |                    |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | NON-MTU RCS, ENGINE 1/2: CONTROL STAND TRANSFER |          |                                         |                    |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                 |          |                                         |                    |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Norm.    | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                 |          |                                         |                    |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name | Abt. TDE |            |        |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                             | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. X00031757 | Blatt<br>Sheet 179 |
|         |                      |              |            |      |          |            |        |                                                                                                                |                                                                                                  | Referenz-Nr./Reference-No.                      |          | -----                                   |                    |

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

013R001F

RUPLAN

! GETRIEBE OHNE WENDETEIL  
ANTRIEBSART WJ/CPP  
GEARBOX WITHOUT REVERSING COMPONENT  
PROPULSION TYPE WJ/CPP

NICHT VERWENDET  
NOT USED

VORGABE  
AUS  
DEMAND  
OUT

VORGABE  
EIN  
DEMAND  
IN

FREMD RCS  
ANGESCHLOSSEN  
NON-MTU RCS  
CONNECTED

VORGABE AUS  
(SIMULIERT)  
DEMAND OUT  
(SIMULATED)

OR

! GETRIEBE MIT WENDETEIL  
ANTRIEBSART WJ (MIT SPÜLFUNKTION)  
GEARBOX WITH REVERSING COMPONENT  
PROPULSION TYPE WJ (WITH FLUSHING FUNCTION)

VORGABE  
SPÜLEN  
DEMAND  
BACK FLUSH

VORGABE  
AUS  
DEMAND  
OUT

VORGABE  
EIN  
DEMAND  
IN

FREMD RCS  
ANGESCHLOSSEN  
NON-MTU RCS  
CONNECTED

VORGABE AUS  
(SIMULIERT)  
DEMAND OUT  
(SIMULATED)

OR

! GETRIEBE MIT WENDETEIL  
ANTRIEBSART FPP  
GEARBOX WITH REVERSING COMPONENT  
PROPULSION TYPE FPP

VORGABE  
GLEICHLAUF  
DEMAND ENGINE  
WISE ROTATION

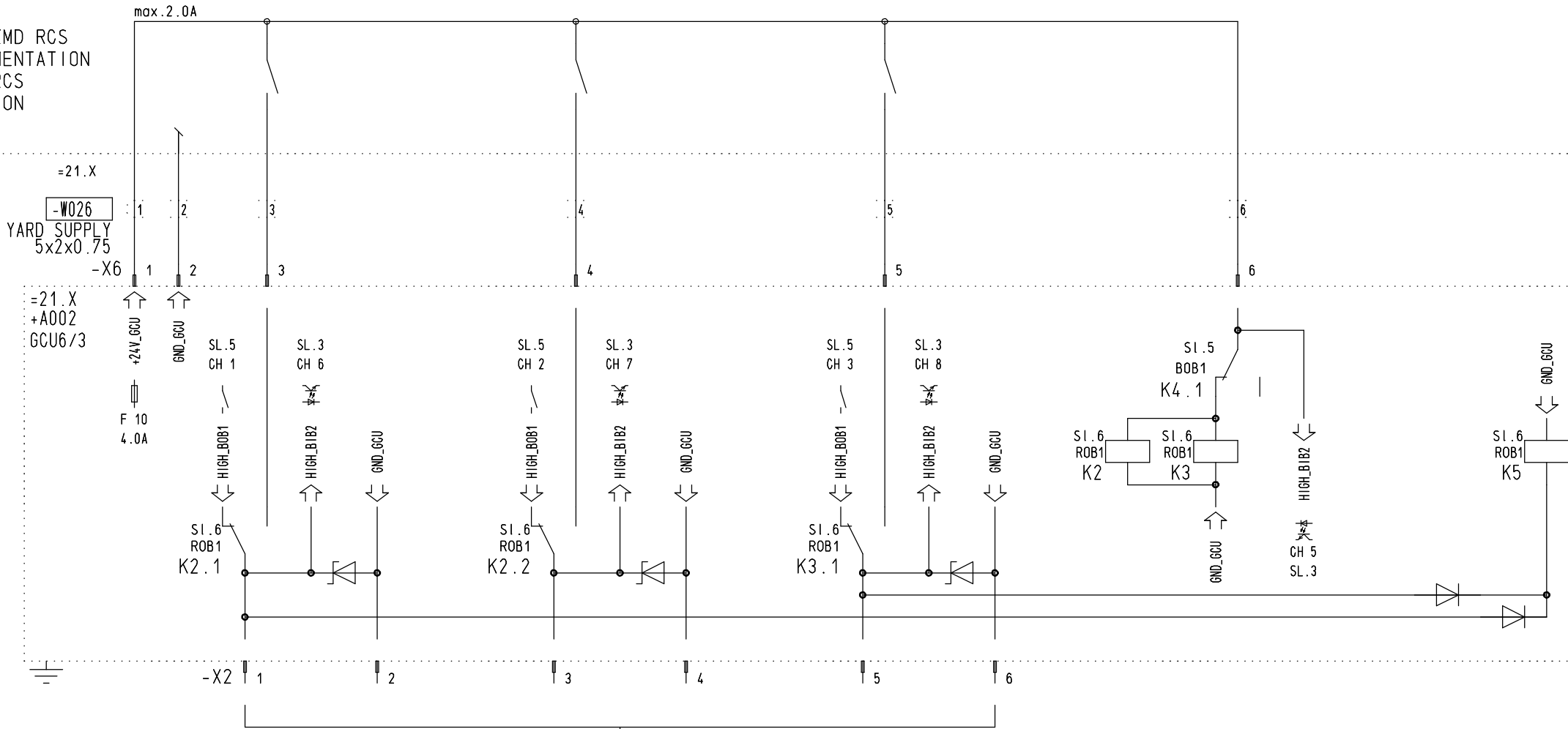
VORGABE NEUTRAL /  
AUSKUPPELFREIGABE  
DEMAND NEUTRAL /  
DISENGAGE RELEASE

VORGABE  
GEGENLAUF  
DEMAND COUNTER ENG.  
WISE ROTATION

FREMD RCS  
ANGESCHLOSSEN  
NON-MTU RCS  
CONNECTED

VORGABE NEUTRAL  
(SIMULIERT)  
DEMAND NEUTRAL  
(SIMULATED)

=37 .X.1  
+A011  
YARD ! PRINZIPVERKABELUNG SCHNITTSTELLE FREMD RCS  
DETAILIERTE BESCHREIBUNG SIEHE DOKUMENTATION  
PRINCIPLE WIRING INTERFACE NON-MTU RCS  
DETAILED DESCRIPTION SEE DOCUMENTATION



VERKABELUNG SIEHE SCHEMA X00031435  
WIRING SEE SCHEMA X00031435

|         |                      |              |            |      | CAD      | Datum      | Name   |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gedr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Änderung / Revision  | Auftrags-Nr. | Datum/Date | Name |          |            |        |

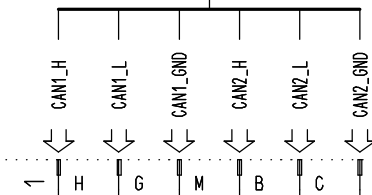


Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

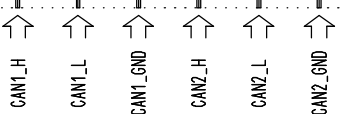
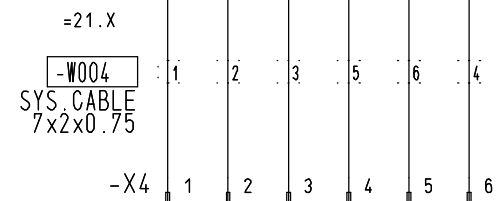
|                                                                                     |          |                               |           |
|-------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| FREMD RCS, MOTOR 1/2: KUPPLUNGSSTEUERUNG<br>NON-MTU RCS, ENGINE 1/2: CLUTCH CONTROL |          |                               |           |
| Auftrag/<br>Projekt                                                                 | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                          |          | -----                         |           |
|                                                                                     |          | Blatt 180<br>Sheet 180        |           |

DIN A3





NOTE: DEPENDING ON THE SCOPE OF DELIVERY, THE "REMOTE SERVICES" SYSTEM CAN BE CONNECTED HERE. REPLACE PLUGS -X5 AND -X6 (FIELD BUS FORWARDING) WITH CABLE -W191.1 (AO BZ00E50201858) FOR ENGINE 1 AND WITH CABLE -W191.2 (AO BZ00E50201859) FOR ENGINE 2. FOR MORE DETAILS SEE DOCUMENTATION SYSTEM "REMOTE SERVICES"



DIN A3

Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

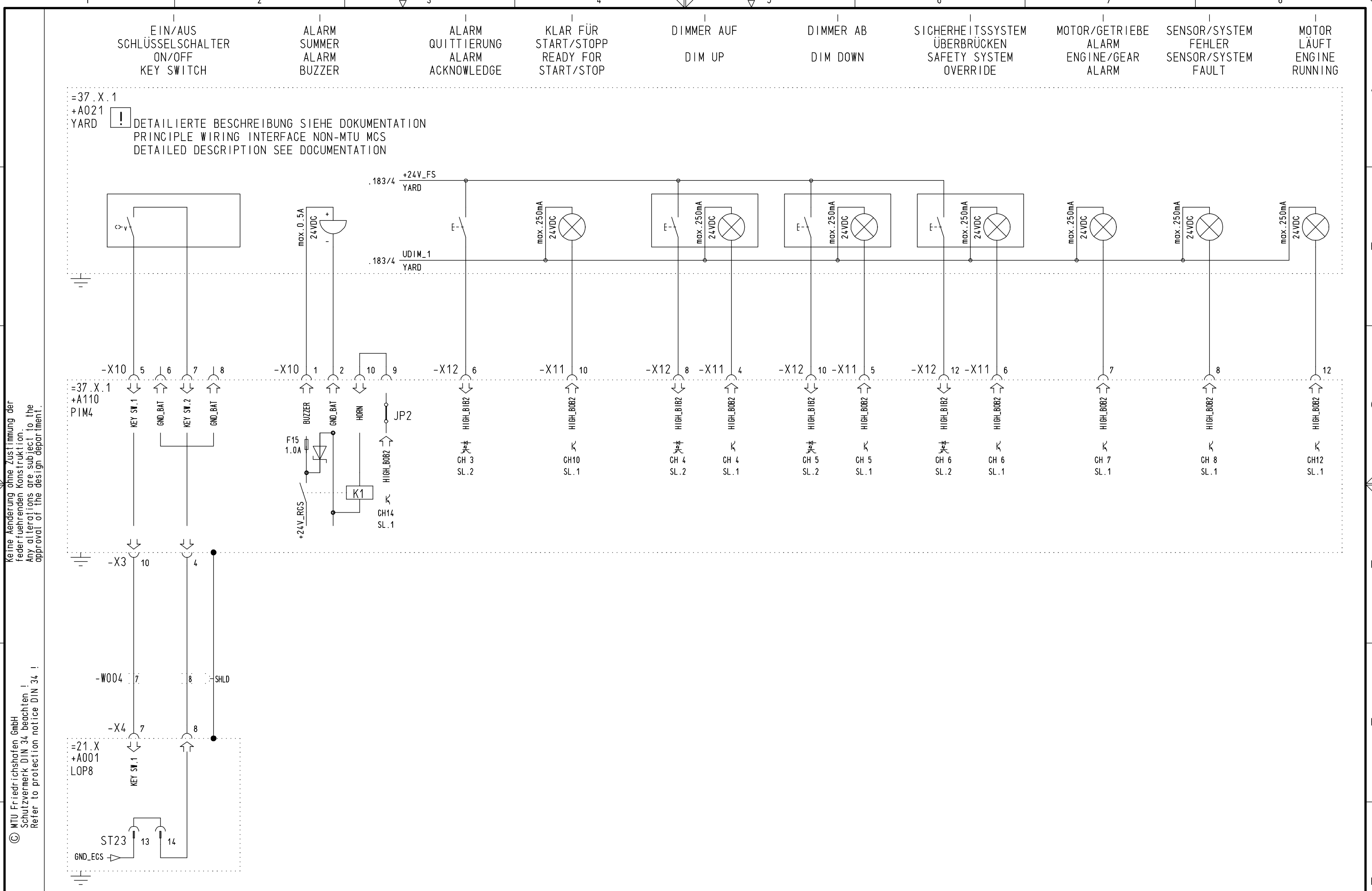
© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |      |          |            |        |
|---------|----------------------|--------------|------------|------|----------|------------|--------|
|         |                      |              |            |      | CAD      | Datum      | Name   |
|         |                      |              |            |      | Bearb.   | 26.10.2018 | SAUTER |
|         |                      |              |            |      | Gepr.    | 26.10.2018 | BOLL   |
|         |                      |              |            |      | Norm.    | 26.10.2018 | SAUTER |
| i       | Verkabelung geändert | Freigegeb.   | 05.11.2018 | Boll | Abt. TDE |            |        |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name |          |            |        |



Benennung/Title  
SCHEMA VERKABELUNG  
ANLAGENAUTOMATION  
WIRING DIAGRAM  
PLANT AUTOMATION

|                                                                                                                |          |                               |           |
|----------------------------------------------------------------------------------------------------------------|----------|-------------------------------|-----------|
| FREMD MCS (PIM4), MOTOR 1/2: START/STOPP/NOTSTOPP<br>NON-MTU MCS (PIM4), ENGINE 1/2: START/STOP/EMERGENCY STOP |          |                               |           |
| Auftrag/<br>Projekt                                                                                            | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Referenz-Nr./Reference-No.                                                                                     |          | -----                         |           |

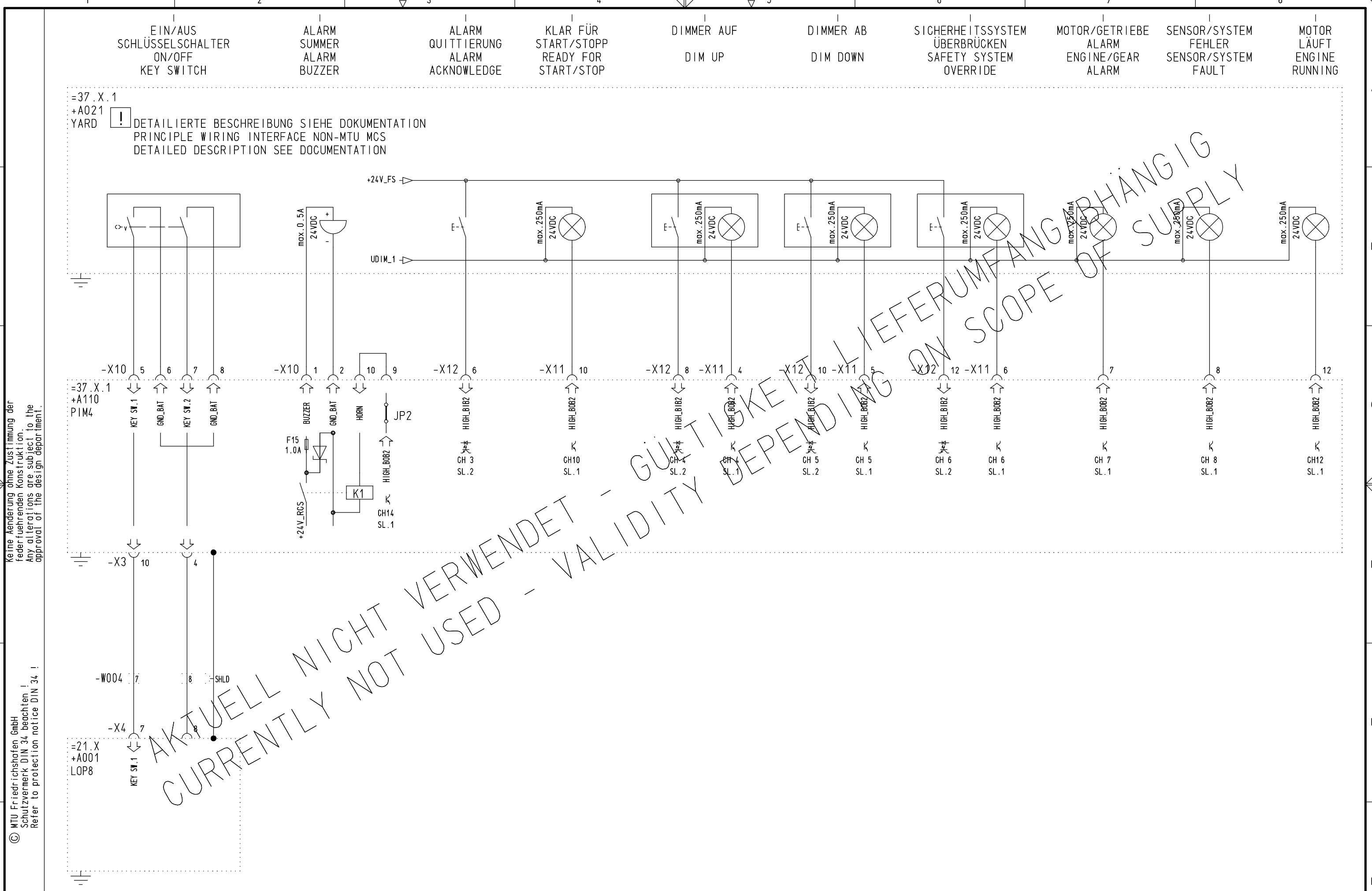


Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

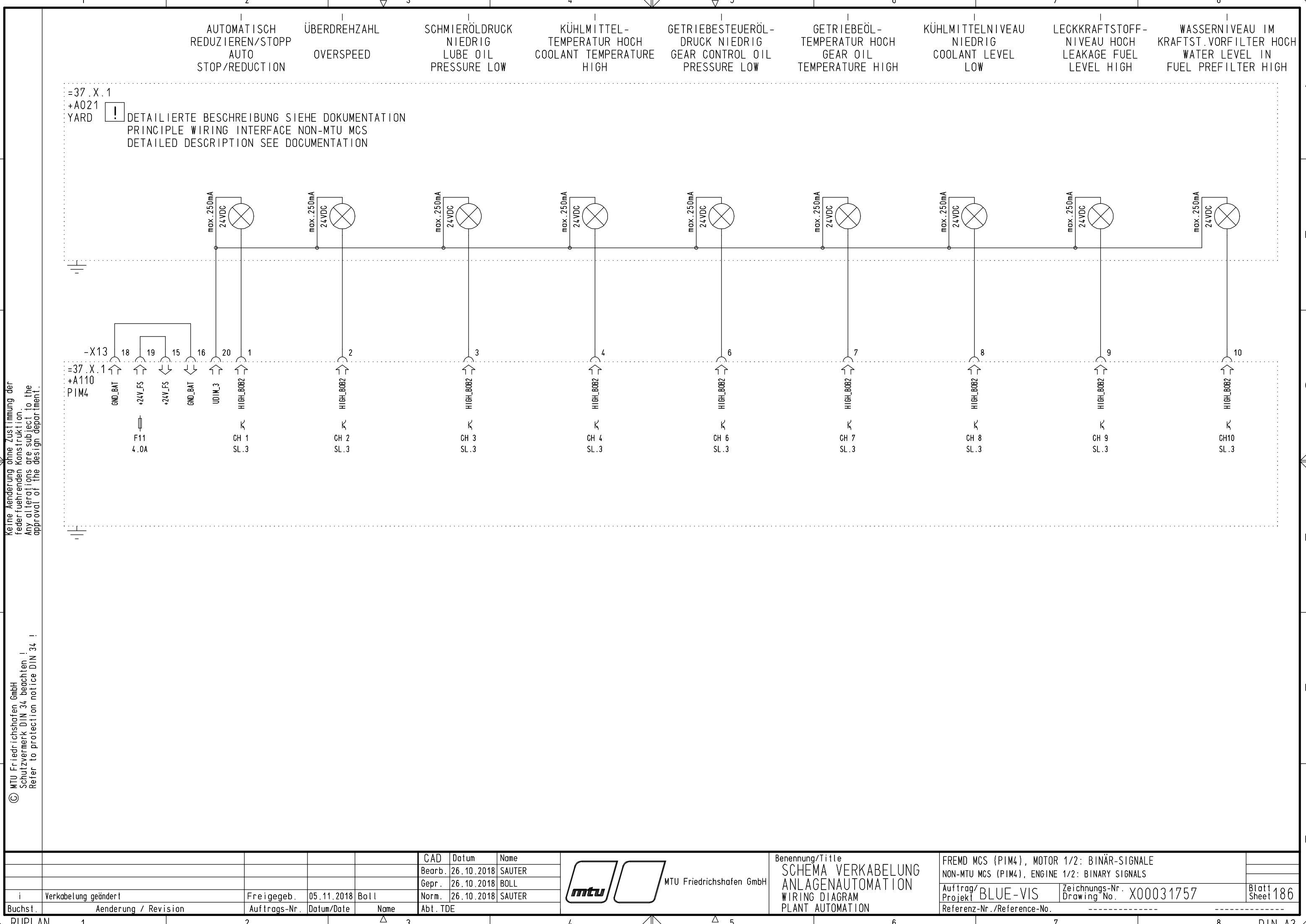
© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |              |            |        |            |        |                                                    |  |                                                                                                  |                             |                                                                                                                          |                                         |
|---------|----------------------|--------------|------------|--------|------------|--------|----------------------------------------------------|--|--------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|         |                      |              |            | CAD    | Datum      | Name   | <div>mtu</div> <div>MTU Friedrichshafen GmbH</div> |  | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION |                             | FREMD MCS (PIM4), MOTOR 1/2: BINÄR-SIGNALE (LOP_KEY_GND)<br>NON-MTU MCS (PIM4), ENGINE 1/2: BINARY SIGNALS (LOP_KEY_GND) |                                         |
|         |                      |              |            | Bearb. | 26.10.2018 | SAUTER |                                                    |  |                                                                                                  |                             |                                                                                                                          |                                         |
|         |                      |              |            | Gepr.  | 26.10.2018 | BOLL   |                                                    |  |                                                                                                  |                             |                                                                                                                          |                                         |
|         |                      |              |            | Norm.  | 26.10.2018 | SAUTER |                                                    |  |                                                                                                  |                             |                                                                                                                          |                                         |
| Buchst. | Aenderung / Revision | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE   |        |                                                    |  |                                                                                                  | Auftrag/Projekt<br>BLUE-VIS |                                                                                                                          | Zeichnungs-Nr.<br>Drawing No. X00031757 |
|         |                      |              |            |        |            |        |                                                    |  |                                                                                                  | Referenz-Nr./Reference-No.  |                                                                                                                          | Blatt<br>Sheet 184                      |





|                               |                      |  |              |            |            |          |                                                                                                                |                                                                                                  |                                                             |        |                              |  |
|-------------------------------|----------------------|--|--------------|------------|------------|----------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------|------------------------------|--|
|                               |                      |  |              | CAD        | Datum      | Name     |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | FREMD MCS (PIM4), MOTOR 1/2: BINÄR-SIGNALE (PIM4_KEY_GND)   |        |                              |  |
|                               |                      |  |              | Bearb.     | 26.10.2018 | SAUTER   |                                                                                                                |                                                                                                  | NON-MTU MCS (PIM4), ENGINE 1/2: BIN. SIGNALS (PIM4_KEY_GND) |        |                              |  |
|                               |                      |  |              | Gepr.      | 26.10.2018 | BOLL     |                                                                                                                |                                                                                                  |                                                             |        |                              |  |
| i                             | Verkabelung geändert |  | Freigegeb.   | 05.11.2018 | Boll       | Norm.    |                                                                                                                |                                                                                                  | 26.10.2018                                                  | SAUTER | Auftrag/<br>Projekt BLUE-VIS |  |
| Buchst.                       | Aenderung / Revision |  | Auftrags-Nr. | Datum/Date | Name       | Abt. TDE |                                                                                                                | Referenz-Nr./Reference-No.                                                                       |                                                             |        |                              |  |
| RUPLAN 1 2 3 4 5 6 7 8 DIN A3 |                      |  |              |            |            |          |                                                                                                                |                                                                                                  |                                                             |        |                              |  |

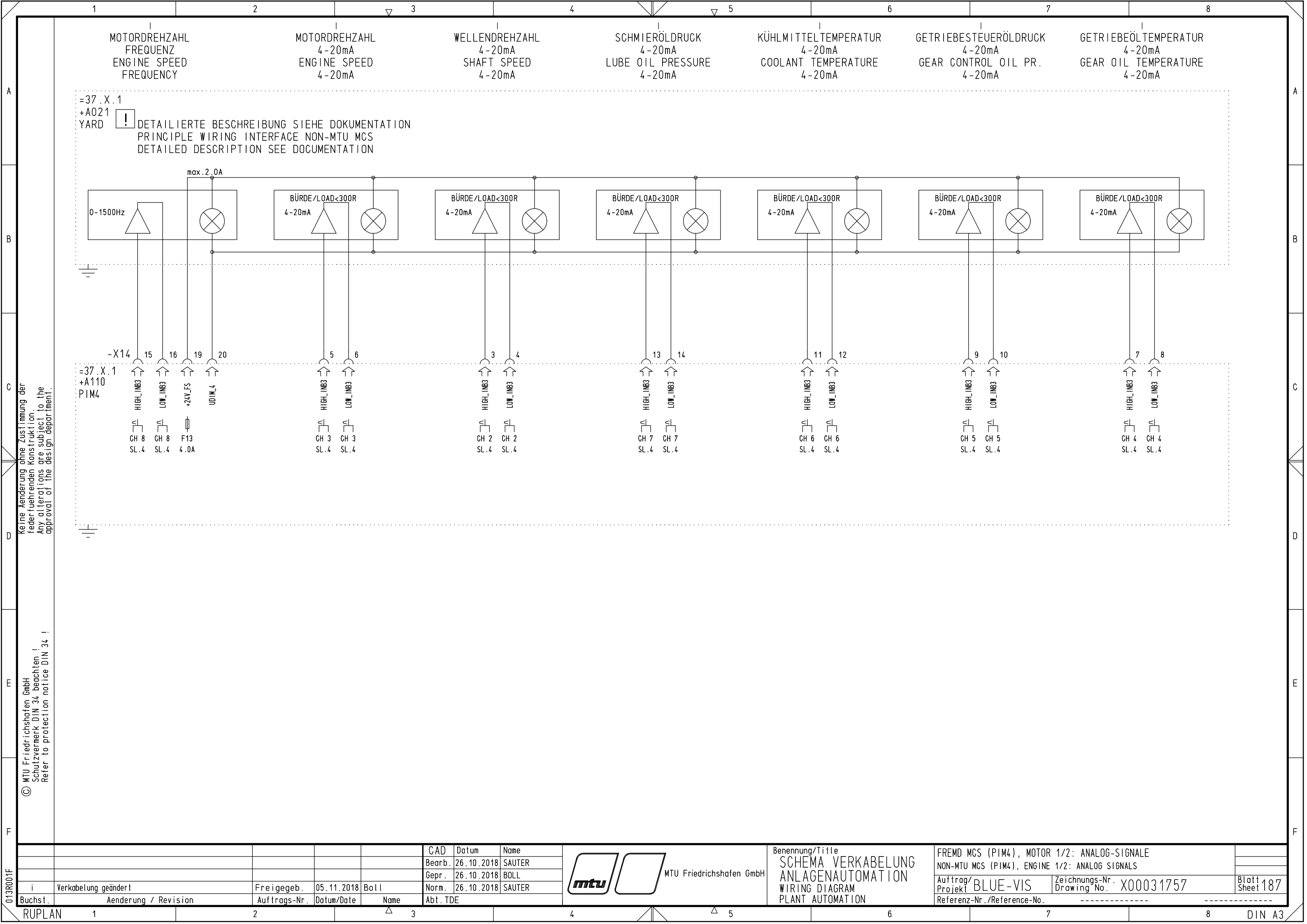


Keine Änderung ohne Zustimmung der  
federführenden Konstruktion.  
Any alterations are subject to the  
approval of the design department.

© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

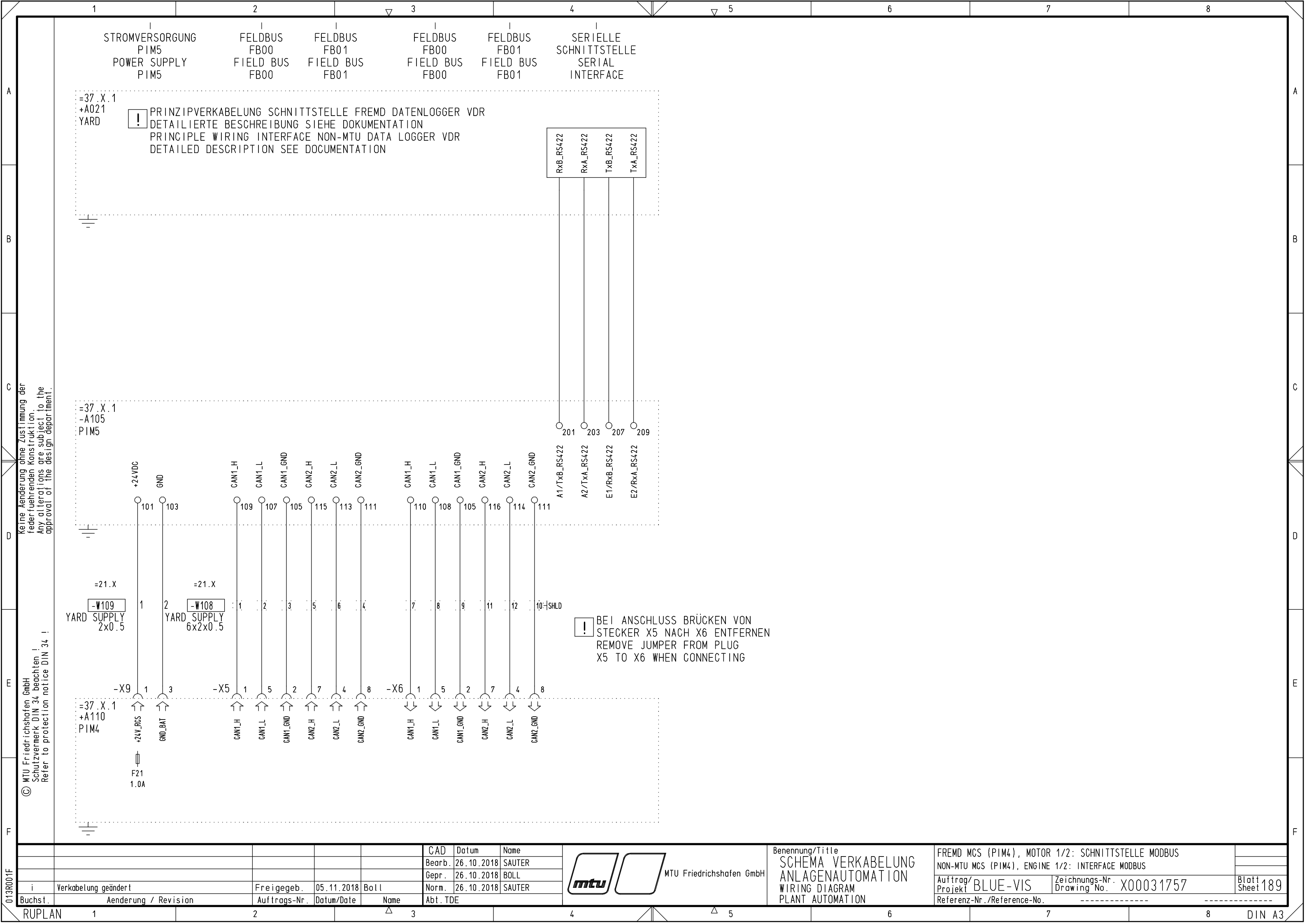
|         |                      |  |  |              |            |        |                                                                                                                |                                                                                                  |                                                |          |                               |           |
|---------|----------------------|--|--|--------------|------------|--------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------|----------|-------------------------------|-----------|
|         |                      |  |  | CAD          | Datum      | Name   |  MTU Friedrichshafen GmbH | Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | FREMD MCS (PIM4), MOTOR 1/2: BINÄR-SIGNALE     |          | Blatt<br>Sheet 186            |           |
|         |                      |  |  | Bearb.       | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  | NON-MTU MCS (PIM4), ENGINE 1/2: BINARY SIGNALS |          |                               |           |
|         |                      |  |  | Gepr.        | 26.10.2018 | BOLL   |                                                                                                                |                                                                                                  |                                                |          |                               |           |
|         |                      |  |  | Norm.        | 26.10.2018 | SAUTER |                                                                                                                |                                                                                                  |                                                |          |                               |           |
| i       | Verkabelung geändert |  |  | Freigegeb.   | 05.11.2018 | Boll   |                                                                                                                |                                                                                                  | Auftrag/<br>Projekt                            | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. | X00031757 |
| Buchst. | Änderung / Revision  |  |  | Auftrags-Nr. | Datum/Date | Name   | Abt. TDE                                                                                                       |                                                                                                  | Referenz-Nr./Reference-No.                     |          | -----                         |           |

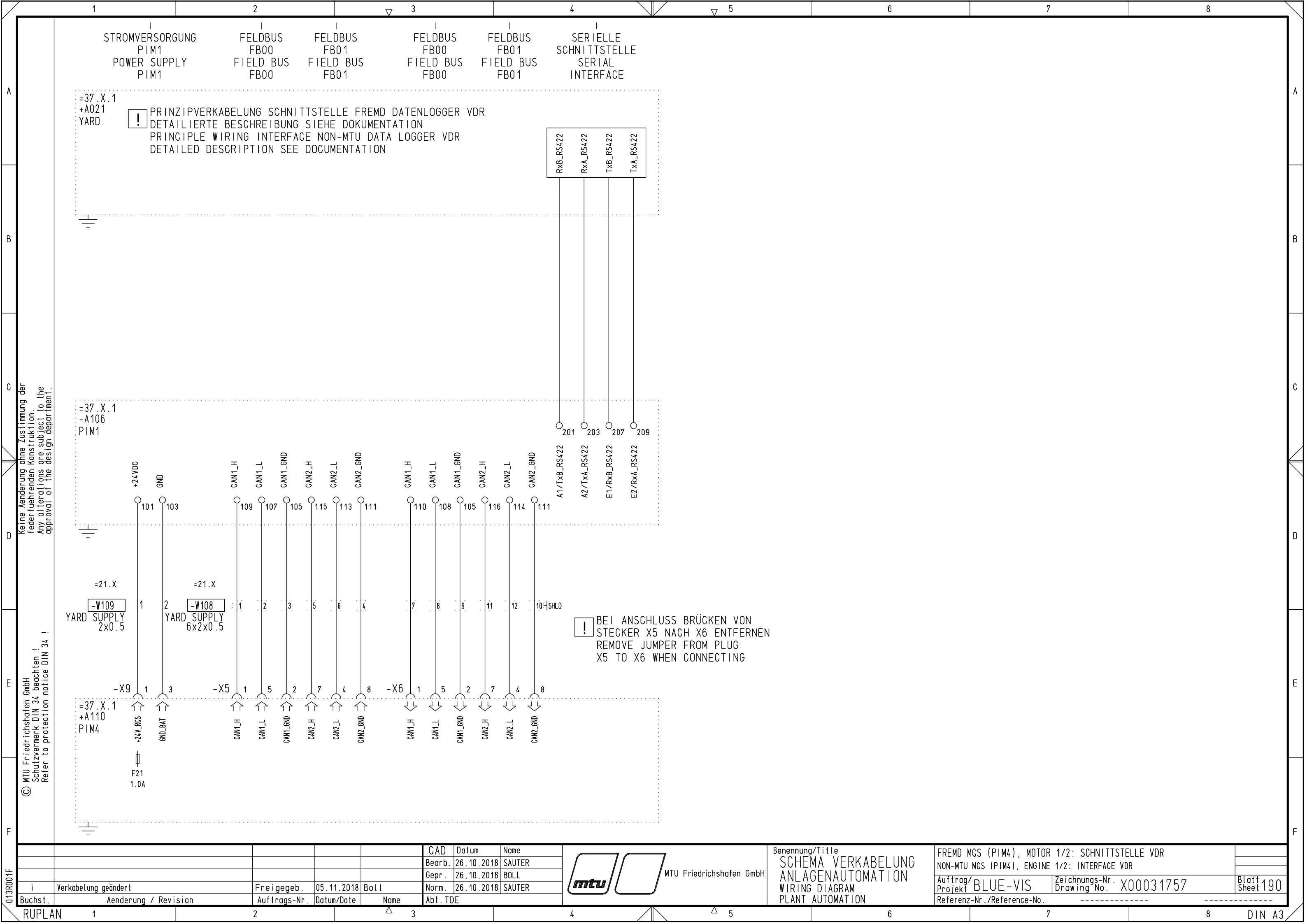




© MTU Friedrichshafen GmbH  
Schutzmerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|        |   |   |                                                                                         |                                                                                         |   |   |   |        |
|--------|---|---|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---|---|---|--------|
| RUPLAN | 1 | 2 |  3 |  5 | 6 | 7 | 8 | DIN A3 |
|--------|---|---|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---|---|---|--------|

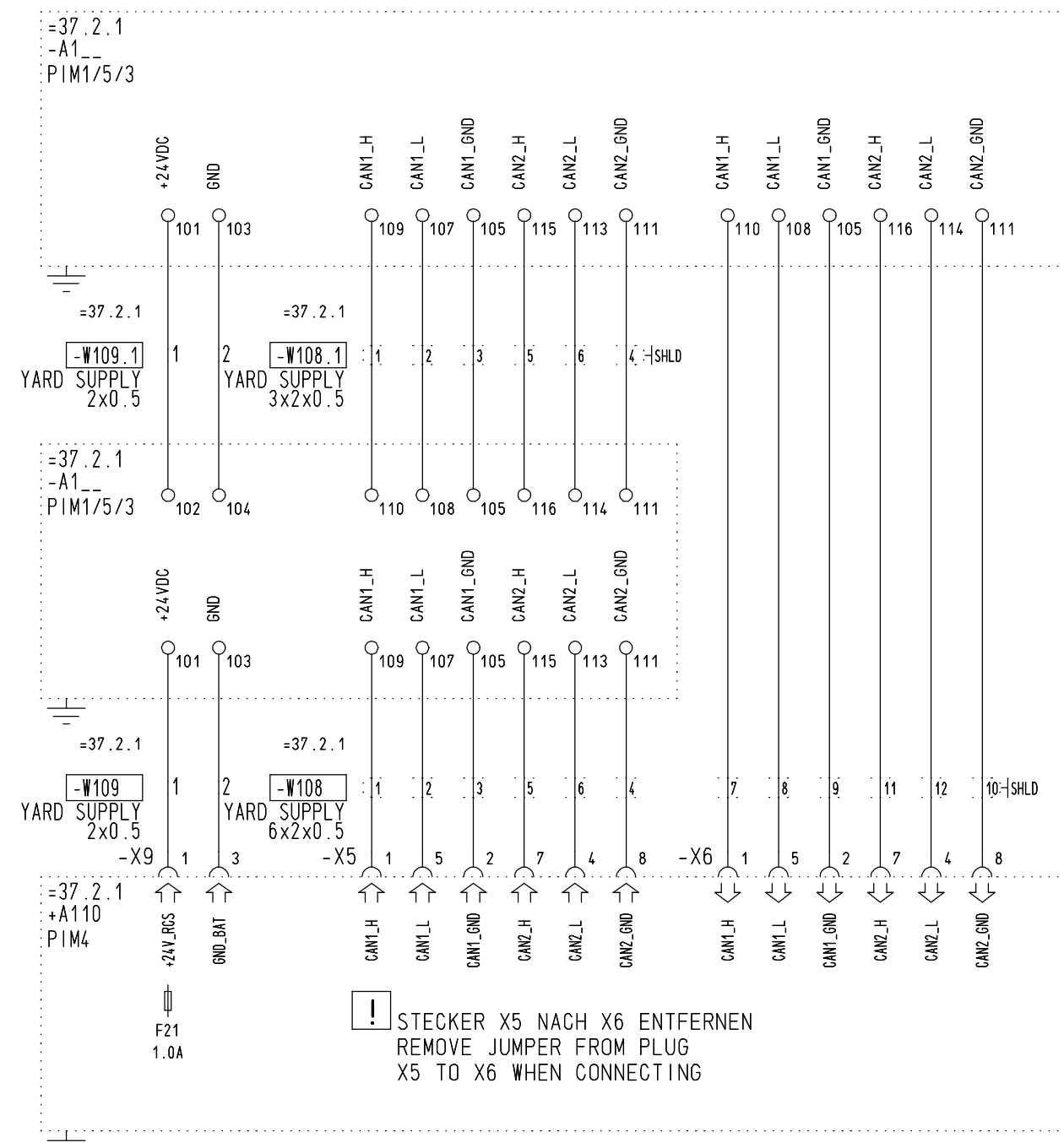




HINWEIS: HIER KANN ABHÄNGIG VOM LIEFERUMFANG DAS SYSTEM "REMOTE SERVICES" ANGESCHLOSSEN WERDEN. DAZU WIRD AM LETZTEN PIMx UND KABEL -W108 (ÜBER STÜTZPUNKTLEMMEN) DAS KABEL -W191.2 (AO BZ00E50201867) ANGESCHLOSSEN. WEITERE DETAILS SIEHE DOKUMENTATION SYSTEM "REMOTE SERVICES"

NOTE: DEPENDING ON THE SCOPE OF DELIVERY, THE "REMOTE SERVICES" SYSTEM CAN BE CONNECTED HERE. CONNECT THE CABLE -W191.2 (AO BZ00E50201867) TO THE LAST PIMx AND CABLE -W108 (VIA TERMINALS).  
FOR MORE DETAILS SEE DOCUMENTATION SYSTEM "REMOTE SERVICES"

WEITERE SCHNITTSTELLENMODULE (PIM1/5/3)  
SIND SINNGEMÄSS ANZUSCHLIESSEN  
ADDITIONAL INTERFACE MODULES  
MUST BE CONNECTED ANALOGOUSLY



MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

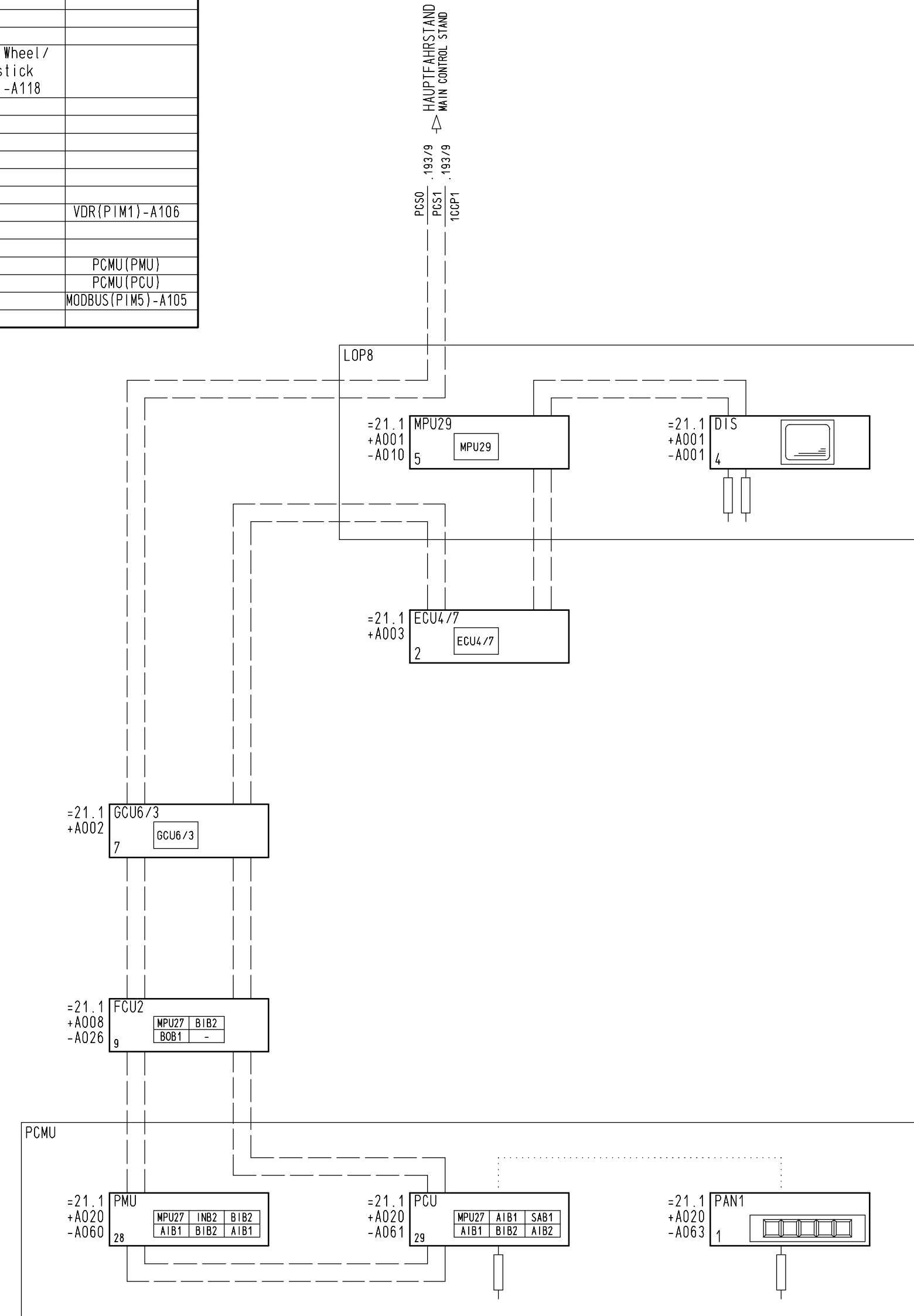
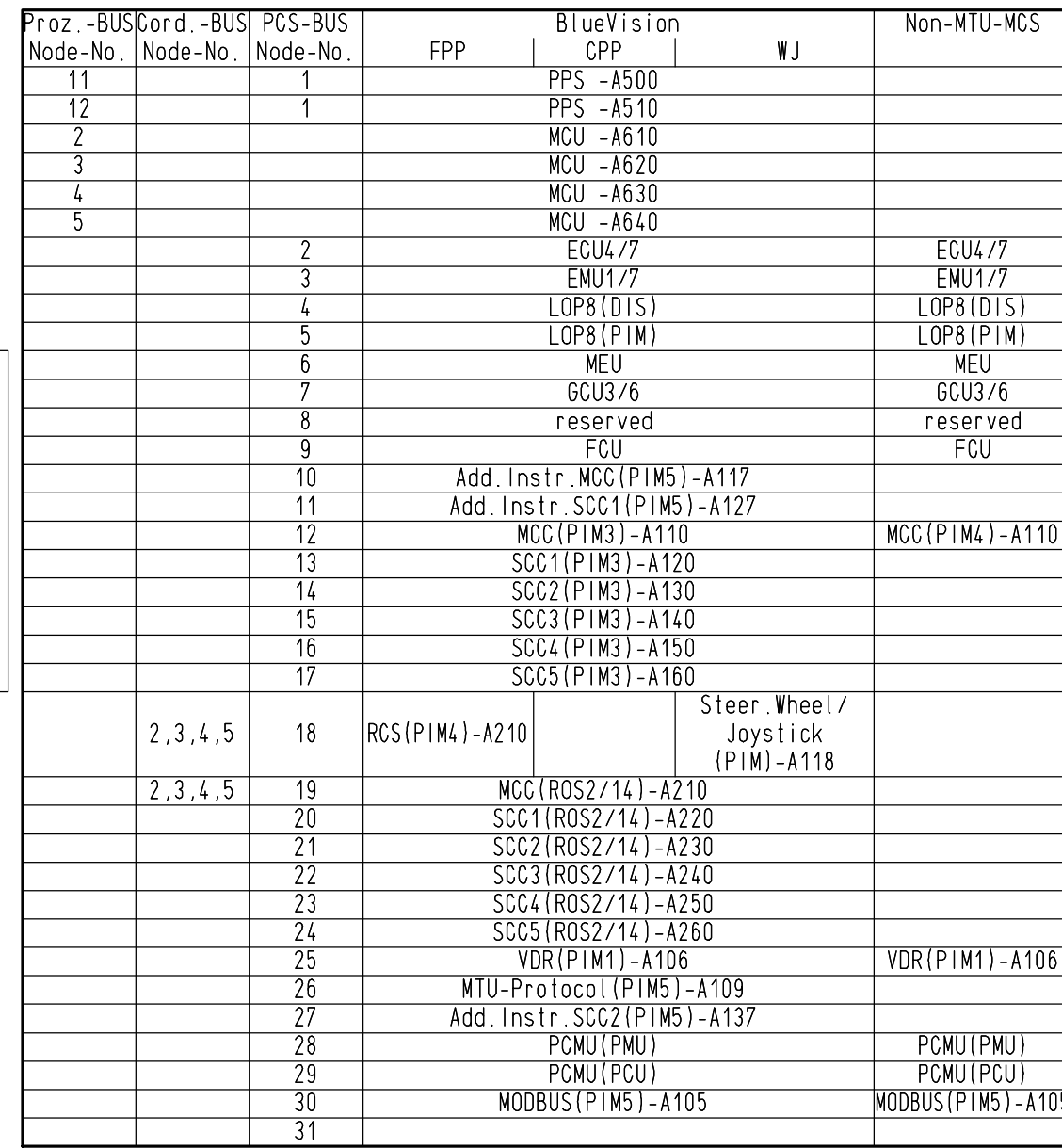


MTU Friedrichshafen GmbH

|                                                                                                  |                                                          |                               |                |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------|----------------|
| Benennung/Title<br>SCHEMA VERKABELUNG<br>ANLAGENAUTOMATION<br>WIRING DIAGRAM<br>PLANT AUTOMATION | PRINZIPVERKABELUNG MOTOR 1/2: ZUSÄTZLICHE SCHNITTSTELLEN |                               |                |
|                                                                                                  | PRINCIPLE WIRING ENGINE 1/2: ADDITIONAL INTERFACES       |                               |                |
|                                                                                                  | Auftrag/<br>Projekt                                      | Zeichnungs-Nr.<br>Drawing No. | Blatt<br>Sheet |
|                                                                                                  | BLUE-VIS                                                 | X00031757                     | 191            |
|                                                                                                  | Referenz-Nr. /Reference No.                              |                               |                |



BEISPIEL FÜR  
GRUNDUMFANG  
EXAMPLE FOR  
BASIC SCOPE



© MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !



|                            |          |                               |
|----------------------------|----------|-------------------------------|
| BUSTOPOLOGIE, MOTORRAUM    |          |                               |
| BUS TOPOLOGY, ENGINE ROOM  |          |                               |
| Auftrag/<br>Projek         | BLUE-VIS | Zeichnungs-Nr.<br>Drawing No. |
|                            |          | X00031757                     |
| Referenz-Nr./Reference-No. |          | Blatt<br>Sheet                |
|                            | -----    | 192                           |

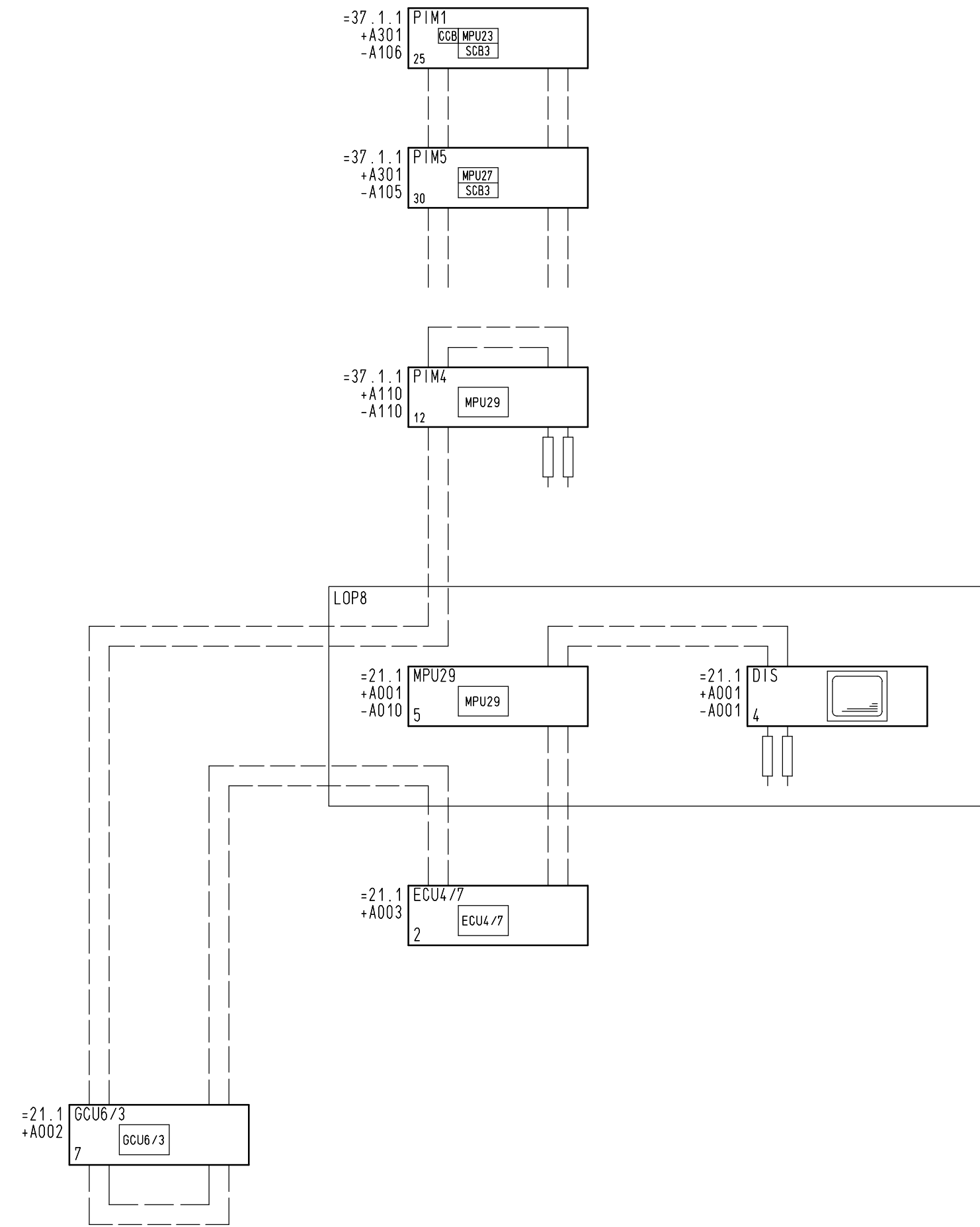








BEISPIEL FÜR  
GRUNDUMFANG  
EXAMPLE FOR  
BASIC SCOPE



MTU Friedrichshafen GmbH  
Schutzvermerk DIN 34 beachten !  
Refer to protection notice DIN 34 !

|         |                      |   |   |              |            |        |                                                                                                                |   |   |                            |    |                                  |                               |           |                |     |
|---------|----------------------|---|---|--------------|------------|--------|----------------------------------------------------------------------------------------------------------------|---|---|----------------------------|----|----------------------------------|-------------------------------|-----------|----------------|-----|
|         |                      |   |   | CAD          | Datum      | Name   |  MTU Friedrichshafen GmbH |   |   | Benennung/Title            |    | BUSTOPOLOGIE, FREMD MCS (PIM4)   |                               |           |                |     |
|         |                      |   |   | Bearb.       | 26.10.2018 | SAUTER |                                                                                                                |   |   | SCHEMA VERKABELUNG         |    | BUS TOPOLOGY, NON-MTU MCS (PIM4) |                               |           |                |     |
|         |                      |   |   | Gepr.        | 26.10.2018 | BOLL   |                                                                                                                |   |   | ANLAGENAUTOMATION          |    |                                  |                               |           |                |     |
|         |                      |   |   | Norm.        | 26.10.2018 | SAUTER |                                                                                                                |   |   | WIRING DIAGRAM             |    |                                  |                               |           |                |     |
| i       | Verkabelung geändert |   |   | Freigegeb    | 05.11.2018 | Boll   |  MTU Friedrichshafen GmbH |   |   | Auftrag/Projekt            |    | BLUE-VIS                         | Zeichnungs-Nr.<br>Drawing No. | X00031757 | Blatt<br>Sheet | 196 |
| Buchst. | Änderung / Revision  |   |   | Auftrags-Nr. | Datum/Date | Name   | Abi. TDE                                                                                                       |   |   | Referenz-Nr./Reference-No. |    |                                  |                               |           |                |     |
| RIPL AN | 1                    | 2 | 3 | 4            | 5          | 6      | 7                                                                                                              | 8 | 9 | 10                         | 11 | DIN A2                           |                               |           |                |     |